CHAPTER 1

Engineering Application Module 1 – Civil Structures

Historical and Societal Influences

Explanatory Note: The term civil structures relates to all engineered structures in our world. Due to the limited scope of this book it is impossible to look at the development of all civil structures, so bridges are the only ones considered.

Historical Developments in Civil Structures

The bridge is an important engineered structure. Transport would not have developed as easily without the development of the bridge. In the following section, bridges are grouped according to their design, and their development traced. There is an immense variety of bridges around the world and there is limited scope in this book to cover them, therefore only an overview of bridge development has been given.

Beam Bridges

The word beam comes from the Old English word for wood. A beam is simply a member that is supported in such a way that the supports do not carry any longitudinal forces. Beams may be loosely grouped into two types; if the beam is supported at both ends then it is a *simply supported beam*, whereas if it is only supported at one end it is a *cantilevered beam.* A beam may be wood, metal or stone. A beam does not have to be solid; it may be a tube, and a truss is a type of beam. Thus a variety of truss bridges fall under the category of beam bridges.

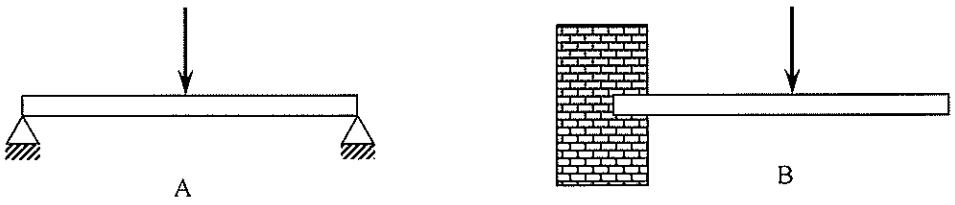


Fig. 1.1: *Two beams are shown above.* A *represents a simply supported beam with a single point load, while* B *represents a cantilever anchored to a wall with a single point load.*

The Greek historian Herodotus wrote of a multi-span beam bridge in Babylon (present day Iraq) in the 5th century BC. Each span was only 1.52 m long but a total length of 200 m was attained by having 100 stone piers, each linked by a separate beam.

Early beam bridges often used timber, but because timber rots and is attacked by borers and termites there are few left.

In 55 BC Julius Caesar built a 550 m long wooden beam bridge that incorporated 50 spans; construction took ten days.

By 1570, Italian architect, Andrea Palladio had developed a truss girder bridge. The idea had come from timber roof trusses used in Gothic churches. One of his timber trusses was used to span the River Cismone and it was approximately 30 m long.

Early in the 19th century, English engineer James Warren developed a truss girder bridge that would be extensively utilised by railway engineers. One example is the Crumlin Viaduct (1857) in South Wales. This carries the Taff Vale railway 67m above the River Ebbw over a span of 45.7 m.

It was in America, however, that the truss girder was developed to its modern day form. Engineer Ithiel Town developed a truss girder bridge with criss-cross diagonals in 1820. This was suitable for light road traffic but not sturdy enough for railways, which required the development of a truss, built up of triangles (the strongest geometric shape). Wooden trusses designed by William Howe (1840) and Caleb and Thomas Pratt (1844) fulfilled this requirement.

In 1847, Squire Whipple developed the iron truss, a design that carried on to the twentieth century. More importantly, he developed a way to determine the force in each member of the truss. Through mathematics, it was now possible to determine how to build safer and more cost effective bridges.

In 1849, the famous English engineer, Robert Stephenson, developed the Bowstring Girder. This consists of an arch (the bow) and a horizontal tie (the string) that constrains the bow from spreading. It is a beam and is used with a support at either end. This bridge, in Fig. 1.2, is like two Sydney Harbour Bridges, but unlike our famous landmark the bowstring girder, like all beams, does not place any lateral thrust on the supports. The Sydney Harbour Bridge, being an arch bridge, does. Stephenson designed the bowstring girder for his High-Level Bridge to cross the River Tyne at Newcastle. There are six spans, each 38 m long, with the roadway running under the railway.

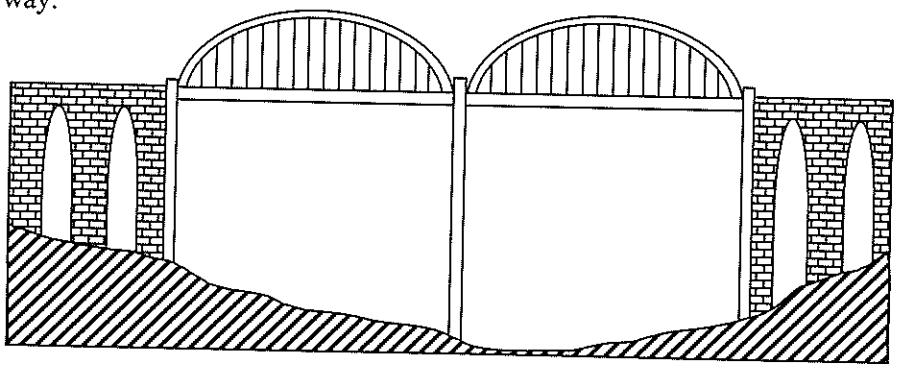


Fig. 1.2: *A bowstring girder (or tied arch). It may look like an arch bridge but the abutments carry no lateral thrust, only vertical forces.*

In 1850 Stephenson collaborated with others to complete another important type of beam bridge. The Admiralty had requested a high bridge that would not impede shipping. Suspension bridges had too much flex for railways, so Stephenson, Eaton Hodgkinson and Sir William Fairburn developed the Brittania Bridge. It carried the railway over the Menai Strait in North Wales, and consisted of rectangular wrought iron tubes, or tunnels, so that trains could run through them. The bridges consisted of two end spans each 70 m and two central spans each 140 m. The tubes were joined as they passed through the pylons, creating two tunnels, one for each track. One of the

greatest problems with this structure was that the wrought iron plates that made up the tubes tended to buckle. Many models were made and tested until the cellular box beam was developed to alleviate this problem.

In 1867, Heinrich Gerber built the first balanced cantilever bridge over the River Main in Germany. The balanced cantilever consists of two side spans that extend past a central pylon to form cantilevers.

The first large-scale balanced cantilever bridge for rail was the Forth Bridge in Scotland. It has two main spans that are 521 m long (at the time the longest in the world) and overall the bridge is around 3.2 km long. It was completed in 1890 after a construction time of eight years. It consists of over 51,000 tonnes of steel. The main spans are constructed as a simply supported steel truss 106 m long, connecting the cantilevers.

In 1879 the Tay Bridge (on the same rail line as the Forth Bridge) was built. This consisted of 85 lattice girder spans creating a bridge over 3 km in length. The girders were supported by slender cast iron columns. The 13 central spans were termed 'high girders' because they were mounted above the railway line to provide clearance to shipping. In the last week of 1879 a train and the high girders were blown off into the river below. Investigations revealed that the designer, Sir Thomas Bouch, had not allowed enough tolerance for wind and that the columns and had been filled with substitutes, such as wax, to save on recasting. This incident is a good example of the responsibility that engineers have.

1948 saw the development of one of the most commonly used beam bridges today, the box girder bridge. This is a tube with a rectangular cross section stiffened by a series of internal walls, creating a box. Bridges may use a box girder on each edge of the roadway, or there may be a central girder with the roadway forming cantilevers on each side. Originally made of steel, box girders are now also made from reinforced or pre-stressed concrete.

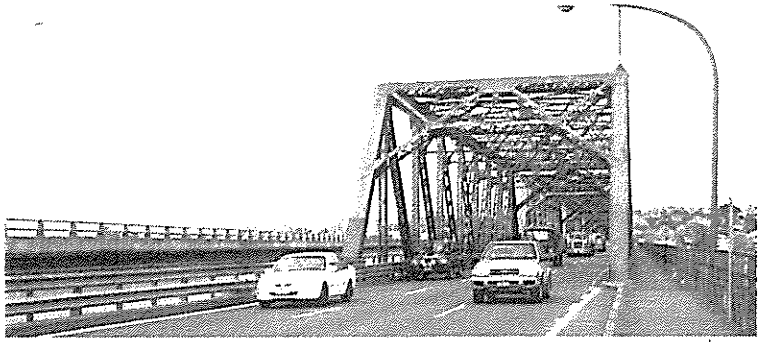


Fig. 1.3: *Cars exiting the northern end of Tom Ugly's Bridge over the Georges River (NSW). It utilises a truss style beam while the new bridge alongside uses box girder construction.*

After WWII most of the medium bridges (150-300 m span) in Germany and Austria had been destroyed. The steel box girder was the solution to this vast rebuilding project. Most road and freeway road bridges seen today utilise the box girder design.

Arch Bridges

The use of an arch for bridge construction dates back thousands of years. Nowadays the arch bridge may come in a variety of forms.

Masonry Arch Bridges

The Stone Arch bridge was extensively used by Roman engineers from 200 BC to 400 AD. The Romans built bridges that were to last for many years, and many still exist today.

The oldest Roman bridge is the Ponte Rotto in Rome. Built in 179 BC, it has since been rebuilt and altered. Another old Roman bridge is Ponte de Augusto at Rimini over the River Marecchia, built in 20 AD. The piers are squat. As the bridge was built, a pier would have an arch on one side only, so to stop the pier falling over it was enlarged in cross section. This problem was solved by building an arch on either side of the pier at the same time. In the building of aqueducts, the Romans learned to use lighter piers using this method

The biggest Roman bridge span was 42 m on a four span viaduct in Narni, on the Via Flaminia. They dammed the river to make construction easier.

After the fall of the Roman Empire in 400 AD, Europe entered the Dark Ages, a period of little learning. Much information on Roman technology and engineering was lost. Medieval bridges were not up to the standard of Roman bridges. London Bridge was typical of these. Built to replace a small timber bridge over the River Thames, it was completed in 1209 after 30 years of work. Its 276 m length consisted of 20 small spans of differing shapes and sizes ranging from 7.5 to 10.4 m in length. The bridge actually had houses built on it and it blocked three-quarters of the width of the River Thames. So great was the blockage that at half-tide there was a difference of 1.5 m in the water level on opposite sides of the bridge, causing a problem for navigation.

The Renaissance followed the Middle Ages. Learning and technological development began again. Bridge development started in earnest. Medieval bridges used pointed arches, which resulted in a lot of material between the arch and the roadway but Renaissance designers developed lower arches to alleviate this. A fine example of a low-rise Renaissance bridge is the Rialto Bridge in Venice, completed in 1591 after three years construction time.

Renaissance engineers further developed the low arch bridge to incorporate narrow piers and an arch with a small rise. Jean Perronet designed the Pont de la Concorde, completed in 1791 and still standing today. It consists of five, 31 m arches which only rise 4 m. Low arches place large loads on the piers but this is offset by the adjacent pier pushing in the opposite direction, but the load on the abutments is very high. This elegant bridge, however, with its narrow piers has a roadway level with the bank and only obscures 35% of the river.

If an arch bridge is to be built over a low bank river on flat ground, the challenge is to try to attain a very low rise. Jean Perronet achieved this, as did the English engineer Isambard Kingdom (I.K.) Brunel. He designed a brick arch bridge at Maidenhead to carry the Great Western Railway over the River Thames in 1837. Because of the low banks, the bridge was built with two 39 m spans with a rise of only 7.3 m. The public and pundits alike thought the bridge would collapse, and as a joke Brunel left the wooden supports in place to make people think the bridge needed them. After a year the supports were destroyed in a storm and, of course, the bridge remained in place.

Only then did Brunel explain that the supports had been clear of the bridge for most of the time. The bridge still stands today, carrying far heavier loads than Brunel ever envisaged.

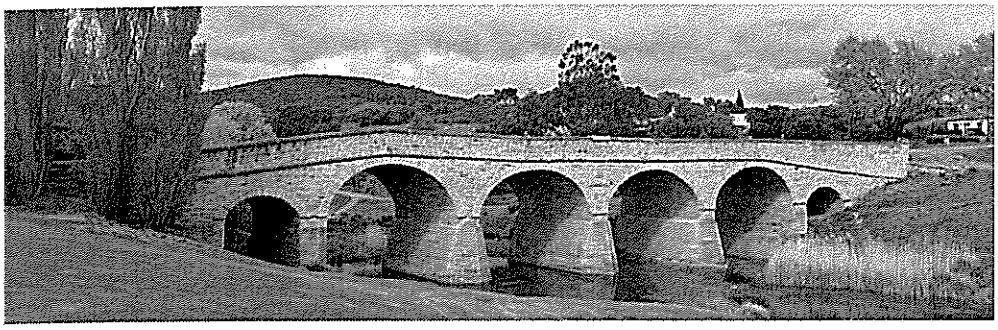


Fig 1.4: *The Richmond Bridge in Tasmania is the oldest bridge in Australia; it is a sandstone arch bridge. (http://en.wikipedia.org/wiki/File:Richmond_Bridge_Panorama_Restitch.jpg)*

Cast Iron Arch Bridges

Although some societies had entered the Iron Age about 1500 BC, European civilisations could not yet melt iron and cast it. The Chinese could in 600 BC, but Europe didn't achieve this feat until the 14th century. Cast iron was not widely used until the Industrial Revolution, which commenced around 1750.

Cast iron has similar properties to stone. It is strong in compression and, although weak in tension, it can be cast as an open frame thus reducing weight over a similar size stone block. It is cheaper to cast iron than to carve stone.

The first cast iron bridge was the Coalbrookdale Bridge built over the River Severn at Shropshire in 1779. It consists of five cast iron ribs, forming a 30 m semi-circular arch. Interestingly, it used joining methods now more often associated with woodworking. Joints were constructed using tenons and dovetails, as new technology for joining cast iron was not yet developed. Steel was not used, as it was not yet cheaply made.

Thomas Telford built a cast iron bridge over the same river at Buildwas in 1796. This bridge, which had a span of 40 m, used a segmental arch and used half the iron of the Coalbrookdale Bridge.

Steel Arch Bridges

The first noteworthy steel arch bridge was the St Louis Bridge, which crosses the Mississippi River. Designed by James Eads, it was completed in 1874. Steel was now a viable bridge material thanks to Henry Bessemer's converter developed in 1856, which produced relatively cheap steel in large quantities. Ead's bridge consists of three arches, each with a span of 152m, with a railway running over the arches and a roadway running over the railway. Eads devised special caissons or devices used to sink piers to the river bed without damming the whole river, to reach the bedrock 40m below the river level.

The Hells Gate Bridge, completed in 1917, crosses East river at New York. It has a suspended roadway, which means that the roadway does not travel over the arch but is actually suspended under it. Suspended roadway arch bridges look similar to

bowstring girder bridges but differ because an arch bridge pushes horizontally as well as vertically on the supports, whereas the bowstring girder supports only carry vertical loads.

The Hells Gate Bridge is a railway bridge carrying four railway lines over a span of 298m connecting the New York, New Haven, Harrford and Pennsylvania Railroads.

The most famous Australian bridge, the Sydney Harbour Bridge, was commenced in 1924 and completed in 1932. This bridge finally linked the North Shore with Sydney City. The span of the bridge is 503 m, making it longer than Hells Gate Bridge. It is also wider, carrying 8 traffic lanes, two railway lines and 2 footpaths, with a total deck width of 49 m. It was built from either side of the harbour while anchored to the shore, finally meeting in the middle at a pin joint. It is long enough to cross the Mississippi at St Louis in one single span, as opposed to three spans for the St Louis Bridge.

The Sydney Harbour Bridge is not the largest span steel arch bridge. That honour goes to the Bayonne Bridge in New York, deliberately built 600 mm longer, to out do the Sydney bridge. It, however, only carries traffic lanes.

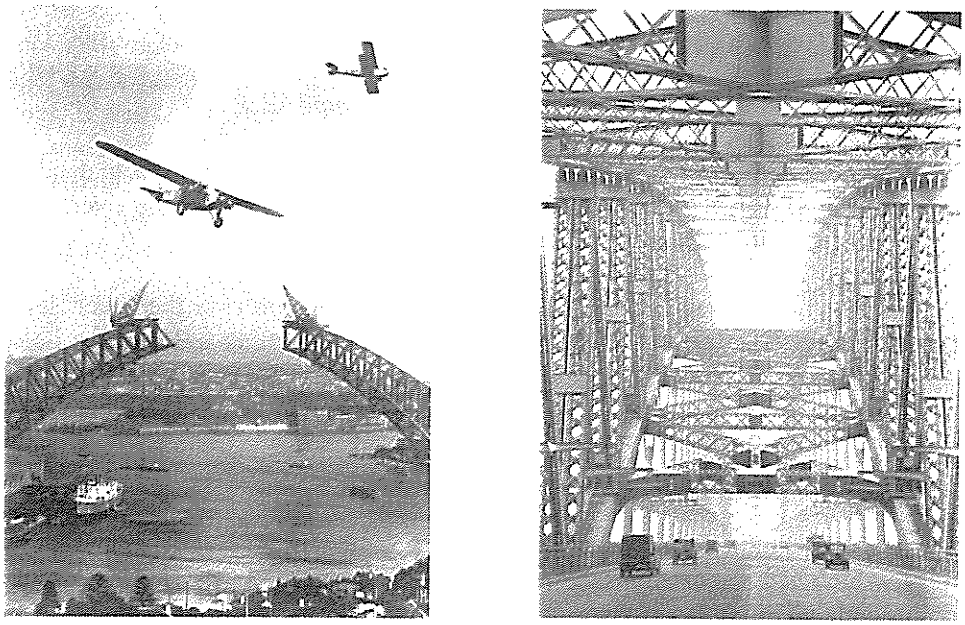


Fig 1.5: *The Sydney Harbour Bridge; left, during construction and right, the day after the 1932 opening. (Photographs reproduced with permission, State Library of NSW).*

Concrete Arch Bridges

Concrete is now the most commonly used material in bridge design. It is very strong in compression so it is ideal for an arch. With steel reinforcing, it is suitable for beam or cantilever use. Concrete, however, is subject to shrinkage. As a concrete bridge sets (this can take a whole year) the bridge shrinks and without some compensatory feature to deal with this it may crack and fail. The solution is a third hinge. Most steel arches are continuous throughout with a hinge at each end. Robert Maillart, a Swiss engineer, was the first to realise the possibility of the third hinge in the centre of the bridge to allow for shrinkage. In 1901 his first three-hinged, reinforced concrete bridge was built over the River Inn in Switzerland.

In 1904, Eugene Freyssinet came up with the idea of pre-stressed concrete but could not manufacture it until high tensile steel wire was developed after World War II. Then he designed five bridges over the River Marne in France at Esbly, Annet, Tribardos, Changis and Ussey. They are flat arches with a span of 74 m.

To achieve the required strength, pre-stressed, concrete ducts were run through the lower arches. These had wires passed through them which were pulled tight after the arch was constructed. The compression greatly enhanced the strength of the bridge and reduced the probability of tensile failure.

The Gladesville Bridge, built in 1966 in Sydney, is a good example of a concrete arch bridge.

Suspension Bridges

Suspension bridges support the deck (roadway, railway) on steel cables strung between support towers. The cables are in tension. Since tension members do not buckle they can be thin and thus light, offering lightweight and longer spans than other types of bridges.

In the Stone Age, before 4000 BC, suspension bridges were simply built by having tension members, vines or creepers, slung across a river or ravine with wooden slats connecting the two to form a walkway. Many movies have featured adventures on these primitive suspension bridges over steep ravines.

The oldest suspension bridges are in China. One in the Yunnan Province, built in 1470, with a span of 69 metres, uses wrought iron chains. In Europe wrought iron was not available in commercial quantities until the 19th century, when the modern suspension bridge started to develop.

Early suspension bridge designs suffered from a lack of lateral stability. The remedy for this was to use a stiffened deck hanging from the suspension cables with wire ropes. To produce a horizontal deck the cables ran over two towers and were anchored at their ends. The Austrian Bishop, Fausto Veranzio proposed this design in 1595, but it was James Finlay, of Pennsylvania, who in 1801 built such a bridge. It was only 21.3 m long.

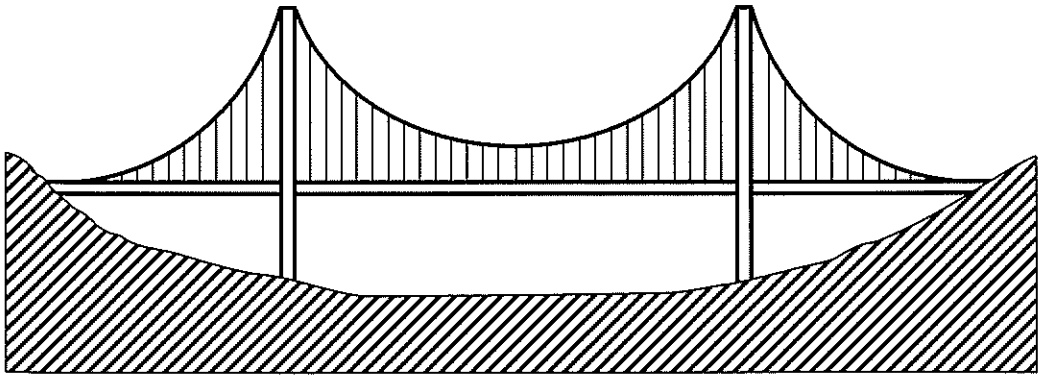


Fig 1.6: *The basic form of the suspension bridge devised by James Finlay.*

Thomas Telford produced the first successful large-scale suspension bridge; the Menai Strait Bridge, completed in 1826 in Wales. The towers were 47 m high, and the 16 chains that comprised the suspension members were made of wrought iron. These chains consisted of 3 m bars linked by 75mm diameter pins. The chains were dipped in linseed oil to preserve them. The 9 m roadway was timber, 176.5 m long and hung from the chains by iron rods 1.5 m apart. In 1896 the roadway was replaced by metal and wrought iron chains were replaced by steel in 1936.

In 1829 a committee called for submissions for a bridge across the Avon Gorge at Clifton. Brunel submitted four designs. His favoured design was a suspension bridge, using the cliffs as anchors. The judge, Thomas Telford, rejected all of the designs, so the committee asked Telford to submit a design. His submission was so poor that the committee rejected his design and called for a further round of submissions. Brunel's design for a 214 m suspension bridge with Egyptian style piers and an abutment on one side was successful. The design set Brunel's career on a prosperous path, even though the bridge itself was not built until 1864, eight years after Brunel's death.

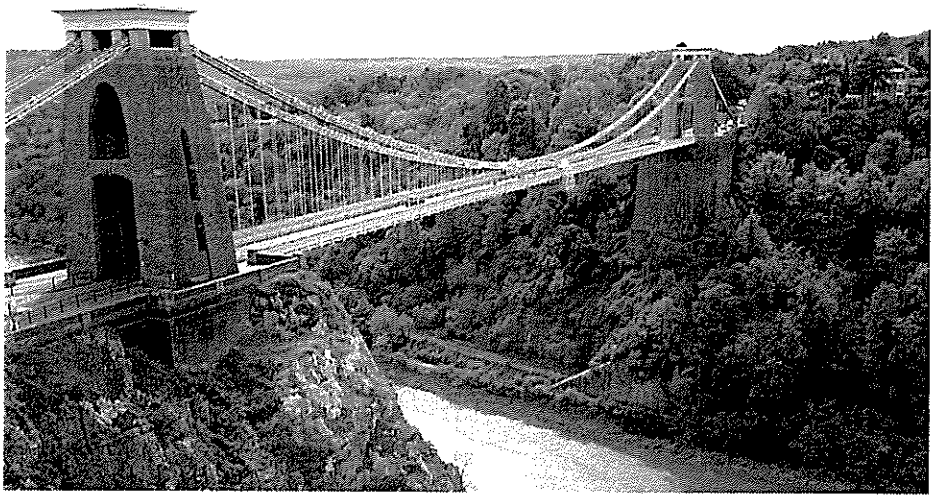


Fig. 1.7: *Brunel's Clifton suspension bridge; the "chains" are made with wrought iron plates.*

John Roebling was primarily responsible for the introduction of wire cables for suspension bridges instead of chains. He designed suspension bridges over the Niagara Gorge in 1855 and over the Ohio River at Cincinatti in 1867 but his true masterpiece was the Brooklyn Bridge. The Brooklyn Bridge has a span of 486 m and was the first to use steel wire cables. Work on the bridge began in 1869 and shortly afterwards Roebling died of tetanus when his foot was crushed by a ferry. His son, Washington Roebling took over. By 1876 the 82.5 m masonry towers were complete and they set about producing the four 400 mm diameter cables, which housed 1932 km of wire. With no rock to anchor the cables, large masonry gravity anchorages were built at each end of the bridge. These had a mass of 44,000 tonnes. The bridge was opened in 1883.

After the completion of the Brooklyn Bridge, American engineers built suspension bridges, each of which was longer and narrower than the last. In 1940 the Tacoma Narrows Bridge over Puget Sound in Washington State went just too far. At the time it was the third longest suspension bridge in the world, with a span of 853 m yet it was only 11.8 m wide and was stiffened by 2.4 m plates along each side. It was designed

to withstand winds of up to 200 km/h. On November 7th, four months after it opened, a wind of 67 km/h occurred. The deck began to oscillate in vertical waves of 9 m height. After three hours, the waves stopped and the deck started to twist through 90 degrees until several suspension cables snapped and 300m of the centre span fell into the river. Following this the side spans started to "dance" until they too, fell into the river.

After this monumental failure, the American engineers returned to their previous practice of building heavier and more rigid suspension bridges. This rigidity was achieved by adding a thick layer of concrete to the deck and placing deep stiffening trusses on the sides. This seemed the best way to deal with side loads caused by wind.

Twenty six years later British engineers came up with a better solution. The Severn suspension bridge is an important milestone in bridge design. Originally designed as a heavy, stiff bridge with a span of 988 m, designer Gilbert Roberts experimented with a fin-shaped box-section deck in a wind tunnel. This shape proved less susceptible to side loadings from wind so the deck could be made lighter without the stiffness previously required. The new deck was only 3 m deep. A 30% saving was made in the amount of steel used. The only fault was a vibration at low wind speeds. This was alleviated by alternately inclining the suspension cables.

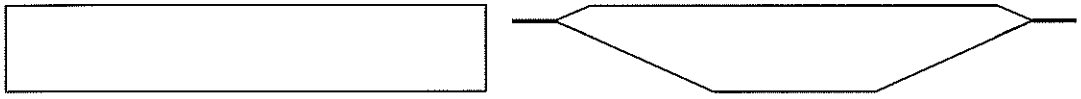


Fig 1.8: *The cross section of the Severn bridge deck on the right shows a more aerodynamic shape than the conventional rectangular box on the left.*

Another type of suspension style bridge is the *cable-stayed* bridge. This differs from true suspension bridges in that the roadway is supported by cables attached directly to the vertical columns. In Australia one of the most recognisable cable stayed bridges is the Anzac Bridge (originally called the Glebe Island Bridge). Made from reinforced concrete and steel cables, with the two tall concrete towers at each end, it dominates the Sydney skyline.

Opening Bridges

It is often inconvenient or excessively costly to build a bridge over a waterway high enough not to impede the flow of water traffic. In England during the 19th century, the Admiralty often set the height so that shipping would not be impeded. In Sydney there are three different types of opening bridges.

a) *The Pyrmont Bridge* at Darling Harbour used to be a roadway but now carries a footpath and monorail. It opens by rotating on a turntable to turn through 90 degrees and allow ships with high masts to pass through the opening. It is the world's last example of an early DC electric powered opening bridge.
b) *The Spit Bridge* in the northern suburbs of Sydney opens by raising a cantilevered opening section of the bridge through an angle of 90 degrees to allow watercraft through. Since it is on a busy arterial road route it is only allowed to open at certain times.

c) *Ryde Bridge,* in the mid western suburbs of Sydney, opens by the centre section being raised up by a gear system on a set of runners.

All of these bridges are variations on beam bridges. The Pyrmont Bridge is a type of balanced cantilever. Opening bridges tend to slow up road and waterway traffic and are only used where there is no other option.

Engineering Innovations in Civil Structures and Their Effect on People's Lives

Invention is the development of something entirely new. Innovation means making an alteration to something that has already been invented, and improving upon it.

Innovation is an important part of design and engineering, for through it engineers improve on an original concept. The suspension bridge was an invention, but the many variations that have improved it are innovations. Some innovative steps in bridge design along with their effects are listed below. It should be noted that prior to the Industrial Revolution water transport was the most important form of transport so keeping rivers open to water traffic was essential. Only with the advent of railways did this start to change

Summary.

- From 10,000 BC – primitive beam and suspension bridges – transport across rivers and ravines.
- Greeks perfect beam bridges – improve transport for human and animal drawn traffic.
- Roman engineers – arch bridges – revolutionary design – more stable and secure – improved transport – arch did not impede water traffic as much as early beams.
- The fall of the Roman Empire, 5th century – backward step – medieval arch bridges, not as efficient in design and impede water traffic, e.g. London Bridge.
- 16th century – the truss girder improves beam bridge design – longer spans – greater safety – less pylons.
- The Renaissance – improvement in arch bridge design – back to Roman standard – improved clearance for water traffic – lower arches
- James Finlay, early 19th century – modern suspension bridge – long spans – little impediment to water traffic.
- The bow-string girder, 19th century – embraced by railway engineers – cheap and simple
- 20th century – first box girder – useful design for freeway construction – freeways improve road transport especially for road freight – contributes to railway decline.
- First modern balanced cantilever bridges – limited impediment to water vehicles – effective use of materials.
- The Brooklyn Bridge – first suspension bridge to use steel cables – improved safety.
- Concrete used as a building material for arch bridges. After WWII – most used building material - alters bridge design significantly.
- The Sydney Harbour Bridge completed 1932 – cuts travel times from the north to the city – speeds development, northern Sydney

- Tacoma Narrows Bridge failure – shows danger of poor design – leads to safer suspension bridges
- Severn Suspension Bridge – streamlined cross section to deal with wind.
- Akashi Kaikyō suspension bridge, opened in 1998, becomes the longest span bridge in the world with a span of 1991 m.

Construction and Processing Materials Used In Civil Structures Over Time

Bridge development has been facilitated by the development of different materials that have proved suitable in bridge construction. The following is a list of materials used in bridges.

Timber was used in the earliest bridges, which may have been as simple as a log between two banks. Originally used in its natural state (a tree), it has been developed to having pieces of timber cut and fabricated into a truss to form a beam bridge. Timber was also used as the deck in early suspension bridges. Many old bridges in country NSW are timber. A good example is the Gundagai rail bridge. It is the longest timber truss bridge remaining in NSW. Timber offered ease of manufacture and was a readily available resource. Its main disadvantage is that it rots away.

Rope was used, by early cultures, on early suspension bridges. Just like timber, it rotted away and could only carry small loads.

Stone is a far more permanent building material than timber or rope. Many Roman arches were made from stone and still stand today. Stone is strong in compression yet weak in tension, so it is perfect for use in arch bridges. Not until cast iron and steel were developed was stone replaced for arch bridge construction. Now another ceramic, concrete, is used.

Bricks are sometimes used in bridge manufacture. Although they have similar properties to stone they are manufactured from clay and not chiselled to shape from natural stone. Brunel's low arch, Maidenhead Bridge is an example of a brick bridge. Often bricks were used in making viaducts that led to a bridge. See Fig. 1.2.

Cast iron became readily available during the Industrial revolution, and by the late 18^{th} century it was used in bridges. Like stone, it is weak in tension but strong in compression. Unlike stone it can be melted and cast, thus saving on time and also introducing the concept of pre-fabrication. Cast iron bridges could be made of frames which provided a strength equal to stone but with greatly reduced weight.

Wrought iron was used in early suspension bridges, by Thomas Telford and others, in the chains from which was suspended the deck. Wrought iron was an unreliable material due to the fibrous structure present in the ferrite, which tended to weaken it and make it difficult to tell if this affected the whole of the wrought iron. This lack of reliability in structure (compared to steel wire) was the reason that suspension bridges were limited in length. When wrought iron was replaced with steel, suspension bridges could be built much longer.

Steel was mass-produced in 1856 by Henry Bessemer and it changed bridge design. The steel arch became a viable option; the St Louis Bridge has a roadway over steel

arches while the Sydney Harbour Bridge has a suspended roadway below the arch. Steel is equally strong in both tension and compression compared to stone and cast iron; so steel arches differ in construction from masonry arches. Steel cables proved valuable in suspension bridges as greater spans could be achieved, and at the turn of the century steel was used to create reinforced concrete. Eventually, with the development of high tensile steel, it was possible to produce pre-stressed and post-stressed concrete.

Concrete was not used in bridge construction until the shrinkage problem was addressed by the third hinge, and due to its weakness in tension it was only used when reinforced with steel. Reinforced concrete has been developed into pre-stressed concrete and post-stressed concrete to improve strength and performance. Concrete is the bridge construction material of the 20th century with most new bridges using concrete of some type.

Stainless Steel is now being used in certain bridges, often pedestrian bridges. The Marina Bay Bridge in Singapore uses 2205 duplex stainless steel for its tubular spiral stainless steel members. It is 280 metres long and 6 metres wide.

Environmental Implications of Materials used in Civil Structures

Since all building materials are derived, at some stage, from the earth, there will always be some impact when the raw material is gathered. The following information describes the environmental impact of the use of some bridge construction materials.

Timber's use as a source of heat for steam power, and as a building material, resulted in deforestation of the surrounding area. As that area ran out of trees they were sought elsewhere. Trees take centuries to grow and old growth forests have never recovered from the onslaught.

Stone does not magically appear in the shapes needed. Ancient and medieval cultures had to mine stone and they did this by digging large quarries. These scarred the landscape and had a severe impact on natural fauna and flora.

Bricks require vast amounts of clay and shale to manufacture, so large pits were dug to find these materials. Often the brick works were alongside the pit. The environmental impact of such pits was similar to that caused by stone quarries. When the pit was no longer used it was filled in and then reclaimed. Subsidence is often a problem on old brick pit sights.

Cast and wrought iron were true products of the Industrial Revolution. They were one of many products requiring the combustion of fossil fuels. Before the industrial revolution our impact on the environment was relatively small, so in the 300 years since, humans have had a greater impact than our predecessors had over 100,000 years. Cast and wrought iron required iron to be mined and smelted. This required, not just a mine, but also processing plants and transport facilities, which involved railways. All these associated industries had a huge impact on the landscape, and we are starting to feel the legacy of such impacts today.

Steel falls into a similar category to cast and wrought iron. However its impact is even greater due to its proliferation through more industries. Steel has allowed longer bridges to be built, using more steel and greater processing. Steel helped to open access to other areas and reduce transport times so it was possible to gain resources from places that previously would have been unviable. Steel is also required for locomotives and motor vehicles so the demands have increased exponentially over time.

That said, longer bridges with fewer piers have less impact on waterways, so by comparison with earlier bridges, are less likely to interrupt the delicate marine ecosystem that lives in the waterway.

Concrete, in its various forms, is a popular material in bridge building. Like other bridge construction materials, it requires vast amounts of minerals to manufacture it, and obtaining these has an adverse environmental impact. Also, some would argue that it is visually unappealing. Like steel, however, it has resulted in bridges, which have a reduced impact on waterways, compared with earlier bridges.

Engineering Mechanics

Truss Analysis

Since the development of the truss it has become one of the most important engineered structures. Trusses are based on the use of the strongest geometric shape, the triangle. Unlike polygons with more sides, a triangle cannot pivot at its joints to deform so it is a very rigid structure when made of timber, steel, etc. Trusses take advantage of this by taking any shape and triangulating it to maximise strength.

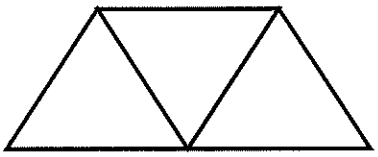


Fig 1.9: *A simple Warren truss is made up of equilateral triangles.*

A truss is essentially a form of simply supported beam and usually is supported at either end on supports which provide reactions to the applied loads acting on the truss. The types of supports are shown in Table 1.1.

Reactions at the Supports

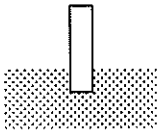
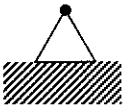
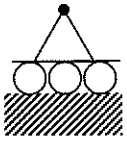
Solving any problem involving a truss usually involves finding the reactions at the supports of the truss. When doing this it is helpful to follow this sequence,

1. When the diagram given is overly complex, draw a free body diagram showing all loads and dimensions.
2. Determine the horizontal and vertical components of any angled forces.

3. Take moments about the non-roller support first (as you always know the direction of the roller supports reaction) to determine the reaction at the roller support.
4. Once this reaction is found then use ΣF_V and ΣF_H to find the vertical and horizontal reactions at the pin support.
5. Find the total reaction at the pin support.

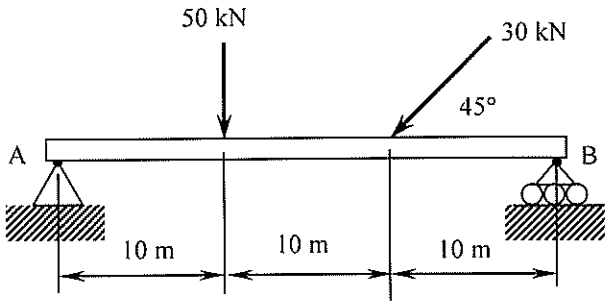
Note: If all loads are vertical or horizontal then the procedure is greatly simplified.

Table 1.1: *Types of supports for beams and trusses.*

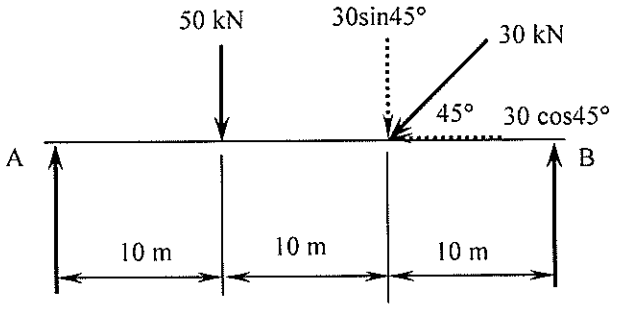
Support	**Image**	**Free Body Diagram**	**Attributes**	**Reason for Use**
Fixed			Can support vertical and horizontal loads and moments.	To firmly anchor a structure
Pin or hinge			Can support vertical and horizontal loads. Free to rotate	To provide a non-moving support for a bridge.
Roller			Can only provide reactions normal to the surface. Free to rotate	To allow side movement for expansion in a truss.

Example 1.1

Find the reactions at the supports for the beam shown below.



We will first take moments about point A as it is a fixed support and we must break up the inclined force into vertical and horizontal components.



$$\circlearrowleft^{+} \Sigma M_A = 0$$

$$0 = 50\times10 + 30\sin 45^\circ \times 20 - 30R_B$$

$$30R_B = 500 + 424.26$$

$$R_B = \frac{924.26}{30}$$

$$R_B = 30.8\ \text{kN} \uparrow$$

Now to find the reaction at A we must add the vertical and horizontal forces. This will find us the components of R_A that we must, in a vectorial manner, add together to find R_A.

$$+\uparrow \Sigma F_V = 0$$

$$0 = 30.81 - 50 - 30\sin 45^\circ + R_{AV}$$

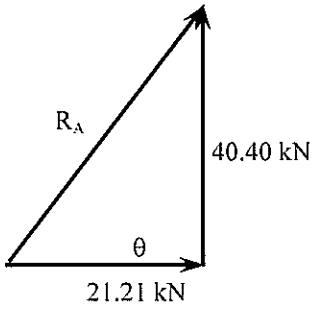
$$R_{AV} = 40.40\ \text{kN} \uparrow$$

$$\rightarrow^{+}$$

$$\Sigma F_H = 0$$

$$0 = -30\cos 45^\circ + R_{AH}$$

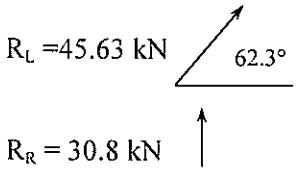
$$R_{AH} = 21.21\ \text{kN} \rightarrow$$



$$R_L = \sqrt{40.40^2 + 21.21^2} \qquad \tan\theta = \frac{40.40}{21.21}$$

$$R_L = 45.63\ \text{kN} \qquad \theta = 62.3^\circ$$

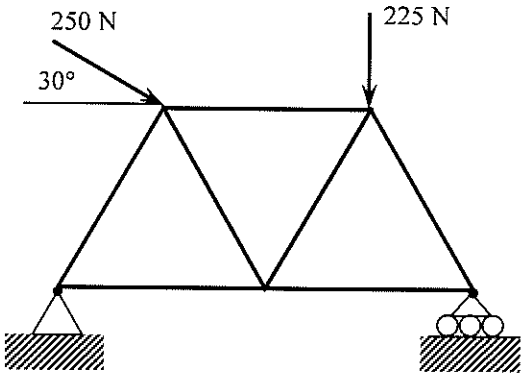
So we can now state that the reactions at the supports are:



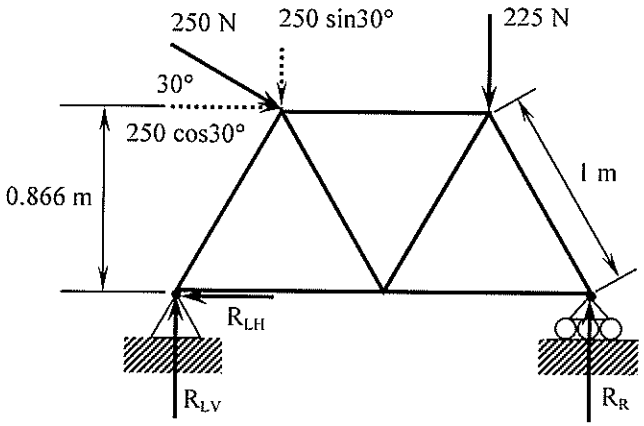
Note that the directions must be given, because forces are vector quantities.

Example 1.2

A Warren truss shown below is subjected to two loads. Find the reactions at the supports.



We must now break up the angled force into vertical and horizontal components, and also draw in our reaction forces. Since a Warren truss is made up of equilateral triangles, all sides are equal. As such we do not need actual dimensions and we can consider the sides to be 1 unit long. We can then find the height of the horizontal component of the truss.



$$\circlearrowleft^{+} \Sigma M = 0$$

$$0 = 250\cos 30^\circ \times 0.866 + 250\sin 30^\circ \times 0.5 + 225 \times 1.5 - 2R_R$$

$$2R_R = 187.49 + 62.5 + 337.5$$

$$R_R = \frac{587.49}{2}$$

$$R_R = 293.75\ \text{N} \uparrow$$

Now that we have determined R_R we can find R_{LV} and R_{LH}.

$$+\uparrow \Sigma F_V = 0$$

$$0 = 293.75 - 250\sin 30^\circ - 225 + R_{LV}$$

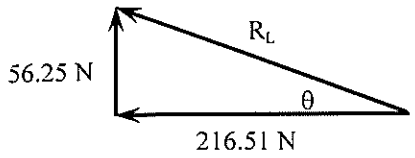
$$R_{LV} = 56.25\ \text{N} \uparrow$$

$$\xrightarrow{+} \Sigma F_H = 0$$

$$0 = 250\cos 30^\circ - R_{LH}$$

$$R_{LH} = 216.51\ \text{N} \leftarrow$$

Now we can determine the total reaction R_L by vectorially adding the components:



$$R_L = \sqrt{56.25^2 + 216.51^2}$$
$$R_L = 223.70 \text{ N}$$
$$\tan\theta = \frac{56.25}{216.51}$$
$$\theta = 14.56^\circ$$

The reactions at the supports are:

$R_L = 223.7$ N 14.56°

$R_R = 293.75$ N ↑

Analysing the Truss Members

Although a truss is considered as a beam to determine the supports, the frame that is the truss also contains forces. The framework allows the truss to span greater distances than a simple beam, and it also has better capacity to support greater loads. There are two methods available for finding the reactions at the supports of a truss, the Method of Joints and the Method of Sections. In both methods we must state the direction of the force in the truss members, by classing it either as tension or compression. Tension forces will stretch the members, while compressive force will tend to shorten the members.

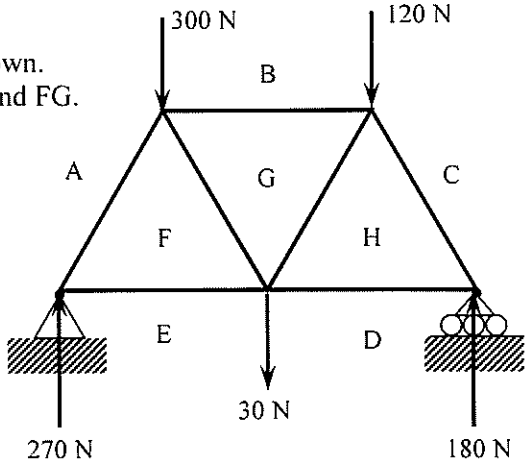
For the sake of simplicity, let us assume that each joint of the truss is pinned, and can rotate in relation to another. By doing this, we can ignore any moment at the joint. In actuality truss joints are usually welded, riveted or bolted in such a way that rotation is prevented, so the joints actually have resistance to rotation which complicates the truss analysis, which is why we simplify it for calculations.

The Method of Joints

The Method of Joints is a process used to find the forces in truss members by analysing each joint as a concurrent force system. One starts at one end and works through to the other. Example 1.6 outlines the process of Method of Joints.

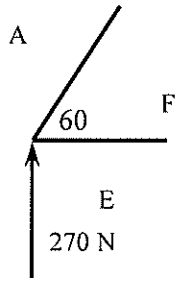
Example 1.3

A truss is loaded and produces reactions as shown. Determine the force in members AF, EF, BG and FG.



The truss has already been labelled using Bows notation, which involves labelling the spaces between forces and members and naming the force according the letter on either side. This is not always done; often the joints themselves are labelled.

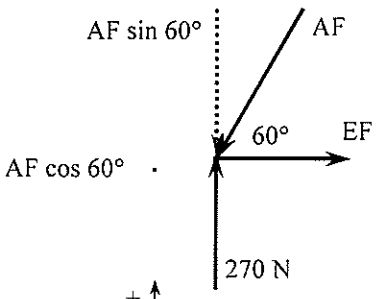
To find the force in members AF and EF, look at joint AEF.



It can be seen that we have a three-force system that must be in equilibrium or the system will fail. It follows that if a total mechanical system is in equilibrium, as our truss is, then all of its components are also in equilibrium. Procedure 1 is shown to the left, while Procedure 2 is shown on the right.

• Procedure 1

This method involves breaking the AF into horizontal components and then finding the sum of the vertical and horizontal forces, remembering ΣF_V and ΣF_H is equal to zero.



$+\uparrow \quad \Sigma F_V = 0$

$0 = 270 - AF \sin 60^\circ$

$AF = \dfrac{270}{\sin 60^\circ}$

$AF = 311.77 \text{ N (compression)}$

$\longrightarrow +$

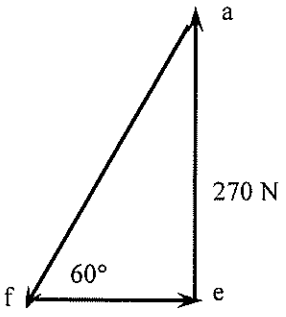
$\Sigma F_H = 0$

$0 = EF - 311.77 \cos 60^\circ$

$EF = 155.88 \text{ N (tension)}$

• Procedure 2

This method involves considering the three forces as they are and drawing up a force triangle to solve the values for AF and EF.



$\tan 60^\circ = \dfrac{270}{EF}$

$EF = \dfrac{270}{\tan 60^\circ}$

$EF = 155.88 \text{ N (tension)}$

$\sin 60^\circ = \dfrac{270}{AF}$

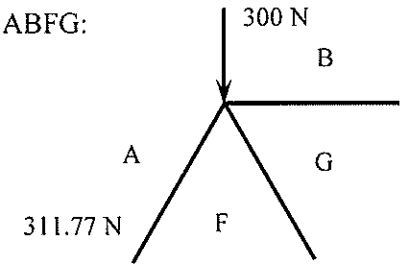
$AF = \dfrac{270}{\sin 60^\circ}$

$AF = 311.77 \text{ N (compression)}$

The same answers are derived from both methods. The tension and compression are determined by looking at the force and the way it travels. In the case of EF the force is moving to the right and away from the joint, stretching the member tension.

Procedure 2 can be done graphically, and in fact on any joint with more than three members when Procedure 2 is used it must be solved graphically since the force polygon will have more than three sides and trigonometry cannot be used. However such a joint can be solved mathematically using Procedure 1. It should be noted however that for joints with more than three forces, the graphical method is significantly quicker than the analytical method in procedure 1

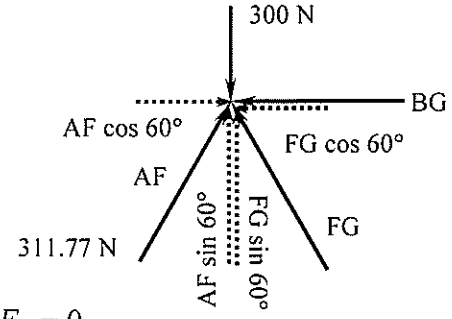
Consider joint ABFG:



Since we have found AF this joint can now be solved as we have two known members and two unknown members. It would not be possible to go directly to this joint at the start, as one would have three unknown members. Let us once again look at the two procedures.

• Procedure 1

As at the previous joint we shall break up all inclined members into horizontal and vertical components, the sum of the vertical and horizontal forces.



$+\uparrow \Sigma F_V = 0$

$0 = 311.77\sin 60^\circ - 300 + FG\sin 60^\circ$

$FG = \dfrac{30}{\sin 60^\circ}$

$FG = 34.64$ N (comp)

$\xrightarrow{+}$

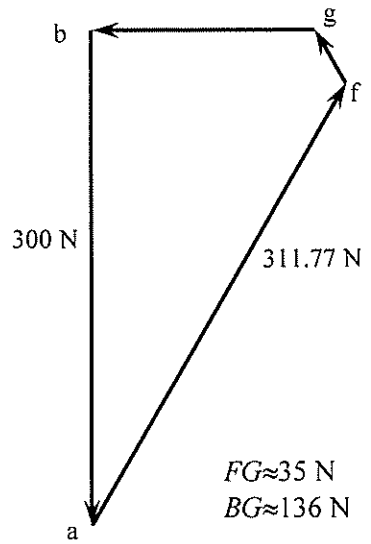
$\Sigma F_H = 0$

$0 = 311.77\cos 60^\circ - 34.64\cos 60^\circ - BG$

$BG = 138.54$ N (comp)

• Procedure 2

Since there are four forces we cannot use sin, cos or tan for determining the values so we must do it graphically. To do this we must draw a scale diagram.



$FG \approx 35$ N

$BG \approx 136$ N

FG=34.64 N (comp)
BG=136 N (comp)

The accuracy of a graphical solution depends on the scale and precision to which it is drawn. Use as large a scale as possible to improve the accuracy of your answer.

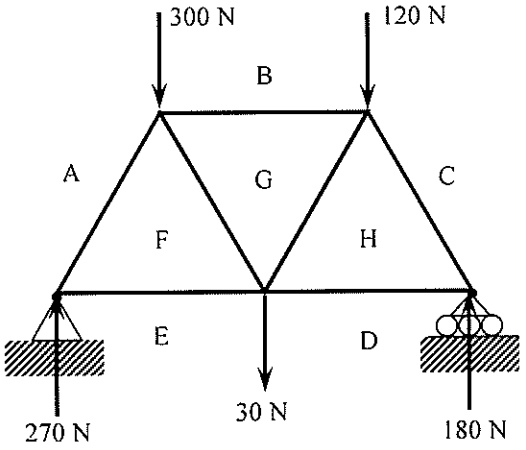
As with the previous joint we were able to determine the nature of the force due to the direction each travels. The Method of Joints solution will continue across the truss to find the force in all the members.

The Method of Sections

Often it will not be necessary to find the force in all members of a truss, perhaps only one or two. If these members are on the ends Method of Joints will be fine, but if they are in the middle of the truss, the Method of Joints is cumbersome and slow. The Method of Sections allows us to find the force in a member in the centre of a truss directly.

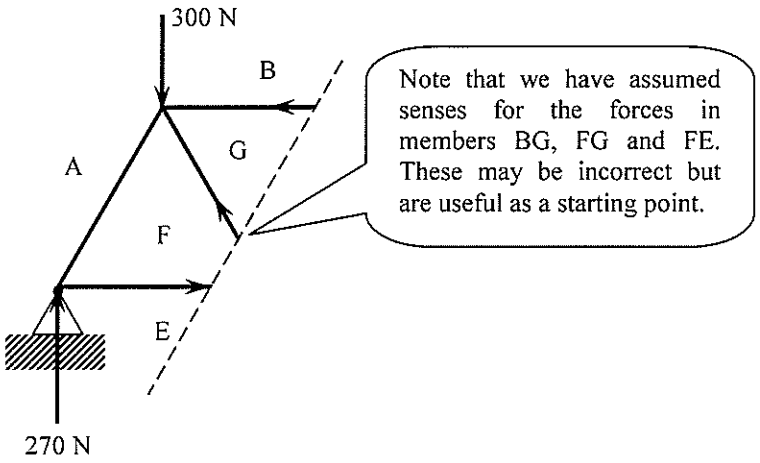
Example 1.4

A truss is loaded and produces reactions as shown. Determine the force in members BG, FG and FE. Assume each side of the truss is 1m long.



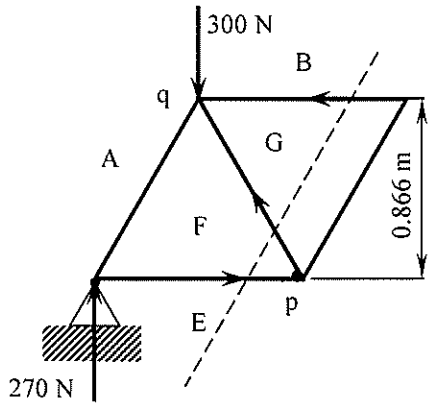
The principle of this method is based on the fact that if the whole truss is in equilibrium, then any part, or component will also be in equilibrium. If we 'cut' the truss through at least two members and 'remove' a portion, the system that remains must also be in equilibrium.

To solve this Method of Sections problem pass a section plane through the truss that cuts through the members we are trying to find. We then consider the cut members as external forces of one half of the truss.



We now consider the left hand side of the truss only. The forces FE, FG and BG are balancing the two external forces. We will use a combination of the three equations for equilibrium (i.e. $\Sigma F_H=0$, $\Sigma F_V=0$ and $\Sigma M=0$).

To find the force in BG we must take moments. There are five forces on the truss that create moments and three are unknowns, so we must take moments about a point that eliminates FG and FE, so we need to find a point that these two both pass through. Although that point is on the half of the truss we removed, we can still take moments about that point since we can take moments about any point in space. We know the position of this point and it is the point we need. We shall call it point p.



$$\circlearrowleft^{+} \Sigma M_p = 0$$

$$0 = (270 \times 1) - (300 \times 0.5) - (BG \times 0.866)$$

$$0.866BG = 270 - 150$$

$$BG = \frac{120}{0.866}$$

$$BG = 138.56 \text{ N (compression)}$$

We can see our assumed sense for BG was correct and the member is in compression, as the force tends to squash it.

Now find the force in member FG. Since it is the only unknown with a vertical component it is possible to use $\Sigma F_V=0$ to solve this problem. Note that the vertical component of FG is FGsin60°.

$$+\uparrow \Sigma F_V = 0$$

$$0 = 270 - 300 + FG\sin 60^\circ$$

$$0.866FG = 30$$

$$FG = \frac{30}{0.866}$$

$$FG = 34.64 \text{ N (compression)}$$

We can now find FE and since it is the last unknown we can either take moments about point ABFG (point q) or we can find the sum of the horizontal components of the forces. Both methods are shown below. Note that the horizontal component of FG is FGcos60°.

$$\circlearrowleft^{+} \Sigma M_q = 0$$
$$0 = (270 \times 0.5) - (FE \times 0.866)$$
$$0.866FE = 135$$
$$FE = \frac{135}{0.866}$$
$$FE = 155.88 \text{ N (tension)}$$

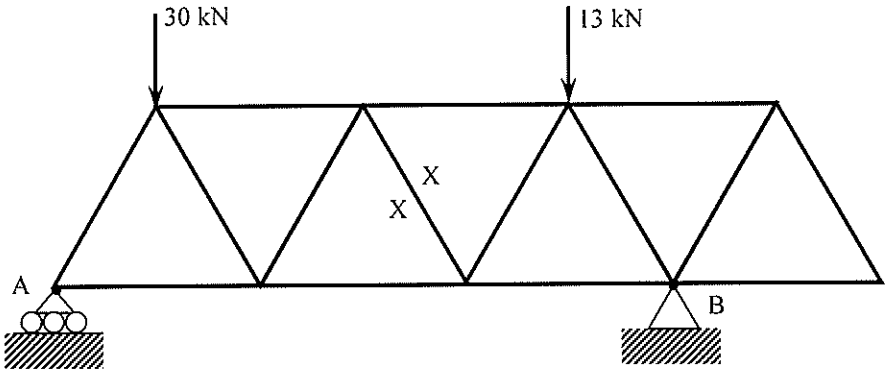
$$\longrightarrow^{+}$$
$$\Sigma F_H = 0$$
$$0 = FE - 138.56 - 34.64\cos 60^\circ$$
$$FE = 138.56 + 17.32$$
$$FE = 155.88 \text{ N (tension)}$$

It is apparent that it is simpler to merely find the sum of the horizontal components however in some problems this may be difficult.

Although the Method of Sections may not seem any simpler, on a truss like the one in Example 1.5, it will be much simpler to find member XX using Method of Sections than it would be Method of Joints. Let us see the method.

Example 1.5

Find the force in member XX of the truss below.

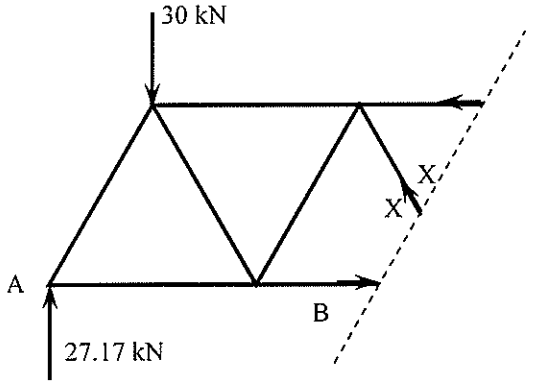


Since we have not been asked to find the reactions at the supports we will only find one, then consider that side of the truss to find XX.

Let us find the reaction at the left support (R_A) by taking moments about B. This will be a vertical force due to the roller support.

$$\circlearrowleft^{+} \Sigma M_B = 0$$
$$0 = -(13 \times 0.5) - (30 \times 2.5) + 3R_A$$
$$3R_A = 6.5 + 75$$
$$R_A = \frac{81.5}{3}$$
$$R_A = 27.17 \text{ kN}$$

We have found the reaction at support A, so we shall now cut the truss in half through member XX and consider the left side. Remember we guess the senses of the forces in the cut members.



To find the force in XX we shall find the sum of the vertical forces.

$$+\uparrow \Sigma F_V = 0$$

$$0 = 27.17 - 30 + XX \sin 60^\circ$$

$$XX = \frac{2.83}{0.866}$$

$$XX = 3.27 \text{ kN (comp)}$$

As you can see, this is quite a quick method of finding the force in a member located away from the ends of the truss.

Bending Stresses Induced by Point Loads

Whenever a non-axial (eccentric or transverse) force acts on a beam there will usually be some bending that takes place. This force tends to create a bending stress that is a measure of the beam's internal resistance to bending. In this section we discuss shear force and bending moment, how to graph these quantities, and then determine the bending stress (or internal resistance to bending) of the beam.

Concept of Shear Force and Bending Moment

In order for us to understand the behaviour of a loaded beam we must introduce the concept of shear force and bending moment. Shear force at any point on a beam may be defined as the algebraic sum of all external forces to one side of the beam. The shear force is actually the reaction, at a given point along a beam of the material to being sheared apart by the external forces.

Bending moment at a given point along a beam is equal to the total moment developed at that point by the external force system. If we sum moments at a given point along a beam that is in equilibrium, the value will be zero. Therefore to find bending stresses we have to only find the moment set up in one section of the beam.

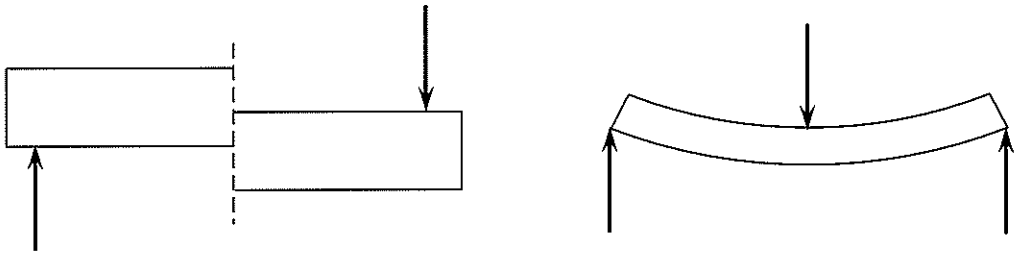


Fig. 1.10: *Diagrammatic representation of shear (left) and bending (right).*

Sign convention

Sign convention is always a hot topic for engineers in relation to shear force and bending moment, particularly for shear forces. In this book, the following sign convention is used for positive shear forces and bending moments.

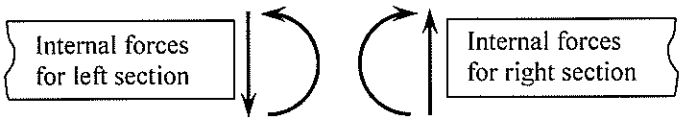


Fig 1.11: *The positive senses for the internal reactions for shear force and bending moments.*

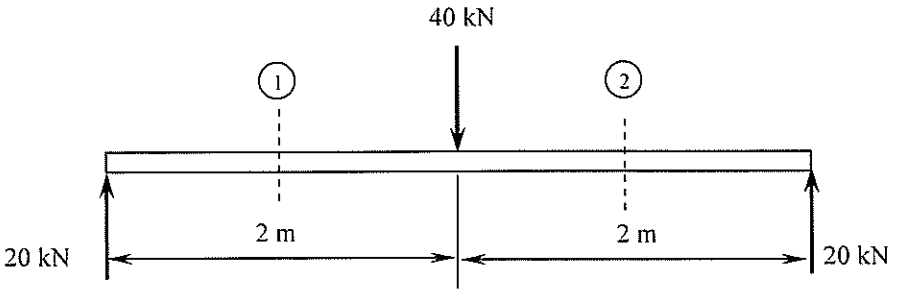
Some publications use a different convention for shear force but this type is considered simplest, particularly when drawing shear force diagrams.

Calculating Shear Forces

This involves progressively working your way along a beam summing the external forces to determine the shear force.

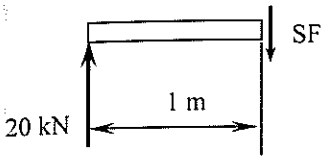
Example 1.6

Find the shear forces and bending moments at the labelled cross sections of the beam below.



First find the sum of the vertical forces that exist to the left of the section. Remember that since the shear force is equal to the sum of all external forces we shall make it the subject of the equation. Although the downward shear force is positive we shall consider downwards-external forces to be negative.

Shear Force At Point 1

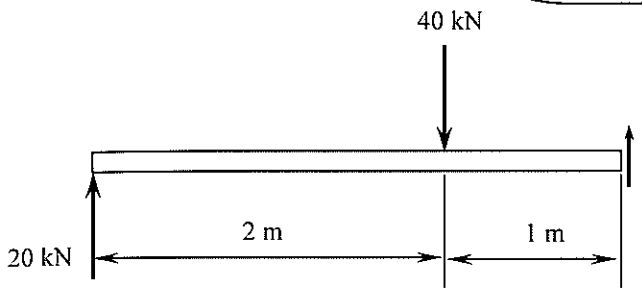


$$+\uparrow \Sigma F_V = 0 = 20 - SF$$

$$SF = 20 \text{ kN (positive SF)}$$

The positive SF value means our assumption that the SF is acting downward was correct.

Shear Force At Point 2



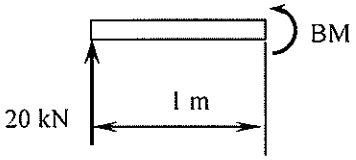
$$+\uparrow \Sigma F_V = 0 = 20 - 40 + SF$$

$$SF = 20 \text{ kN (negative SF)}$$

The positive SF value means our assumption that the SF is acting upward is correct. Therefore, by our sign convention it is a negative SF.

Now find the bending moment at the points shown on the diagram.

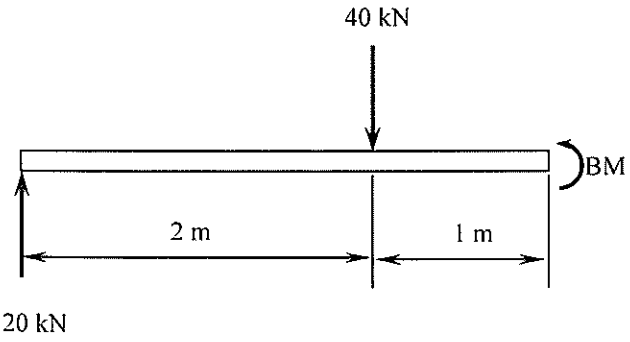
Bending Moment At Point 1



$$\oplus \circlearrowleft \Sigma M_1 = 0 = 20 \times 1 - BM$$

$$BM = 20 \text{ kNm}$$

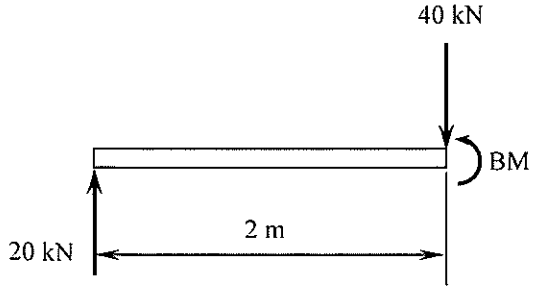
Bending Moment At Point 2



$$\oplus \circlearrowleft \Sigma M_2 = 0 = 20 \times 3 - 40 \times 1 - BM$$

$$BM = 20 \text{ kNm}$$

Now we have determined the bending moments at the same points as we did for the shear force. However these are not necessarily the maximum values. Engineers are concerned to find the maximum shear force and bending moment, because that is where the stresses will be the greatest. In this beam the maximum bending moment occurs at the 40kN force.



$$\oplus \circlearrowleft \Sigma M_{centre} = 0 = 20 \times 2 - 40 \times 0 - BM$$

$$BM = 40 \text{ kNm}$$

So the bending moment rises from zero at one end of the beam up to 40 kNm at the centre then back to zero at the other end, while the SF is 20 kN between the left end and the centre and –20 kN from the centre to the right end.

It is better to make bending moment calculations at points where there are applied forces, particularly when there is more than one.

Shear Force and Bending Moment Diagrams

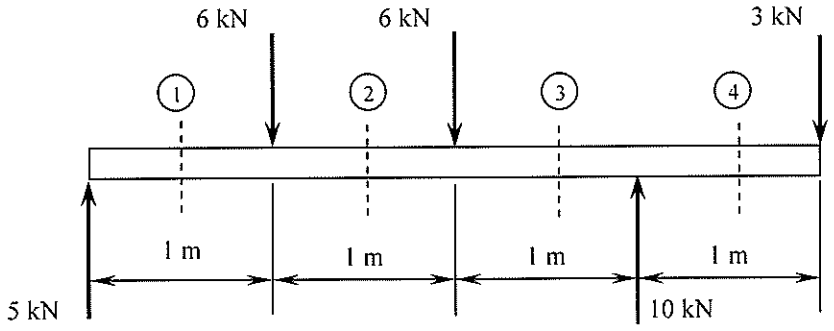
It is often useful to place the shear force and bending moment values onto a graph to show the nature of these values across the length of the beam. It is here that sign convention plays an important part.

Shear Force Diagrams

This is a graphical representation of the distribution of shear forces across a beam. There are two ways of determining the shear force diagram, the long way and the short way. The long way never fails, while the short way works best with point loads, which is what we are considering at the moment. Example 1.10 outlines both methods.

Example 1.7

Determine the shear force diagram for the beam shown below.



a) Method 1: Start at the left and add the forces at the numbered points and at the point loads. (Note upward forces are assumed positive)

SF at left reaction (5 kN): $\Sigma F_V = 0 = 5 - SF$

$$SF = 5 \text{ kN (positive SF)}$$

SF at point 1: $\Sigma F_V = 0 = 5 - SF$

$$SF = 5 \text{ kN (positive SF)}$$

SF at 6kN force: $\Sigma F_V = 0 = 5 - 6 + SF$

$$SF = -1\,\text{kN} = 1\,\text{kN} \text{ (negative SF)}$$

SF at point 2: $\Sigma F_V = 0 = 5 - 6 + SF$

$SF = -1\,\text{kN} = 1\,\text{kN}$ (negative SF)

SF at 2nd 6kN force: $\Sigma F_V = 0 = 5 - 6 - 6 + SF$

$SF = -7\,\text{kN} = 7\,\text{kN}$ (negative SF)

SF at point 3: $\Sigma F_V = 0 = 5 - 6 - 6 + SF$

$SF = -7\,\text{kN} = 7\,\text{kN}$ (negative SF)

SF at right reaction (10kN): $\Sigma F_V = 0 = 5 - 6 - 6 + 10 - SF$

$SF = 3\,\text{kN}$ (positive SF)

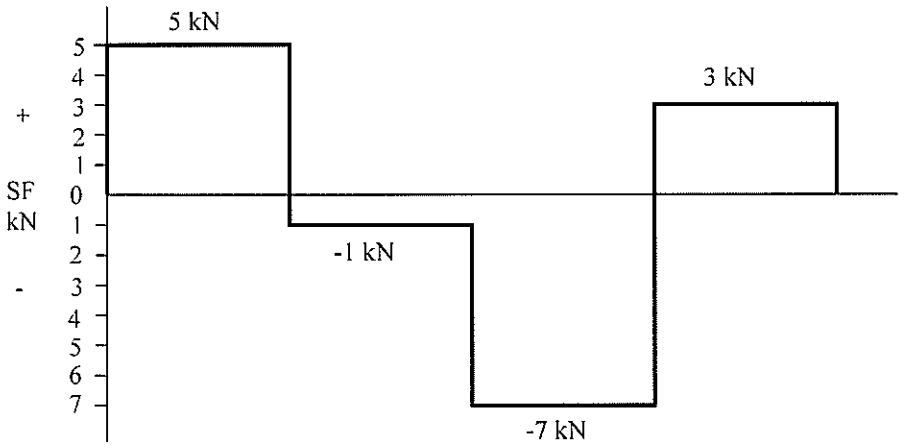
SF at point 4: $\Sigma F_V = 0 = 5 - 6 - 6 + 10 - SF$

$SF = 3\,\text{kN}$ (positive SF)

SF at right end: $\Sigma F_V = 0 = 5 - 6 - 6 + 10 - 3 + SF$

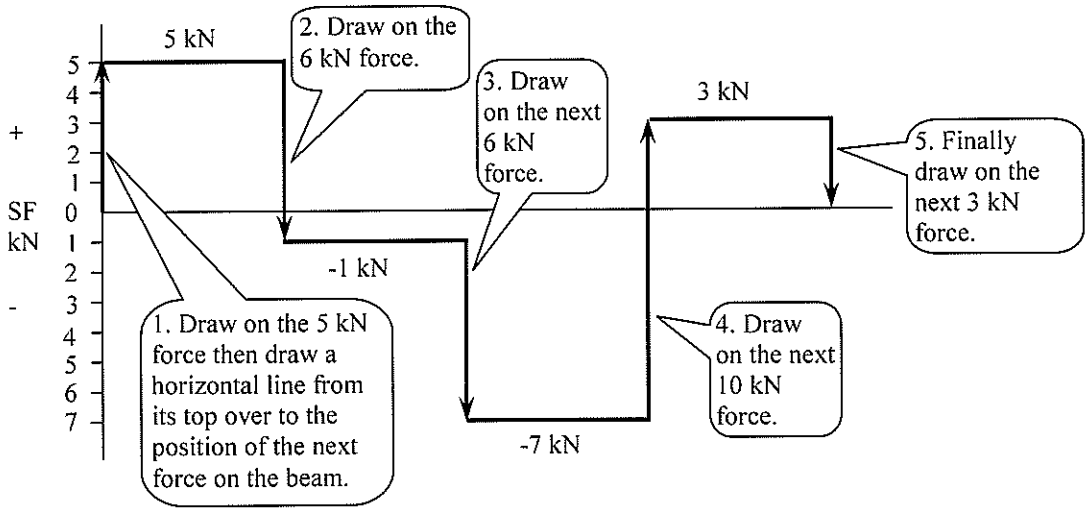
$SF = 0\,\text{kN}$

Now that we have the figures we can draw up a diagram and plot the values.



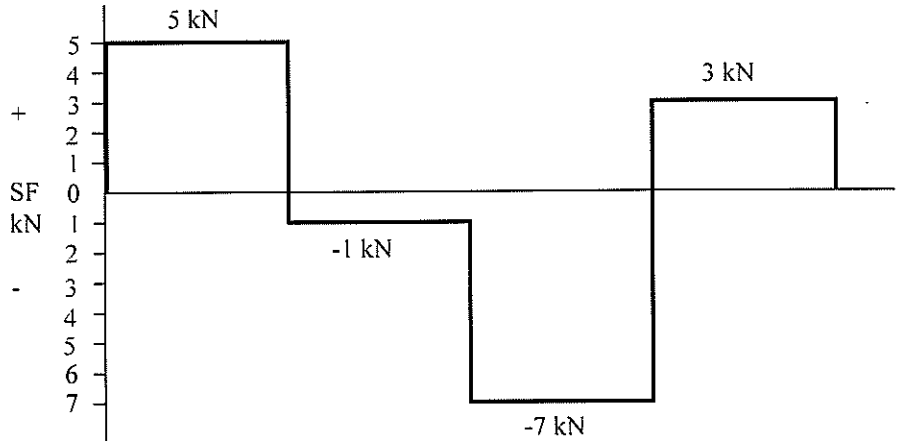
b) Method 2: Solve the problem using the "Follow the Force Rule"

This simple method involves drawing on all the external forces and linking them with horizontal lines. The result is a SF diagram that is very easy to determine.



Note that the next force will be drawn from the level that the previous force has reached and that the forces are merely drawn in the position in which they exist on the beam. When you draw a force, remember to draw the horizontal line over to the position of the next force. Finally, there is no need to put arrowheads on the SF diagram; they are only there so you can see the connection to the external forces. The final SF diagram on the next page is exactly the same as the one you would get in summing up forces.

The "Follow the Force" method is the reason we use the sign convention listed, otherwise this method must be used backwards, which is likely to cause errors. In most cases this is the simplest and quickest method to use.

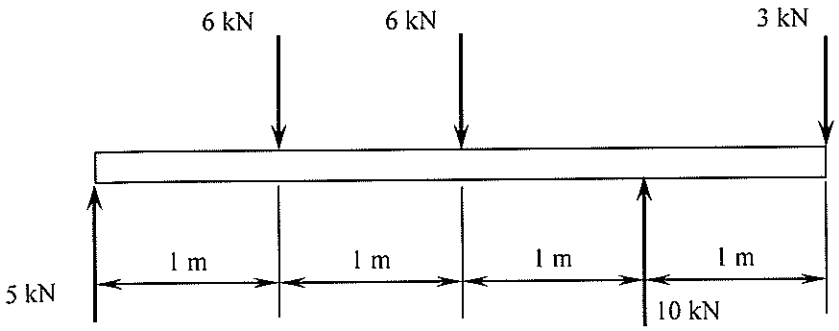


Bending Moment Diagrams

Like shear force diagrams there are two ways to determine the bending moment diagram for a beam. One method is simpler than the other, but does require the SF diagram to be drawn already.

Example 1.8

Determine the BM diagram for the beam used in Example 1.7.



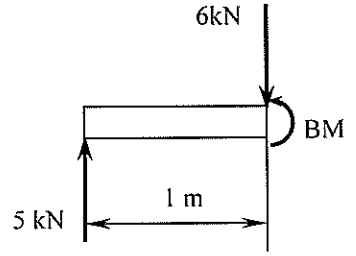
When calculating bending moments we take moments about the applied forces and reactions, as this is where there is a change in value. In between the forces, the bending moment will be a straight line (when dealing with point loads).

a) Method 1: summing up moments about active and reactive forces.

At 5 kN force:

$$\oplus \Sigma M_{5kN} = 0 = 5 \times 0 - BM$$
$$BM = 0 \text{ kNm}$$

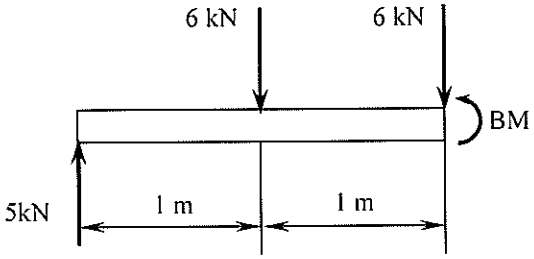
At 6 kN force:



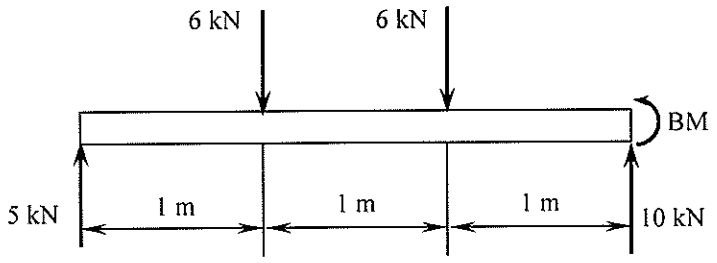
$$\oplus \Sigma M = 0 = 5 \times 1 - 6 \times 0 - BM$$
$$BM = 5 \text{ kNm}$$

At next 6 kN force:

$$\Sigma M = 0 = 5 \times 2 - 6 \times 1 - 6 \times 0 - BM$$
$$\oplus \; BM = 4 \text{ kNm}$$



At 10 kN force:

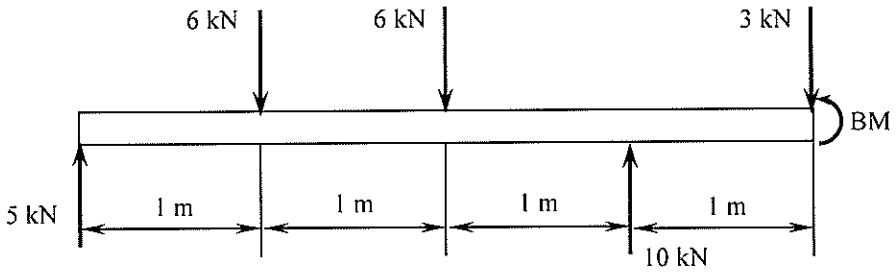


$$\oplus \Sigma M = 0 = 5\times 3 - 6\times 2 - 6\times 1 + 10\times 0 - BM$$

$$BM = -3 \text{ kNm}$$

$$BM = 3 \text{ kNm (negative BM)}$$

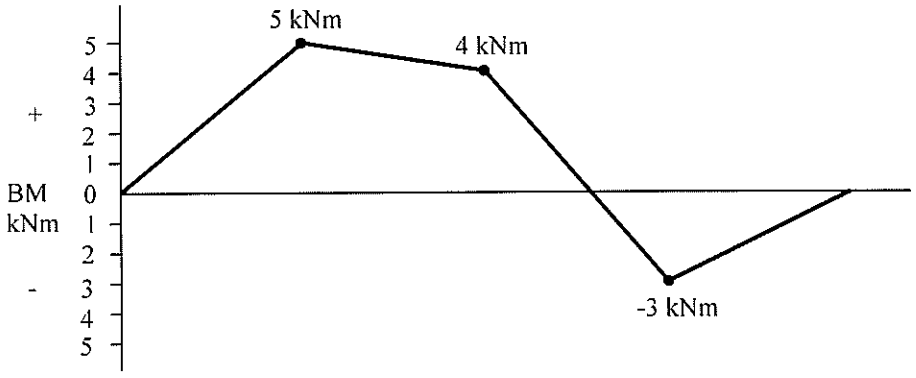
At 3kN force:



$$\oplus \Sigma M = 0 = 5\times 4 - 6\times 3 - 6\times 2 + 10\times 1 - BM$$

$$BM = 0 \text{ kNm}$$

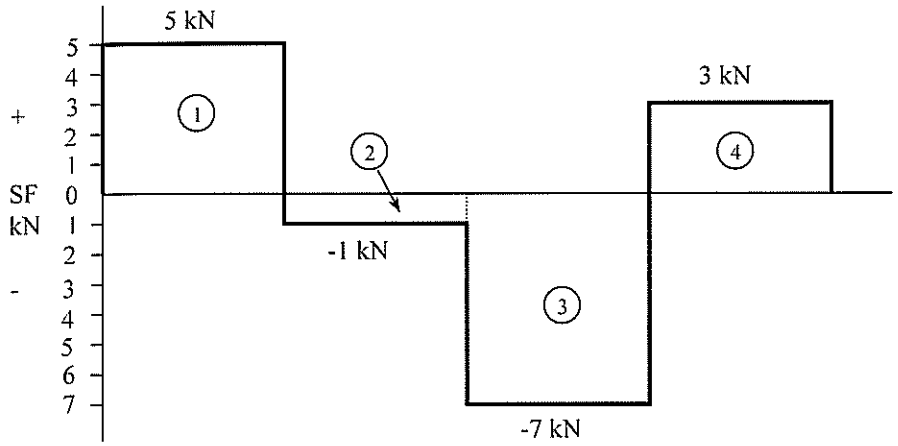
Now that we have all the relevant values, we can draw the bending moment diagram on the axes below.



From this diagram it is possible to tell that the beam predominately tends to "sag", that is, flex in a way that the ends flex upwards, while a part tends to "hog", that is flex so the middle will flex upwards.

b) Method 2: area under the SF diagram

To determine the BM diagram in this way we must work out the area under the curve from left to right of the SF diagram. Negative areas are subtracted from positive areas to give a cumulative total. The SF diagram has the areas labelled 1 to 4. The area is found by multiplying the value of the SF (height) by the length that the beam would be, in this case all rectangles have a length of 1m.



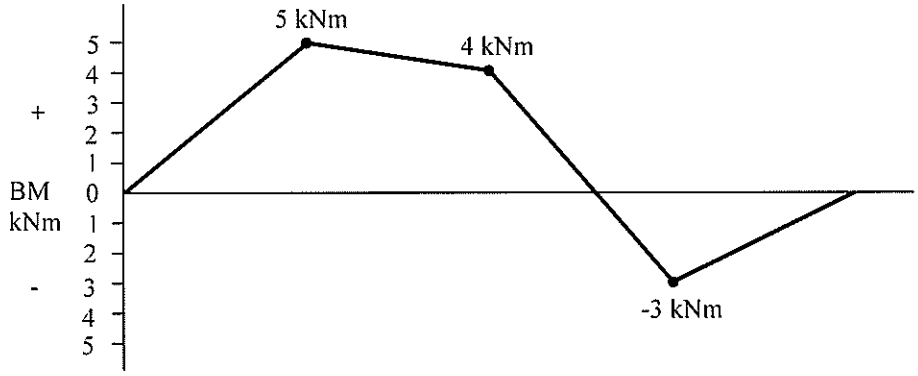
BM at 6 kN force = Area 1 = 5×1 = 5 kNm

BM at next 6 kN force = Area 1+Area 2 = 5+(-1×1) = 4 kNm

BM at 10 kN force = Area 1 +Area 2 + Area 3 = 5+(-1)+(-7×1) = -3 kNm

BM at end = Area 1 +Area 2 + Area 3 + Area 4 = 5+(-1)+(-7)+(3×1) = 0 kNm

The same values have been attained as for the previous method, but more quickly. We can now plot these values to form the same BM diagram as the one above.



Method 2 is the better and quicker option if a SF diagram has already been drawn.

Hints for Drawing Shear Force and Bending Moment Diagrams

1. If possible, draw both SF and BM diagrams below one another and either follow the scale given, or use a suitable one for magnitudes on the diagrams.

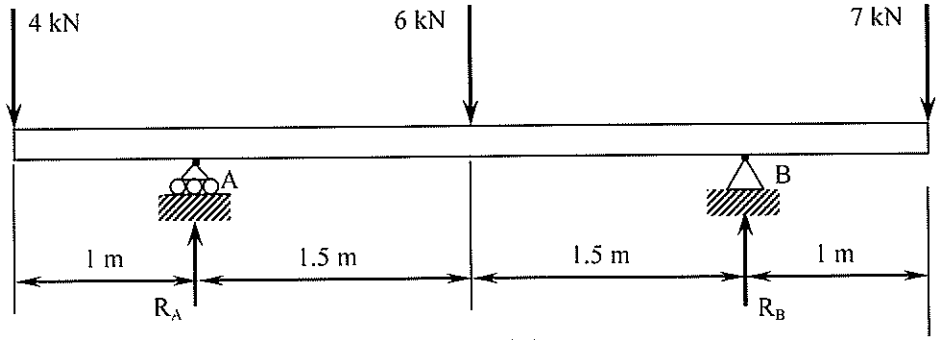
2. Draw a FBD (Free Body Diagram) of the beam showing all known loads (watch for the weight of the beam). Calculate the reactions and show them on the FBD.
3. Draw the S.F. diagram, with reference to the FBD, by starting at the left hand end, plot upwards for upward forces and downwards for downward forces. This is called the *Follow the Force* rule.
4. Find the positions of zero S.F., for these will be the positions where the BM is a maximum i.e. SF = 0, BM = maximum).
5. Forces going down produce positive bending moments (sag); forces going up produce negative bending moments (hog).
6. Bending moments are always zero at the ends of a beam (unless a couple acts at the end e.g. a cantilever beam).
7. Calculate the B.M. either by adding moments or adding areas at the following significant positions:
   - at the supports;
   - at each point load;
   - at points where the SF is zero (this point is where the maximum BM should be).
8. Between the significant positions listed in 7, draw suitable lines according to the following: if the beam is unloaded the B.M. varies linearly (i.e. it forms straight lines)
9. Point loads produce sharp changes in the S.F. diagram, but merely slope changes in the B.M. diagram.

Point 4 is a useful check. If you refer to our example you will note that at the maximum BM the SF diagram travels across the zero axis; this is a useful thing to remember as a check when drawing BM diagrams.

We will do one more example to reinforce our knowledge.

Example 1.9

Draw the shear force and bending moment diagram for the beam when it is loaded as shown below.



First determine the reactions at the left and right support.

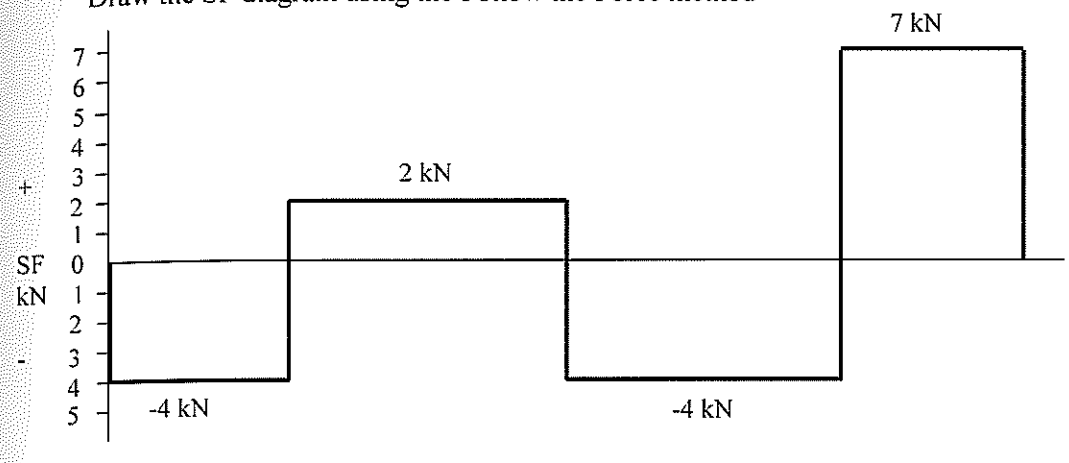
$$\circlearrowleft \Sigma M_A = 0 = -(4\times 1)+(6\times 1.5)+(7\times 4)-3R_B$$

$$R_B = 11 \text{ kN}$$

$$\Sigma F_V = 0 = -4-6-7+11+R_A$$

$$R_A = 6 \text{ kN}$$

Draw the SF diagram using the Follow the Force method



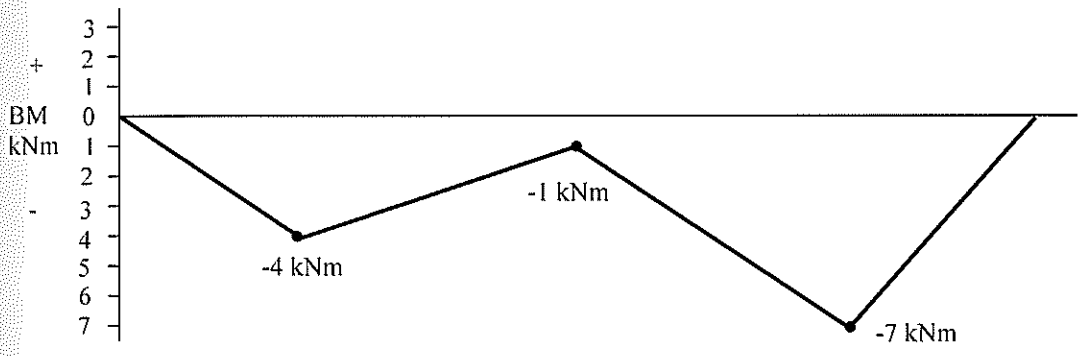
To draw the BM diagram we must find the cumulative area from left to right.

At both ends BM = zero

At 6kN force BM = -4+(2×1.5) = -1 kNm

At R_A BM = -4×1 = -4 kNm

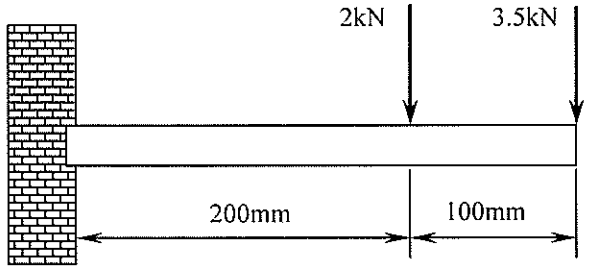
At R_B BM = -1+(-4×1.5) = -7 kNm



This example is a good illustration of how easy these problems can become when we use the methods above. Now consider an example with a loaded cantilever beam.

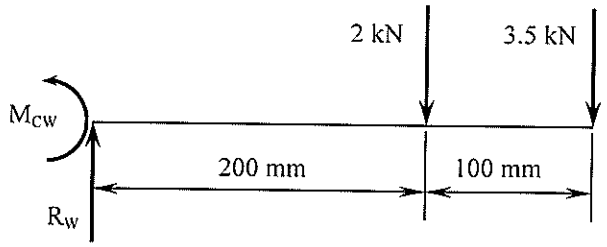
Example 1.10

The cantilever shelf in a shop is loaded as shown. Determine the shear force and bending moment diagrams.



With a cantilever there is only one fixed end, so that end must provide a force couple to stop the beam falling or rotating. The force will be equal in magnitude but opposite in direction to the sum of the vertical forces, while the reactive couple will equal the

sum of all couples but have the opposite direction. Let us now draw a free-body diagram (FBD) and determine the reactions.



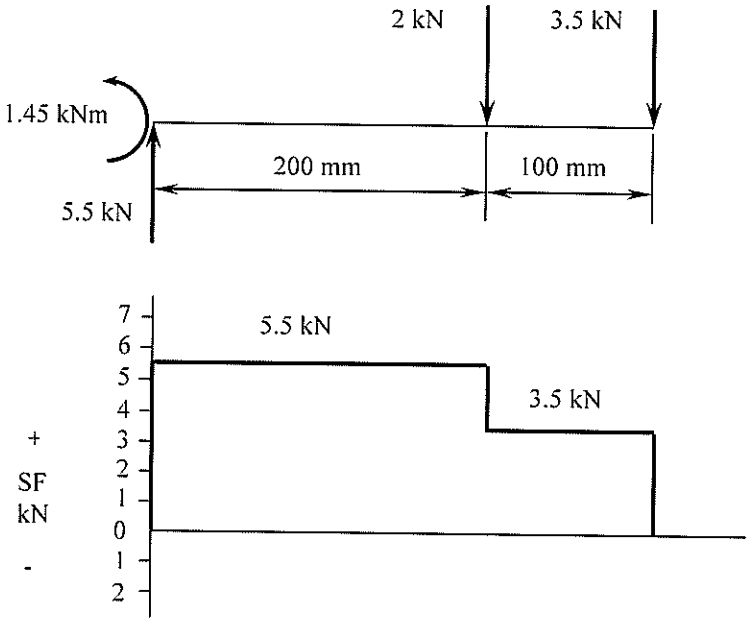
Now for the couple. It would be simple to just take moments about this point and have the couple reacting in the opposite direction. Although this works out, it is conceptually incorrect (as outlined in Volume 1 p. 62: a couple can only be balanced by another couple). Since the vertical reaction is in essence two vertical forces together (2 kN and a 3.5 kN), we must add two couples together; one couple is 2 kN times 0.2m the other is 3.5 kN times 0.3m.

Therefore: $\Sigma M_W = 0 = 2 \times 0.2 + 3.5 \times 0.3 - M_{CW}$

$M_{CW} = 1.45 \text{ kNm}$

Now that we have the reactions we can produce the SF and BM diagrams. The FBD has been reproduced above the SF and BM to clarify the method.

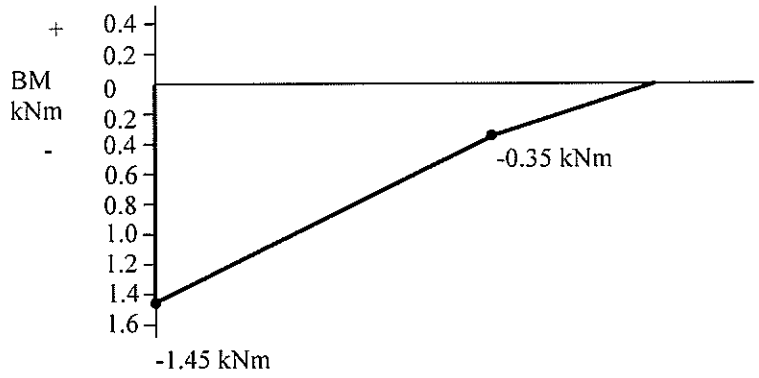
Cantilever beams do not lend themselves as well as simply supported beams to the quicker methods for solution. With this cantilever we are dealing with the right section where the SF is positive if going up (see p. 24).



The BM diagram is still created using the area under the SF diagram. However the BM diagram will be negative since cantilevers always have a negative BM diagram. The calculations are shown below.

At 350 N force, BM = 0 At 280 N force, BM=3.5×0.1 = 0.35 kNm

At end, BM = 0.35+5.5×0.2 = 1.45 kNm



Note that the SF diagram for cantilevers can be either positive or negative, but the BM diagram will *always* be negative. This is useful when calculating the area under the SF diagram.

The Neutral Axis and Outer Fibre Stress

When a beam is subjected to bending, not all of the beam undergoes the same types of stress. Let us consider a beam (Fig. 1.12) bent so that the ends are up and the centre down (positive bending), the upper surfaces will be in compression and the lower surfaces will be in tension. A good test of this is to wrap a tissue over a rule and bend it. The tissue on the upper surface will wrinkle while the tissue on the lower surface will tear.

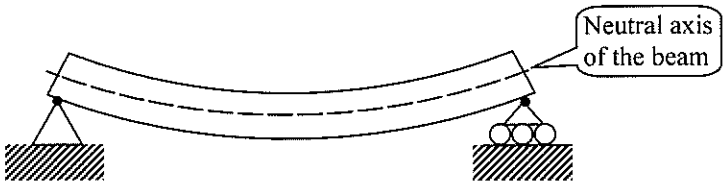


Fig 1.12: *A beam undergoing positive bending. The upper surfaces undergo tension, the lower compression and somewhere in the centre will be the neutral axis with no stress.*

Since the stresses in the beam basically reverse from top to bottom there must be a portion of the beam undergoing zero stress. This is called the neutral axis.

It follows that the greater the distance from the neutral axis the greater the stress present. In the case of a concrete beam, failure occurs on the lower surface as the tensile stress is greatest here and concrete is typically weak in tension. This is circumvented by placing steel rods in the concrete to reinforce it and take the tensile load.

It follows that there must be some relationship between stress due to bending, the bending moment present, and the distance of the section under consideration from the neutral axis.

Bending Stress Calculation

The relationship of bending stress (σ_b) to certain characteristics of the beam my be expressed mathematically as follows:

$$\frac{M}{I} = \frac{E}{r} = \frac{\sigma_b}{y}$$

M = bending moment at section (Nm)
I = second moment of area (m^4)
E = Young's Modulus for the material (Pa)
R = radius of curvature (m)
σ_b = bending stress at section (Pa)
y = section's distance from neutral axis (m)

The most used form of this relationship is the formula below,

$$\sigma_b = \frac{My}{I}$$

Radius of Curvature and Second Moment of Area – Or What are r and I?

The radius of curvature (r) is a measure of the curve the beam takes on when loaded. The smaller the value, the more bowed the beam is. It is not constant along the beam and is affected by the bending moment at that point of the beam.

The second moment of area (I) is a measure of the resistance the beam's cross-section offers to bending. The formula for the second moment of area for most cross-sectional shapes is very complex. To simplify discussion in this book, the second moment of area will be given. To help you understand what effect the second moment of area has on the body's resistance to bending, do the investigation below.

Investigation: Get two pieces of timber with the same cross-sectional dimensions and place one flat and one on its edge. Place weight on both. Which has the greatest resistance to bending?

I-beams are used extensively in civil structures. An I-beam has a cross section with greater depth than breadth, and it also places a majority of the material at the outer fibres where bending stress will be the greatest. This efficient shape uses the web to control shear stresses, the flanges to control bending stresses and also will be lighter than a solid beam with the same cross sectional dimensions.

Bending Stress and How to Calculate it.

Now that we know what "I" represents we can start to determine the bending stress in beams.

Example 1.11

Find the bending stress in the lower surface of a beam that has a depth of 300mm, a BM of 25kNm applied and the second moment of area is 3.4×10^{-4} m^4. Consider the neutral axis to be at the midpoint of the cross-section.

$$\sigma_b = \frac{My}{I}$$
$$\sigma_b = \frac{25000 \times 0.15}{3.4 \times 10^{-4}}$$
$$\sigma_b = 11024911.76 \text{ Pa}$$
$$\sigma_b \approx 11 \text{ MPa}$$

So the stress present in the lower surface of the beam is approximately 11MPa.

Example 1.16

A steel I beam has a second moment of area of 1.67×10^{-4} m^4, and a Young's Modulus of 206GPa. If the applied moment is 12kNm, then what will be:
a) The stress in the surfaces of the beam if y = 140mm, and
b) The radius of curvature of the beam?

We shall solve a) first:

$$\sigma_b = \frac{My}{I}$$
$$\sigma_b = \frac{12000 \times 0.14}{1.67 \times 10^{-4}}$$
$$\sigma_b = 10059880.24 \text{ Pa}$$
$$\sigma_b \approx 10.06 \text{ MPa}$$

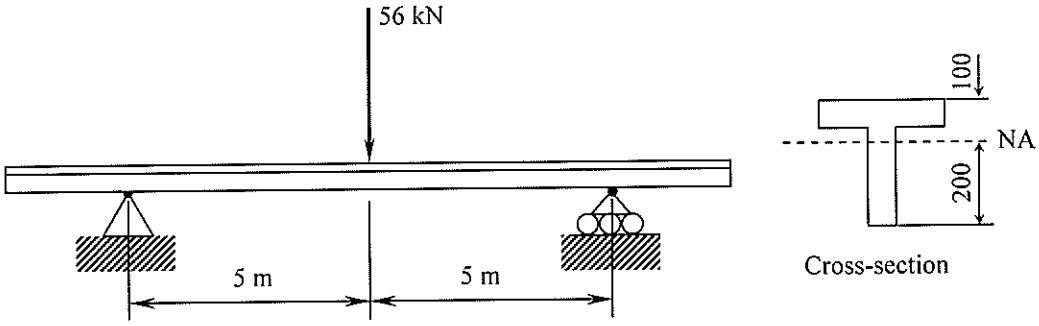
Now solve b):

$$\frac{E}{r} = \frac{M}{I}$$
$$r = \frac{EI}{M}$$
$$r = \frac{(206 \times 10^9) \times (1.67 \times 10^{-4})}{12000}$$
$$r = 2866.83 \text{ m}$$

Example 1.12

A beam is loaded as shown below. Find:

a) The bending stress at the applied force in the upper and lower surfaces of the beam shown ($I = 4.6 \times 10^{-4}$ m^4);
b) The Value for Young's Modulus if the beam has a radius of curvature of 230 m.

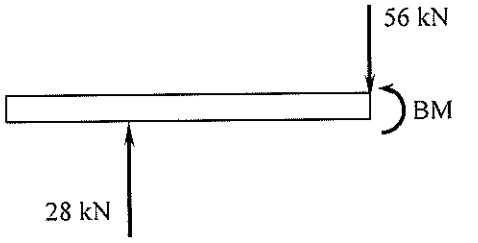


This cross section differs from the previous I beam as it is not symmetrical about two axes. As a result the neutral axis (NA) is not halfway between top and bottom.

Let us solve part a):

First we must find the bending moment that exists at the load. Since the load is midway between the reactions then each reaction must be half of the load. Therefore both reactions are 28 kN vertically up.

So the BM is:



$$\circlearrowleft + \quad \Sigma M = 0 = 28 \times 5 - 56 \times 0 - BM$$
$$BM = 140 \text{ kNm}$$

For the upper surface

$$\sigma_b = \frac{My}{I}$$
$$\sigma_b = \frac{140000 \times 0.1}{4.6 \times 10^{-4}}$$
$$\sigma_b = 30434782.61 \text{ Pa}$$
$$\sigma_b \approx 30.43 \text{ MPa}$$

For the lower surface

$$\sigma_b = \frac{My}{I}$$
$$\sigma_b = \frac{140000 \times 0.2}{4.6 \times 10^{-4}}$$
$$\sigma_b = 60869565.22 \text{ Pa}$$
$$\sigma_b \approx 60.87 \text{ MPa}$$

The stress in the lower surface will be twice the amount of the stress in the upper surface, as the lower surface is double the distance away from the neutral axis. If this was a material weak in tension, such as concrete, there would need to be reinforcing done on the lower surface, or the section of the beam would be changed.

Now we shall solve part b).

$$\frac{E}{r} = \frac{M}{I}$$
$$E = \frac{Mr}{I}$$
$$E = \frac{140000 \times 230}{4.6 \times 10^{-4}}$$
$$E = 7 \times 10^{10} \text{ Pa}$$
$$E = 70 \text{ GPa}$$

With such a figure for E, it is likely that that beam is made of an aluminium alloy. If the alloy was steel with a higher value for E, then the radius of curvature would be *larger*, since the beam would not bend as much. The smaller the radius of curvature, the more a beam has bent.

Uniformly Distributed Loads

Unlike a point load, a uniformly distributed load (UDL) is a load that is uniformly spread across a beam. If you are to stand on a beam on one foot and allow all your weight to be supported by that one foot, then your weight is concentrated as a point load. However if you lie face down on the beam your weight will be uniformly spread across most of your length, hence your weight will be spread out.

Good examples of uniformly distributed loads are wind blowing against a building, a fish tank on a shelf, and perhaps best of all, the weight of a beam itself. When a beam is placed on supports the beam itself has mass that acts on the supports. We usually neglect this since it simplifies the problem and the engineer has already designed the supports to take this load in addition to any further external loads.

If a beam is uniform it means that its mass is evenly distributed across its length. Fig. 1.15 (a) shows a beam resting on supports. If it has a mass of 1000 kg and is 10 m long then it is considered to have a mass of 100 kg per metre (kg/m). Now the UDL of this beam will be the weight, so if the total mass was 1,000 kg then the weight is 10,000N or 100 kN, so the UDL will be 10 kN/m. Fig 1.15 (b) shows how we represent the UDL of the beam.

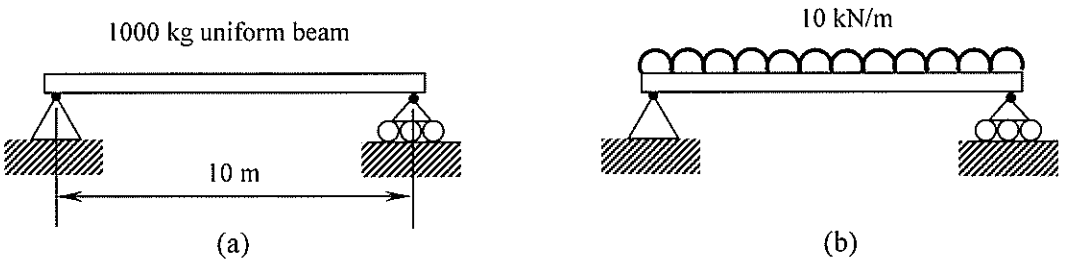


Fig 1.13: *The same beam: (a) shows it simply with its mass listed, (b) shows it with its weight as a UDL.*

A UDL will tend to produce a different shape in shear force and bending moment diagrams, while you are not expected to be able to draw them for the course you must be able to recognise them. In a shear force diagram the angled lines replace the horizontal lines of a point load SF diagram. In the shear force diagram the UDL is represented by the slope of the lines, if the UDL is large the SF diagram will have a large gradient.

Whereas in a BM diagram point loads create angled lines, with a UDL the lines will be curves. This curve will at times be parabolic in nature and a simple UDL with no point loads will bring about pure parabolic curve for the bending moment diagram.

In most UDL situations there will be a mix of point loads and UDLs, which means the key features to check for in the UDL regions are angled lines in the SF diagram and curves in the BM diagram.

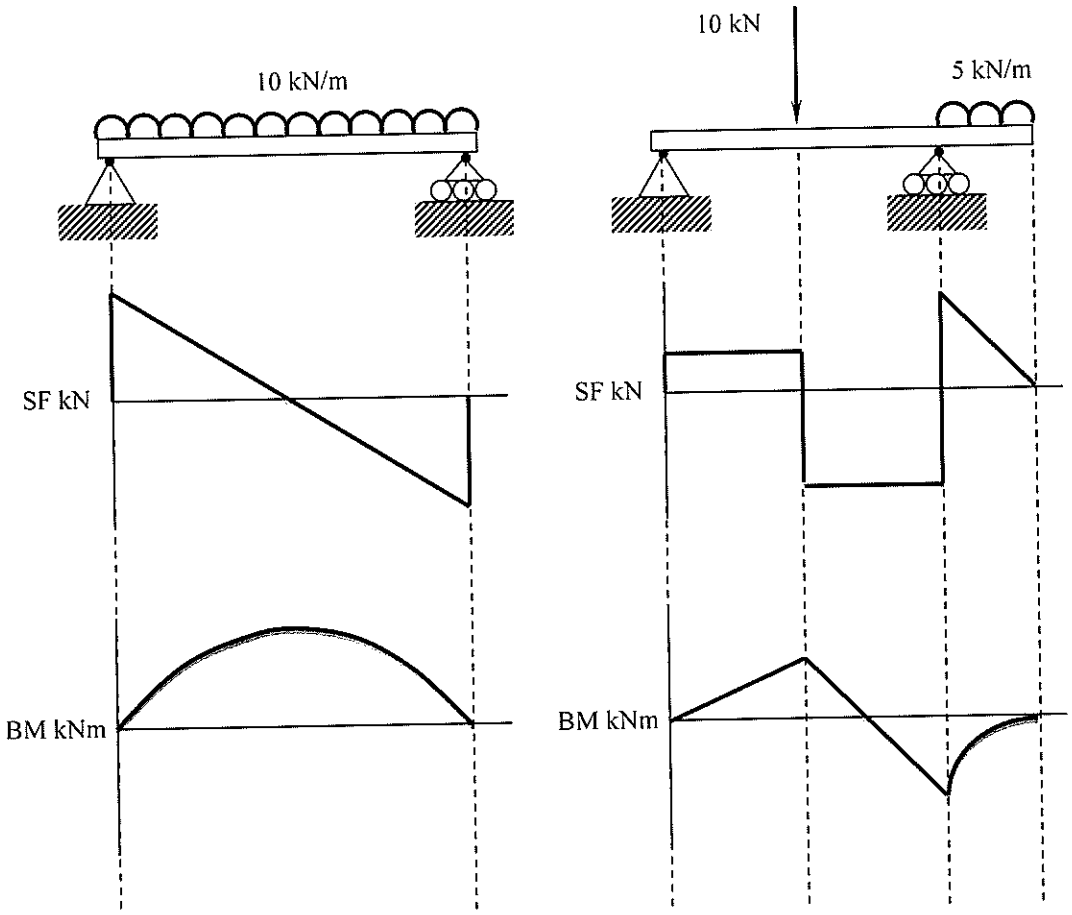


Fig 1.14: *Two beams with UDLs with the resulting Shear Force and Bending Moment Diagrams.*

Stress and Strain

Stress and strain were concepts first covered in Volume 1. To recap what they are: Stress may be defined as a measure of the internal reaction that occurs in response to an externally applied load. This internal reaction is related to the original cross sectional area to quantify the nature of the reaction. The unit for stress is the Pascal (Pa) and one Pascal is 1 N/m^2. The formula for stress is as follows:

$$\sigma = \frac{P}{A}$$

σ = stress (Pa)
P = load (N)
A = cross sectional area (m^2)

Strain is defined as the proportional change in length caused when a specimen is under an axial load. Strain may be found using the following formula:

$$\varepsilon = \frac{e}{L}$$

ε = strain
e = extension (m)
L = original length (m)

Both of these concepts are important in the design of civil structures, as stress and strain are primary factors in material choice and design style.

In mechanically elastic structures such as bridges and buildings, stress is directly proportional to strain.

Shear Stress

Shear stress differs slightly from tensile and compressive stress. It is essentially the same concept but it is a measure of the internal reaction to a shearing force. Although the formula is of the same form it is the value for area that differs. In compressive and tensile stress the area used for computation is the cross-sectional area, whereas with shear stress, the area represents the shear area, which is decided by the type of shearing performed. So the formula becomes:

$$\sigma_s = \frac{P}{A}$$

σ_s = shear stress (Pa)
P = load (N)
A = shear area (m^2)

If the shear plane runs perpendicularly across the object, the shear area will be the cross-sectional area. If, however, the shearing operation punches a hole then the shear area will be the circumference of the hole multiplied by the thickness of the material.

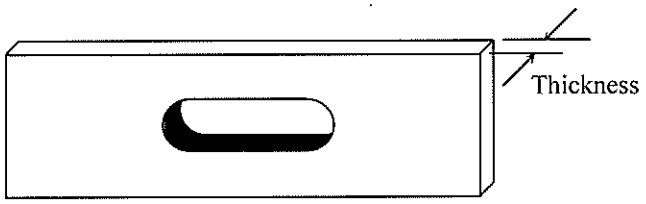


Fig. 1.15: *The shaded portion of the opening shows part of the shear area. This total shear area is calculated by multiplying the perimeter of the shape by the thickness.*

Example 1.13

Find the shear stress present when a circular hole of diameter 30 mm is punched in a plate for use on a bridge, when the plate is 20 mm thick and the punching force is 60 kN.

The shear area is found by finding the circumference of the circle and multiplying that figure by the thickness of the plate.

$$C = \pi D$$
$$C = 3.14 \times 0.03$$
$$C = 0.0942 \text{ m}$$

$$A = C \times d$$
$$A = 0.0942 \times 0.02$$
$$A = 1.88 \times 10^{-3} \text{ m}^2$$

Once the area is found it is possible to find the shear stress.

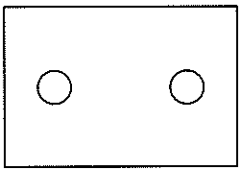
$$\sigma_s = \frac{P}{A}$$
$$\sigma_s = \frac{60000}{1.88 \times 10^{-3}}$$
$$\sigma_s = 31914893.62 \text{ Pa}$$
$$\sigma_s = 31.9 \text{ MPa}$$

So the shear stress set up by punching this hole into the plate is 31.9MPa.

Sometimes a punching operation will punch out the exterior shape and holes at the same time. This is outlined in Example 1.2.

Example 1.14

A rectangular plate for a truss bridge is to be punched from 10mm sheet steel with a force of 60kN. During the punching operation two 25mm diameter holes are also punched. Find the shear stress if the plate has dimensions 150mm by 120mm.



A diagram of the plate is shown above (note the diagram is not to scale). We must find the total perimeter by taking the perimeter of the rectangle and adding it to the circumferences of both circles, then multiplying that total perimeter by the thickness to find the shear area.

$$\text{Total Perimeter} = 2b \times 2h + 2\pi D$$
$$\text{Total Perimeter} = 0.3 + 0.24 + 2(3.14 \times 0.025)$$
$$\text{Total Perimeter} = 0.54 + 0.157$$
$$\text{Total Perimeter} = 0.697 \text{ m}$$

The shear area is now found by multiplying the total perimeter with the thickness of the plate. Hence,

$$\text{Shear Area} = \text{Total Perimeter} \times \text{Thickness}$$
$$\text{Shear Area} = 0.697 \times 0.01$$
$$\text{Shear Area} = 6.97 \times 10^{-3} \text{ m}^2$$

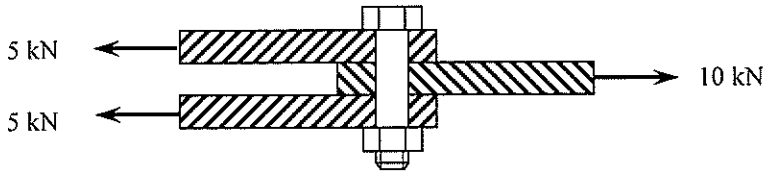
Now we can find the shear stress using the load given and the shear area we just determined.

$$\sigma_s = \frac{P}{A}$$
$$\sigma_s = \frac{60000}{6.97 \times 10^{-3}}$$
$$\sigma_s = 8608321.38 \text{ Pa}$$
$$\sigma_s = 8.61 \text{ MPa}$$

The shear stress in the material is 8.61MPa.

Example 1.15

A bolt is used to hold a coupling together. If the coupling takes a load of 10 kN and the maximum allowable shear stress of the bolt is 30 MPa, determine the minimum diameter of the bolt.



This situation is a classic shear stress problem, for it illustrates the concept of double shear. The arrangement of the coupling means the bolt will try to shear in two places. This means that the area is doubled in the calculations.

$$\sigma_s = \frac{P}{A}$$

$$\sigma_s = \frac{P}{2A}$$

$$\frac{\pi d^2}{2} = \frac{P}{\sigma_s}$$

$$\pi d^2 = \frac{2P}{\sigma_s}$$

$$d = \sqrt{\frac{2P}{\pi\sigma_s}}$$

$$d = \sqrt{\frac{2\times 10,000}{\pi\times 30\times 10^6}}$$

$$d = 0.014567 \text{ m}$$

$$d = 14.57 \text{ mm}$$

This means that the minimum size for the bolt must be 14.57 mm. As no such bolt is readily available the engineer will choose a commercially available size larger than this, such as 16 mm.

Engineering and Working Stress

In Module 3 in Year 11 we discussed stress/strain diagrams. Examples are shown in Fig. 1.16 (stress/strain diagrams are covered in greater detail in this chapter on page 46).

One thing that puzzles most students is why the stress falls away at the end of the diagram for ductile materials. This is because engineers tend to cheat a little with these diagrams; they show *engineering stress*. Engineering stress is calculated with the load always divided by the original cross sectional area, even though the cross section actually reduces just before failure. So this type of engineering does not take

into account the falling cross sectional area. This is done for ease of calculation otherwise the cross sectional area would need to be determined continually, making tensile testing a laborious process. The term used to describe the "real" stress is *true stress*.

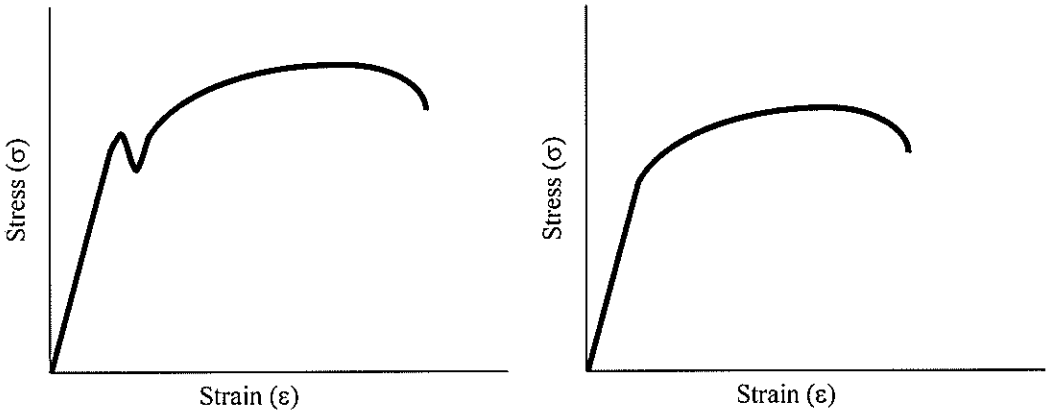


Fig 1.16: *Two stress/strain diagrams. The one on the left shows definite yield points and the other shows progressive yield.*

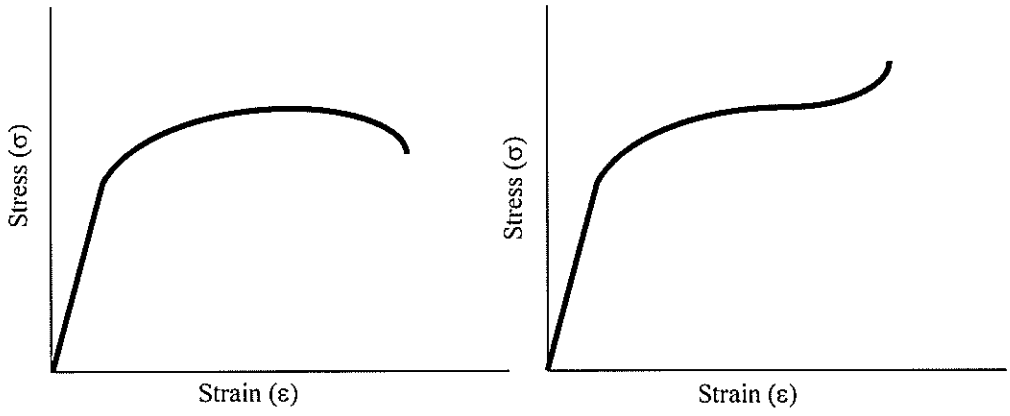


Fig 1.17: *Two stress/strain diagrams, the one on the left shows engineering stress and the other shows true stress.*

Maximum allowable stress, or safe working stress, may be defined as the maximum permissible stress that a material or object will be subjected to when in service. Working stress is usually less than the yield stress and the UTS. The ratio of working stress to one of the other two values is called the Factor of Safety (outlined below) and it is an important consideration in designing equipment. Safe working stress is always less then the elastic limit for a given material.

Related Terms and Definitions

Elastic limit is the stress up to which the strain is elastic. This means that if the load is removed, the object will return to its original shape and size. Beyond this point, permanent deformation occurs.

Yield stress is the stress where there is a marked increase in strain without a corresponding increase in stress. In mild steel, there are definite yield points but most

materials show progressive yield. The yield stress is always greater (though often only by a small amount) than the elastic limit, but less than the UTS.

Proof stress is the amount of stress necessary to bring about a certain amount of permanent strain in a material. Proof stress is used as a measure of yield on materials that do not show a marked yield point (i.e. progressive yield). Usually a set amount of strain is given, for example 0.2%, and the amount of stress necessary to record this level of strain is calculated.

Resilience is the ability of a material to absorb energy, but only up to the elastic limit. The area under the curve in the elastic deformation region represents the resilience of a material. The resilience represents the amount of strain energy that is stored in the deformed object.

Toughness is a measure of the ability of a material to absorb energy. Tough materials are those capable of absorbing strain energy or the energy of an impact, and are the opposite of brittle materials. The toughness found by the area under the stress/strain curve is a different value of toughness to that found in impact testing.

Hooke's Law may be stated as follows: "*Stress is proportional to strain up to the elastic limit*". This means that any increase in stress will bring about a proportional increase in strain up to a given point. English physicist Robert Hooke (1635-1703) studied elastic materials to establish this law. Written mathematically Hooke's Law is:

$$E = \frac{\sigma}{\varepsilon}$$

E = constant
σ = stress (Pa)
ε = strain

This constant is a measure of the materials resistance to stretching up to the elastic limit and is sometimes called the *modulus of stiffness* or *Young's Modulus*.

Young's modulus is actually the more common name for modulus of stiffness, represented by the constant in Hooke's Law. English scientist Thomas Young (1773-1829) determined this modulus and it was named *Young's modulus of elasticity* in his honour. Mathematically it can be written as follows,

$$E = \frac{\sigma}{\varepsilon} = \frac{P/A}{e/L} = \frac{PL}{eA}$$

E = Young's Modulus of Elasticity (Pa)
P = axial force (N)
A = cross-sectional area (m^2)
e = deformation (elongation, compression, etc.) (m)
L = original gauge length (m)
σ = stress (Pa)
ε = strain

Factor of Safety (FofS)

Factor of Safety is the ratio of Safe Working Stress to either the yield stress or ultimate tensile stress depending on the type of material. It is an important consideration in designing equipment.

For ductile materials

$$\text{FofS} = \frac{\text{Yield Stress}}{\text{Maximum Allowable Stress}}$$

For brittle materials

$$\text{FofS} = \frac{\text{UTS}}{\text{Maximum Allowable Stress}}$$

The FofS has a large impact on the design of an object. In buildings and bridges the FofS is usually large to ensure that there is no chance of failure. In aircraft design the FofS is as low as possible due to the problems related to the extra weight of a design. FofS is decided by both the application and the material.

Example 1.16

A bridge is designed to have a factor of safety of 8. If a member is made of steel and the yield stress is 250 MPa, what will be the maximum allowable stress?

$$\text{FofS} = \frac{\text{Yield Stress}}{\text{Maximum Allowable Stress}}$$

$$\text{Maximum Allowable Stress} = \frac{\text{Yield Stress}}{\text{FS}}$$

$$\text{Maximum Allowable Stress} = \frac{250}{8}$$

$$\text{Maximum Allowable Stress} = 31.25 \text{ MPa}$$

To achieve a low maximum available stress the member must be made larger than it would be for a lower FS.

Stress/Strain Diagrams

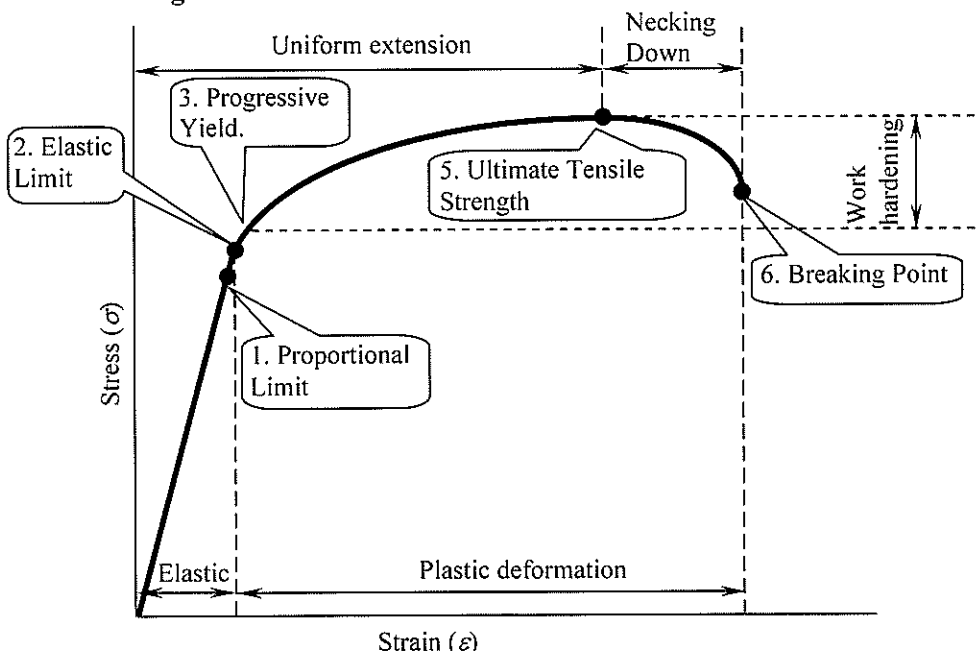


Fig 1.18: *A stress/strain curve for a material showing progressive yield.*

Stress/strain curves are graphs used to compare the stress and strain of a material loaded in a test. Originally a load/extension (or deformation) curve will be produced, but these values are converted to stress and strain respectively. The stress/strain curve will show features such as elastic limit, type of yield (progressive or definite), UTS and breaking point.

Stress/strain curves are very useful for determining the performance of different types of materials and different sized test pieces. A stress/strain curve (typical of 70/30 annealed brass) similar to one from Volume 1 is shown in Fig 1.10.

Engineering Materials

Testing of Materials

We learn a lot about how materials deform and crack by carrying out specially devised tests. Some testing for cracks and cracking can be carried out in situ using X-rays and ultrasound e.g. for cracks/inclusions in welded high pressure gas pipes and aircraft engine components. However, other destructive tests allow us to watch and analyse material behaviour under load.

Specialised Testing of Engineering Materials and/or Systems

Table 1.2: *Non-destructive and destructive tests used for civil structures*

Test	Type	Use
X-ray	Non-destructive	To determine if cavities are present.
Dye penetrant	Non-destructive	To find small cracks in the surface by placing a dye on the surface, applying a developer and then examining the surface after cleaning. (UV light may be used)
Ultrasonic	Non-destructive	Ultrasonic pulses are used to determine if cavities are present.
Tensile	Destructive	Used to determine the tensile strength of materials used. Test piece is stretched and load and extension are recorded.
Compressive	Destructive	Used to determine the compressive strength of materials used. Test piece is compressed and load and deformation are recorded.
Transverse	Destructive	This test is used to determine a materials performance when undergoing bending and shear. Outlined in more detail below.
Torsion	Destructive	Torsion tests are done on material to see how they will cope with twisting forces (couples).

Often, when examining the materials suitable for use in civil structures, engineers will need to perform various tests, some non-destructive, others destructive. Table 1.2 lists some tests used in civil structures.

Non-destructive testing

There are a variety of non-destructive tests that can be used. For civil structures, X-ray testing, dye penetrant methods and ultrasonic testing are of primary importance.

X-ray testing

X-ray testing for engineering operates on the same principles as medical x-rays. X-rays are passed through an object and expose a photographic film on the opposite side. If there is a cavity in the material the x-rays will pass through that area more easily and thus more x-rays will reach the film. At that point the film will show a darker patch and reveal a possible cavity in the material. Nowadays x-ray testing is often linked to computerised systems that allow 3D imaging of fractures and imperfections in the component.

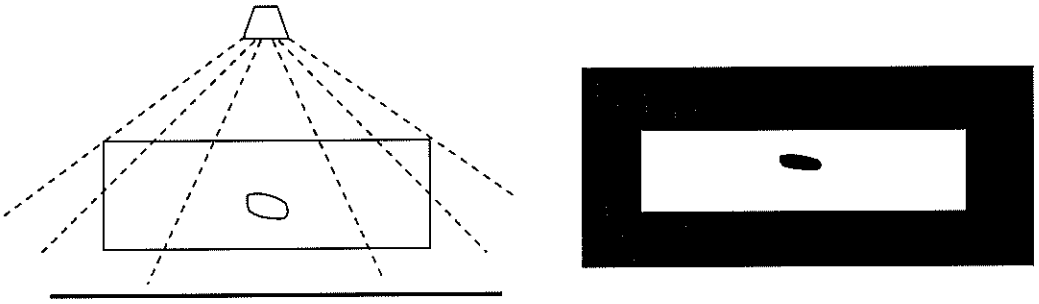


Fig. 1.19: *X-ray testing, the figure on the left shows the x-rays passing through an item to the photographic film, while the figure on the right shows the resulting exposed film with the cavity darker.*

Dye penetrant testing

This testing procedure is used to determine if cracks are present in metallic items. The surface is prepared by removing any oil or surface contaminants in the suspect area. The suspect area is heated and a dye is then placed over the surface and excess is wiped off. A developer (usually a white powder) is sprinkled over the area and then the object is cooled. If a crack is present the dye will be squeezed out of the crack by the contraction of the cooling item and reveal the presence of a crack. UV light may be used to maximise the chances of detecting very fine cracks.

Ultrasonic testing

This system works similarly to x-ray testing, but high frequency radio waves are used instead. A transmitter sends ultrasonic waves through a component and normally these pass through to the other side of the component and are reflected back to the sender. The reflected signals are recorded by a display. The height on the display is proportional to the thickness. If a cavity is encountered the sound wave will be reflected early and the display will show a lower reading.

Transverse test

The transverse test is useful for determining how a material will perform when bent. It is particularly useful for materials which are difficult to tensile test. The material is supported at either end and a load is applied on the opposite side in the centre of the two supports, or at positions a third and two-thirds between the supports. In both cases a measure is taken of the bending load and of the total deflection at rupture. The data is converted into stress and strain for analysis.

Testing of Concrete

Slump Test

The slump test is used to check that the concrete is of appropriate fluidity for casting. The concrete is cast into a conical mould, open at the top and the bottom. The mould is placed on a board then the concrete is poured in. Upon removal of the mould the conical wet concrete that is left behind, should slump slightly according to set specifications. If it simply collapses it is to wet, if it breaks in half and crumbles then it is too dry. If the concrete does not have the right consistency there may be difficulties with its casting.

Compressive Test

Concrete is strong in compression and after casting various specimens will be taken and subjected to compression tests after specified time intervals, e.g. 30 days. If the concrete fails any of the tests, then the material was a faulty mix and may need to be removed and re-cast, a very expensive and time consuming option.

Crack Theory

Stress concentration can have a large impact on the behaviour of a material or object in service. At the turn of the century much effort was put into trying to explain why some ships broke in half in heavy seas, when theoretically they were capable of the loading. Professor Inglis, in 1913 showed that the reason for this was that certain parts of the ship's structure e.g. doors and sharp corners tended to concentrate stress at a much higher level than if the irregularity had not been there.

In 1920, a gentleman by the name of A. A. Griffith published a paper that explained that the reason that the practical, mechanical strength of a material is lower than its theoretical mechanical strength is due to irregularities in the surface that cause stress concentrations and then cracks. Essentially it is the same concept that Inglis proposed, but on a much wider scale, because it was not just confined to ships.

The way cracks form and grow is closely linked to the applied stress, the Young's Modulus of the material and the strain energy present.

Strain Energy

Work and energy are important concepts. We first looked at these in Volume 1 (pp. 101-104). Work occurs when a force causes motion and work is closely linked to energy. Let us imagine you lift a 10 kg mass up 1 m, the work done is 100 J and the mass now has potential energy of 100 J; your work has become potential energy. Now let us suppose you compress some metal in a vice. You have done work but where has it gone? The metal doesn't have potential energy, as it is still at the same height as before and no kinetic energy as it isn't moving; instead it has gained *strain energy*.

Strain energy is a close relative of potential energy and is a measure of the amount of energy stored in a material that has undergone some strain, whether it be compressive,

tensile, shear, bending or torsion. The strain energy per unit volume of a material is determined using the formula below.

$$SE = \tfrac{1}{2}\sigma\varepsilon$$

SE = strain energy/unit volume (J/m^3)
σ = stress (Pa)
ε = strain

If we can write strain in terms of stress and Young's modulus then the formula becomes:

$$SE = \tfrac{1}{2}\sigma\varepsilon$$

$$SE = \tfrac{1}{2}\sigma\left(\frac{\sigma}{E}\right) \qquad \left(\because \varepsilon = \frac{\sigma}{E}\right)$$

$$SE = \frac{\sigma^2}{2E}$$

We can say that strain energy is proportional to the square of stress and inversely proportional to Young's Modulus. That means if stress is increased then strain energy increases, and if Young's modulus for a material is high, strain energy will be less than for a material under equal stress with a lower Young's Modulus.

So why is strain energy important? When a crack begins to grow, the material is releasing its strain energy, and thus the strain energy present in a material will have a large impact on the formation and propagation of cracks in the material. Larger strain energy means more energy to release and greater chance of crack propagation.

Crack Formation and Growth

How do cracks form and grow? The answer is neither brief nor simple. A material has two methods of failure, brittle or ductile failure, and the structure of the material determines how it will fail. Metal alloys such as mild steel undergo ductile failure up to a point, so they extend in length until brittle failure takes over. Brittle materials like concrete and glass do not undergo ductile failure; they undergo immediate brittle failure. Crack formation and growth is a method of brittle failure.

Theoretically glass and other ceramics should be very strong in tension. Unfortunately they never achieve this good tensile performance because there are tiny micro cracks, micro gas bubbles and minute crystallites on the surface that concentrate stress at one point and from this grows a crack. Since these are brittle materials it is a healthy breeding ground for the crack to grow i.e. propagate through the material. So, how does a crack grow and why doesn't it stop?

Fig 1.20 (a) shows a set of atoms in a material with a perfect linking of the atomic planes, however (b) shows that an imperfection is present in the top layer. This will tend to concentrate twice the stress at the next bond, as it must do the work of two bonds. By (c) the crack is travelling through the bonds as they break and the remaining bonds become more and more highly stressed (d). As this continues the likelihood of the crack growing increases and eventually the crack will travel right through the material.

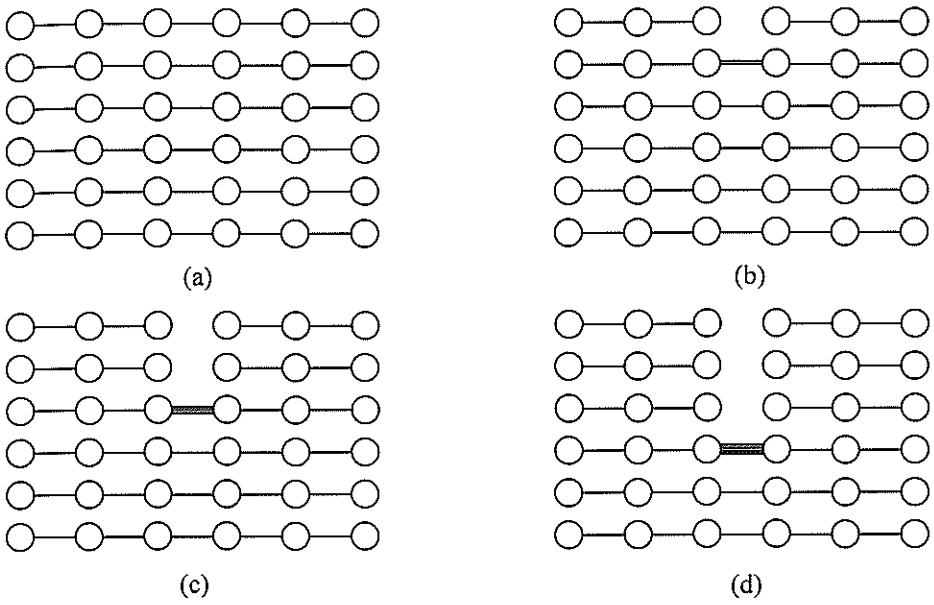


Fig 1.20: *The growth of a crack at the atomic level*

Strain energy is closely linked to the growth of cracks. When a crack forms, the strain energy held in the entire object is retained but it is released from the area adjacent to the crack. This released strain energy is now concentrated at the bottom of the crack, thus increasing the strain energy at the tip, so when cracks get larger they grow faster and faster.

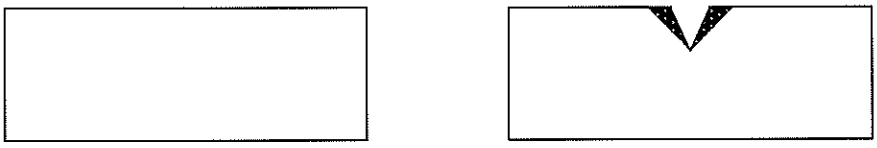


Fig. 1.21: *The left figure is one strained but without a crack, the right figure shows the growth of a crack and the release of strain energy around the crack.*

Critical Crack Length

There is a critical length that a crack must get to before it proceeds through a material. This is a particularly important factor in design of components in civil structures that use brittle materials e.g. concrete. Once this crack length is exceeded under the load that has created it, a crack will travel through a material until failure occurs.

While determining critical crack length is beyond the scope of this course it is crucial in the design of many objects, for a material will fail once this length is exceeded, even if the stress applied is lower than that which the material can theoretically allow. While students will not be expected to manipulate the critical crack length formula we can say that critical crack length is directly proportional to Young's Modulus, and inversely proportional to the applied stress. This means for a given stress a material with a higher Young's Modulus should have a longer critical crack length.

The longer the critical crack length is the less likely it is that a material will fail. As stress increases the critical crack length will decrease, so to compensate we want a material that has a high stiffness (Young's Modulus). One such material is mild steel; along with the fact that it is cheap and relatively easily shaped and fabricated this is part of the reason for its wide use.

Failure Due to Cracking

Failure due to cracking as outlined above is a brittle fracture mechanism. As a consequence some materials are more likely to crack than others. In accordance with a material's propensity to crack is its critical crack length; the more brittle a material, the shorter its critical crack length will be. Once the critical crack length is exceeded then failure is inevitable if stress levels are maintained.

One material that is greatly affected by cracking in bridges is concrete. Being a brittle material it has a far shorter critical crack length than steel, so engineers must take account of this in relation to stress levels that are placed on the material. Reinforced concrete has been developed in an attempt to overcome the fact that when a concrete beam bends, its lower surface is placed in tension and thus risks cracking. By placing steel rods in the concrete, in this area, the danger of a crack forming and growing is greatly reduced.

Repair and/or Elimination of Failure Due to Cracking

The next step is how to prevent cracking, and if it does occur, to repair it and stop it from occurring again.

In the event of crack formation we are aware that stress concentration plays a large role. Once a crack grows, the stress concentration increases, so if a crack forms, what can be done?

If the material in question is metallic then one option to repair a crack is through welding. This solution has problems, however, because although it will repair the crack, microstructural changes will appear around the weld that may further weaken the material, and the weld may be a point of stress concentration. After welding, it may be necessary to heat treat the area to avoid microstructural problems.

In polymeric materials it may be possible to use adhesive technology to repair cracking. If the failure is in a thermoset and adhesives are not available, replacement is the answer. However, for thermoplastic polymers, adhesives are not the only solution. Another option is polymer welding, which offers strengths close to the parent material.

Ceramic materials such as stone and concrete are more difficult to repair and often need to be replaced in the event of crack formation

But how do we eliminate cracks from forming? One method is to design the item without sharp corners, which tend to concentrate stress at a point. Another solution is to place interfaces within a material. An interface is an area within a material, weaker than the surrounding area that runs perpendicularly to the expected growth of the crack. When a crack travels through the material it reaches the interface but is

blocked from passing it. By placing the interface at the correct position it is possible to avoid a crack getting to the critical length for the material.

Ceramics

Structure and Property Relationships/Applications

Ceramics are a diverse field of materials that are used extensively in civil structures. The structure that the ceramic takes tends to have a considerable effect on its appearance and behaviour. Below is a list of ceramics used in civil engineering, their structures, properties and their uses.

Stone

Stone may come in various forms. In Sydney many civil structures used sandstone in the early days of European settlement because of its availability. Sandstone is a sedimentary rock that forms by the build up of layers of sediment over millions of years. The pylons at either end of the Sydney Harbour Bridge are not sandstone but granodiorite. Granodiorite is a rock with a structure between that of granite and diorite. These three are examples of igneous rocks, which form by the cooling of lava from volcanic activity. Slate was used on many early buildings for roof tiles. It is a metamorphic rock, which is either an igneous or sedimentary rock that has metamorphosed into another form because of heat and pressure.

Sedimentary rocks, such as sandstone, characteristically tend to have a "grainy" appearance, while igneous rocks tend to have greater cohesion. Igneous and metamorphic rocks are dense because of their formation from the liquid state or compaction under pressure.

All stone exhibits similar mechanical properties; it is weak in tension, strong in compression and low in toughness, i.e. brittle. This brittleness made it easy to chisel but meant that the stone structure had to be made so it was loaded in compression. Sedimentary rocks such as sandstone characteristically tend to have a "grainy" appearance, while igneous rocks tend to have greater cohesion. Igneous and metamorphic rocks tend to be very dense due to the formation from the liquid state or compaction under pressure.

Glass and Cement

The structure/property relationships of these two materials are outlined in the next two sections.

Glass

Glass has found immense use in many civil structures. Originally used sparingly in openings on stone buildings, glass now dominates the architecture of many modern buildings. Glass is an amorphous solid. That is, it does not have a regular crystalline pattern like metals. In essence it is a very viscous liquid. This structural feature makes glass transparent, which provides it with its primary function. This feature, however, comes at a cost because the amorphous structure makes glass unfavourable to shaping by force as slip of the atoms cannot occur. Therefore, shaping must be done at

elevated temperatures where the viscosity of the glass is reduced. Glass has low toughness, so it is brittle, and is weak in tension, which is why windows often break on impact. Glass shows almost perfect brittle fracture under tensile testing.

Common window glass is a mixture of silica (SiO_2) soda (Na_2O) and lime (CaO). The soda overcomes devitrification (where the glass crystallises) and the lime makes the glass non-water soluble. These glasses soften at temperatures above 850°C and are relatively cheap to make. As mentioned earlier, these glasses exhibit weakness in tension and are also brittle. This can be a problem in some civil structures where a window may be subjected to shock impact loads. To overcome this inherent weakness in tension *toughened glass* was developed.

Toughened glass is a heat treated glass that increases the toughness of a glass pane. The glass pane is heated then the outer surfaces are cooled quickly by blasts of cold air. This leaves the surface cold but the interior hot. As the interior cools it contracts and places the outer surface into compression, so to cause a tension failure in one of the surfaces the compressive force must be overcome first.

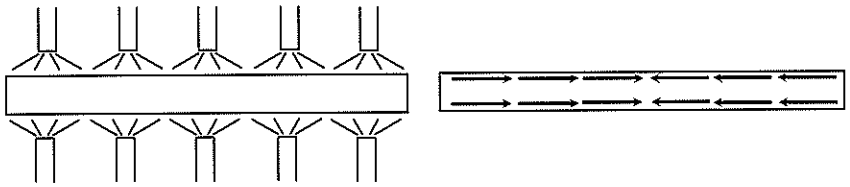


Fig. 1.22: *Toughened glass being produced with blasts of cold air at the surface. The outer layers are then placed in compression, which strengthens the glass.*

Cement

Cement is a ceramic material formed by complex reactions when alumina, soda and lime are reacted at high temperatures. Cement can come in two forms hydraulic cement, that which will harden underwater and non-hydraulic that which will not harden underwater. Portland cement, which is an hydraulic cement, is commonly used in civil structures. Like many other ceramics it is strong in compression but weak in tension, while also exhibiting low toughness. It has very similar properties to many types of stone but has the advantage of being able to be cast to almost any shape.

Cement and concrete are different although lay people use the two terms interchangeably. Cement is a ceramic material while concrete is a composite that in part consists of cement.

Cement powder is a mix of various materials. The steps for producing cement powder are listed below.

1. Limestone and shale are both crushed then mixed together, then pass through grinding mills into a kiln.
2. In the kiln, at an approximate temperature of 1480°C, the mix fuses and forms *clinker*. The clinker is now ground and stored ready for the next step.
3. The clinker is then mixed with gypsum (up to 5%) which slows down the setting time once water is added.
4. The clinker/gypsum mix is then reground and mixed to the very fine cement powder and stored in a moisture-free environment, ready for use.

Setting of Cement

When Portland cement is mixed with water a complex set of chemical reactions occur that tend to form a silicate gel. Essentially the setting involves both the evaporation of water and the formation of these silicates. This silicate gel accounts for about half the mass of the set cement and binds the other constituents together. The gel provides the strength of the set cement. These chemical reactions are exothermic which means they give off heat. This is the reason why setting cement is often hosed down during curing.

It should be noted that although cement (and concrete) dry hard with a day or two, they do not reach full bond strength for many years.

Bricks

Bricks are generally rectangular building blocks that are traditionally made of clay, but nowadays may be made form concrete. Very early construction in Sydney used sandstone blocks cut from quarries, but it is significantly more cost effective to manufacture bricks from clay. The clay is formed into the brick shape and then fired to set the clay to that permanent shape. Bricks may be pressed, recognised by being solid, or extruded where upon they will have a number of holes through them. Bricks are still used extensively in housing construction either as brick veneer or as double brick construction. Many large brick wall sections now user "besser" blocks, which are large concrete blocks, with two hollow channels in them.

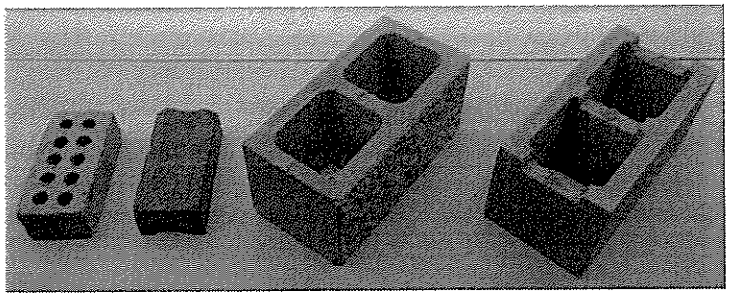


Fig. 1.23: *Four bricks; left is a clay extruded brick, centre left is a clay pressed brick and on the right are two concrete "besser" style bricks with glass fibre added.*

Composites

As we learnt in Volume 1, composite materials are where two or more materials are mixed or used together to take advantage of the best properties of each. There are however natural composites as well, such as timber.

Timber

Timber is a composite in that the cellulose fibres or *tracheids*, are held together by the *lignin*, a natural resin. The timber composite is tracheids in a lignin matrix. It was not until the 1940s that humans mirrored this type of structure to achieve lightweight composites such as fibreglass and carbon fibre.

Timber can come in two forms: softwood and hardwood. These are rather misleading terms though. For example balsa wood is quite soft yet it is a hardwood. A more

correct terminology is to call them pored (hardwood) and non-pored (softwood) timbers. Structurally they differ in that the pored timbers have pores or vessels running through the structure between the tracheids to carry nutrients, while non-pored timbers have a neater, more uniform structure without pores, just the tracheids. Pored timbers come from flowering plants (angiosperms) while non-pored timbers come from the pines and conifers (gymnosperms).

Although there are materials that now favourably compete with timber for use in civil structures, it still offers some desirable properties. It has an excellent specific strength (strength/weight ratio), and reasonable performance in bending. It also has a relatively high Young's modulus. It is, however, adversely affected by weather and also susceptible to attack from pests, such as termites. Most ancient timber structures do not exist today because of this.

Mortar

Mortar is the material used between bricks in building. Prior to the development of Portland cement, mortar consisted of slaked lime ($Ca(OH)_2$), sand and water. This creates a slow hardening paste that hardens, with the lime reacting with carbon dioxide to produce calcium carbonate ($CaCO_3$), which replaces the sand/slaked lime paste with a hard solid. Nowadays mortar contains Portland cement, sand and lime in the ratio of 3:2:1.

Concrete

Concrete and steel are two of the most important engineering materials used today. Concrete is a composite material that consists of cement, sand and aggregate (which is commonly basalt but some blast furnace slag may also be used). The sand fills the gaps between the aggregate while the cement acts as the binder or matrix that holds the sand and aggregate together. The sand and aggregate tend to bulk out the concrete and reduce the cost because of less cement being required. It so happens that they also contribute to the final strength. Concrete offers far greater strength than cement and is cheaper, so it is used extensively. There are some newer concretes that add glass fibre to the mix, to further improve the strength of the finished item.

When concrete is mixed, the ratio is usually 4 parts aggregate, 2 parts sand and 1 part cement.

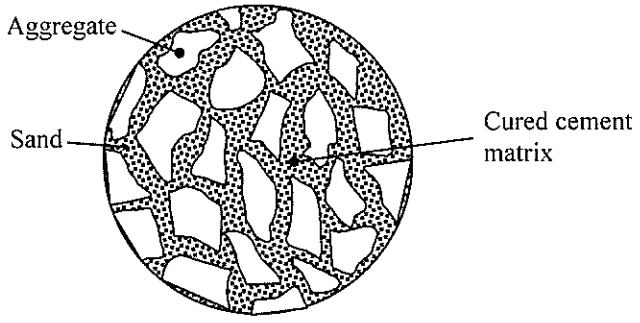


Fig. 1.24: *The microstructure of concrete. The sand fills the void between the aggregate and the cement binds the sand and aggregate together.*

Concrete behaves in a very similar way to stone, as it is strong in compression, weak in tension and has low toughness. Concrete is also fireproof and does not corrode.

The ratio of water to cement in a concrete is decided by two factors, the workability and the final strength. If there is too little water then the concrete will not be fluid enough and will not flow into the mould, while if there is too much, it will flow very well but the final strength will be low and shrinkage may become a problem. The ratio of water to cement is usually 0.45 to 0.55.

Reinforced Concrete

Because of concrete's inherent weakness in tension there may be problems when a concrete beam sags as the lower surfaces, when placed in tension, cause cracks to form. To overcome this, concrete may be reinforced with rods or steel mesh that takes the tensile load and makes the concrete more resistant to failure.

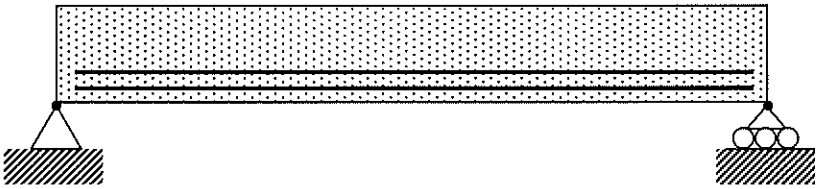


Fig. 1.25: *A reinforced concrete beam, the rods are placed on the tension side of the beam.*

Pre-stressed Reinforced Concrete

To improve the performance of reinforced concrete, two methods have been developed that place the concrete member into compression prior to loading, so for tensile failure to occur the compressive stress must first be overcome.

Pre-tensioned (pre-stressed) concrete is created when the concrete is cast over a series of steel rods or cables that have been tensioned prior to pouring. Once the concrete sets and cures the cables are released,. As they try to return to their unstrained state, they set up compressive stress in the concrete beam.

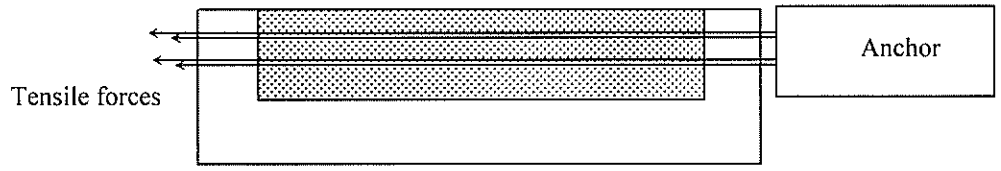


Fig. 1.26: *A pre-tensioned pre-stressed concrete set up. The concrete is cast over the tensioned rods. Release of these will place the concrete in compression.*

Post-tensioned (post-stressed) concrete is formed when concrete is cast with tubes running through the slab. After setting and curing, wires are pulled through the slab and anchored to plates at one end and tensioned at the other. This tensioning tends to compress the slab and place it into a state of compressive stress. Once the required tension is gained, the wires are anchored to the slab at the tensioning end. The process is completed when the cables are left in this state and cement slurry is injected into the tubes to stop corrosion of the steel.

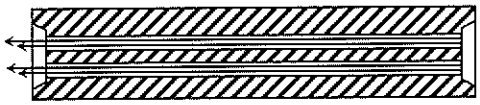


Fig. 1.27: *A post-tensioned pre-stressed concrete slab. Tension wires are passed through the tubes and are locked into place at the free end and the slab is placed in compression.*

Spalling (Concrete "Cancer")

Reinforced concrete is not without its faults, one of which is spalling, also called concrete cancer. This happens when the reinforcing steel corrodes. When steel corrodes it expands which causes the concrete to crack, hence making the structure unstable and dangerous. Careful design and not placing reinforcing too close to the surface can reduce spalling.

The steel can corrode due to the presence of salt, air and water. This is why it is essential to ensure a correct water ratio, and to vibrate the concrete into position to reduce porosity.

Asphalt

Also called tarmacadam (or tarmac), this is the most widely used surfacing for roads in Australia. Asphalt is a composite with hard aggregate (often crushed blast furnace slag or basalt) and bitumen as the matrix. It is advantageous for roads as it is tough and crack resistant due to the bitumen, yet hard wearing due to the exposed aggregate. It is also impervious to contamination by oil. Asphalt is laid hot and when it cools the bitumen solidifies. It can deal with slight movements in the road base better than concrete can, because of its toughness.

Laminates

Laminates are materials that consist of varying materials sandwiched together. Examples of laminates are listed below.

Plywood is a laminate that is useful where the grain structure of timber causes weakness. It consists of layers of timber with the grain arranged at 90° to each successive layer.

Laminated Veneer Lumber (LVL) is similar to plywood in appearance but the veneers are not arranged at 90° to the adjacent one, hence each veneer runs along the beam. LVLs have become popular in construction; they are stronger than an equivalent piece of solid timber, and are less susceptible to shrinkage or warping.

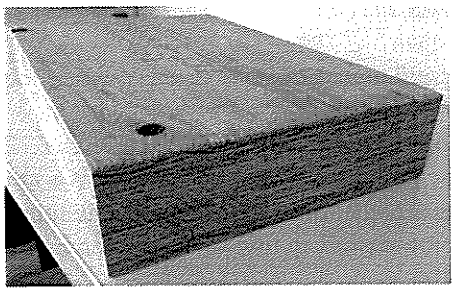


Fig. 1.28: An LVL offcut from a standard beam.

Laminated glass is used when a shatter-resistant glass is needed. Two layers of glass are passed through rollers that compress a polymer sheet (usually polyvinyl butyral) lying between them. This creates a glass that will not shatter, making it more resistant to breaking into shards after impact.

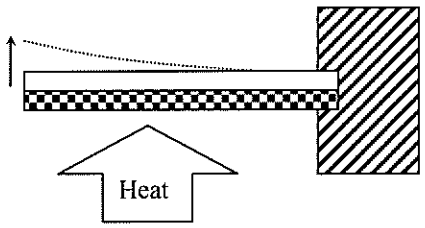


Fig. 1.29: *When heat is applied to a bimetallic strip one side will expand more than the other and deflect from centre.*

Bimetallic strips use two metals back to back. One metal will have a different thermal expansion rate to another so as it heats up it will deflect from its neutral position. This makes it highly useful for thermostats and protection circuits in gas systems.

Geotextiles

Geotextiles are woven polymers (usually polypropylene or polyester) or ceramic fibres that are used for a variety of purposes in civil construction. One application for geotextiles is for stabilising road base. When a road is constructed the asphalt is laid over a fine crushed material called road base. Over time the road base will often become contaminated by the clay, rock or soil it is resting on. To prevent this, geotextile sheet is placed between the subsoil and the road base, thus stabilising the road surface and making it less likely to form potholes.

Geotextiles are also used as filters for drainage systems. The NSW railways use geotextiles in drainages systems so the drains under the tracks do not fill up with silt and then clog. Geotextiles are also used to stop the road base around the drainage pipes from contaminating the ballast upon which the tracks rest. Geotextiles are also used on construction sites to stop dirt in water runoff from getting into storm water drains.

Corrosion

Corrosion may be defined as the chemical deterioration of a material. Although we often think of corrosion as only occurring in metals, ceramics and polymers also deteriorate or degrade, usually involving the breaking down of the material by chemical means such as the ultraviolet deterioration of a polymer.
Metallic corrosion involves the deterioration of metals or metal alloys, and is basically the reverse of the refining process. In nature, most metals exist as chemical compounds. Refining them produces the pure metal, while corrosion progressively returns them to the combined state.

Corrosion involves complex chemistry. The following section is an overview. More information may be found in references at the end of this chapter.

Oxidation and Reduction

Oxidation occurs when the metal loses electrons, and it occurs at the anode. Oxidation = loss.

Reduction is the consumption of electrons and occurs at the cathode. It is the opposite to oxidation thus reduction = gain.

An acronym to remember this is OILRIG (Oxidation Is Loss, Reduction Is Gain)

Some metals are more likely to be anodic while others are more likely to be cathodic. Cathodic metals tend to be more stable and less affected by corrosion. Table 1.3 lists metals and their reaction when immersed in seawater.

Dry Corrosion

Dry corrosion occurs through chemical reactions of metals or alloys with gases, in furnaces at high temperatures. Examples include fire tubes in steam locomotive boilers, water tubes in power station boilers and metal components in heat treatment furnaces. A principal cause is a reaction of the metal with oxygen and other molecules in flue gases.

Wet Corrosion

Wet corrosion occurs when a metal is placed into a fluid, usually an electrolyte. An electrolyte is a solution containing ions which carry electric charges.

Uniform attack

If a metal is placed in an electrolyte, some parts will become anodic while others will become cathodic. The locations of the cathode and anode will continually change, resulting in uniform corrosion. Steel, under certain circumstances, will form a uniform layer of rust as it corrodes.

Galvanic Attack

This type of corrosion comes in three forms,

Galvanic corrosion occurs when dissimilar metals are placed together in the presence of a corrosive environment. Different metals have a greater affinity to corrode than others; so one metal will become cathodic, while the other becomes anodic. Table 1.3 is useful for determining which metal will become the anode or cathode.

Consider a steel pipe with a copper sleeve over it. The steel pipe under the sleeve will corrode as copper is more cathodic than steel. Composition cells also occur microstructurally, for example ferrite in steel will become the anode, while cementite will become the cathode.

Concentration cells occur when there is a difference in concentration of the electrolyte. A good example is the car door shown in Fig 1.29.

Table 1.3: *The Galvanic series in Seawater*

Anodic	Magnesium
	Zinc
	Galvanised Steel
	Aluminium Alloys
	Mild Steel, Cast iron
	Stainless Steel (active)
	Tin
	Aluminium Bronze
	Copper
	Nickel base alloys (passive)
	Stainless Steels (passive)
	Silver
	Titanium
	Graphite
	Gold
Cathodic	Platinum

In the car door, the trapped water will become stagnant and the bottom of the pool will become low in oxygen (O_2), while the surface will have higher levels of oxygen. This differential in oxygen levels sets up a concentration cell and corrosion will occur. High O_2 will create a cathode while the area of low O_2 will become the anode. So the lower part of the door will lose electrons and deteriorate, while the cathode gains electrons and rust layers will build up. The result will be the eventual destruction of the door from the inside out. This is why a plastic coating on the outside does not stop the rust problem.

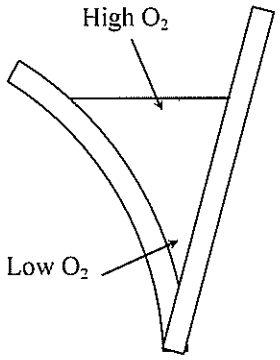


Fig. 1.30: *The bottom of a car door can hold water in dust and dirt. When this occurs a concentration cell develops and corrosion begins at the water surface.*

Concentration cells take a particularly damaging form as *crevice corrosion* in structures. Like the car door example above, crevice corrosion is where an electrolyte fills a crevice and we get different oxygen levels between the top and bottom of the crevice. Crevice corrosion can be problematic on any metal structure where joints may crate crevices.

Stress cells are the result of high residual stress in parts of a metal object. These areas of high stress tend to become anodic while the lower stress areas become cathodic.

Often a stressed metal will corrode when an unstressed piece in similar conditions will not corrode as quickly.

Grain boundaries in metals tend to be a location of high stress and thus a metal with a fine grain is more liable to corrode internally than a coarse grained metal. This situation differs from a composition cell in the microstructure of steel, as stress corrosion at grain boundaries can occur in a pure metal where all grains have the same composition.

Corrosion – is it all bad?

Corrosion is always going to occur in metals. The effect corrosion has can be horrendous and so a lot of time and money is spent combating it. "Corrosion" is most often associated with steel and rust, and it is a problem that engineers have not yet overcome.
Some metals like steel will form a porous layer (rust) that allows corrosion to continue. Others, like aluminium, form a protective film that makes the metal passive and inert. Metals in this situation display a condition called *passivity*. Passivity is a condition where a metal that should be quite reactive appears inert. Metals exhibiting passivity have a corrosion product that is a non-porous tenacious film that protects the underlying metal from further corrosion. The film is self-healing, so when scratched it immediately reforms. Examples of passive metals are aluminium, titanium and chromium, while a passive alloy is stainless steel.

Rust is simply the name given to the corrosion product of steel. It is the antithesis of the corrosion products on passive metals. It is porous and flaky and as it grows and flakes it continually exposes fresh corrodible metal beneath. This is why most steel structures must be protected from the atmosphere or corrosive environments to ensure longevity.

Protecting Civil Structures

Most steel civil structures such as bridges use one of two protective mechanisms to stop corrosion. One is paint on the surface, e.g. The Sydney Harbour Bridge. Another method is to galvanise the steel pieces. Galvanising involves dipping the steel part into molten zinc, which then covers the steel and protects it from corrosion. To be technically correct the zinc coating is not passive, but very slowly corrodes away. Once it has corroded away after many years, the steel will then rust, but until that time it is readily protected.

To enhance the corrosion resistance of galvanised steel, other products like Zincalume® have been developed. Zincalume® coats the steel in a zinc/aluminium alloy which offers improved protection. Colorbond® steel uses a Zincalume® base and adds a polyester coating and then baked on colour top coat, to produce a decorative and durable steel sheeting product.

Some sheet aluminium alloy applications use Alclad™, which is duralumin (aluminium/copper alloy) coated with pure aluminium. The passive aluminium protects the more corrosive-susceptible alloy. Aluminium alloy window frames are normally anodised which involves creating a thick oxide layer to further enhance the oxide aluminium naturally forms.

Cathodic Protection

It is possible to protect civil structures and other objects by a process called cathodic protection. Here the object to be protected is made the cathode and hence protection from corroding.

Sacrificial anodes, these are blocks bolted to a metal such as steel which would normally corrode, but by using more reactive materials for the sacrificial anodes such as zinc, aluminium or magnesium, the steel becomes cathodic to the anode. Sacrificial anodes find extensive use in boat hulls to stop the steel hull corroding in the harsh marine environment.

Impressed Current Protection Systems (ICCP), uses a current to reverse the standard electrical flow associated with corrosion, and by reversing the current flow the structure or object, usually steel, becomes the cathode. It finds use in pipelines and long structures where sacrificial anodes would be ineffective over such a large surface.

Communication

Orthogonal drawing

There is a plethora of drawing texts available that should serve all needs. Please refer to the *Additional References* section to find some suitable drawing references. To augment your personal research, it is strongly suggested that you attain copies of previous Engineering Science orthogonal drawing questions and solutions.

In nearly every HSC students will be asked to complete some form of assembly orthogonal drawing. Students must consult previous HSC exam questions on this to get a feel for doing assembly drawings of various difficulty levels.

Threads in Orthogonal Drawing

Threaded features such as bolts and nuts are drawn using specific AS1100 standards. Below is a brief guide to drawing bolts and nuts, further consultation should be made to the AS1100 handbook.

When dealing with threads the size details of the thread are normally given as follows:

M10×1

- M represents metric, so we know it is a metric thread,
- 10 in this case is the diameter of the thread; and,
- 1 is the pitch of the thread.

So a thread described as M12×1.5 is a metric thread of diameter 12 mm and a pitch of 1.5 mm. The numbers are important when we are drawing threads to AS1100 standards because the size of the bolt head and nut is decided by the thread's diameter and the pitch decides the thread depth.

The AS1100 standard dimensions for hexagonal head bolts, hexagonal nuts and flat washers are outlines in Fig. 1.31.

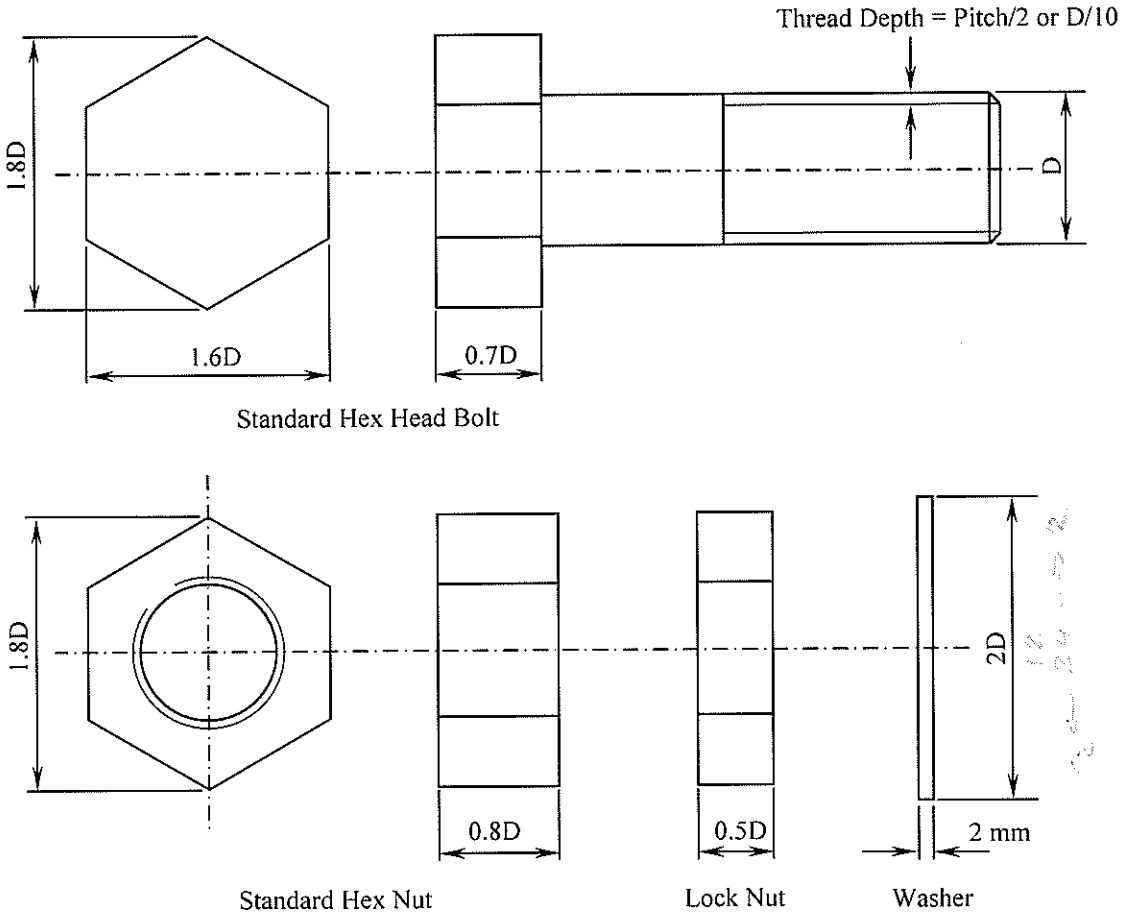


Fig 1.31: *AS1100 standard dimensions for screw thread representations. Note that the washer's thickness drops to 1 mm for thread diameters below 8 mm. Pitch/2 is the preferred way to calculate the thread depth.*

Orthogonal Drawing Features

Counter bores are used in engineering to allow a bolt head to be recessed below the face of an object. This is done by creating a lager diameter hole at the top of an existing hole through an object. Countersinking also allows a recessed head. Both of these features and their standard dimensioning methods are shown in Fig. 1.32.

Computer Graphics

Computer Aided Drawing

CAD software has created massive advances in engineering communication. A software package such as PTC Creo, which is used by quite a number of high schools, is also used by large engineering companies like Toyota, Nokia and Raytheon. The software is not just a drawing program, it also allows features to be tested, allows engineers to roll back to previous designs, allows worldwide collaboration and also allows engineering material and mechanical analysis. All of this makes CAD an indispensible feature in Engineering.

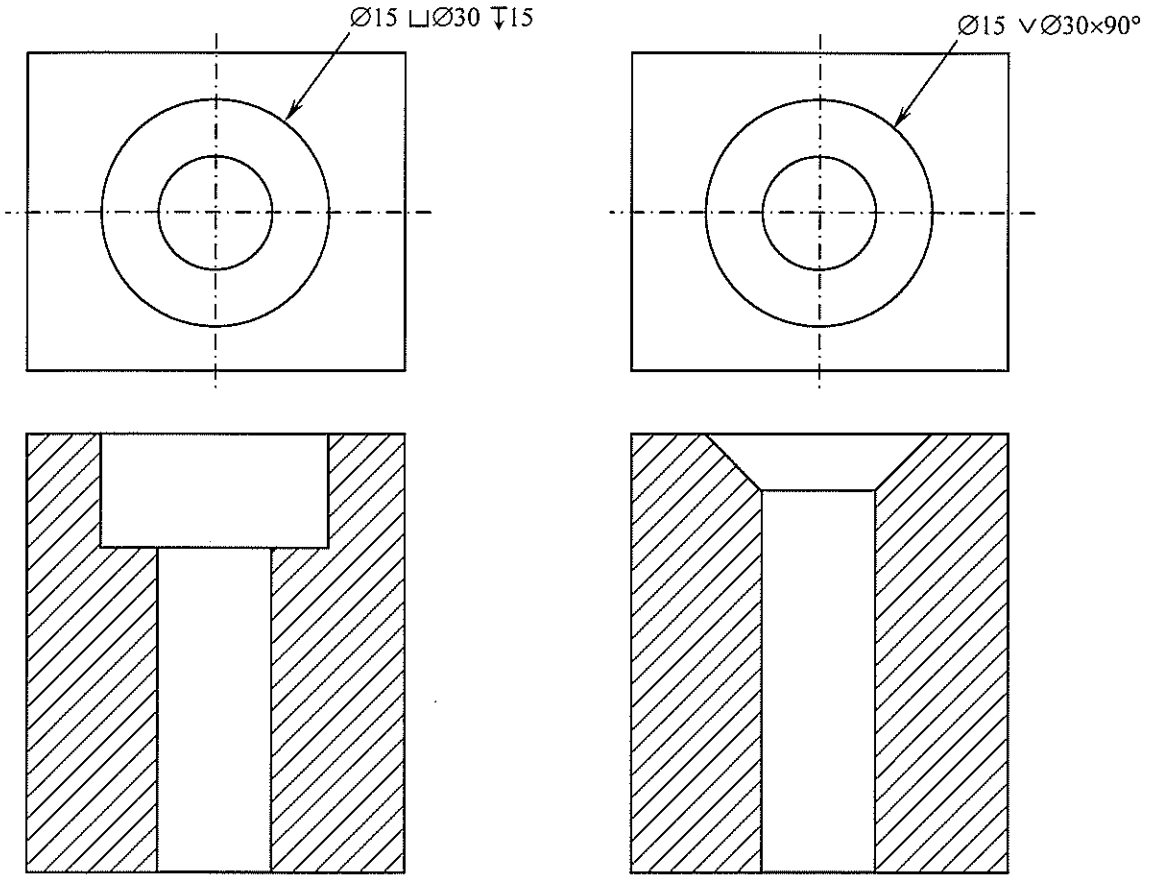


Fig. 1.32: *Sectioned views of a counter bore hole and countersunk hole. Dimensioning standards for them are shown.*

Other CAD packages include but are not limited to, SolidEdge, AutoCAD, TurboCAD and Vectorworks. All of these along with Creo have their own way of approaching engineering communication but their common goal is to facilitate effective communication of engineering ideas, in a way that is time consuming and difficult to replicate with manual drawing techniques.

Problem solving Applications

There is now a range of applications about that allow the analysis of force systems. Some of these, such as Autodesk ForceEffect, are marketed for tablets so they are available for the engineer on the move. Others offer different analysis methods. But all offer something that twenty years ago were the preserve of high level computing. Now a smartphone can allow truss analysis.

Engineering Report

In Volume 1 of this book the Engineering Reports were limited to a fairly simple approach using a comparison of engineered items. For the HSC course, however, we need to lift the reports to a higher level, using a more formal structure.

HSC Online provides an excellent range of guides for engineering report writing here: http://hsc.csu.edu.au/engineering_studies/reports/

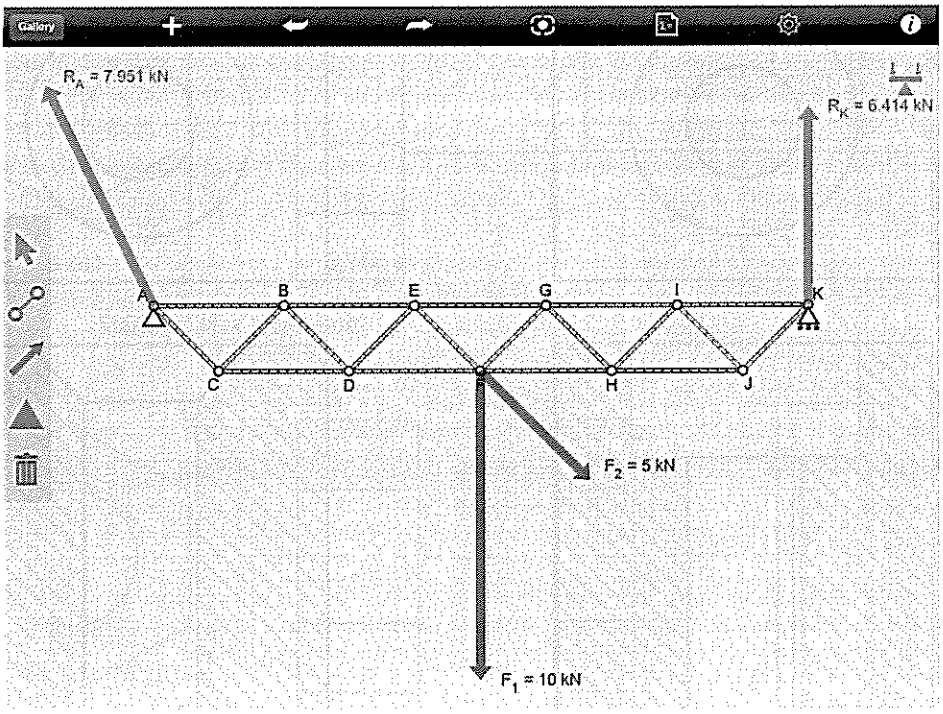


Fig 1.33: *A truss analysed using Autodesk ForceEffect.*

Below is a list of headings that your engineering report could be based on.

Title page: This should list the title of the report and your or your group's name. A title page should be visually appealing. Make sure it presents your work well without overdoing it.

Synopsis: This is a summary of the content and aim of the report. The synopsis should provide enough information to give an overview, or outline, of the primary points of the report.

Introduction: The introduction sets the scene for the report by giving the subject, purpose and parameters of the report. You may also include a background for the report such as some history of the topic.

Methodology: This is the description of the method or process used to carry out the work required for the report. If you have had to conduct experiments then this section is where you outline the equipment and procedure used.

Results: It is here that the outcome of the investigation or experiment is recorded. It should include figures, graphs and perhaps tables that have allowed you to gain the results. Do not load up the results sections with a lot of calculations, pictures or drawings. It is better to place these in an appendix at the end and then refer to the appendix.

Conclusion: This is the resolution of the report, the judgement made from the evidence in the results section. It may be the choice of which alloy steel is best for reinforcing concrete, or which metal tube is best for bicycle manufacture.

Recommendation: After the report is concluded a recommendation may be made as to the final choice, such as, "The most appropriate tubing for the bicycle is an air hardening steel, such as Reynolds 853."

Acknowledgements: If somebody has helped you in any way in the preparation of your report, their names should be included here. This section is a good place to add names of people who have helped you obtain information from companies you contacted. Give these people the credit they deserve. It is unethical to use other people's work without due acknowledgement.

References: If you gain any information from books or journals then they should be included here. At university this will become an exceedingly important task, as to claim work that is not yours, is academic misconduct and may result in expulsion. Use a similar style for referencing to the method below, which is used by many tertiary institutions,

> Author Name/s. (Year), *Title of Book*, Location, Publisher. For example,
>
> Copeland, Paul L. (2011) *Engineering Studies – The Definitive Guide*. Sydney, Anno Domini 2000

Academic fraud or plagiarism is when you claim as yours work that is actually somebody else's. If you use somebody's work exactly as written you must quote him or her, while if you use his or her work as a reference then you must list it as a reference. With free online resources like Wikipedia it is often appealing to just cut and paste work into your document. This is fraudulent and should not be done. Make sure the report is your own work.

Moreover as digital citizens you must learn to be discerning consumers of the information you read. SAs outlines in Volume 1 you must use Wikipedia and similar resources sensibly, any online information that is not validly referenced must be treated with suspicion.

Appendix: Finally, at the very end, include any appendices, with calculations, drawings etc. This means that the readers do not have to read this information but it is there if required.

Presentation – Or… "Does this look OK?"

Many people will give very different views on how to present your report. These are some tips that could be useful.

- When you write headings do not use all capitals. They are hard to read. For example, try to read "LAUGH" from a distance, then try to read "laugh" with the font enlarged to a similar size.
- Underlining is out of date. Underlining affects the readability of words as it cuts through descenders (the bottom of letters that run under the line) and it

tends to run all letters in the word together. A better option is to use *italics* or **bold** letters. Moreover nowadays underlining indicates a hyperlink.

- Fancy fonts or typefaces may look good on a party invitation but they are often harder to read. They also tend to make your report look too informal.
- Don't write the main body of your report in a font any larger than 12 point. Larger fonts look as if you are trying to pad out work that is insufficient.
- When choosing a font for the body of the report choose a serif font. A serif font is one with little tails on the end of the letters like the Times New Roman font used in this book. Sans serif fonts like Arial are not as easy to read in large blocks of writing. They are, however, good for headings and now find extensive use in websites where serif fonts look "fussy", particularly on modern LCD screens.
- When you write make sure you use a formal writing style. That means not using colloquialisms and limiting your use of contractions ('do not' instead of 'don't'). Do not write in the first person, i.e. do not write "I have conducted an experiment..."; rather say "An experiment was conducted...". Ensure that all sentences are in the same tense.
- In most word processors, many words that are spelt correctly will be listed as incorrect by the spell checker because spell checkers are automatically set to American spelling. Try to set the computer to British spelling (not Australian as the Australian spell checker is often, incorrectly, the same as American), and watch for words like 'aluminum' and 'airplane' in American texts. In Australia we say 'aluminium' and 'aeroplane'.
- Make sure you use the correct punctuation for the sentence; for example, a semi colon (;) is used when a sentence changes subject, not at the end of a sentence prior to a list.

Additional References

This list of useful references allows students to carry out further investigation. Any highly recommended book or website has two stars next to it.

Engineering Mechanics and Hydraulics

**Gordon, J. E., (1978), *Structures – or Why things don't fall down*, Da Capo Press, New York.
Ivanoff, V., (1996), *Engineering Mechanics*, McGraw Hill, Sydney.
**Schlenker, B. and McKern, D., (1976), *Introduction to Engineering Mechanics*, Jacaranda Wiley, Milton.

Engineering Materials

Askeland, D., (1989), *The Science and Engineering of Materials*, Chapman and Hall, London.
DeGarmo, E. P., Black, J. T. and Kohser, R. A., (1988), *Materials and Processes in Manufacturing 7th Edition*, Macmillan Publishing Company, New York.
**Gordon, J. E., (1974), *The New Science of Strong Materials – or Why you don't fall through the floor*, Penguin, Middlesex.
**Higgins, R., (1997), *Materials for the Engineering Technician 3rd Edition*, Arnold, London.
Schlenker, B., (1974), *Introduction to Materials Science*, Jacaranda Wiley, Milton.

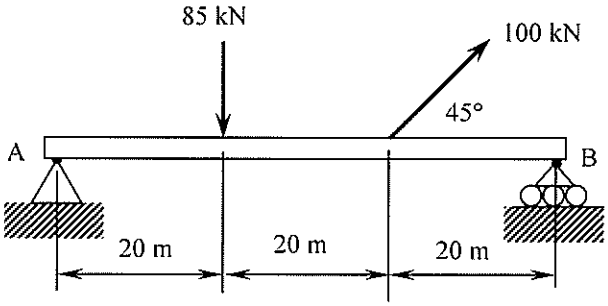
Smith, W., (1990), *Principles of Materials Science and Engineering*, McGraw Hill, New York.

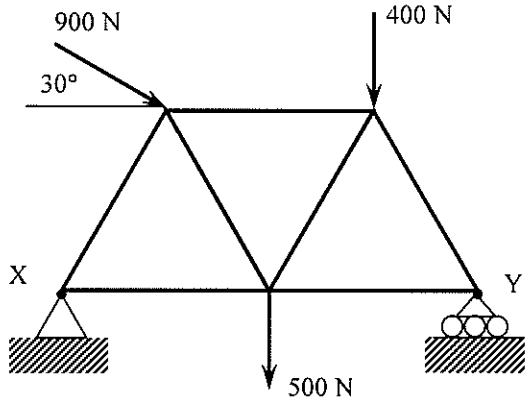
Communication

**Boundy, A., (1991), *Engineering Drawing*, McGraw Hill, Sydney.
Boundy, A. & Hass, I. L., (1996), *Graphics 2000 Book 2*, McGraw Hill, Sydney.
Mullins, R. K. & Cooper, D. A., (1990), *Programmed Technical Drawing Book 3*, Longman Cheshire, Melbourne.
Park, A, Dodds, K and Bland, S., (1989), *Technology Drawing*, Longman Cheshire, Melbourne.

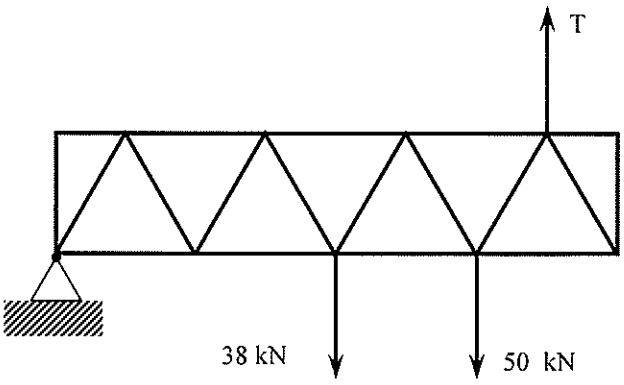
Review Questions

1. Explain what a beam is.
2. Explain the difference between a tied arch and an arch.
3. Why are suspension bridges usually lighter than equivalent span arch bridges?
4. What made cast iron appealing as a bridge building material?
5. What is significant about the Brooklyn (USA) Bridge in terms of suspension bridge design?
6. Why was the mass production of steel an important development in bridge design?
7. Why is concrete so popular in bridge construction?
8. Why do bridges not use roller supports at both ends?
9. Find the reactions for the following beam and truss?

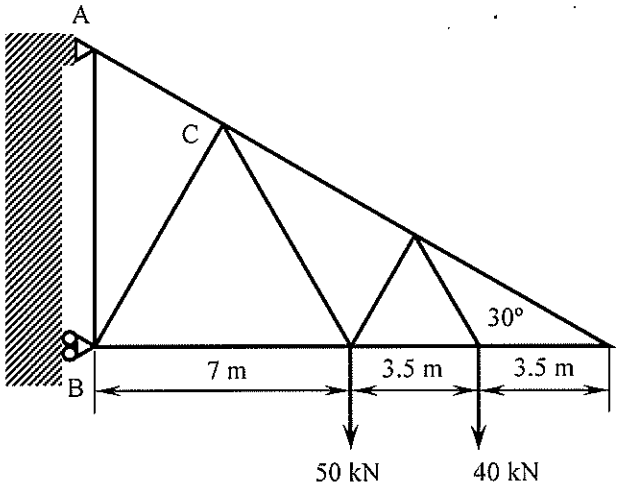




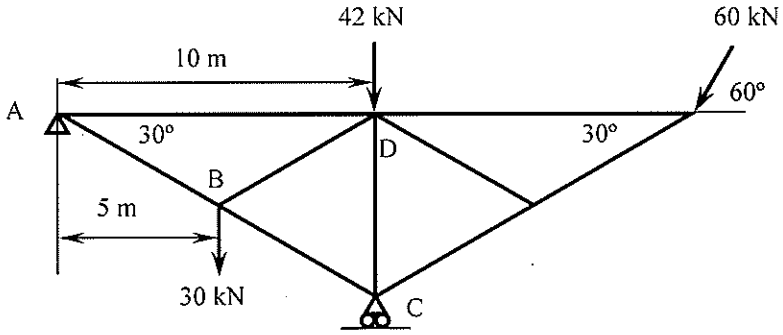
10. A truss for a gantry is anchored at one end and supported by a cable on the right hand end. Find the tension in the cable and the reaction at the support.



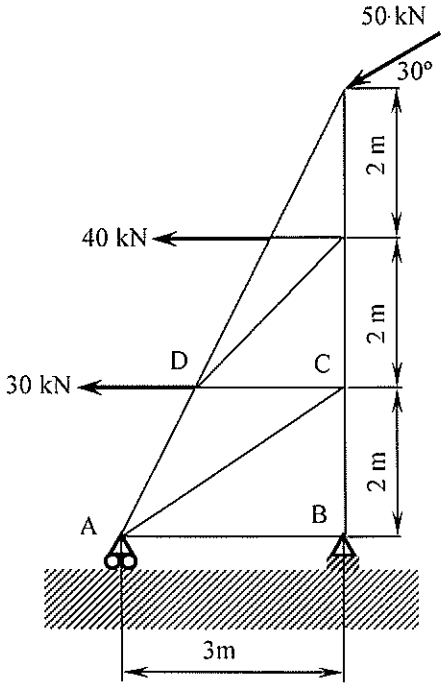
11. A triangular frame, shown below, is loaded as shown. Find the reactions at the supports and the force in member AB.



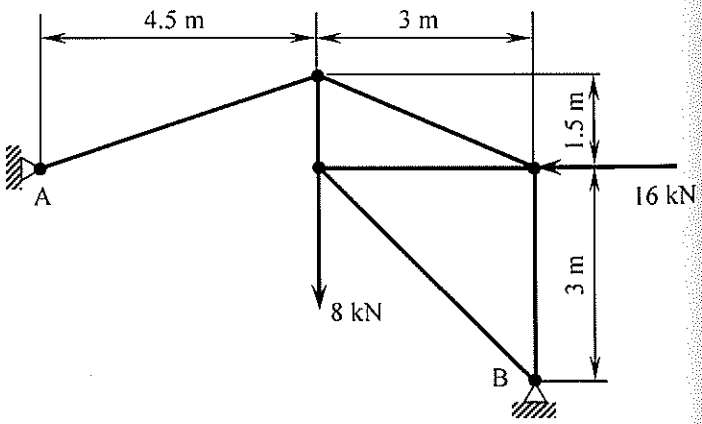
12. Find the reactions at the supports for the truss shown, and the force in members BC and BD.



Question 12

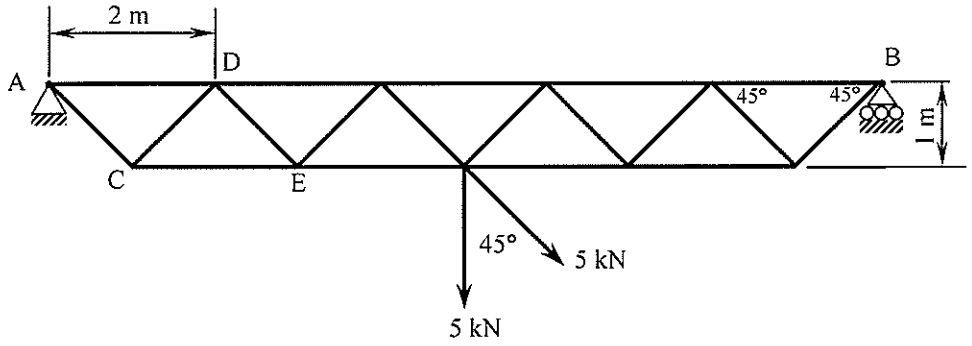


Question 13

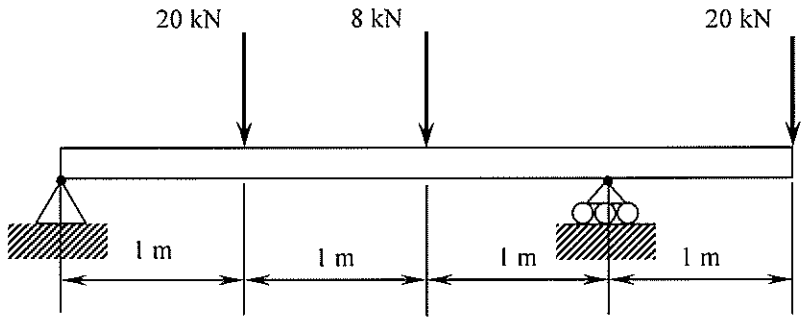


Question 14

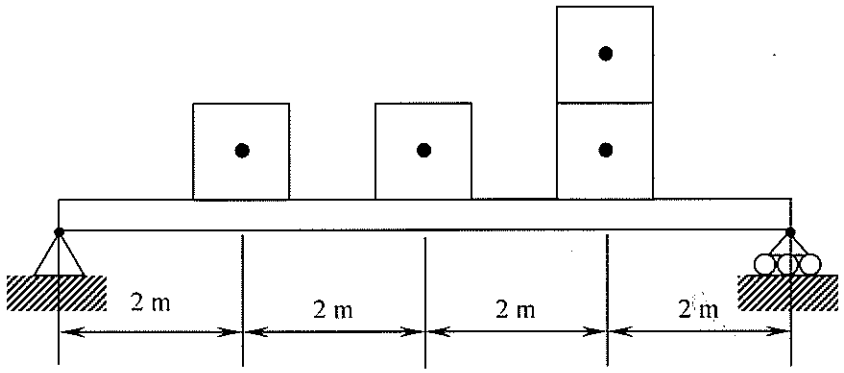
13. Find the reactions at the supports and the force in member DC.
14. Determine the reactions at A and B for the frame shown.
15. Shown below is a 10 m truss for a roof gantry. Determine the reactions at A and B and then determine the magnitude and nature of the forces in DE and CE.



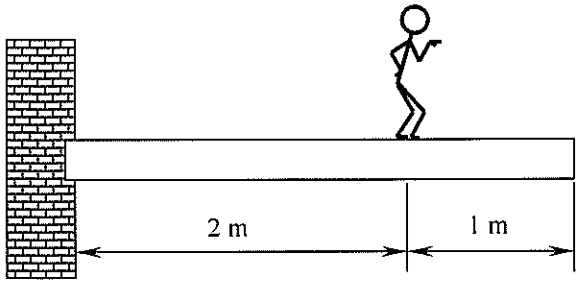
16. Produce the shear force and bending moment diagrams for the beam below.



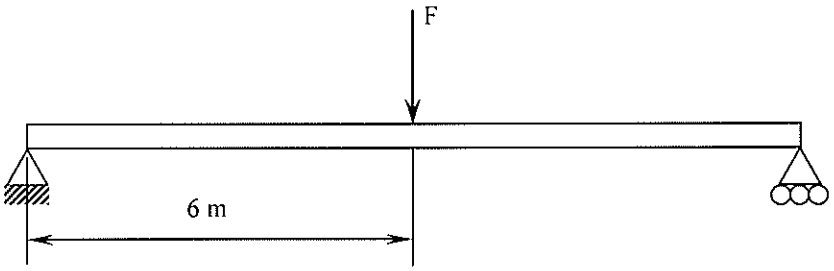
17. Draw the shear force and bending moment diagram for the beam below if each box has a mass of 25 kg (Ignore the weight of the beam).



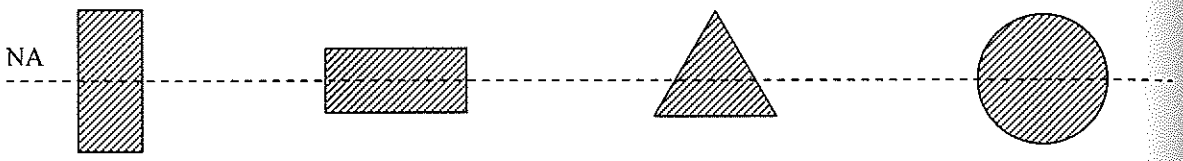
18. Draw the shear force and bending moment diagrams for the cantilever if an 80 kg person is standing 1m from the free end.



19. An engineer is planning a 12 metre beam that will be arranged as shown and is yet to decide the cross sectional shape (all cross sections have the same area). If the maximum permissible bending stress in the beam is 25 MPa, find the maximum load that the different cross sections will allow.

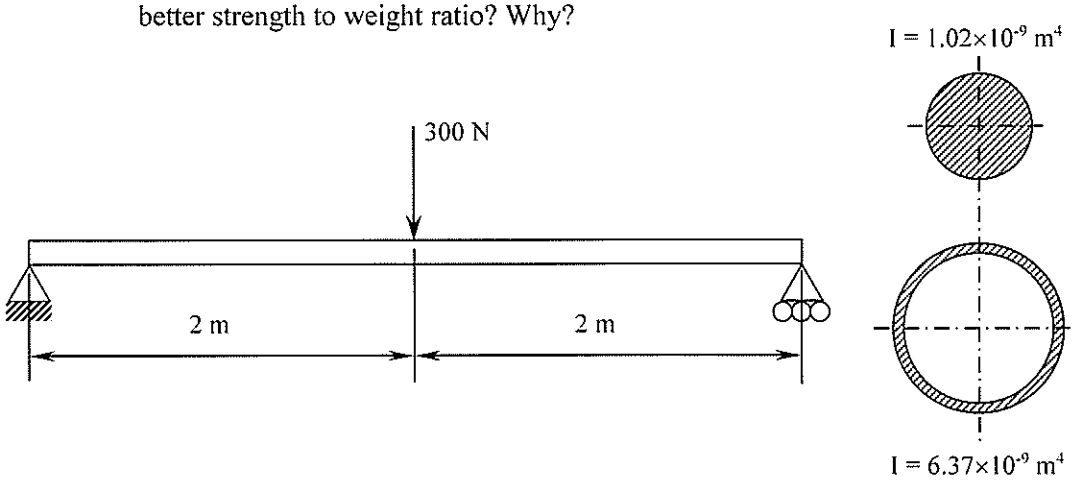


Cross sections



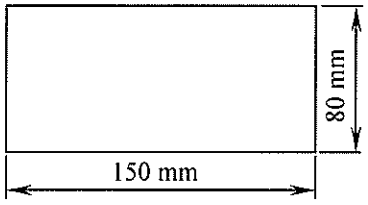
B×D = 50 ×175 mm	B×D = 175 ×50 mm	B×D = 142.15 ×123.10 mm	∅ = 105.5 mm
$I = 2.23\times10^{-5}$ m^4	$I = 1.82\times10^{-6}$ m^4	$I = 7.37\times10^{-6}$ m^4	$I = 6.09\times10^{-6}$ m^4
y = 87.5 mm	y = 25 mm	y = 41.03 mm	y = 52.75 mm

20. An engineer has the choice of either ∅12 mm solid bar or ∅25 mm tube (1.2 mm wall thickness). Find the maximum bending stress in the beam for each cross section when the beam is loaded as shown. Which material will have a better strength to weight ratio? Why?

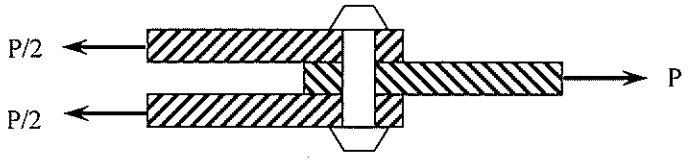


21. A steel bridge cable carries a load of 200 kN. What will be the tensile stress if the cable has a diameter of 20 mm?
22. A punching machine can produce a shear force of 5kN. If a plate with holes in it is to be punched in one operation, and the plate is 100mm by 70mm, with two 15mm diameter holes in it, what will be the maximum thickness of the plate? (Plate shear stress is 1MPa).
23. A bolt used in a bridge is subjected to a 40kN shear force perpendicular to its axis. Determine the shear stress if the bolt has a diameter of 18mm.

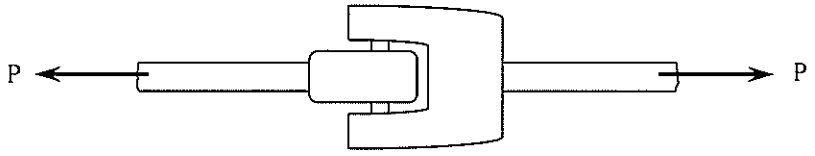
24. A shape, as shown below, is punched from a 5 mm thick steel plate by a 20 kN force. What will be the maximum shear stress induced in the plate?



25. Explain engineering stress.
26. What is Hooke's law?
27. What is proof stress?
28. The ultimate shear strength for steel is 360 MPa. A mild steel pin is used in the coupling shown below to hold the joint of a truss together. If the diameter of the pin is 10 mm what will the maximum load permissible be if a factor of safety of 6 is used?



29. A coupling on a bridge is shown below. The device is subjected to a shear load of 20 kN, and the maximum shear stress allowed in the bolt is 500 MPa. What will the minimum diameter of the bolt be?



30. Explain the procedures of X-ray, dye penetration and ultrasonic testing.
31. Why is the critical crack length of a material important?
32. Why is cracking more likely to occur once a crack has started to form?
33. What is an interface in relation to cracking?
34. Define glass.
35. What materials are used to make glass?
36. What is toughened glass?
37. What is the difference between cement and concrete?
38. Explain what mortar is.
39. What is the reason for reinforcing concrete?
40. Describe the two types of pre-stressed reinforced concrete.
41. Explain what asphalt is.
42. How does laminated glass differ from normal glass?
43. What is a geotextile?
44. Describe what is meant by the term corrosion.
45. Explain the difference between oxidation and reduction.
46. Explain the three types of galvanic corrosion.
47. Describe what crevice corrosion is and explain why it is a problem for civil structures.
48. How may steel structures be protected from corroding?

CHAPTER 2

Engineering Application Module 2 – Personal and Public Transport

Historical and Societal Influences

Historical Developments in Transport Systems

Cycle development

The bicycle is an interesting machine to have developed. Prior to it, nothing existed that operated on the principle of balanced travel and it has only extended to the motorcycle. The car is merely a progression from the horse drawn cart, while the truck is a development of the old bullock teams. **The bicycle, however, marked the beginning of a new style of transport.**

Early Machines

The invention of the bicycle has been credited to a Frenchman, le Comte de Sivrac, who introduced his machine, the *célérifère*, at the Palais Royal Gardens (Jardin du Palais Royale), in Paris, in 1791. He put wheels on what had been a child's toy, the rocking horse.

The célérifère was renamed the *velocipede* or dandy-horse and efforts were made to improve the appearance by making it resemble animals such as lions, horses or even dragons. In 1816, a German, Baron Karl von Drais von Sauerbron improved the original velocipede by fitting a steerable front wheel, armrests, a padded seat and even a primitive rear wheel brake. His machine was called a *draisienne* and was still propelled by walking it along the ground. The draisienne became popular, especially in England, where it became known as the hobby-horse.

The method of human propulsion on the early machines was hardly ideal. The introduction of mechanical propulsion was sure to come. In 1821 an Englishman, Lewis Compertz, designed a rack and pinion (a cog meshing with a toothed rack) system so the arms could be used to assist the feet pushing along the ground. It was not a perfect solution but definitely an improvement.

Pedal Power

A Scottish blacksmith, Kirkpartick Macmillan, developed a pedal drive in 1839. He used a pair of hanging stirrup pedals, attached to long arms, that connected to cranks on the rear wheel. It was similar in principle to the locomotive drive, with the backwards and forwards motion of the pedals converted into rotation by the cranks on the back wheel. Gearing was determined by the size of the rear wheel; the larger the wheel, the higher the gear. Macmillan's machine, sometimes called Macmillan's Velocipede, weighed just over 30 kg. It was archaic in concept and construction but it now enabled a rider to travel without having to push with his feet on the ground.

Rear wheel drive was not to become popular on bikes for a long time. The next stage of bicycle development was front wheel drive. In 1861 Monsieur Brunel took his draisienne for repairs to a coach maker, Pierre Michaux. Michaux's son, Ernest, suggested that the machine could be improved by fitting cranks, with pedals on the ends, to the front wheel so that it could be pedalled.

By 1867, at the Paris Exhibition, this new bicycle, the velocipede (the cynics called it "boneshaker") had became, an established mode of transport. Michaux opened a factory and employed 300 workers to produce these bicycles. Soon three-wheeled, four-wheeled, tandem and even triplet versions of the velocipede appeared. Schools for riding sprang up, and organised races were introduced.

Lady cyclists on "boneshakers" were among the first women to cast skirts aside and appear in public showing stocking covered legs. "Bloomers" for cycling, based on baggy pantaloons were popularised by Mrs Amelia Bloomer in America. In 1874 Englishman James Starley, famous in the cycling world, patented the first 'lady's' bicycle. It was ridden side-saddle with one pedal. Starley also invented the spoke, radiating at an angle from the wheel hub.

The "Old Ordinary" (Penny-Farthing)

As with the car the speed of bicycles has been something that has motivated new designs. The problem with the velocipede was that it moved along only one wheel circumference per turn of the pedals, because there was no variable gearing. Before gears, the solution was to increase the front wheel diameter so that, for each rotation of the wheel, the bike went further. Wheels became larger and larger with the only limitation being the length of the rider's leg. The new bicycle had a large front wheel up to 1.52 m (5 feet) in diameter and a small rear wheel. Released in 1870, it became known as the *Old Ordinary* or *Penny Farthing* (two British coins). In retrospect the Old Ordinary was not a particularly feasible design. It was unstable and when braking heavily it usually sent the rider over the handlebars. Descending hills was a thrilling, if not dangerous, experience as the pedals span wildly. To counter this final problem, footrests were fitted high on the front forks so that the rider could free his/her feet from the pedals. Although they were tricky to ride, they were the fastest things on two wheels until the chain driven 'safety' bicycle appeared.

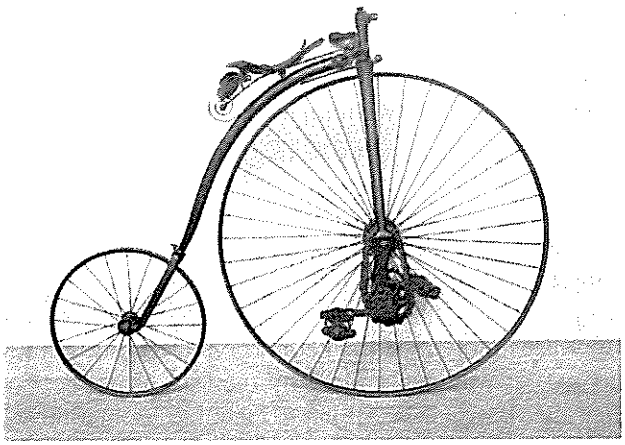


Fig. 2.1: *The Kangaroo bicycle mixed the design of a Penny Farthing with an early chain drive in an attempt to reduce the front wheel size.*
(http://en.wikipedia.org/wiki/File:Kangaroo_Bicycle_Rev.jpg)

The Safety Bicycle

Attempts to make the front wheel smaller, and therefore the bicycle safer, were defeated by the lack of a feasible solution at the time. A reliable bicycle chain was a major invention needed to develop a safe successor to the Old Ordinary. James Starley gave us the makings of the modern bicycle when he developed the Rover 'safety' bicycle of 1885. Starley had already designed a chain-driven bike for Rover in 1880. It had linkage steering of the front wheel, which was still larger than the rear wheel. His new design brought together some of the main elements of the modern machine: geared chain drive, wheels of equal size, direct steering, inclined forks, and the diamond-shaped frame. These safety bicycles were a vast improvement on the Old Ordinary. Not only were they easier to mount and ride, they allowed improved gear ratios, which made them faster but requiring less effort to ride.

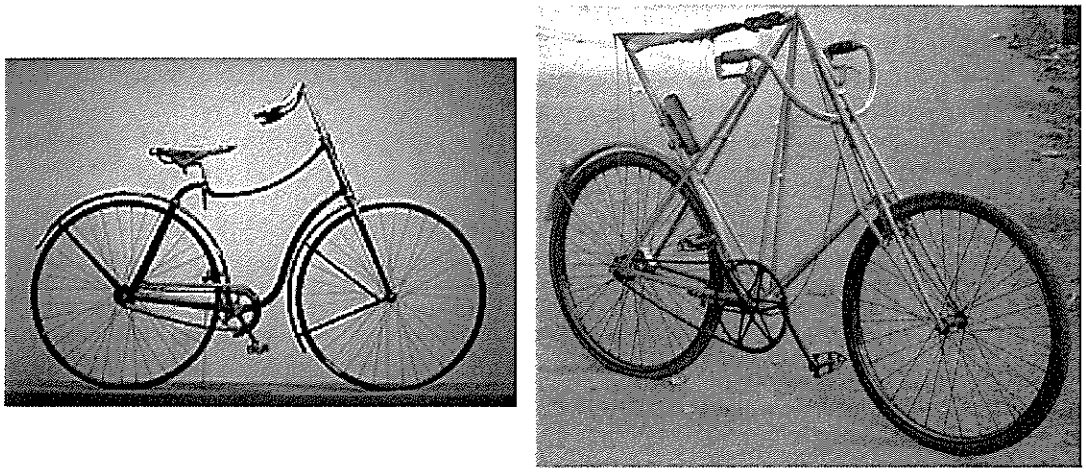


Fig 2.2: *An 1886 Rover Safety bicycle (left). A Dursley Pederson bike from 1898 is shown right. The Pederson bicycle used multiple small tubes to create a truss both longitudinally and laterally. Note the hammock style saddle.*

Initially there was some resistance to the new design but another development was about to make the "Rover Safety" the number one design. In 1888 a Belfast veterinary surgeon, John Boyd Dunlop invented the pneumatic cycle tyre. Prior to this, all tyres were solid rubber. With pneumatic tyres the bicycle was more comfortable so the rider could travel even faster. Sensing the commercial value of this, Dunlop went into the tyre manufacturing business, and within a few years, the pneumatic tyre had completely replaced the solid tyre as standard equipment on bicycles and on the new inventions, cars and trucks. The Dunlop Company had become the largest in the tyre business.

Mass Production

By the late 1800s the bicycle had ceased to be a novel status symbol and was becoming a cheap and practical form of personal transport. Mass production of the bicycle began, and by 1914 the practical utility bicycle was being assembled from some 300 separate components. The majority of these were manufactured by specialist firms. Many cycle companies were the core of the fledgling automotive industry, for example Morris and Riley and Peugeot.

Up to the Present

Through the 1900s bicycle frame design did not change a great deal. The frame angles were altered and different alloys were used to reduce weight. By the 1960's, high strength, lightweight aluminium alloy frames were used and new gearing systems were developed. Nowadays we see very base end bicycles made from mild steel tubing, entry level road and mountain bicycles using aluminium alloys and high end bicycles now use carbon fibre reinforced polymers.

Big developments occurred in the 20th century with gearing and braking. By the 1990's the mountain bicycle had demanded wider range gearing and the simple 10 speed derailleur system expanded out to take in first 7 speed rear clusters with now 10 speed rear clusters being the norm for high end bicycles. Front derailleurs have remained two speed for most road bicycles and 3 speed for mountain and touring bicycles. Hub gears are rare in Australia but are used extensively on city and folding bicycles through Europe. Derailleur gear systems are more efficient than hub gears but require more maintenance. Currently the "fixie" is very fashionable, using only single speed chain drive, in a modern road bike frame.

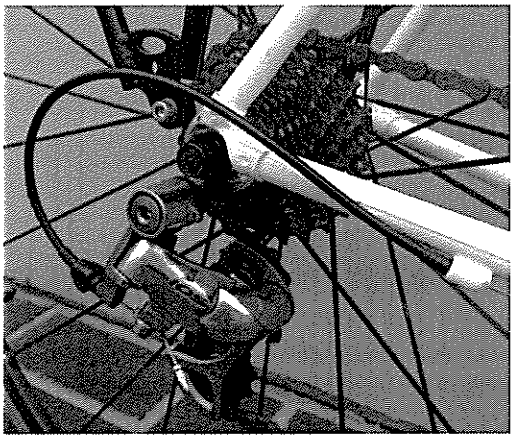


Fig. 2.3: *The rear derailleur from a modern road bike, here the derailleur moves the chain left to right across 10 different cogs ranging in size from 12 teeth to 25.*

Brakes have improved with cast aluminium alloys now allowing dual pivot calliper brakes with good braking performance and also lightweight for weight conscious road bicycles. Early mountain bicycles used cantilever brakes, which were predominately replaced by V brakes and now disc brakes; both cable and hydraulic discs are used on mountain bicycles.

Modern Bicycles

Mountain Bikes

Initially the term "mountain bike" was marketing exercise; simply a standard diamond frame bike that had been more heavily constructed to stand up to off road riding. Mountain bikes typically have smaller wheels than road bicycles (ETRTO 559 mm) with knobbly tyres. Such tyres are not ideal for road riding because of the high rolling resistance. However entry-level mountain bicycles have become the bicycle of choice for recreational cyclists. These typically use aluminium alloy frames (such as 7000 series). High-end mountain bicycles are popular with off road riding and various

competitive events exist for such cycling. Here carbon fibre reinforced frames dominate.

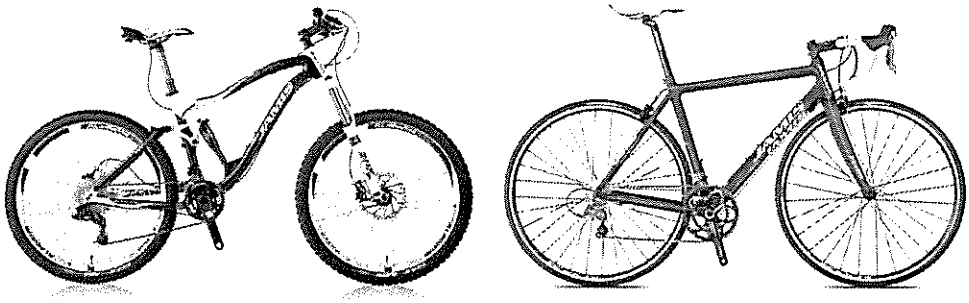


Fig. 2.4: *A dual suspension mountain bicycle left and a road bicycle on the right. Both have carbon fibre reinforced polymer frames.*

Most high-end mountain bicycles use dual suspension designed to cope with the rough terrain.

Road Bicycles

The modern road bicycle typically uses an aluminium alloy frame for entry-level bikes and then carbon fibre reinforced polymers moulded frames are used with more expensive models. They have large diameter (ETRTO 622 mm) but narrow wheels and tyres. Road bicycles rely on some flex in the frame to provide improved pedalling response and some release from road vibrations, but require rigidity as well so pedalling forces are not lost. Originally steel and alloy steel frames offered this perfect compromise, but desire for lighter weights led to aluminium alloys which tend to have less flex because aluminium alloys have no definite fatigue limit so frames are built with little or no flex. Carbon fibre reinforced polymers offer the ride of steel but are lighter than aluminium alloys so they are now the frame material of choice. Their durability, however, in terms of long term ownership is questionable.

Moulton Bicycles

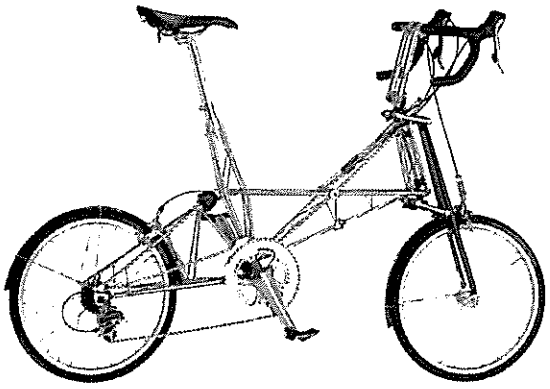


Fig 2.5: *The Moulton NS Double Pylon, 304 stainless steel tubing silver soldered with almost 200 individual soldered joints. It has rubber suspension front and rear.*

Alex Moulton's bicycles pioneered the use of small wheels with high-pressure tyres. Prior to this small wheel bicycles were simply children's bikes. Moulton proved that small wheels with high-pressure tyres were the equals of large wheeled bicycles.

However they had a rougher ride so he fitted suspension as well. Moulton bicycles were very popular in the 1960s but as the craze waned Moulton moved into high quality small volume bicycle production and his designs now use a complex stainless steel space frame design, which is light, strong but quite expensive.

Folding Bicycles

There are a wide variety of bicycles that took the small wheel concept pioneered by Alex Moulton and developed them into bikes that can fold for easy commuting and transport. One of the best examples of a folding bicycle is the Brompton bicycle. Within 20 seconds a Brompton can be folded to a package less than a quarter its original size. Although the Brompton is clearly designed with city riders and commuters in mind, its devotees have toured all over the world on them.

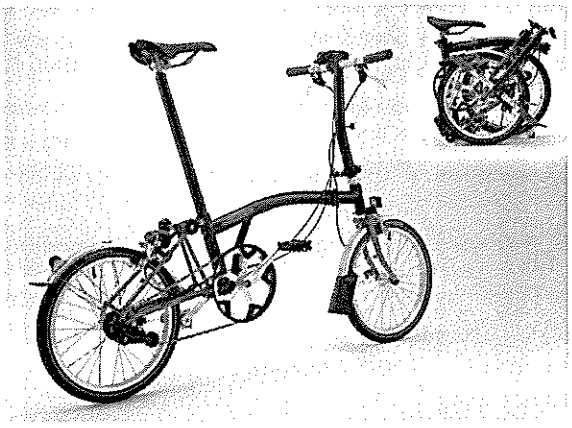


Fig 2.6: *Brompton folding bicycle, unfolded and folded.*

Recumbent Cycles – Or Riding Reclined

The concept of the recumbent bicycle (and tricycle) is a sound engineering principle. The human body is not really designed for the arms to support a person's weight. Yet the common or upright bicycle requires that part of the body weight is supported by the arms. The bicycle saddle, developed many years ago, is not designed to support all of the rider's weight. If one sits vertically, without putting weight on the arms, the bottom will soon get sore.

Recumbent bicycles

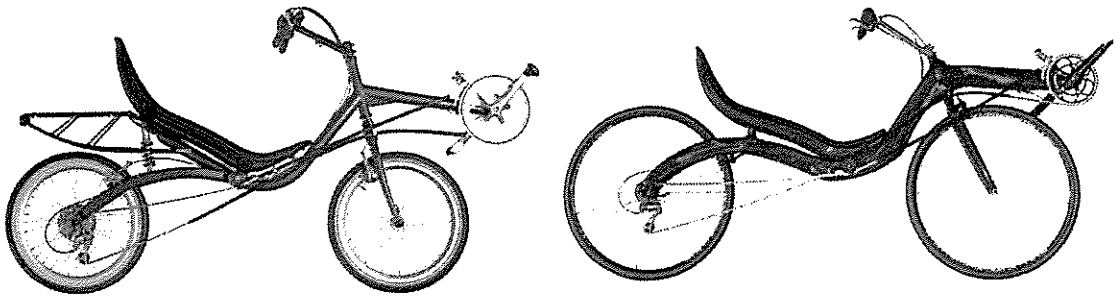


Fig. 2.7: *An M5 Shockproof cro-moly steel frame recumbent bicycle with dual suspension and a CFRP frame M5 high racer recumbent (photo courtesy http://flyingfurniture.com.au/)*

The recumbent bicycle is one solution. The larger, more comfortable seat, has a backrest and is reclined like a car seat. The pedals are out in front. The rider's arms carry no weight. A softer, larger seat makes for greater comfort and lower frontal area because it is reclined. The machine is more aerodynamic and finally, if the rider falls off, he/she has less distance to fall. Conversely they are heavier, more costly, and harder to balance, especially initially.

The world of recumbent bicycles is highly diverse and covers many different styles, materials and manufacturing methods.

Recumbent Tricycles

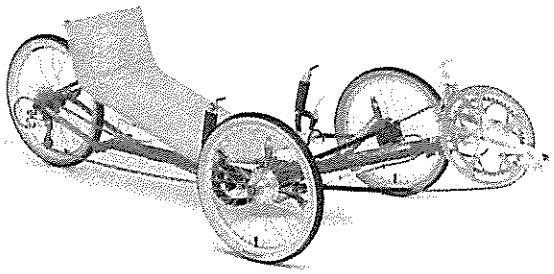


Fig. 2.8: *Greenspeed GLR road racing trike (picture courtesy Greenspeed, www.greenspeed.com.au)*

The upright tricycle is primarily a child's toy or a load carrier. With the high centre of gravity they have a tendency to roll over. One of the disadvantages of a bicycle is the need to balance it and the rider must allow the bike to wander slightly from side to side in order to maintain balance. The tricycle is a solution to this, but, it is highly susceptible to rolling at moderate speeds. The recumbent tricycle has a low centre of gravity so rolling is far less of a concern, and there is no balance issue.

Because of their extra mass, tricycles tend to be a slower on hills but on the flats and downhill they are equally as quick as any bicycle. They can also corner as quickly as any bicycle. Being low, means that they have an aerodynamic advantage in head and cross winds. They are better in wet conditions than the bike because their design makes them more stable when cornering. But they are heavier and more costly than an equal quality upright bicycle.

Car Development

Introduction

In Chapter 2 of Volume 1 we looked at the development of the motor car by looking at some key designs. Books and books have been written about the development of the car. The author suggests that when considering car design and the impact of the engineer, it is appropriate to consider key designs such as the Ford Model T, the Volkswagen Beetle, the Citroën DS and the BMC Mini. Naturally other cars have advanced car design, but these cars significantly changed car design, or they were big sellers that effected society in some way.

Car Propulsion.

It is nearly 12 years ago since the author wrote the first edition of this book and at the time hybrid technology seemed to be on the march. Yet twelve years on hybrid cars have barely made an impact, with Toyota/Lexus and Honda being the prime drivers with other companies proposing but yet to release hybrid vehicles. Instead we have seen the petrol engines improved with direct injection designs, and we have seen diesel engines given advanced engine management and injection systems that have vastly improved their performance while maintaining their economy. All this has meant that hybrids have gained little traction in the market, and the long proposed fuel cell cars seem even further away.

Train Development

So far, all of our forms of transport have been personal transport. We shall now turn our attention to a public transport system. In the following section some important developments in trains will be discussed.

From the 14th to the 18th centuries the train was in use but horses powered it. Many wagons were hauled up out of mines using "pit ponies", and this practice continued in Australia until the 1950s. Originally rails were timber but iron rails were introduced in 1728 in England. In 1803 the first public railway was opened for hauling freight; it was just over 15km long and ran from Wandsworth to Croydon in Southern England. This was followed in 1806 by the first public passenger line with a horse-drawn train in South Wales. In Australia the very first railway was in Tasmania and was actually convict powered. The train ran on rails, but instead of a steam engine, the convicts ran alongside the train, pushing it.

Steam Trains

In 1803 the very first steam locomotive was constructed by Richard Trevithick for the Coalbrookdale Ironworks. He built a second locomotive the following year and it was apparent that this machine could do more work than a horse at a speed of 8 km/h. Trevithick used his high-pressure steam engine as opposed to competing atmospheric steam engines that were used for driving pumps and machinery. This heavy locomotive, however, broke up the rail line, so it was abandoned.

In 1812 the first locomotives to conduct regular work were designed by John Blenkinsop, for a colliery in Yorkshire. They differed from current trains as the wheels and tracks had interlocking teeth. Blenkinsop felt that smooth rails would slip too easily. Trevithick had proved this was not the case, but Blenkinsops's system was to reappear, in a modified form, for mountain railroads.

George Stephenson, a mine mechanic, invented a locomotive in 1814, and it was to be the first step in his work of developing railways. Stephenson was a railway man through and through. He envisaged a railway system criss-crossing the country, carrying freight and passengers alike. In 1823 he was appointed as the engineer to the Stockton and Darlington railway. This 16 km line ran from a colliery to the port of Stockton-on-Tees, and was opened in 1825. His locomotive, *Locomotion,* was designed with his son, Robert. It was the first locomotive to haul on a public line and was capable of speeds up to 25 km/h.

Stephensons' *Rocket* is a famous locomotive and it had an immense impact on locomotive design. The future of profitable railways rested in providing quick passenger transport; something that was previously unavailable. In 1829 a series of trials were held to determine the best locomotive design. George and Robert Stephenson entered their *Rocket*, which had an innovative boiler design; 25 tubes carrying water ran through the firebox, more efficiently creating steam. Along with an improved exhaust system, The *Rocket* was twice as fast as rival designs; it could pull a 14 tonne train at 46 km/h.

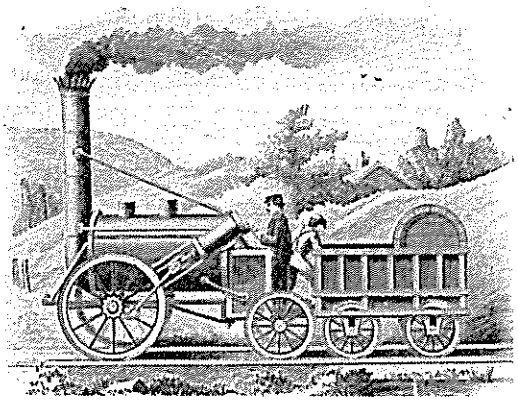


Fig 2.9: *Stevenson's Rocket of 1829*

The *Rocket* started an epoch in train design. From that moment, rail transport expanded, as it offered speedier transport than horse-drawn services. Naturally the challenge was to get trains to go faster and faster, but along with such increases in speed came difficulties in stopping the vehicles. Mechanical brakes were difficult to use, but in 1869 George Westinghouse patented an air brake design that was to become widely used. Operation centred on the brakes being *held off* by the air pressure. When the air pressure was released the brakes were applied. This made it safer if the train lost its air pressure, than if the system operated in the reverse fashion.

Another big advance in steam locomotive design was the development of the compound steam engine developed by Anatole Mallet in 1876. Instead of used steam merely being given off, it was first used in a small cylinder, then used in a larger cylinder, so the steam was used twice. Following "compounding", there was superheating, which was initially adopted in Germany in 1898. The moisture content of the steam produced by the boiler was reduced by increasing the temperatures, hence increasing the efficiency of the motor.

If one looks closely at railway design one may notice that railways neither are very steep, nor do they have very tight curves. Tight curves are difficult for trains to negotiate, while steep grades mean they lose traction. While cars can comfortably, even if a little slowly, ascend a grade of 1 in 6, trains are lucky to climb grades higher than 1 in 25. The solution to both of these problems is to produce large sweeping curves and to use tunnels and cuttings to pass over hilly terrain.

The standard steam locomotive has one or two large driving wheels, with many smaller wheels that are carried as pivoting bogies to balance the train at front and back. The train often carries a tender carriage behind it containing water and coal, which has no drive wheels under it. The problem with this is that a lot of the weight is

not carried over the drive wheel, so on steep grades, the wheels will slip. Two types of locomotives that avoided this were the Garret and the Shay.

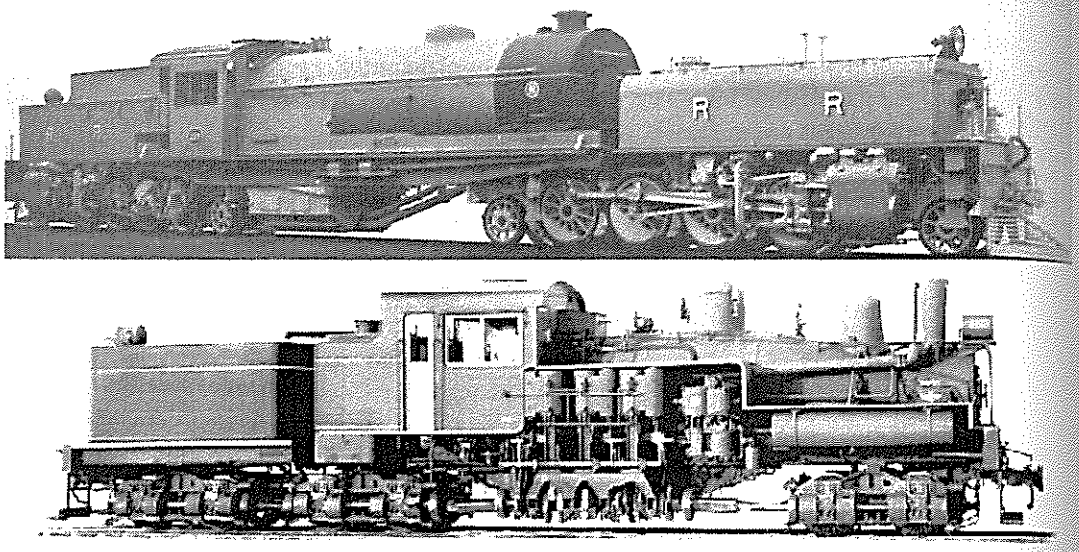


Fig. 2.10: *A Rhodesian railways Garratt locomotive (top) and the clever and specialised Shay locomotive (bottom).*

The Garratt locomotive was made in three articulated parts; the middle was the boiler and cab while the front and back carriages were the tenders for water and coal respectively. Unlike other trains, the bogies on the tenders were driven, so you there were more driving wheels and therefore less chance of wheel spin. Garratt locomotives were very large, yet could deal with tight curves. The main disadvantage was the wear on the pipes that carried steam to the tenders at either end.

The Shay locomotive was a truly fascinating design. The boiler is offset to the left side of the locomotive to allow a set of three vertical pistons on the right hand side. These pistons moved up and down and were connected to a crankshaft that ran along the right hand side of the locomotive. Mounted to the crankshaft were pinions that engaged with bevel gears on every wheel. This meant that every wheel was a driving wheel, as the left side wheels were connected to the right hand wheels by axles. The crankshaft had to have universal and sliding joints to allow the bogie to turn, and the tender also had drive wheels underneath to improve traction. The main advantage was that each wheel was driving so wheel spin was less likely, and it was designed to deal with tight curves. The disadvantage was that the design was slow, with a maximum speed of 25km/h and it suffered a lot of wear in the universal and sliding joints.

Four Shay locomotives were used in NSW, on a private railway line that ran into the Wolgan Valley to ferry refined shale oil from the refining plant. At the time of operation (1909-1911, then intermittently up to 1932), rail transport was the only option available for freight. It was also the main source of income for the residents of the township of Newnes, now a ghost town. The railway was a miraculous feat, with tight curves and 1 in 22 grades and only two tunnels. Although the line is gone it is possible to walk along the old track that was the line. One tunnel is now a home to glow-worms, a tourist attraction for the Lithgow area.

Gauges – to be wide, or not to be wide; that is the question!

Gauge refers to the distance between the tracks. By the 1830s it was a hotly contested issue. Originally the gauge of lines was 4 foot 8½ inches, or approximately 1.44m. This odd size came about, as that was the gauge of the lines that coal mines used with the wagons pulled by pit ponies. There is also some evidence that this strange size dates back to the distance between the wheels of Roman chariots. All railway engineers simply used this accepted gauge, except Isambard Kingdom Brunel. We discussed Brunel's bridge designs in Chapter 1. He was responsible for the Great Western Railway (GWR) in England. Brunel felt that a wide gauge of 6 feet (1.83m) offered distinct advantages, and he was right. The wider gauge makes a train more stable and less likely to roll, as its wheels are further apart. It does, however, make tight curves even more difficult.

Unfortunately he was the only person to adopt this gauge and as more railways sprang up around England it became clear that the wide gauge days were numbered. In Australia, the eastern states could not agree on the gauge for their railways. NSW used the standard 4'8½" (1.44 m) gauge, while Victoria used a 5'2" (1.57 m) gauge and Queensland used a narrow 3'6" (1.07 m). This ridiculous mismatch was only rectified in the 20th century, when the gauges in Victoria and Queensland were adjusted to the NSW standard.

Electric Trains

The first real challenge to the dominance of steam was the electric train. With reliable electric motors and power stations, both available by the 1880s, electric trains were a viable option to steam. One clear advantage was that they were less polluting. The power station still polluted but it was away from the rail line and was much more efficient in converting coal to the power source (electricity) than a large number of steam engines They were also quieter. The disadvantage is that they need infrastructure in place to provide power, because an electric train cannot use just any rail line.

The fist electric train design was shown by Werner von Siemens at an 1879 Berlin exhibition, and by 1883 public electric lines were opened. The electric train offered great advantages with its underground rail lines. This was an answer to congestion in cities. In 1890 the first electric underground railway was opened in London and it has since become an essential part of the city's transport. In Sydney the underground stations of Museum and St James were opened in 1926. Town Hall and Wynyard stations followed in 1931, and with the completion of Circular Quay station in 1956 the City Circle was created. In the London Underground, electric trains were essential. Electric trains have been used on suburban railway lines since the late 1920s in Sydney, while outlying areas like Wollongong, Newcastle and Lithgow did not gain electric trains until the 1980s.

Electric trains need to have power provided. In NSW, the power is provided by copper, overhead wires. Some of the trains' carriages have a pantograph that is connected to the electric motors that run along the overhead wires. In other situations, a third rail, between the other two, is used to provide power. Without these systems, the electric train cannot run, but with them they are a viable option. Many goods trains in NSW are pulled by electric locomotives. It is important to emphasise that ***electric***

trains do not stop pollution; they merely move it from the city to wherever the power station is.

Diesel Trains

The diesel train is the true successor to the steam train. Electric trains ran alongside steam trains in the cities but they could not service outlying areas. The first diesel train ran in 1912 in Germany, but the future for diesel trains lay in the diesel electric arrangement, where a diesel engine drives a large generator that uses electricity to drive electric motors at the wheels. This is a far better system than a gearbox, for large power delivery. Another system uses hydraulics to connect the motor and wheels and this is called a diesel hydraulic drive. A diesel motor drives an hydraulic pump which creates high pressure oil in pipes. The pipes feed hydraulic motors, mounted on the wheels.

By the 1960s in NSW, diesel trains were becoming widespread and by the early 1970s the steam train was gone. Diesel electric trains are extensively used for long distances. In NSW, for example, the diesel electric system is used on the XPT rail services and it is extensively used when hauling freight. Two large diesel electric locomotives haul the Indian Pacific that runs from Sydney to Perth. The Explorer and Endeavour passenger trains, in NSW, use a diesel hydraulic system.

Effects of Engineering Innovation in Transport on People's Lives

It is important to the study of historical topics that students learn how to make notes. Following is a note summary of effects of innovation on people's lives.

Cycles

- The pedal-powered velocipede, 1839, greatly improved the usability of the bicycle.
- Old Ordinary (Penny Farthing) – faster transport than the velocipede but dangerous to ride
- Rover Safety Cycle, 1885 – safer transport with similar speeds
- 1888 – Dunlop's pneumatic tyres
- Early 1900s – mass production of the bicycle
- Freewheeling hubs made cycling far safer for riders
- Internal hub gearing – improved bike as a form of transport
- High strength steel alloys, such as Reynolds 531 – bikes became lighter.
- Recumbent bikes – better comfort but outlawed from racing – stalled their development
- Lightweight aluminium alloys & reliable derailleur gears – improved traditional design of Safety Cycle.
- After WWII – cycle usage declined as cheap cars available – subsequent pollution not much considered.
- Suez oil crisis, 1950s – forced many people back to bikes – cycle development took off again
- Moulton bike sparked cycling craze in UK – suited image of the swinging sixties.

- BMX (Bicycle Motor Cross) – small wheels – off road racing – popular with children
- 1980s – rise of the mountain bike – grew into the most popular bike – many specialised components developed.
- Recumbent bikes and trikes grew in popularity in the 1990s to create a niche market – good for long-distance touring – easier on the body
- 1990s and on – more exotic materials – greater weight savings – improved performance

Trains:

- Steam train – 19th century alternative to the horse and cart for transport.
- Railways systems developed world-wide, 19th century – ability for people to travel across country for first time- meant a boom for many small towns.
- Electric trains, 20th century, in urban environments, reduced pollution compared to steam trains.
- Train, important tool in WWI and WWI – supplies, movement of troops.
- After WWII, diesel train started to appear – by 1960s starting to replace steam – less pollution, greater reliability, quicker times
- More electric rail networks – further improvement of air quality.
- High-speed trains greatly cut transport times, e.g. English Channel Tunnel train faster from London to Paris than plane.
- Today, some see well-designed electric trains and light rail systems as answer to traffic congestion.

Construction and Processing Materials Over Time

Cycle

Over the years various materials have been used in cycle construction.

Timber

Timber was used in early bicycles because of a lack of suitable alternative materials. Frames were made from timber pieces that were shaped, while timber was also used for wheel construction.

Iron

Iron was used initially as a tyre on wooden wheels and in early frames, but it was replaced by steel frames and rubber tyres.

Steel

This was produced in large, cheap quantities after Henry Bessemer developed his converter in 1856. From this point, cycle frames began to use steel construction and by the 1870s steel wheels with thin steel spokes were used. Steel tubing offered good strength with a relatively low weight. Although not as light as some materials, tubing was lighter than solid rods. Cold drawn steel tubes offered the best strength for cycle frames.

Usually steel frames were joined by brazing with lugs. The lug was a small joint piece into which the tubes slid. They were then brazed in place. Only cheaper frames were welded. With newer welding techniques, however, most plain carbon steel frames are welded.

Steel was used extensively in brake and gear construction up to the 1970s when aluminium alloys started to encroach, in an attempt to save weight. Steel is still extensively used in chains and gear clusters and on many cheaper bikes all components are steel, because it offers reasonable strength, ease of fabrication and it is cost effective.

Alloy steels (as used in frames)

The term *alloy steel* refers to steels with deliberately added materials in addition to iron and carbon. Reynolds 531 was a manganese-molybdenum steel that offered a better strength to weight ratio than plain carbon steel. It was first introduced in 1935. Reynolds 531 was used in racing bikes for many years.

Chromium-molybdenum steels are also used and these once again offer superior strength to weight ratios than do plain carbon steels. Whereas Reynolds 531 was really only suitable for brazing (with lugs), chromium-molybdenum (called Cro-Mo) steels can be welded. With modern production techniques, this is an advantage, as the TIG and MIG welded frame is quicker, lighter and cheaper to fabricate than a brazed steel lug frame.

Fig 2.11: *"Steel is real", this road bicycle uses a Reynolds 631 frame, with similar weight to aluminium alloy frames, but with a more flex giving the famed ride of steel. However these steels are now more difficult to work than aluminium alloys in most mass production environments now. Note the contrast of old school leather saddle and the CFRP front forks.*

Reynolds and True Temper have released air hardening steels such Reynolds 631, 853 and True Temper OSX. The problem with welding or brazing a tube of steel that is cold rolled is that, around the weld joint, the steel is annealed and softens. Air hardening steels harden in still air to maintain strength around the welded joint. The

result is an extremely strong frame with strength to weight ratios approaching many aluminium or titanium alloys.

Stainless steel

This high alloy steel was not typically used for frame construction, excepting for some Moulton bikes, but is now offered for frames in the form of a martensitic ageing stainless steel, with Reynolds 953 being an example. This stainless alloy offers UTS values up to 2000 MPa. Stainless steels are also used in the manufacture of components such as cables and pins for brakes and gears. Its corrosion resistance makes it quite desirable for these applications.

Aluminium alloys

These are now widely used in cycle construction. Reynolds offers a 6061 alloy (heat-treatable Al/Si/Mg) and a 7005 alloy (heat-treatable Al/Zn). 6061 offers excellent corrosion resistance and is more workable than other heat-treated alloys. 7005 offers greater strength but greater density than 6061. Aluminium is light but considerably weaker, so tubes are often not circular but oval, to increase the resistance to bending. Aluminium must be welded, usually using Tungsten Inert Gas (TIG) or Metal Inert Gas (MIG) methods, and offers the advantage of being relatively corrosion resistant.

Aluminium alloys are also used in the manufacture of brake and gear parts because of their lightweight nature. Brake levers and arms, derailleurs and hubs use aluminium alloys.

Titanium alloys

To a lesser extent than aluminium alloys, titanium alloys, such as 3% Al/2.5% V/Rem. Ti are finding favour in cycle manufacture, not just in frames but also in gear componentry. Titanium components and frames are usually very expensive and in frames, at least, they offer little over steel alloys like Reynolds 953, or carbon fibre reinforced polymers. Titanium must be welded using the TIG method and this must be done very carefully or there may be weld cracking.

Titanium sprockets offer a viable alternative to steel cogs but they come at a high cost and, as such, are only used on the best racing bikes.

Carbon Fibre Reinforced Polymer (CFRP)

Carbon fibre reinforced polymer is an excellent alternative to aluminium alloy or alloy steel frames. It has an excellent strength to weight ratio that is very desirable on racing bikes. The frame is moulded as opposed to fabricated and additional parts are usually placed on the frame with an adhesive.

Carbon fibre is tough and strong. When it fails, however, it happens suddenly and often catastrophically, with little or no plastic deformation to warn of imminent failure. It has become the material of choice for all mid to high level road bicycles and high end mount bicycles.

Rubber

This material was partly responsible for the growth of cycling. Solid rubber tyres were used to replace iron tyres. It was lighter and provided moderate springing. In 1888, John Boyd Dunlop developed the pneumatic tyre that made cycling even more pleasant. Today, synthetic rubber is still used in tyre construction. Rubber has also been used for suspensions on bikes. On Moulton bikes it is used for both rear and front suspension. It has the advantage of being light and also being self-damping, so no damper (a.k.a. shock absorber) is needed.

Polymers

Polymers are greatly used in the manufacture of cycles and they have found use in applications where their flexibility is a great advantage. Polymer sheaths are placed over cables, and are also used in pedal construction. They offer flexibility, are lightweight and show good resistance to deterioration caused by the weather and UV light. They are also used in cycle lights and finishing pieces.

Environmental Effects of Transport Systems

Cycles

Cycling is often looked on as one of the many solutions to environmental problems, such as pollution and the greenhouse effect. Cycling, as a form of transport, is one of the most efficient ways to travel; that is, the energy expended is used more usefully than many other forms of transport. It is non-polluting and human power is a renewable energy source.

It is perhaps not as popular as it should be as many adults abandon cycling once they get a driver's licence and the weather becomes a major issue for travel. Many people also fear riding on roads because of the dangers posed by motor vehicles, drivers who are unsympathetic to cyclists, and the absence of good cycle paths.

Trains

Trains have also had negative impacts on the environment. The construction of railways has resulted in tree felling and the levelling of land. Many railway lines also incorporate tunnels and cuttings which, when blasted, can have an adverse impact on the surrounding areas.

Steam trains produced excessive pollution, with an efficiency of only 5 to10 %. In many developing countries steam engines are still used but run on oil which produces even more smoke than coal did previously.

Electric trains also cause pollution, as the electricity must be produced somewhere! In NSW, that is usually in coal power stations, which are major contributors to the greenhouse effect. The electric train, however, does take the pollution away from urban areas where air quality is always a problem.

Diesel trains, although more efficient than steam trains, still pollute. Newer diesel locomotives, however, emit less smoke than the older ones, and modern computerised engines ensure minimum pollution for maximum power.

For passenger and freight transport, however, trains are better environmentally than road vehicles. Railway lines are narrower than freeways so have less of an impact, and they can haul many people or goods without congesting roads. Electric railways can move vast amounts of people in the city without pollution or traffic congestion.

Environmental Implications From the Use of Materials in Transport

Below is a list of different materials used in both private and public transport. The effects of using such materials are given.

Forests have been greatly affected by large scale transport developments. *Timber* was used for railway sleepers and as a fuel source for some steam locomotives. Such uses resulted in the clearing of large areas of forest. We are now feeling the legacy of this with global warming and local climate change. The removal of these forests has also had an impact on native fauna by reducing their habitat.

Steel has been the main transport material since 1856. It has found use in both plain, carbon and alloyed forms in all types of transport. The thirst for steel has seen the establishment of large steelworks, such as those at Port Kembla (NSW), which often affect the local atmosphere with the large amount of pollutants produced in working and refining the steel. Steel production requires large amounts of iron to be produced, which means iron ore must be mined. These mining operations are usually open-cut mines which involve large open pits with machinery digging down to extract the ore. As well as iron-ore, coal and limestone, are needed for the fuelling of the blast furnaces to produce the iron. Coal also has to be mined and refined to coke.

Cast iron has had a similar impact to steel but is used to a lesser extent nowadays. Essentially similar metallurgically to the iron from the blast furnace, it still requires vast amounts of iron-ore and coal and coke.

Aluminium has increased in use greatly in the last half of the 20th Century. Like iron, it is refined from an ore, bauxite, which is mined in an open cut manner, also affecting the local landscape and environment. Aluminium is refined from the ore using electricity, which in Australia, primarily comes from coal-fired power stations. The use of coal greatly contributes to greenhouse gases and as a result has a detrimental effect on our atmosphere. Aluminium refining also produces fluorine gas, a major polluter in the past, which is now controlled from reaching the atmosphere.

Polymer usage in transport systems has exploded since WWII, because polymers offer lightweight transport machinery which improves fuel efficiency. To protect the environment, the extensive use of polymers must be backed up by the recycling of old equipment, as polymers greatly contribute to landfill.

Engineering Mechanics

Simple Machines

Mechanical Advantage (MA)

The mechanical advantage of a machine is a measure of how it helps the user. Mechanical advantage in a mechanical machine is the ratio of load to effort and is found by the formula,

$$MA = \frac{L}{E}$$

MA = mechanical advantage
L = load
E = effort

The higher the mechanical advantage, the lower the effort must be for a given load. If the mechanical advantage is below one, then we have a mechanical disadvantage, which means the effort is greater than the load. Mechanical advantage is not a constant value as friction and other losses in a system affect it, meaning the theoretical mechanical advantage may be considerably higher than the practical mechanical advantage.

Velocity Ratio (VR)

The velocity ratio is the ratio of the distance the effort moves to the distance the load moves in a mechanical system. VR is represented by the formula,

$$VR = \frac{d_E}{d_L}$$

VR = velocity ratio
d_E = distance the effort moves
d_L = distance the load moves

The higher the VR the greater the distance that the user must move. Unlike MA, VR is not affected by friction and system losses. If a machine is perfectly efficient (which is not practically possible) then MA will equal VR. However the lower the velocity ratio the greater the effort that is required.

Efficiency (η)

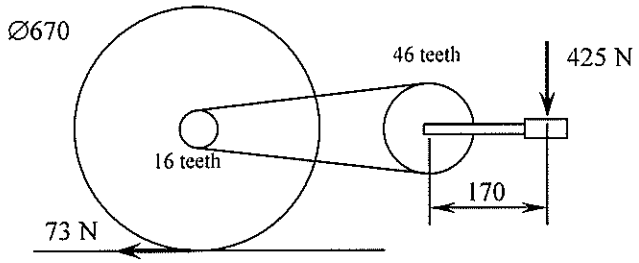
An ideal machine is one that is 100% efficient. That means all energy put into the machine is used. In reality, however, this never occurs. There is always some type of energy loss (usually as a result of friction) that results in the efficiency being below the ideal 100%. In the case of levers there may be friction in the pivots or the lever may bend slightly. VR is always the same irrespective of efficiency, since there is no change in the distances the effort and load move. But MA is effected by a less than ideal efficiency; thus the MA will always be less than the VR for machines with efficiencies below 100%. The percentage efficiency is found by the following formula,

$$\eta = \frac{MA}{VR} \times 100$$

η = percentage efficiency
MA = mechanical advantage
VR = velocity ratio

Example 2.1

A single speed bicycle drive system is shown below. The tractive force at the wheel and the force on the pedals have been established experimentally. Determine the MA and VR of the bike drive system and then its efficiency.



First calculate the MA.

$$MA = \frac{L}{E}$$

$$MA = \frac{73}{425}$$

$$MA = 0.1718$$

To determine the VR we must be aware that the bicycle's gearing means the rear wheel will rotate multiple times for one rotation of the pedals.

$$VR = \frac{d_E}{d_L}$$

$$VR = \frac{d}{D \times N_{rev}}$$

$$VR = \frac{2 \times 170}{670 \times \frac{46}{16}}$$

$$VR = 0.1765$$

Now we can calculate the efficiency.

$$\eta = \frac{MA}{VR}$$

$$\eta = \frac{0.1718}{0.1765}$$

$$\eta = 0.9731$$

$$\eta = 97.31\%$$

If the drive system was poorly maintained, the efficiency would be effected as would the MA, but the VR would remain constant. The only way to change the VR is to alter the gear ratios, crank sizes or wheel size. Derailleur gears are commonly used to alter the VR of the bicycle to allow for different conditions, such as hill climbing versus fast descents.

Example 2.2

A simple screw jack is being temporarily used to raise cars for tyre changes. The thread has a pitch of 4 mm and the handle has a radius of 200 mm. If the jack has an efficiency of 50%, determine the effort required to lift the front corner of a car, which requires a load of 6 kN.

Solution

The pitch is important here because one rotation of the handle moves the scissor arms in the pitch, which is 4 mm, so each rotation of the handle moves the arms inwards by 4 mm. Therefore the pitch is the distance the load moves, and the circle described by the handle is the distance the effort moves.

$$VR = \frac{d_E}{d_L}$$

$$VR = \frac{2\pi r}{Pitch}$$

$$VR = \frac{2\pi \times 200}{4}$$

$$VR = 314.16$$

Now we can calculate the MA using the efficiency and the VR.

$$\eta = \frac{MA}{VR}$$

$$MA = VR \times \eta$$

$$MA = 314.16 \times 0.5$$

$$MA = 157.08$$

Finally calculate the effort using the MA formula

$$MA = \frac{L}{E}$$

$$E = \frac{L}{MA}$$

$$E = \frac{6000}{157.08}$$

$$E = 38.2 \text{ N}$$

We can see the jack is the complete opposite to the bicycle drive, very large MA and hence very large VR, which means your hands will move a long way operating the jack. But you are able to lift a very large object. The low efficiency is caused by friction in the mechanism, which is why the jack stays up and doesn't unwind.

Modern cars are usually provided with a scissor action jack, which is a form of screw jack, but the VR varies as the angle of the arms changes, making calculation much more challenging.

Static Friction

Static friction is friction that relates to objects that are either not moving, or are at the point of moving. If the object is moving then we are dealing with kinetic friction, which is not part of this course. We will be discussing only static friction.

The Concept of Friction and its use in Engineering

Friction is the enemy of efficiency, or the ratio of input power to output power, yet without friction, life and especially transport systems, would be very difficult. Imagine the riding of a bicycle down the road. Without friction, the first corner one turned would see the front wheels slide out and the rider lying on the ground. Stopping would also be impossible without friction. Yet at the same time friction in the drive system saps the bike of power and thus lowers efficiency.

The simple fact is, friction is essential and one must make advantage of it in traction and braking situations. Friction must be limited, however, in power transmission by lubrication and through designs that minimise internal friction like roller bearings.

Coefficient of Friction (μ)

The coefficient of friction is the ratio of the friction force compared to the normal reaction. So the formula is:

$$\mu = \frac{F_F}{R_N}$$

μ = coefficient of friction
F_F = frictional force
R_N = normal reaction

Normal Reaction (R_N)

The normal force is something that creates a great deal of confusion. It is not the weight force, which is an important point to understand. The normal reaction is a perpendicular reaction, provided by the surface on which the object is resting.

Fig. 2.12 shows the normal reaction in various situations, as well as the weight force. What is the characteristic of the weight force? What is the characteristic of the normal reaction?

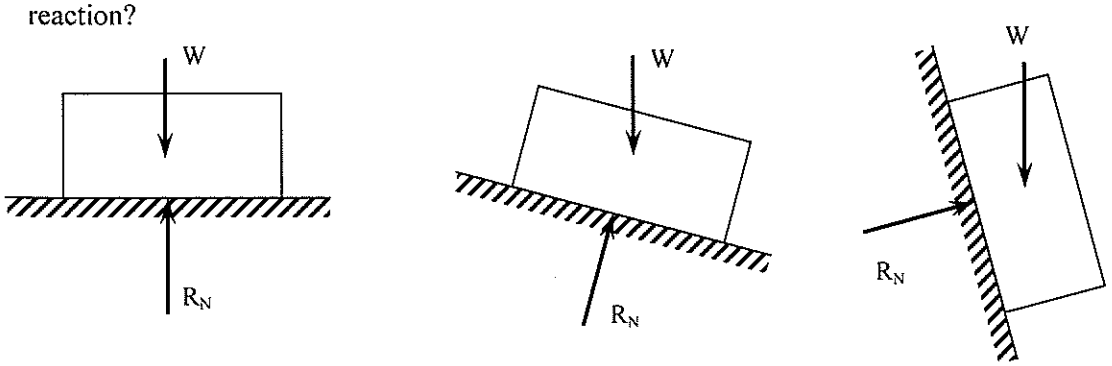


Fig 2.12: *The normal reaction is always perpendicular to the surface upon which the object rests.*

Note that the normal reaction is not equal in magnitude to the weight force in all applications. In some cases the applied forces will tend to lessen or increase the normal reaction. Fig. 2.13 shows two situations of a 10 kg box with inclined applied forces.

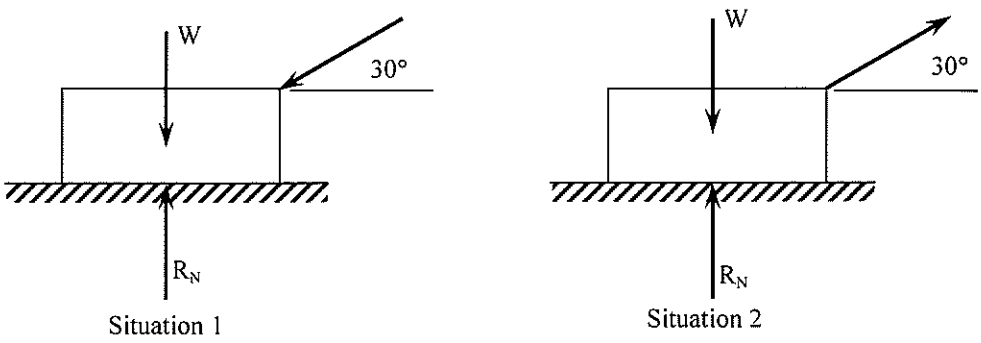


Fig 2.13: *Two 10kg boxes, one being pushed (situation 1), the other dragged (situation 2).*

In situation 1, R_N will be greater than the weight force, because of the vertical component of the applied force acting down. In situation 2, R_N will be less than the weight force as the vertical component of the applied force goes up. This is an important concept, because the larger R_N is, the higher the friction will be so it is often better to drag an object than to push down on it when sliding it.

Friction force (F_F)

Two things determine the friction force, or the frictional resistance, the force resisting the tendency towards motion between surfaces. They are the coefficient of friction between the surfaces, and the normal reaction. The higher the coefficient of friction, the higher the frictional force will be. If, however, the surfaces have a low coefficient of friction then a higher normal reaction is needed to get a high frictional force. For example, in a car the normal reaction is controlled by the mass, so good friction between the tyres and the road is dependant on the coefficient of friction between the surfaces.

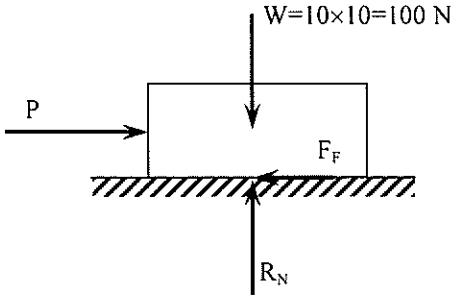
The formula for frictional force is merely a rearrangement of the formula for the coefficient of friction.

$$F_F = \mu R_N$$

Example 2.3

What will be the frictional force present when a 10 kg box is pushed just to the point of sliding by force P if $\mu = 0.5$?

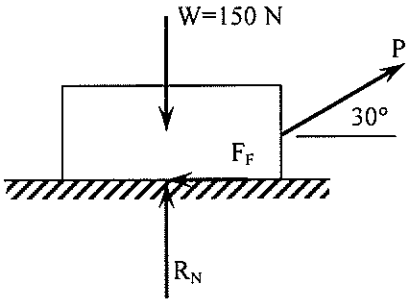
Since the applied force is vertical, R_N will equal the weight force, so $R_N = 100$ N



$$F_F = \mu R_N$$
$$F_F = 0.5 \times 100$$
$$F_F = 50 \text{ N}$$

Example 2.4

A 15 kg box is to be pulled by a force as shown. What will be the minimum value of the force to get the box to the point of sliding if $\mu = 0.25$?



The trick in this problem is that the normal reaction (R_N) will not be equal in magnitude to the weight, since the upward force is partially taking the weight. So R_N will be equal in magnitude to the weight force minus the perpendicular component of the force P.

First find R_N by adding the vertical forces.

$$\Sigma F_V = 0$$
$$0 = P_V + R_N - W$$
$$0 = P\sin 30^\circ + R_N - 147$$
$$R_N = 150 - P\sin 30^\circ$$
$$R_N = 150 - 0.5P \quad \text{——————} (1)$$

Now find the sum of horizontal forces

$$\Sigma F_H = 0$$
$$0 = P_H - F_F$$
$$0 = P\cos 30^\circ - \mu R_N$$
$$0 = 0.866P - 0.25R_N \quad \text{——————} (2)$$

Now substitute equation ① into equation ②.

$$0 = 0.866P - 0.25R_N$$
$$0 = 0.866P - 0.25(150 - 0.5P)$$
$$0 = 0.866P - 37.5 + 0.125P$$
$$0 = 0.991P - 37.5$$
$$P = \frac{37.5}{0.991}$$
$$P = 37.84 \text{ N}$$

Limiting Friction

Limiting friction is the frictional resistance that exists just as motion is about to occur. Static frictional resistance increases up to a maximum at the point of limiting friction. After this point is passed, the frictional resistance falls and the moving surfaces exhibit kinetic friction.

Angle of Static Friction (ϕ_S)

The box in Fig. 2.14 shows that there are four forces acting on the box. If we add vectorially the frictional force and the normal force, the resultant of those two will leave three forces on the body. The angle that force makes with the normal reaction is the angle of static friction.

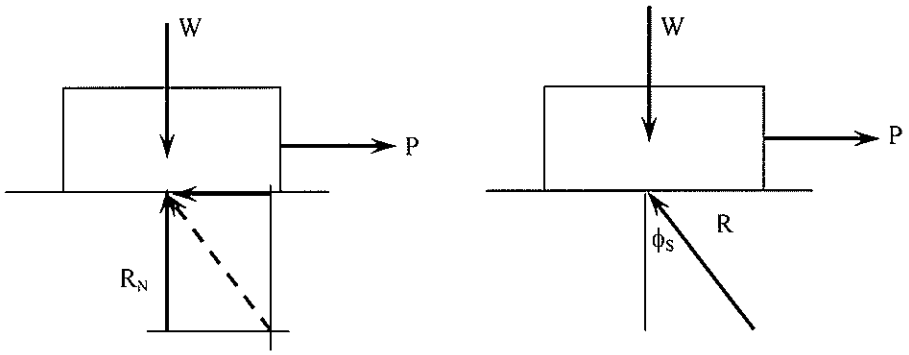
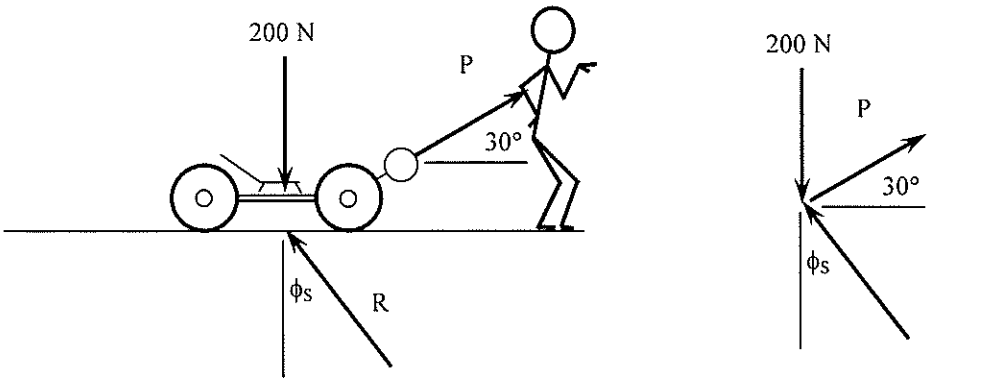


Fig. 2.14: *With the normal reaction and frictional force added together, there are three forces acting on the body, and the angle of the resultant to the vertical is the angle of static friction.*

Why this angle is so useful is that at the point of sliding $\tan\phi_s = \mu$. It is then possible to find the angle of static friction, and then the force required to put an object to the point of sliding, much more easily, particularly with inclined forces. If this system is in equilibrium, these forces will be concurrent and this problem may be solved using the triangle of forces.

Example 2.5

The wheels of a recumbent tricycle have seized, so the rider must drag it home. Find the force P required to put the 20 kg trike to the point of sliding if μ=0.35.

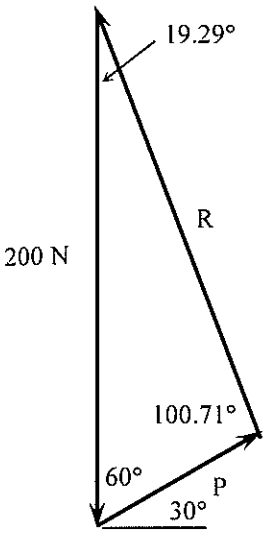


Space diagram

Free body diagram

Since $\tan\phi_s=\mu$ then $\phi_s = \tan^{-1}\mu$, therefore $\phi_s = \tan^{-1}0.35=19.29°$

Now draw a force polygon of the three forces on the box. Since it is not yet moving the system must be in equilibrium. Then use the sine rule to find R.



$$\frac{a}{\sin A}=\frac{b}{\sin B}$$

$$\frac{200}{\sin 100.71^\circ}=\frac{P}{\sin 19.29^\circ}$$

$$P=\frac{200\times\sin 19.29^\circ}{\sin 100.71^\circ}$$

$$P=67.24\text{ N}$$

Therefore the force required to put the box at the point of sliding is 65.90 N. This problem could also be solved graphically by drawing the force diagram to scale.

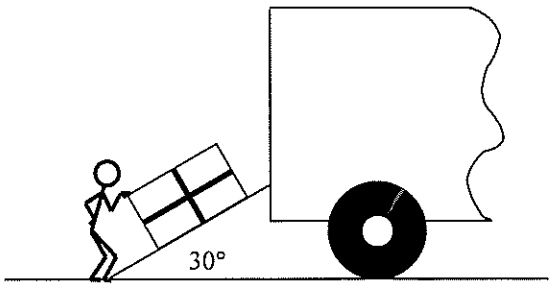
The mathematics involved in this method is far shorter than the mathematics required for Example 2.4. For this reason, many prefer to solve frictional problems using the angle of static friction.

Inclined planes

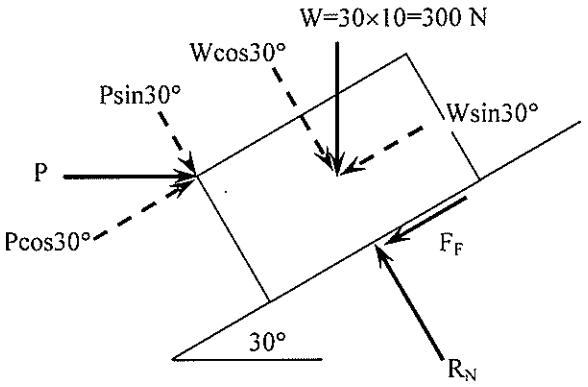
Frictional problems are well demonstrated by inclined planes and the angle of static friction is also relevant. Below is an example of the effect of friction on a box on an inclined plane.

Example 2.6

A 30 kg furniture box rests on a ramp that runs from the ground to the back of the truck. Determine the horizontal force P, the removalist needs to apply, to put the box at the point of moving, if the coefficient of friction between the box and ramp is 0.4.



First draw a free-body diagram of the box.



Add forces parallel to the plane and forces perpendicular to the plane.

$$\overset{+}{\nearrow}\Sigma F_{par}=0$$

$$0=P\cos 30°-300\sin 30°-F_F$$

$$0=0.866P-150-0.4R_N$$

$$150=0.866P-0.4R_N \qquad \text{(1)}$$

$$\overset{+}{\nwarrow}\Sigma F_{perp}=0$$

$$0=R_N-P\sin 30°-300\cos 30°$$

$$R_N=0.5P+259.81 \qquad \text{(2)}$$

Substitute equation (1) into equation (2)

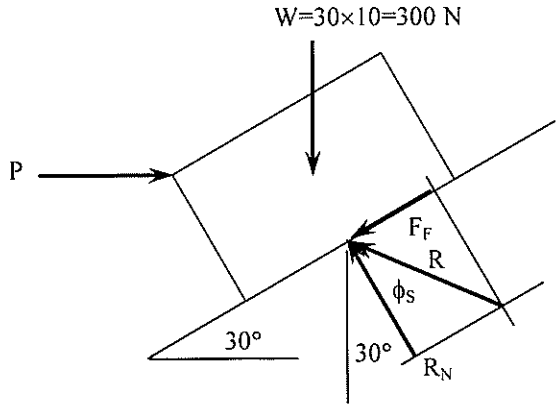
$$150=0.866P-0.4(0.5P+259.81)$$

$$150=0.866P-0.2P-103.92$$

$$0.666P=253.92$$

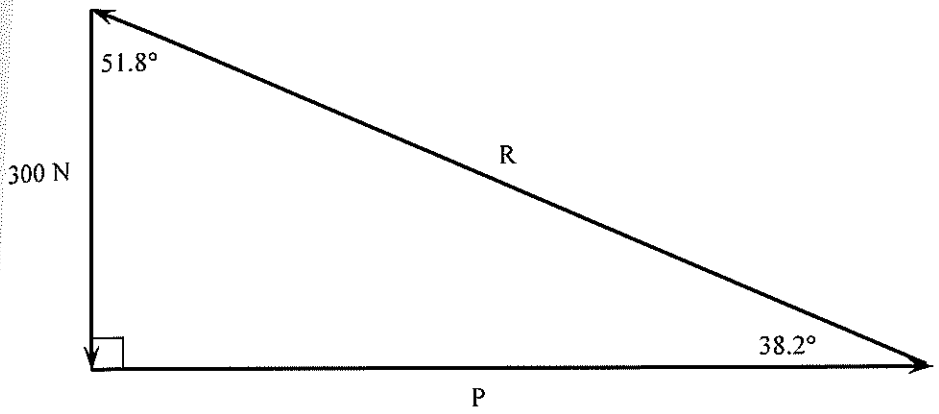
$$P=381.25\text{ N}$$

This problem could also be solved using the angle of static friction as is shown below.



Now $\phi_s = \tan^{-1}0.4 = 21.8°$

Draw the force polygon and use the trig ratios (since it is a right angle triangle) to find P. Note that the angle of friction is 21.8° from the normal reaction, but the normal reaction is inclined at 30° to the vertical, so the angle the reaction makes with the vertical is 21.8°+30° = 51.8°.



$$\tan 51.8° = \frac{P}{300}$$

$$P = 300 \times \tan 51.8°$$

$$P = 381.25 \text{ N}$$

The second method is far easier than the first.

Angle of Repose

If an object is placed on a flat surface with no net force acting upon it, it will not move, i.e. it is in equilibrium. If the surface is raised at an angle to become an inclined plane (Fig 2.16) the weight force will have two components, one acting down the plane ($W\sin\theta$) and one acting perpendicular to the plane ($W\cos\theta$). As the angle increases $W\sin\theta$ will increase, while $W\cos\theta$ decreases. At 45° they will be equal. At a point the value of $W\sin\theta$ will become larger than the frictional force opposing motion and the object will slide down the plane. At that point of limiting friction, F_F will equal $W\sin\theta$ and R_N will equal $W\cos\theta$.

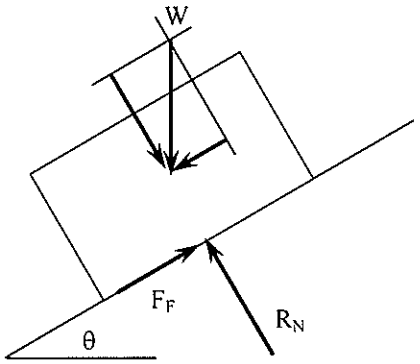


Fig. 2.16: *A box resting on an inclined plane*

Mathematically we can express it as follows:

$$\mu = \frac{F_F}{R_N}$$

$$\mu = \frac{W \sin\theta}{W \cos\theta} = \tan\theta$$

$$\tan\theta = \tan\phi \ (\because \mu = \tan\phi)$$

$$\therefore \theta = \phi$$

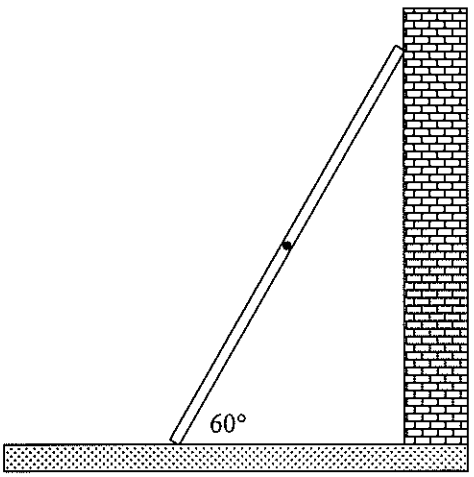
Therefore at the point of limiting friction, there will be an angle where the angle of static friction will equal the angle of inclination of the plane. This angle is called the *angle of repose*. The angle of repose is important in the design of equipment where items must slide of flow.

Friction with Ladders

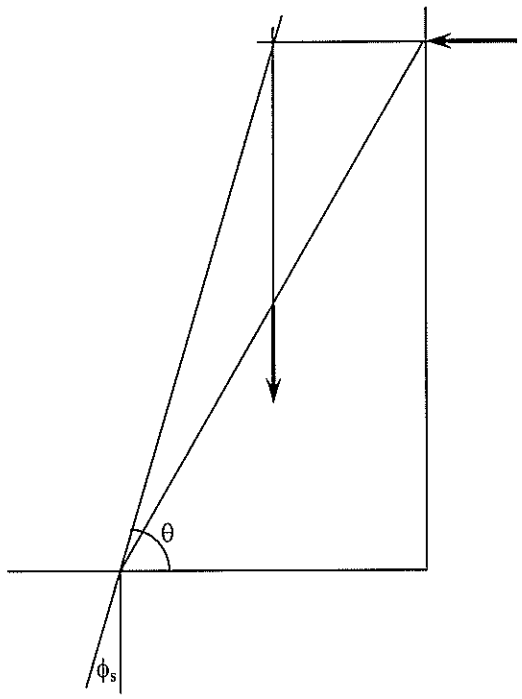
Ladders are a good example of friction action. In most ladder problems the calculation is simplified by considering the wall to be smooth (i.e. friction free). In reality this isn't true and some problems may provide a coefficient of friction for the ladder and the ground and also the ladder and the wall. The example below details

Example 2.7

A uniform 4 m ladder is shown; it rests against a smooth wall and a rough floor. If the ladder has a mass of 10 kg, determine the coefficient of friction needed between the ladder and the ground so the ladder will not slip. Consider the wall to be smooth.



Because the wall is smooth the reaction at the wall will be perpendicular to the wall, so the direction is known. The weight will be vertical which leaves only the third force, the reaction at the ground. We can see this is a three force rule problem, so we can determine the ground reaction's direction using the three force rule.



By measuring the angle of the line representing the third force's line of action we can determine that θ is approximately 73°. Therefore ϕ_s will be 17°.

$$\phi_s = \tan^{-1}\mu$$
$$\mu = \tan\phi_s$$
$$\mu = \tan 17°$$
$$\mu = 0.31$$

This problem can also be solved analytically but the author has chosen to model the quicker graphical solution.

Work, Energy and Power

The concepts of energy, and power were first investigated in Volume 1 (pp. 101-104). Various forms of energy, work and power were introduced and now these concepts will be reviewed.

Work

Energy and work are closely related concepts. It is prudent to discuss work prior to discussing energy. Work in engineering terms is a vastly different concept from the layperson's use of the term work. Work in engineering occurs when a force causes motion, which may either be a bike moving down a road or deformation caused by a tensile or compressive load. Hence, work is done when a force causes a bike to move, and it is also done when we squeeze the bikes brake pads against the rim for braking. If no motion/deformation occurs, no work is done. The value for work is found by multiplying the value of the force by the displacement as the object moves, and is shown by the formula below.

$$W = Fs$$

W = work (J)
F = force (N)
s = displacement of the object (m)

The unit for work is the Joule (J). One Joule is equivalent to one Newton moved one metre. Since force and displacement are vector quantities, so is work. As a vector there can be negative and positive work. Positive work occurs when work is done in the direction of the applied force and negative work occurs when the work is done in the opposite direction to the applied force.

If the force doing work is at an angle to the horizontal, and the object is moving horizontally, the only part of the force doing work is the horizontal component. Hence we need to multiply the horizontal component of the force by its displacement. Another form of work may be done on the box when the vertical component causes small deformations. This work is usually very small and therefore ignored.

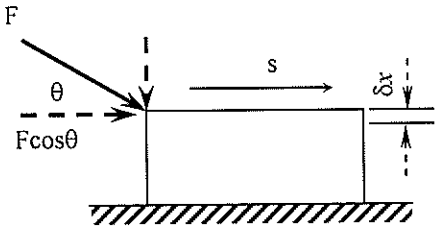


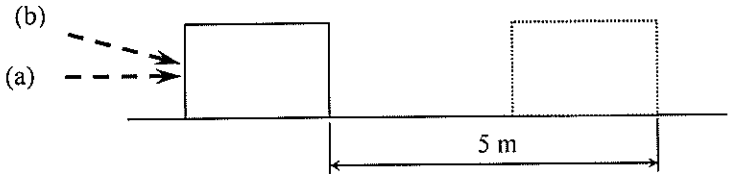
Fig 2.15: *The work done on this box is the product of the displacement (s) and the horizontal component of the force (Fcosθ). Another type of work is the deformation δx that may be caused by the vertical component of F.*

$$W = Fs\cos\theta$$

W = work (J)
F = force (N)
s = displacement of the object (m)

Example 2.8

A box containing a bicycle is slid 5 m by a 200 N force as shown on a smooth surface. What is the work done if the force is (a) horizontal and (b) if it is at an angle of 15° above the horizontal? (Ignore any box deformation.)



(a) $W = Fs\cos\theta$

$W = 200 \times 5 \times \cos 0$

$W = 1000$ J

(b) $W = Fs\cos\theta$

$W = 200 \times 5 \times \cos 15$

$W = 965.93$ J

Therefore the inclined force means less work is being done, as some work is expended in trying to deform the box. If the surface had friction present, then the vertical component would increase the frictional resistance and result in even less work being done.

Energy

In Volume 1, module 3 (pp. 101-104) the concept of energy was discussed. Here we will specifically look at two types of mechanical energy, potential energy and kinetic energy. Energy and work are also closely linked, and the unit for energy is also the Joule.

Potential Energy

Potential energy may be considered as stored energy with the potential to do work. For example, consider a brick weighing 3 kg, held 2 m above the ground. The potential energy will be a measure of its ability to do work from that position. If the brick is dropped from a height of 1 m into soft ground, when it hits, it will do work, i.e. splash mud and make an impression. The mud cannot move unless work is done on it. This work comes from the potential energy that the brick possessed before it fell.

The formula for potential energy (PE) takes into account the mass, gravity and height above a frame of reference.

$$PE = mgh$$

PE = potential energy (J)
m = mass of object (kg)
g = acceleration due to gravity (ms^{-2})
h = height (m)

Kinetic energy

Kinetic energy is energy that a body possesses as a result of its motion. If a body has any velocity then it will possess an amount of kinetic energy which is proportional to the mass, and the velocity squared. Small variations in velocity, therefore, have a greater effect than small variations of mass. The kinetic energy will be equal to the work required to stop the object.

The formula for kinetic energy (KE) is as follows:

$$KE = \tfrac{1}{2}mv^2$$

KE = kinetic energy (J)
m = mass of object (kg)
v = velocity of object (ms^{-1})

PE v's KE

These two quantities are closely linked, as can be seen in transportation devices such as cycles. Fig 2.16 shows a simple drawing of a folding bicycle and rider. As the bike rolls down the slope, its KE will increase, but this will be at the expense of the PE, as energy cannot be created or destroyed but merely converted from one form to another. The reverse is also true; if the bicycle is travelling along a flat road and coasts up a hill, the loss in KE when it stops, will be equal to the PE it has gained. The only way further PE can be gained in climbing the hill is by the rider doing work on the pedals.

In reality the loss in PE and the gain in KE is never exactly the same. The reason is that frictional resistance and air resistance do negative work that opposes motion. A small amount of noise energy would also be given off.

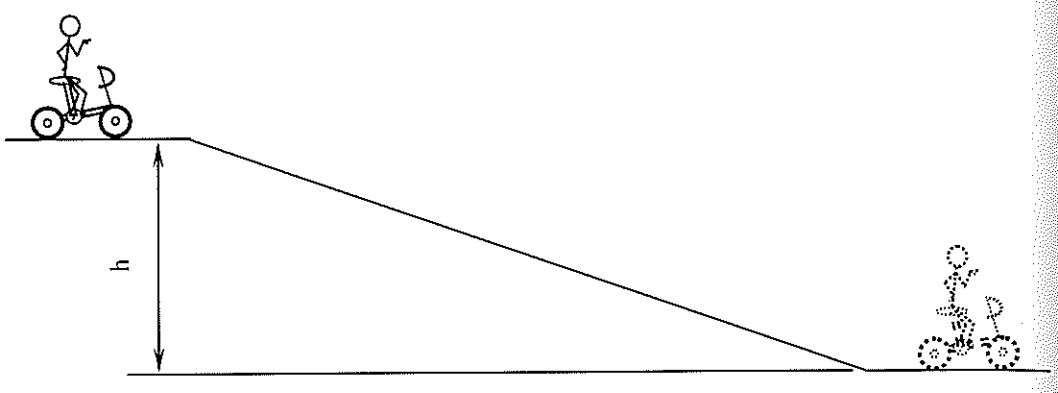


Fig. 2.16: *A folding bicycle and rider is on a road at the top of a hill. They possess potential energy because of the height above the lower portion of the road. This potential energy is lost as they coast down the hill, until at the bottom all the potential energy has become kinetic energy.*

For any object, the initial PE and KE, combined with any positive or negative work will be equal to the final PE and KE combined. Mathematically this may be written as:

$$PE_i + KE_i \pm W = PE_f + KE_f$$

Consider the tricycle example in relation to numbers in Example 2.6.

Example 2.9

The folding bicycle in Fig 2.16 has a mass of 12 kg and the rider weighs 80 kg. What will be the velocity of the bicycle after it coasts to the bottom the hill, if h = 20m and friction and air resistance is ignored?

There are two ways to solve this problem.

Method 1 involves calculating the PE and then determining the velocity.

$$PE = mgh$$
$$PE = 92 \times 10 \times 20$$
$$PE = 18400 \text{ N}$$

$$KE = \tfrac{1}{2}mv^2$$
$$v^2 = \frac{2KE}{m}$$
$$v = \sqrt{\frac{2KE}{m}}$$
$$v = \sqrt{\frac{2(18400)}{92}} \quad (\because KE = PE = 18400 \text{ J})$$
$$v = \sqrt{400}$$
$$v = 20 \text{ ms}^{-1} = 72 \text{ km/h}$$

Method 2 involves algebraic methods to simplify the problem initially.

$$
\begin{aligned}
PE &= KE \\
mgh &= \tfrac{1}{2}mv^2 \\
gh &= \tfrac{1}{2}v^2 \\
v^2 &= 2gh \\
v &= \sqrt{2gh} \\
v &= \sqrt{2\times 10\times 20} \\
v &= \sqrt{400} \\
v &= 20 \text{ ms}^{-1} = 72 \text{ km/h}
\end{aligned}
$$

Method 2 is really the better method as one is less likely to make mistakes in writing down letters, and no calculation errors can slip through until the end.

The final velocity of 72 km/h is fast and somewhat unrealistic. In reality, friction and air resistance would do negative work to lower this figure.

Now we can make the example a little more realistic is we consider a combination of air resistance, friction and rolling resistance causing some negative work. It is only a little more realistic because air resistance is actually a function of velocity so as the bike goes faster the air resistance increases.

Example 2.10

The same 12 kg bicycle and 80 kg rider are at the top of a 20 metre high hill and it is 300 m to the bottom. If the combined resistance to motion is 40 N, determine the velocity of the bicycle after it coasts to the bottom of the hill.

$$
\begin{aligned}
PE_i + KE_i \pm W &= PE_f + KE_f \\
mgh + 0 - Fs &= 0 + \tfrac{1}{2}mv^2 \\
92\times 10\times 20 - 40\times 300 &= \tfrac{1}{2}\times 92\times v^2 \\
46v^2 &= 18400 - 12000 \\
v &= \sqrt{\frac{6400}{46}} \\
v &= 11.80 \text{ ms}^{-1} \\
v &= 42.46 \text{ km/h}
\end{aligned}
$$

This is a more realistic figure for a bike coasting down the hill than in the simpler example. It can be seen the distance the bike travelled was necessary to calculate the negative work. Alternatively the problem could have given the angle of the hill and the distance travelled. The height is then found using trigonometry.

Vehicles loss of Kinetic Energy – Or Energy Wasted v's Energy Used

When a vehicle is moving it has kinetic energy. The brakes must do negative work to retard the vehicle. Currently, on cycles, only friction brakes are available. They convert kinetic energy into heat energy. This system is really a waste of the energy the rider has expended climbing the hill initially. Cars can use engine braking where the motor resists the motion of the car when the accelerator is released, while trucks and buses may have compression braking or retarders, using magnetic methods, to slow a vehicle. All these methods mean that lost kinetic energy is lost for good.

When braking, trains can use the electric motors at the wheels to produce electricity by using the motors, as generators. In an electric train the current is passed back into the overhead lines, while in a diesel electric train the current is fed back through large resistors. Electric cars can also use a similar feature. Electric motors driving the wheels can act as generators, which will tend to slow the vehicle and feed the produced electricity into the battery for later use. In these systems the kinetic energy is converted into a useful form, to help brake the vehicle.

Power

Power is defined as the rate at which work is done. The unit for work is the Watt (W), which is equal to one Joule per second. Power is useful in comparing the outputs of different engines in vehicles. For example, two trucks may do 30,000 J of work but if one does it in half the time of the other then it has double the power of the other engine.

$$P = \frac{W}{t} = \frac{Fs}{t} = Fv$$

P = power (W)
W = work (J)
t = time (s)
F = force (N)
s = displacement (m)
v = velocity (ms^{-1})

Torque

Torque should be mentioned because of its links in engines to power. The torque of a motor is the turning moment that the motor produces. The higher the torque figure, the greater the turning moment. Torque is closely related to moments, and torque is found using the same formula and has the same units. Moments relate to statics, while torque is used in dynamics. Torque is a measure of the forces that a given engine can develop and use in moving a load.

$$\tau = Fd$$

τ = torque (Nm)
F = force (N)
d = perpendicular distance from the force to the pivot (m)

When a motor is connected to a gearbox it is the torque, not the power, that is varied by the gearbox. Trucks typically have high torque engines so large loads can be hauled. The power of an engine is proportional to time rate of change of torque.

Power and Torque Figures – or, "Wow that car must be fast!"

Power figures are regularly quoted in transportation publications as an indicator of how effective a motor is. This figure is only one indicator of how a motor performs. Another is the torque figure which indicates the turning moment of the motor.

The power alone does not decide how fast a vehicle will be. A lighter, less powerful car may have a better ratio of power to weight than a heavy car and hence perform better. Gearing also has an impact on performance with the different gear ratios important determinants. A high torque figure means that the vehicle climbs and pulls loads well.

Finally the power figure given for a motor is the power at the crankshaft. After the gearbox, differential and drive shafts are taken into account, the power figure is considerably lower, so power at the wheels may be 50% less than that at the crankshaft. This is because no engineered device is ever 100% efficient and devices like gearboxes often sap considerable power with friction.

Example 2.11

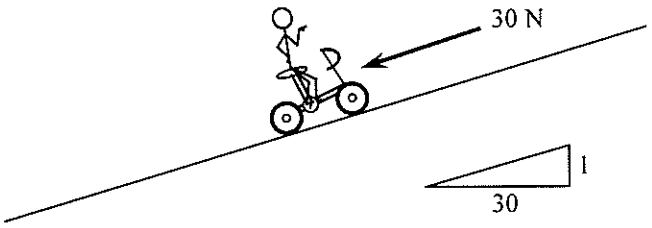
A car is travelling at 110 km/h on a level section of freeway. If the total resistance to motion is 1.1 kN, determine the power the motor must expend to keep the car travelling at this velocity.

Remember we must convert the 110 km/h velocity into 30.56 ms^{-1}.

$$P = Fv$$
$$P = 1100 \times 30.56$$
$$P = 33611.11 \text{ W}$$
$$P \approx 33.6 \text{ kW}$$

Example 2.12

A bicycle and rider (total mass 92 kg) are ascending a hill of grade 1 in 30. Determine the power the rider needs to expend to maintain a velocity of 20 km/h if the total resistance to motion is 30 N.



We must determine the force the rider must provide to overcome both the resistance to motion and the force of his/her mass down the hill, which is $mg\sin\theta$.

The force provided by rider, F_T, against weight and resistance is found by summing forces parallel to the incline.

$$\overset{+}{\nearrow} \Sigma F = 0$$
$$0 = F_T - mg\sin\theta - F_R$$
$$F_T = mg\sin\theta + F_R$$
$$F_T = 92 \times 10 \times \sin 1.91° + 30$$
$$F_T = 60.65 \text{ N}$$

Now determine the power needed with a velocity of 20 km/h, which is 5.56 ms^{-1}.

$$P = Fv$$
$$P = 60.65 \times 5.56$$
$$P = 336.94 \text{ W}$$

Engineering Materials

Testing of Materials

Hardness Tests

Table 2.1: *Hardness Tests*

Hardness Test	*Operation*
Brinell	A hardened steel ball is forced into an object under specified load conditions. The diameter of the ball is dependent on the test piece's thickness. The hardness number is determined by measuring the depth and surface area of the impression and using a formula, which also incorporates the applied load.
Vickers	Here a small square pyramid is forced into a test piece under specified load conditions; the hardness number is derived from a formula that contains the load and area of the indentation. More useful for thin metals than the Brinell test as the loads are lower and the indentor smaller.
Rockwell	This is a simple test that dispenses with the use of a formula. A diamond cone (or sphere) is forced into a test piece under specified load conditions and a reading is displayed on the dial. There are different scales for different loads depending on the material tested. Rockwell hardness testers are the most common hardness testers in schools as they sued to be provided to NSW high schools.
Shore Scleroscope	This test involves a small striker in a tube. The tube is placed over an item and the striker is dropped, the height the striker rebounds from the item is a measure of the hardness. Soft items will absorb more of the energy of the falling striker giving a lower rebound than hard materials

Impact Tests

Impact tests are carried out to determine the notch toughness of a material. This is done by subjecting the material to a concentrated shock load. Tough materials will not break as easily as a brittle material.

There are various ways to perform an impact test. The two most common standard methods are the Izod and Charpy Tests, while a small laboratory system is the Hounsfield balanced impact tester.

The Izod and Charpy Tests are essentially the same except the test piece is held differently. The test involves a large pendulum being raised to a specified height to give it potential energy (PE). Upon release, it loses PE and gains kinetic energy (KE). At the bottom of its swing it will strike the test piece, with some KE absorbed in breaking the test piece. After passing the test piece, the pendulum will swing up in height gaining PE and losing KE, until all energy of swing is used up. The height reached is recorded. The difference between the initial height and the final height is proportional to the energy absorbed when breaking the test piece.

The Hounsfield balanced impact tester is smaller, and used for schools and colleges or in industry, as a transportable machine. It uses two pendulums, with one solid one swinging through the other hollow pendulum. The result is double the impact that would be had with a single pendulum, thus allowing a smaller tester. The energy required to break the material, is merely read from a scale on the pendulum pivot.

The test piece used in all these testers is a small cylinder with a V notch placed in it to concentrate the stress and provide a place to promote crack propagation. In the Hounsfield test, the notched piece is held in the solid pendulum, so the outer pendulum hits both ends. In the Izod test, the notched bar is held vertically, with the upper end free, ready to be struck by the pendulum. The notch is placed so it faces the pendulum. In the Charpy test, the notched bar is held horizontally at both ends. The pendulum strikes it in the middle, with the notch facing away from the pendulum.

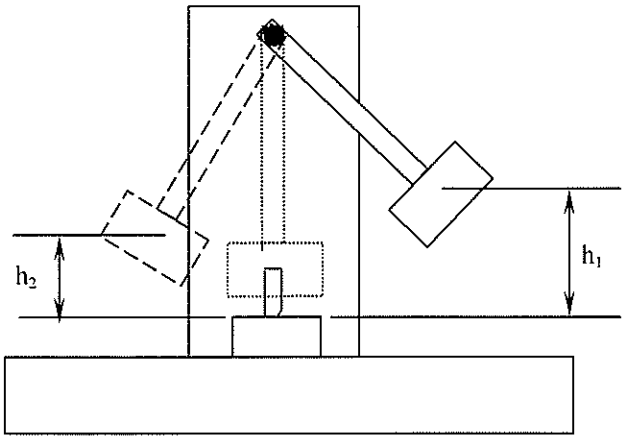


Fig. 2.17: *An Izod test, with the various pendulum positions shown. At the solid position, PE is a maximum; at the dotted position, PE is zero, while at the dashed position, PE is equal to the initial PE minus the energy required to break the notched piece.*

Heat Treatment of Ferrous Metals

The properties of steels can be greatly altered by the way they are cooled after heating. This section outlines some of the possible modifications made to steels through heat treatment.

Annealing

There are two basic types of annealing – process annealing and full annealing.

Process annealing involves the heating of a steel with less than 0.3%C to a temperature usually between 550 and 650°C. The purpose of this is to relieve any stress from distorted grains caused by cold working or deformation. The ferrite grains will reform as unstressed grains, while the pearlite remains in the deformed state

Full annealing involves heating either hypo-eutectoid steels or eutectoid steels into the austenite region at a temperature of about 40°C above the upper critical temperature[1]. The steel is then cooled very slowly, usually in a furnace, with the result being a softer, coarser grained steel than previously existed. All grains will be in an unstressed state.

Normalising

Normalising involves heating a steel up into the austenite region well above the UCT. When the structure is all austenite, it is then cooled in still air. The process takes less time than full annealing and produces a finer grained structure and hence a stronger steel.

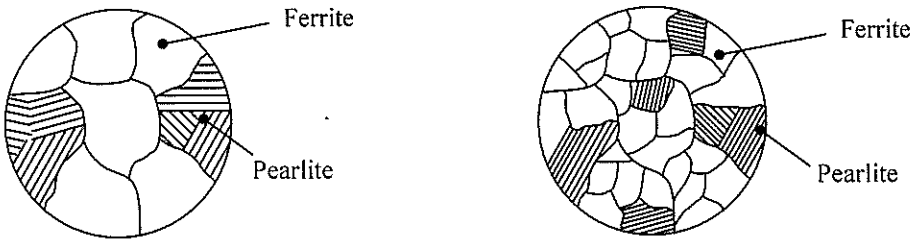


Fig 2.18: *(a) Annealed low carbon steel and (B) normalised low carbon steel. Note the smaller Ferrite grains and finer Pearlite in the normalised steel.*

Hardening and Tempering

The ability to harden steels is one of their most desirable characteristics. This is one of the many reasons why steel is such a versatile material.

Hardening

The concept of hardening steels and the resulting martensite was first covered in Volume 1 (p 110). A brief recap is given here. If a steel is heated until it is austenite in structure and quenched rapidly, the transformation from face centred cubic (FCC) austenite to body centred cubic (BCC) ferrite is not given enough time to occur fully and the steel becomes trapped in between as Body Centred Tetragonal (BCT) martensite. This new structure can be exceedingly hard but quite brittle. Martensite will form in any steel with carbon composition greater then 0.03%. Low carbon

[1] The upper critical temperature (UCT) is a temperature for each steel composition at which the steel fully transforms to austenite. The value UCT is composition dependent and ranges from 910°C for 0.01%C steel then down to 723°C for 0.8% C steel.

martensite is soft and does not become really hard until the carbon composition is in the range of 0.4 and 0.8%.

Air Hardening

If a steel has nickel and chromium added in small amounts (<5%) then it will have air hardening properties. This means that if it is heated to red hot and cooled in still air, martensite will form. Usually molybdenum is also added to reduce brittleness.

Tempering

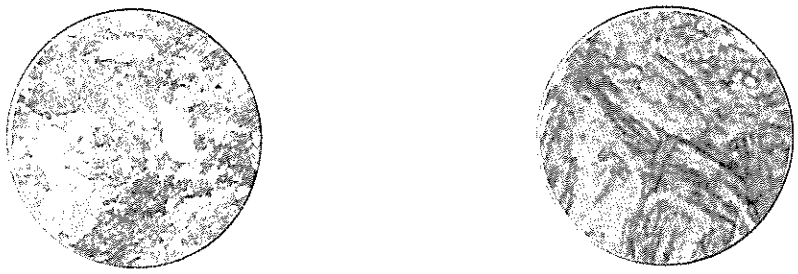
A fully hardened steel has limited usefulness as it can only be used in applications where hardness is the only requirement. If hardness and toughness are both required, a fully hardened steel is no longer useful. It is, however, possible to sacrifice some hardness and gain better toughness through tempering. This involves taking a hardened steel and heating it to a temperature between 200 and 600°C. A low tempering temperature will produce high hardness and moderate toughness while a higher tempering temperature will have the opposite effect.

Although not as hard as a fully hardened steel, tempered steels are still considerably harder than annealed or normalised steels.

Microstructural Changes

One obvious feature of heat treated steels relates to grain size and structure. Annealed steels tend to be soft with moderate strength, in part due to the coarse grain structure that results. Normalised steels tend to have a finer grain structure which results in higher strength.

By hardening a steel, the grain structure is highly stressed and has an acicular appearance, with the result being high hardness and brittleness. Tempered steels tend to have a very fine structure of carbide particles in a ferrite matrix, which gives toughness and moderate hardness.



A B

Fig 2.19: *(A) 0.8% normalised plain carbon steel. (B) 0.8% plain carbon steel hardened to give martensite. (from* Experimental Materials Science, *Gibson and Taylor)*

One structure produced by heat treatment that cannot be produced by normal cooling is *bainite*. Bainite has a structure consisting of very fine cementite particles in a ferrite matrix. Bainite, in fact, is similar to pearlite in overall composition but instead of layers of cementite it consists of tiny particles. Bainite is formed by a process called

austempering where the austentic steel is quenched to around 400°C and held there until that temperature exist throughout, then it is quenched to room temperature. The resulting structure is similar in performance to tempered martensite, but has significantly greater resilience.

Bainite at times occurs when the quench rate is not fast enough to produce martensite as some cementite can form. If quenching is of a moderate speed then there may be a mixture of martensite and bainite. If the quench rate slows further then the result will be fine pearlite.

Surface Hardening of Steels

In some instances we want a hardwearing surface for a steel object, but the toughness of a softer core is desirable so there are a number of processes that harden only the surface but leave the core soft.

Case Hardening

Case hardening involves heating and soaking the steel in a carbon rich atmosphere or medium. The carbon diffuses into the structure at the surface increasing the carbon content and hence the hardness and tensile strength. Any heat treatment of the steel is more challenging now due to the different carbon content between the case and the core. Normalising and hardening of the surface are possible.

Nitriding

Nitriding is a surface hardening process where a special alloy steel is heated in a furnace with gaseous nitrogen present (not atmospheric nitrogen) from the decomposition of a material such as ammonia. The nitriding is performed at 500°C for between 40 and 100 hours and the gaseous nitrogen reacts with aluminium, chromium or vanadium in the steel, depending on the chosen 'Nitralloy' steel. Nitriding offers high hardness in the core, is clean, produces a corrosion resistant surface on the steel and maintains hardness up to 500°C. The initial setup is more costly because special alloys steels are required and if heated past 500°C the hardness from nitriding is permanently lost, so the item must be nitride from scratch again.

Flame Hardening

This process takes a steel of suitable carbon composition ($\geq$ 0.4%C) and applies a flame to a localised area and then quenches it. This process results in a localised area being hardened. The process is mechanised with the flame holder and water jet being the one assembly with the water jet trailing the flame holder.

Induction Hardening

Using the same principle as flame hardening but instead of using a flame an induction coil is used. This increases the temperature in the steel to the desired range for heat treatment, the current is turned off and then the surface is quenched with water to harden it. This process lends itself well to mechanisation and it is used for items such as camshafts in engines

Changes in Properties

There are primarily three ways we can alter the properties of steels. Steels are greatly effected by the addition of carbon, and only small amounts of carbon are needed to achieve the changes. This is because carbon and iron form an interstitial solid solution. In Chapter 3 Volume 1 a chart clearly showed the effect carbon content had on hardness, tensile strength, ductility and toughness. This chart should be reviewed to ensure students understand the clear effect carbon has.

Properties can be further changed by heat treatment; iron's allotropic nature means that by varying the rate of cooling we can bring about vastly different properties in a steel. The hardening and tempering of steel is a process that cannot be repeated with most other alloys.

Finally properties can be further altered by alloying steels. Alloy steels such as stainless steel, high speed steel, HSLA steels and "cro-moly" steels were all outlined in Volume 1. Often by adding alloying elements we bring about improved properties without the detrimental effects of high carbon content. Air hardening steels mentioned earlier in the chapter are only possible with the addition of nickel and chromium, no plain carbon steel can achieve this property.

Manufacturing Processes for Ferrous Metals

When materials are formed, the formative process may have a large impact on the properties of the final article. Below is an outline of some of the effects of manufacturing processes on the materials' properties.

Forging

Forging may be defined as the shaping of a metal through the use of force. Forging may be carried out above the recrystallisation temperature (hot forging) or below it (cold forging or pressing). Forging can take a variety of forms, the simplest type being that which a blacksmith does against an anvil. Forging may draw out a metal while reducing its cross-sectional area (drawing); reduce its length while increasing its cross-sectional area (upsetting), or it may force the metal into dies to take the required shape, as in drop forging.

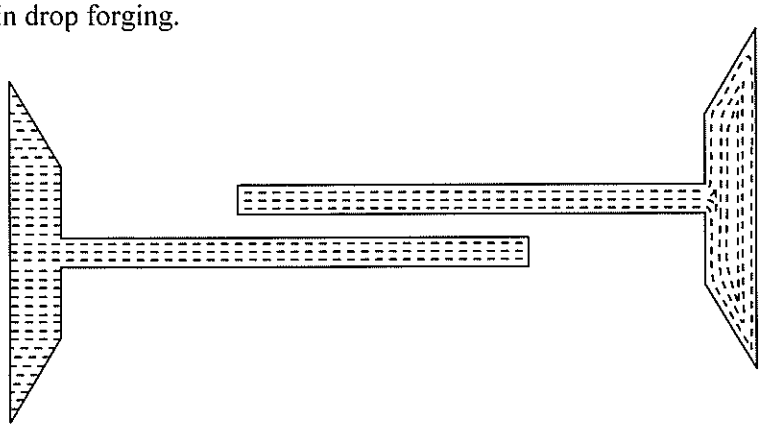


Fig. 2.20: *The macrostructure of two automotive valves are shown. The upper one is forged while the lower one is machined from a hot rolled bar. Note how the grain flows to the shape of the forged valve.*

All of these methods produce grain flow in the metal that follows the shape of the object. This improves the strength of the finished article, as against a machined part where the grainflow may cause planes of weakness.

Rolling

Rolling can be done either above the recrystallisation temperature, (hot rolling), or below the recrystallisation temperature, (cold rolling). The recrystallisation temperature is that temperature above which deformed (stressed) grains in metal will re-nucleate and grow into annealed grains (stress-free). Unstressed grains will not re-nucleate, but they will grow.

Hot rolling

Hot rolling is used extensively in the production of sheets, strips, bars and rods of metal. Ingots of the required metal are passed through successive rollers to produce the required thickness of metal. When passed through the rollers the metal's crystal structure is deformed. Since it is above the recrystallisation temperature however, it recrystallises into an unstressed form.

The *advantages* of hot rolling are:
- less stress on the machinery when compared to cold rolling
- an unstressed finished product.

The *disadvantages* are:
- that the final products are not as dimensionally accurate
- it will have a black oxide layer over the finished product.

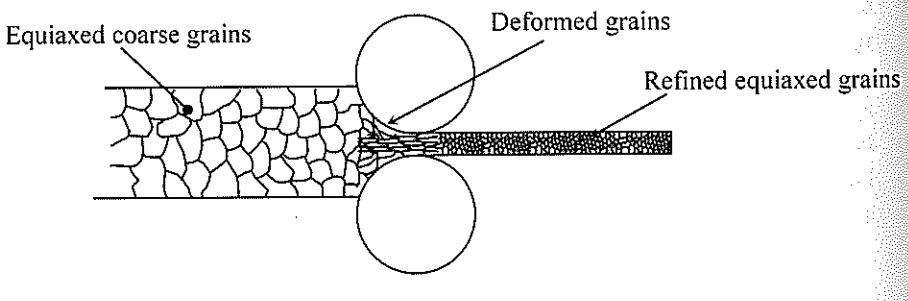


Fig. 2.21: *Hot rolling will produce a fine, but unstressed grain structure*

Cold Rolling

As a result of being carried out below the recrystallisation temperature, cold rolling will produce a slightly different final product. The procedure is essentially the same but the rollers and machinery are more heavily built, as larger forces are required. The finished product will have distorted grain structure that will produce a harder and stronger final product. These properties come at the expense of ductility and malleability.

The *advantages* of cold rolling are:
- a harder final product that is more dimensionally accurate
- a more presentable product because of the lack of oxides.

The *disadvantage* is greater cost because of the heavier machinery needed.

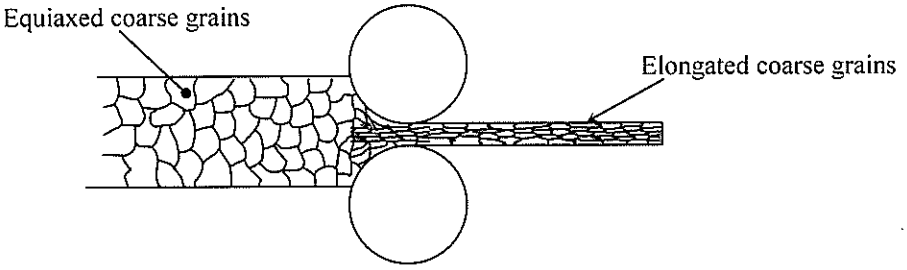


Fig. 2.22: *The final structure has a distorted grain.*

Casting

Casting comes in a variety of forms. In Volume 1 we only looked at Sand Casting, Die Casting and Investment Casting. Now we shall review those and introduce some more.

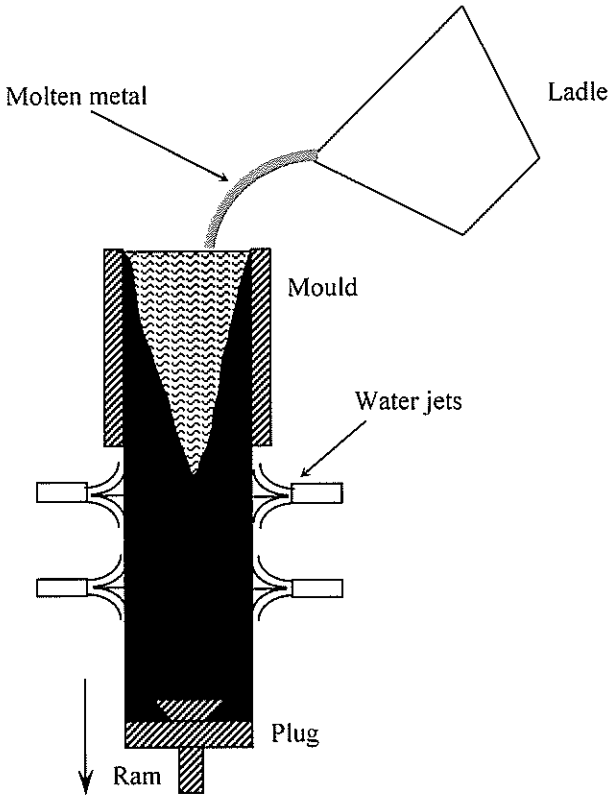


Fig. 2.23: *A method of continuous casting. The water blasts cool the metal as the moving base slides downwards.*

Ingot Casting

Many metals are not cast to their final shape. Often metals are cast into a large block, or ingot, to be shaped into something by rolling, or some other mechanical means. Ingot casting is done by pouring a molten metal into a large tapered metal mould. Upon solidification, the mould is lifted away and the ingot is ready for shipment.

Ingot casting was used extensively, but nowadays it has been replaced by more mechanised methods of continuous casting.

Continuous Casting

This method of casting allows for the rapid production of simple cross-section products like bars and strips. The molten metal is poured into a water-cooled ingot with a sliding bottom. Once the bottom has solidified, the base moves down at a rate that allows the molten metal above it to solidify. The resulting long metal strip is cut to the required length. This casting method is used in large plants because of its rapid speed and cost effectiveness on large runs.

Sand casting

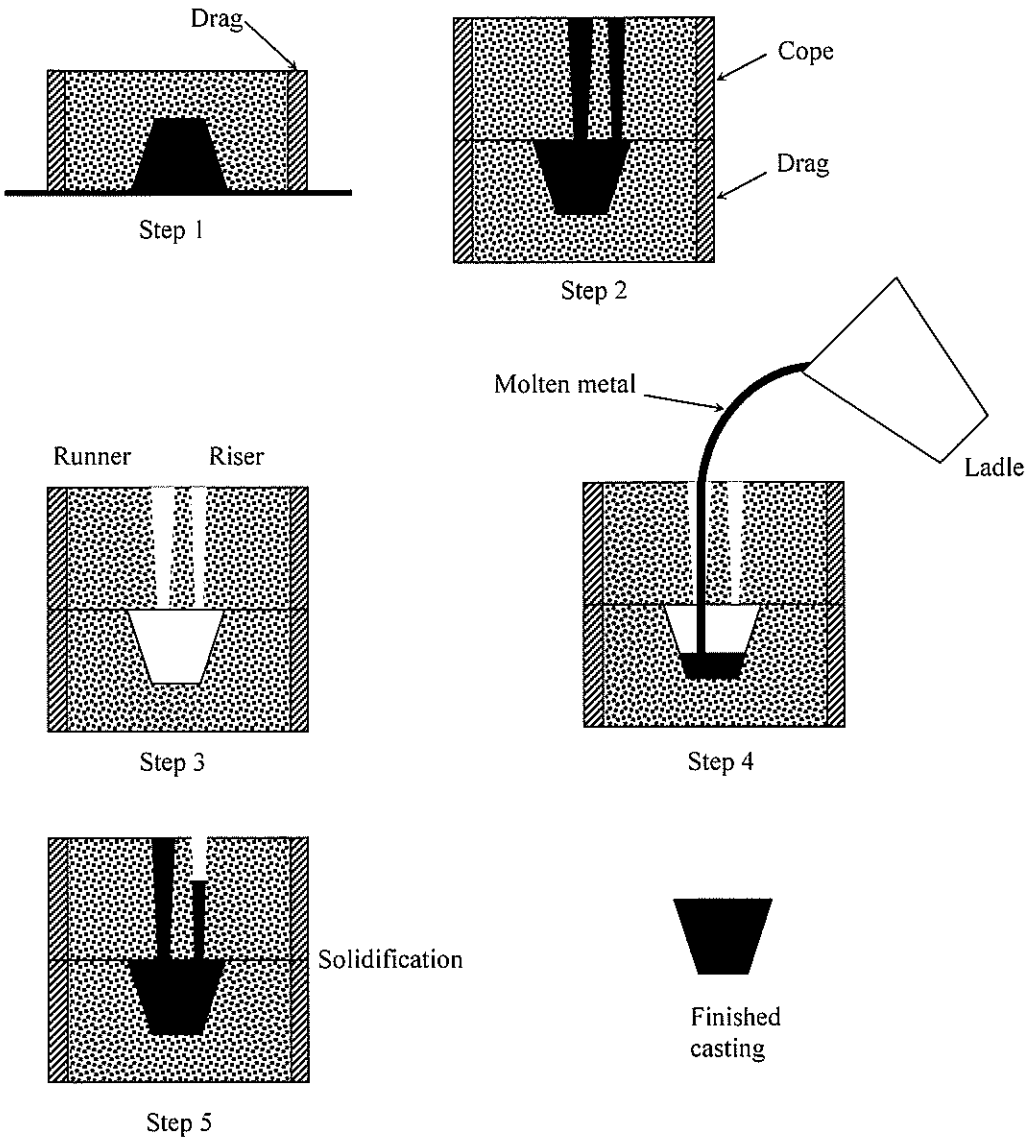


Fig. 2.24: *The steps involved in sand casting*

This is one of the most common casting procedures. Sand, with a binder, so it holds together, known as green sand, is packed around a pattern of the finished product. The

mould is in two halves to allow the pattern to be removed. Once the pattern is removed, and the two halves are assembled a cavity is left for molten metal. Once the metal solidifies, the sand is removed, and reconstituted, ready to be used again. The procedure is outlined below.

In Step 1, the lower box, 'the drag', is placed on a board with the pattern inside. "Green" sand is then packed tightly around the pattern. The drag is inverted and the top box, "the cope", is placed on top. By Step 2, the cope is filled with sand and there are pins for the riser and runners in place. At Step 3, the cope and drag are separated, the pattern, riser and runner pins are removed and the boxes are reassembled. Step 4 sees the molten metal poured into the runner pin until the riser and runners are filled. At Step 5, the casting is allowed to solidify and any shrinkage caused by contraction is alleviated by molten metal from the riser pin running into the casting. Once solidified, the casting is removed and the runner and riser parts are ground off.

The process can be automated, and it finds great use in the automotive fields. Engine blocks and heads are usually sand cast. The advantages of sand casting are cheap and easy castings and good final grain structure. The disadvantages are that the final surface finish is poor and inaccurate. Hence machining is essential for precision parts.

Shell Moulding

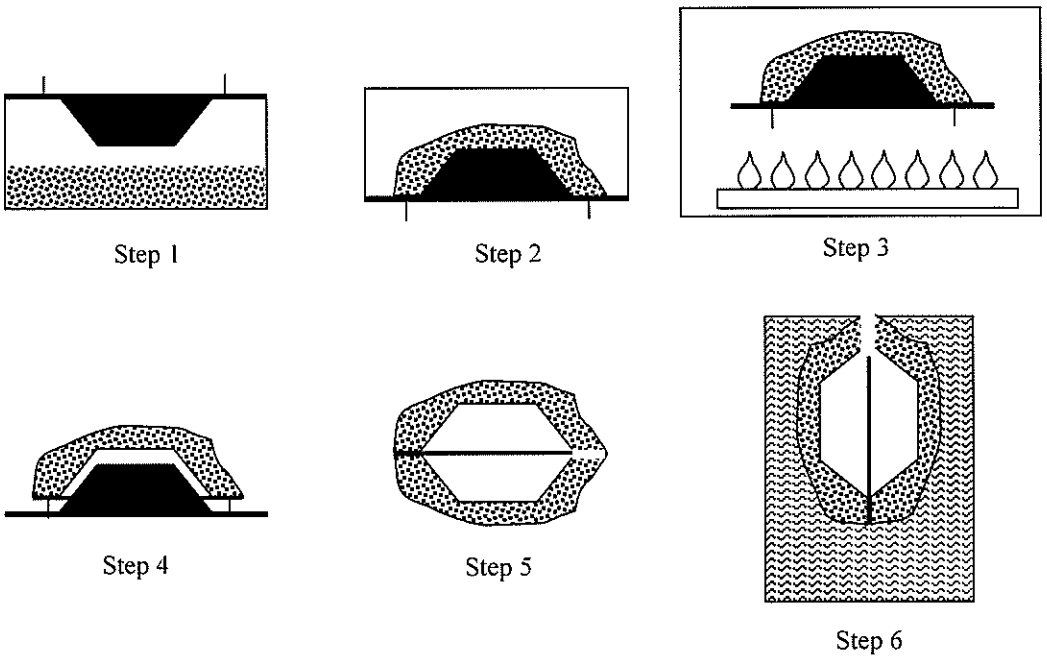


Fig. 2.25: *The steps involved in shell moulding*

This is a close relative of sand casting and utilizes sand as a molten material in a different way. The steps are outlined below.

In shell moulding, a heated pattern plate is placed over a dump box (Step 1). At the bottom of the dump box is a mixture of sand and a thermosetting resin. The dump box is inverted and the sand and resin mix drops over the heated pattern plate (Step 2). The heat from the pattern plate partially sets the resin, holding the sand together. The half mould and pattern plate are then heated in an oven at 315 °C to ensure the resin is

fully cured (Step 3). The cured half mould is then ejected off the pattern by small ejector pins (Step 4). The two half moulds are then placed together; they may be bolted or screwed together (Step 5). Finally the mould is placed in a box surrounded by metal shot, ready to receive the molten charge. Upon solidification of the casting, the mould is separated.

Shell moulding offers closer tolerances than die casting and better surface finish than sand casting. It does, however, cost more than sand casting, as to get high dimensional accuracy, the metal pattern plates are expensive to make.

Centrifugal Casting

This method relies on centrifugal force to spin the metal to the outsides of the mould to create a hollow cylinder. This type of casting is useful in creating pipes. Piston rings for cars can be made this way, with a cylinder produced and the rings then cut off the cylinder.

Permanent Mould Casting (die-casting)

Unlike sand casting, this method involves the use of a permanent mould which is not re-made each time.
(i) Gravity die-casting involves the use of a permanent metal mould, into which molten metal is poured by gravity. Since the mould cannot be broken to remove the cast item, the mould must be able to be separated. The advantages of gravity die-casting on long production runs are that the cost is less than sand casting, and the final casting has a better surface finish. For smaller production runs, however, the permanent steel moulds are too expensive. Many automotive parts, such as pistons, are made using this method.

(ii) Pressure die-casting is a very similar to gravity die-casting, but in pressure die-casting the molten metal is forced in under pressure. The result is a denser casting than is achieved with gravity die-casting. Like gravity die-casting, there is a good surface finish and for high production runs it is quite cost effective. Pressure die-casting is primarily used with lower melting point alloys, such as aluminium alloys and zinc alloys, as dies for high temperature ferrous alloys are expensive. If ferrous alloys are to be cast using this method, it is necessary to use graphite dies. Gearbox casings may be made using this process.

Investment Casting

Also known as lost-wax casting, this method of casting has existed for centuries. Today it is used in the manufacture of high quality castings where dimensional accuracy and high surface finish are required. Fig. 2.26 outlines the process. A pattern of the item is made from wax and a refractory ceramic is then poured over the material and allowed to set (Step 1). The wax is drained out by heating the mould, leaving a cavity that is the perfect replica of the item to be cast (Step 2). This method will also not leave a line around the casting as the mould is not separable like a die cast mould. Molten metal is poured in and allowed to solidify (Step 3), and then the mould is broken away from the cast article (Step 4). Since the mould is destroyed, a new one

must be made each time, so for large runs it can be costly. The final result, however, is worth it, as the final casting has good surface finish and is dimensionally accurate.

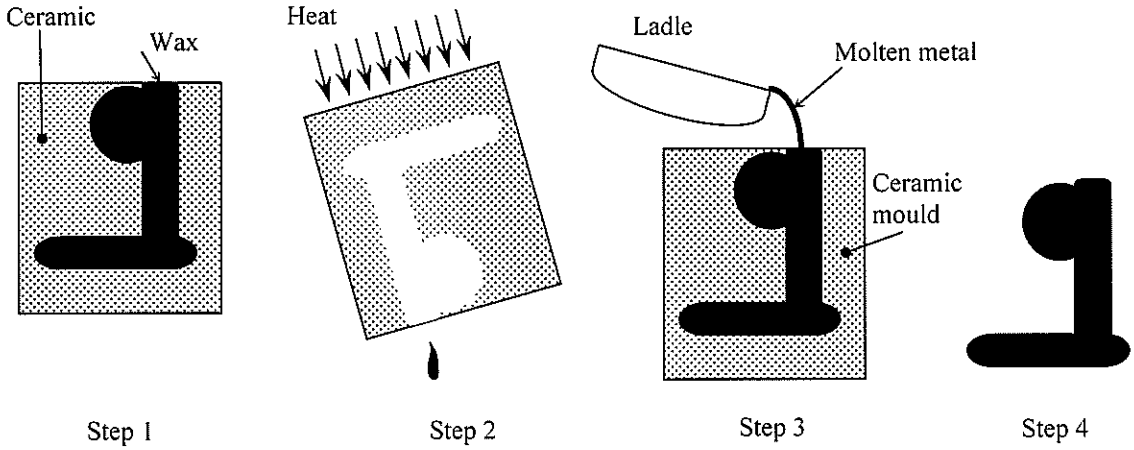


Fig. 2.26: *The investment casting process.*

There are automated types of investment casting used to manufacture rocker arms for automotive engines. It has found extensive use in transport devices Not only are car rocker arms made this way, but turbine blades for jet aircraft are investment cast and drop outs for bicycles may also be made this way. Investment casting is also used to cast metal alloys that are very hard and difficult to machine as there is little need for machining to size, using this method.

The Full-Mould Process

The full-mould process is a close relative to investment casting, but is usually used for one-off items or prototypes. An expendable pattern for the item to be cast, complete with runner, is made from expanded polystyrene (foam). This is then placed in a box and surrounded by sand with small amounts of thermosetting resin. The molten metal is then poured into the foam runner and this melts the foam. Since only 2% of the foam is solid then this solid simply melts to the sand and helps bind it together. The air, that made up the other 98% of the foam, passes through the sand and does not contaminate the casting.

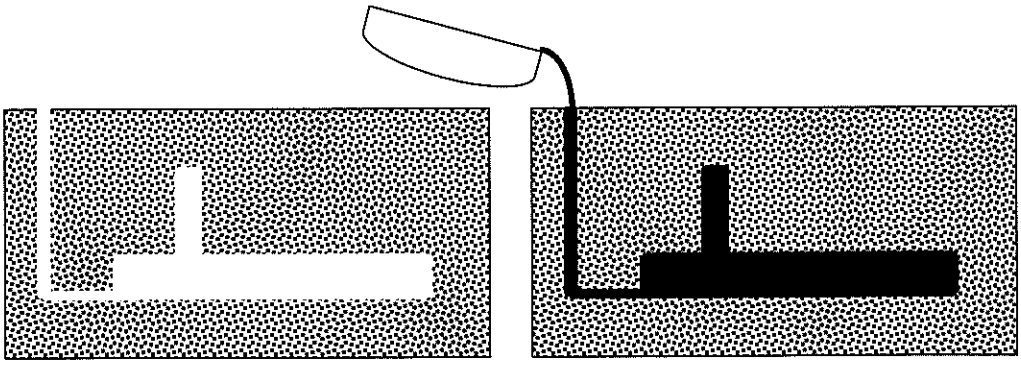


Fig. 2.27: *The diagram on the left is the sand with the foam pattern in place, while on the right the molten metal is poured into the cavity dissolving the foam and leaving the casting behind.*

The full-mould method can be useful in engineering to make a prototype in metal. Suspension systems, for example, may be made like this to assess their design, prior to deciding to mass-produce them, using sand or die-casting.

Extrusion

Direct and Indirect Extrusion

Extrusion may best be likened to squeezing toothpaste from a tube. When the toothpaste comes from the tube it takes the shape of the "die" or nozzle at the end of the tube. The principle is the same with metal. The metal is forced through a die so it takes the shape of the die through which it passes.

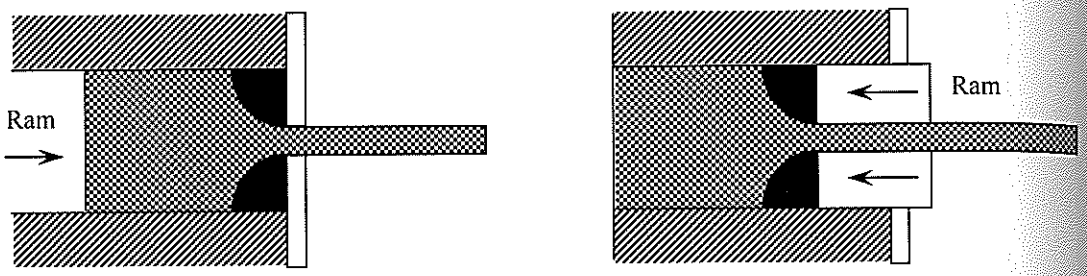


Fig. 2.28: *Direct extrusion is shown to the left and to the right is indirect extrusion.*

Extrusion comes in two forms; direct, where the ram pushes the metal into the die from the other side, and indirect, where the ram and die are the one part. Both are shown in Fig. 2.28. Direct extrusion generally requires more effort so it is used with more ductile materials. Indirect extrusion is used in the extrusion of alloys with lower ductility, but, because the equipment required is more expensive, direct extrusion is used where possible. Both direct and indirect extrusion are hot working processes.

Tubing may be made through extrusion by a mandrel placed in the centre of the die to create a hollow section.

Impact Extrusion

Unlike the other types of extrusion, this is a cold forming process. Impact extrusion involves the use of a hammer impact to extrude a shape. The punch goes into a die and the material blank is forced from the die around the punch. Cans and short tubes are often made using this method.

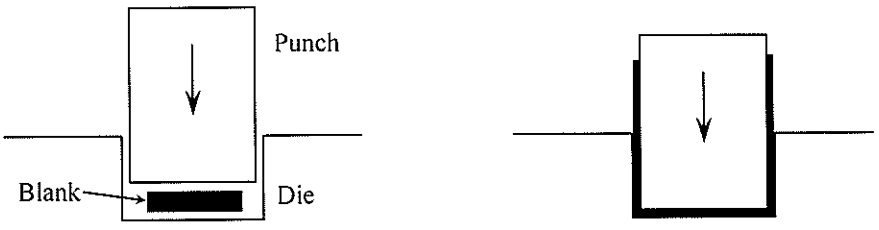
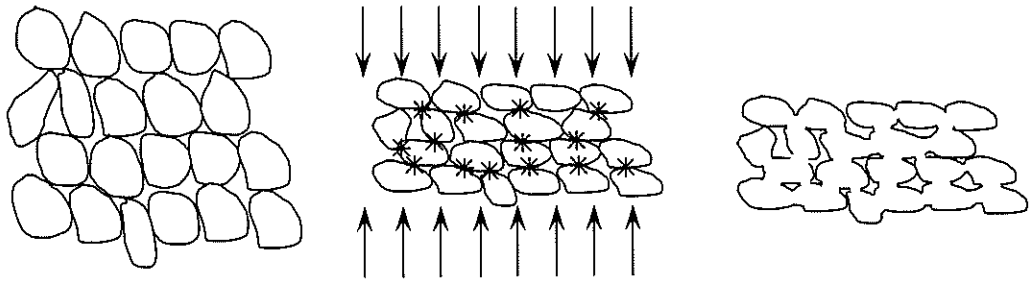


Fig. 2.29: *In impact extrusion, the punch forces the metal to flow out of the die and around the punch.*

Powder Forming

Metal powder forming or powder metallurgy is a process that finds extensive use. The process involves first getting the metal into powder form. This may be done by mechanical disintegration such as grinding, atomizing the metal from liquid form by chemical means, or by electrolytic methods.

Once in powder form, the powders are blended with stearate based dry lubricants, to get the required mix. They are then pressed into a mould to form the shape required. The pressure used is enough to compact the particles together and give the item sufficient strength to be handled. Once pressed, the item is sintered, in a non-oxidising controlled atmosphere furnace, at a temperature to allow atoms to diffuse between grains, producing a homogeneous grain structure (Fig. 2.30). This gives the product its final strength.



A B C

Fig. 2.30: *The steps in powder forming, A shows the powder particles in a free state, B shows them cold welding because of the pressing, and C displays the structure after sintering. Note the grain growth where the cold welding was.*

Products of Powder Forming

The products of powder forming can be grouped into four general groups: porous metals, complex articles, products difficult to machine and composites.

Porous metals are produced by deliberately using larger powder sizes, the result is metal products with porous properties. Such metals can then be impregnated with oil to give a lubricating effect. Such self-lubricating bearings are used in bicycle suspensions and electric motor bearings.

Complex articles are another. Some shapes are simply too difficult or too costly to cast and machine so they are powder-formed instead.

Products difficult to machine are those articles made of materials that are too hard to machine. Metals like tungsten cemented carbides are difficult or nearly impossible to machine so powder forming may be used.

Composites can be readily formed. Many materials will not combine to form an alloy but can be formed together as powders to capitalize on their individual properties. Materials such as copper or silver may be mixed with tungsten or nickel for electric contacts. These will not alloy together so powder forming is used. The copper

provides good electrical conductivity while the tungsten provides resistance to oxidation and wear.

Advantages and Disadvantages

Powder forming allows many articles to be produced that otherwise would be difficult to make. Sintered components are used in automotive gearboxes and perform better than manufactured parts. It is, however, expensive to set up powder-forming facilities and the shapes are still somewhat limited. The finished article is also not as strong as a conventionally produced item.

Welding

Welding processes are summarised in Table 2.2

Welding Structures

When we weld materials we often bring about some significant changes in the parent material. The area around the weld is called the heat-affected zone (HAZ) and in this area the heat of the welding process may cause some changes in the grain structure of the material.

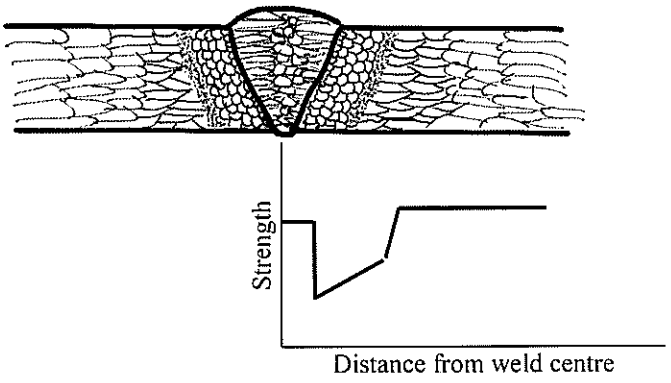


Fig. 2.31: *A cold rolled plate that has been welded. Grain growth around the weld lowers the strength around the weld.*

If we are welding cold rolled plate then in the HAZ the plate will exhibit unstressed grains as the welding process will heat the plates near the weld above the material's recrystallisation temperature. Generally non-ferrous metals will be softened around the weld by the welding process. While mild steel is very easily welded it will also suffer grain growth near the weld, which will soften cold rolled plate. High carbon steels and some alloy steel are more challenging to weld and may need to be slow cooled to stop martensite forming. In some high carbon steels pre-heating may be necessary to stop martensite forming or cracks developing.

Non-Ferrous Metals and Alloys

There are a huge variety of non-ferrous metals and alloys, yet they are still not used as extensively as the various ferrous alloys. They have, however, carved out a substantial niche for themselves, offering diverse properties that the ferrous alloys find

hard to match. While the range of non-ferrous metals and alloys is broad, this section is mainly concerned with the ferrous alloys of aluminium and copper.

Table 2.2: Summary of welding processes.

	Welding method	**Type of process**	**Method**
1	Spot welding	Pressure	Electric current melts metal sheets under pressure joining it in "spots". This process is extensively used in the manufacture of cars and is how most car body panels are joined at their seams
2	Butt welding	Pressure	Metal is butted together at the ends and current melts the metal joining it together. Useful for joining tubes end on.
3	Seam welding	Pressure	Metal is moved through rotating wheels that pass electric current into the metal, melting it, then joining it. Seam welding is used in the manufacture of tubes for cheap bicycles. The flat steel is rolled into a tube and welded at the seam. High quality tubes are however, seamless.
4	Oxy-acetylene welding	Fusion	Metal is melted by oxy-acetylene flame and a filler metal is added. This process is extensively used in repairs and finds greater use in home style DIY projects than in automated manufacture now.
5	Bronze welding	Fusion/ alloying	A flame (usually oxy-acetylene) heats the parent metal and bronze filler metal is added to the joint. Unlike oxy-acetylene welding there is little or no melting of the parent metal. The process resembles brazing except the filler metal is built up at the joint. This process works well for lower temperature projects that cannot have the parent metal melted.
6	Electric arc welding	Fusion	Metal is melted by an electrode, which is also the filler metal, because as the arc is struck the electrode melts into the joint. The temperature of the arc is over 3000°C at the tip. The electrode is covered in a flux to prevent oxidation of the weld metal. Because the electrode is consumable it must be changed periodically, and as it is used the welder must move his/her hand down to maintain the arc
7	Metal inert gas (MIG) welding	Fusion	MIG (Metal Inert Gas) replaces the electrode with a continuous feed wire thus facilitating quicker welding. The flux is replaced by an inert gas which may either be argon or Argoshield, (a mix of argon and carbon dioxide) that protects the weld metal from oxidation when molten. This process is well-suited to automation and is used extensively in many manufacturing industries, such as bicycles and cars.
8	Tungsten inert gas (TIG) welding	Fusion	Similar to MIG, TIG (Tungsten Inert Gas) replaces the continuous feed wire with a tungsten electrode and a filler rod fed by the operator. Like MIG an argon shield is used to protect the weld from oxidation. It is like combining the methods of oxy-acetylene and electric arc. This process is the chosen method for welding materials like stainless steel, aluminium alloys and titanium alloys.
9	Plasma arc welding	Fusion	A gas, such as argon, is passed through an electric arc. The gas ionises electrons and positive ions. This mixture is called plasma. The ions recombine and form a hot flame. It is used with refractory metals such as tungsten and molybdenum.

Aluminium and its Alloys

There is a wide range of aluminium alloys used in transportation systems. They have become more widely used since WWII as a solution to reducing weight in bicycles and cars.

Aluminium

Aluminium is one of the most common metals in the Earth's crust. It was not, however, until the 20th century that it was no longer a precious metal. In nature, aluminium exists as the ore, bauxite and because of its reactivity in elemental form it is never found in nature as pure aluminium. So it was not until a satisfactory method for refining the ore was developed by Hall and Heroult in 1886, that it became an affordable metal.

Aluminium is extremely reactive, has relatively low strength, is ductile, has low specific gravity and is easily fabricated. It is more difficult to weld than ferrous alloys and it is also more costly than mild steel.

What aluminium does offer, is a good strength to weight ratio, corrosion resistance, due to the formation of a tenacious oxide film and good electrical conductivity, only surpassed by gold, copper and silver. In fact, when comparing weight to conductivity, aluminium is a better conductor than copper.

Aluminium is used for overhead electrical wiring, in the manufacture of cooking pots and pans, and for cooking foil. Because of its low strength, most engineering applications are for aluminium alloys.

Aluminium Alloys

Aluminium alloys fall into the loose categories of wrought and casting alloys. The casting alloys are used exclusively for casting while the wrought alloys are for mechanical working.

Wrought alloys: Beyond this classification there are heat treatable and non-heat treatable alloys.

Summary

N.B. For aluminium alloys, the numbering system is such that the first number relates to the family of alloys. The remaining three numbers, shown as xxx for our purposes, describe the varying composition of alloying elements within the families.

Non-heat treatable alloys

1xxx Primarily pure aluminium with small amounts of iron and silicon. Used mainly for sheet metal work.

3xxx Manganese is primary alloying element, provides solid solution strengthening. Used for pressure vessels, chemical equipment and sheet metal work.

5xxx Magnesium is added as the primary alloying element. 5052 (approx. 2.5%Mg, 0.2%Cr) is the most important industrial alloy in this family. Used for sheet metal work, particularly for truck and marine applications.

Heat treatable alloys

2xxx Primary alloying element is copper. Family includes duralumin (2017 – 4 % Cu), which is a well-known alloy, strengthened by solid solution strengthening and precipitation hardening. Primarily for aircraft structures because of high tensile strengths (approx. 4 times that of mild steel).

6xxx In this series there are two primary alloying elements, magnesium and silicon, strengthened by precipitation hardening, good corrosion resistance and strength, used in bicycles frames, truck and marine structures.

7xxx This series, primary alloying element is zinc, but many alloys such as 7075 have additions of magnesium and copper. Alloy strengthened through precipitation hardening. Alloy elements allow denser precipitates; produce a stronger alloy, strengths up to 500 MPa, used in aircraft structures and good quality bike frames.

Casting alloys

Like the wrought alloys the casting alloys are broken up into non-heat treatable and heat treatable. Most aluminium alloys are cast, using either sand casting, gravity die-casting or pressure die-casting. Most of the casting alloys contain silicon, in the range of 5 to 12%, as this tends to produce a lower melting point for the alloy and improves the fluidity of the molten alloy. It also tends to strengthen the alloy. Manganese and copper may also be added to improve strength and hardening qualities.

An important aluminium-casting alloy, used in the transport industry, is the aluminium/silicon/magnesium/iron alloy. This is used in the manufacture of alloy wheels in cars.

Like aluminium wrought alloys, casting alloys use a numbering system, where the primary alloying element decides the number that is allocated. Table 2.3 lists them below.

Table 2.3: *The aluminium casting alloy designations*

Major Alloying element plus additions	**Number**
Aluminium, (99% or greater)	1xx.x
Copper	2xx.x
Silicon with Cu and/or Mg	3xx.x
Silicon	4xx.x
Magnesium	5xx.x
Zinc	7xx.x
Tin	8xx.x
Other elements	9xx.x

Aluminium Lithium Alloys

This alloy deserves mention, because of its use in the manufacture of bicycles. Aluminium lithium alloys create bicycle frames that offer a 100% better fatigue life and 50-100% better strength than 6061 aluminium alloy tubing. Such an alloy is important in the competitive world of cycling. This alloy was first used in the advanced aviation fields, and is still used in some aircraft designs.

Copper, Brass and Bronze

Copper and its two alloys, brass and bronze are immensely important to the engineering world. Bronze, amazingly, has been used by humanity since 3,000 BCE, so when you use a bronze item you have something in common with the ancient Egyptians or Mesopotamians.

Copper

Prior to the Bronze Age, copper was by far the most important metal. Nowadays it is the third most used metal. The "big three" metals are iron, aluminium and copper. Copper finds extensive use in the electrical industries, as its conductivity is second only to silver. It is, however, far more cost effective as a conductor than silver and finds extensive use in railway overhead wiring.

Brass

Brass is an alloy of copper and zinc. The relative amounts of these materials have a large impact on structure and properties. Commercial brasses rarely contain more than 40% zinc as beyond this level of additive, the alloy becomes brittle and is of little use. A list of common brasses is shown below

Cartridge brass also known as 70/30 brass contains 30 % zinc. The alloy is quite ductile and was formerly used in the manufacture of cartridges for bullets, hence the name. The alloy has higher ductility than the copper from which it is mainly composed, which makes it perfect for deep drawing operations.

Standard brass has only 25% zinc and is a good quality, cold working alloy used where the ductility of cartridge brass is not required. It is used for stampings and limited deep drawing.

Muntz Metal, a two phase brass, contains 40% zinc and because of the formation of a brittle phase in its microstructure it is usually hot-worked to shape. It is used in the manufacture of rods and bars. It may also be cast. Tap bodies, for example, are cast using this alloy. Muntz metal can be heat-treated to change its properties.

Naval Brass contains 37% zinc and 1% tin, with the tin adding to the corrosion resistance in seawater, an important consideration in shipbuilding.

High Tensile Brass, also called manganese bronze, is a copper (58%) zinc (≈36%) alloy with small additions (<1.5%) of manganese, aluminium, lead iron and tin. These improve the tensile strength over other brasses, at the expense of ductility. It is used for stampings and pressings, but also for marine propellers and rudders.

Tin Bronzes

The term bronze relates to the tin bronzes as opposed to aluminium bronze outlined further on. In fact, the term bronze is often used to describe copper alloys but the original meaning is a copper/tin alloy. Tin bronzes usually contain tin within the range of 3 to 18 %. A list of common tin bronzes and gunmetals are listed below.

Low tin bronze, with only 3.75% tin is aptly named. It demonstrates good elastic properties and corrosion resistance. It is used for springs.

High tin bronze, with large amounts of tin (18%), is used in heavy load applications, such as slewing turntables on large cranes

Admiralty gunmetal, is a bronze with 88% copper, 10% tin and 2% zinc and some nickel. The zinc makes the alloy more fluid in the liquid state, which suits casting. It is used for pumps, valves and especially marine castings, as it has good corrosion resistance in salt water.

Leaded gunmetal, or red brass, has 85% copper and 5% tin, but also 5% zinc and 5% lead, which reduces ductility and makes it more suitable than Admiralty gunmetal for pressure vessels.

Phosphor bronzes are, bronzes with deliberate amounts of phosphorous added as opposed to trace amounts left over from refining. They tend to have higher tensile strength and corrosion resistance than standard tin bronzes. Importantly, they also have a lower coefficient of friction, that makes them suitable for bearing applications.

Aluminium Bronze

This is a copper alloy with the primary alloying element being aluminium in amounts up to 11%. Aluminium bronzes offer good corrosion resistance, and good tensile strength. Their corrosion resistance sees them used in marine and chemical applications. Casting them is difficult because of the oxidation of the aluminium, so great care must be taken when pouring the molten metal. Aluminium bronze is hardenable through heat treatment, outlined further on.

Structure/Property Relationships

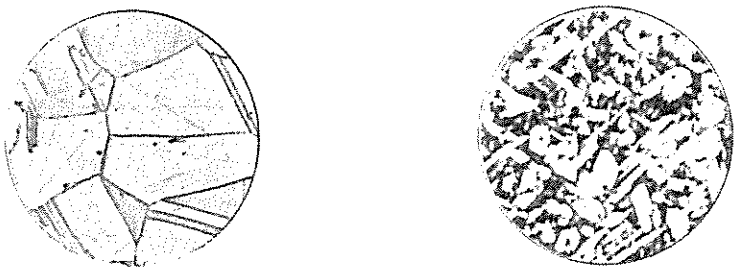


Fig 2.32: *Microstructures for cartridge brass (left) and Muntz metal. [Reproduced with permission from Gibson, J. & Taylor, T. (1979)* Experimental Materials Science.*]*

The microstructure has a large impact on the properties of many non-ferrous alloys. In many cases a second harder phase is often present which reduces ductility, e.g. Muntz metal. The microstructures for Cartridge Brass and Muntz metal are shown above. Fig. 2.32. The white phase, which is not pure copper, is ductile, whilst the dark phase contributes to the brittleness of the alloy.

Heat Treatment of Non – Ferrous Alloys

Annealing is done to relieve internal stress. Annealing, as for ferrous alloys, is done to remove stress that may result from cold working the alloy when producing sheet, plate or bar metals. Each alloy will have an optimum annealing temperature. The result is an equiaxed grain structure similar to that shown in Fig. 2.33 B.

Precipitation hardening is done on duralumin (Al/4%Cu) and other aluminium alloys. The aluminium alloy will, after casting or being hot worked, present as an alloy with a primary phase (α) and a secondary phase (β) usually at the grain boundaries, Fig. 2.33 (A). Two steps follow:

Step 1 – Solution Treatment: The alloy is heated to 530°C until the β phase dissolves to produce a homogenous single phase alloy. The alloy is then quenched to room temperature. The resulting structure will be a single phase microstructure of equiaxed grains of α (Fig. 2.33 B)
Step 2 – Aging: Over time the trapped β phase precipitates out on stress planes within the quenched phase, thus restricting dislocations and strengthening the alloy. This β precipitate is very finely dispersed through the structure (Fig 2.33 C). This process is called *natural aging.* It is possible to accelerate precipitation by reheating the alloy to about 150°C; this is known as *artificial aging*. This must be done carefully because if artificial ageing is done for too long that alloy can be over aged and becomes brittle.

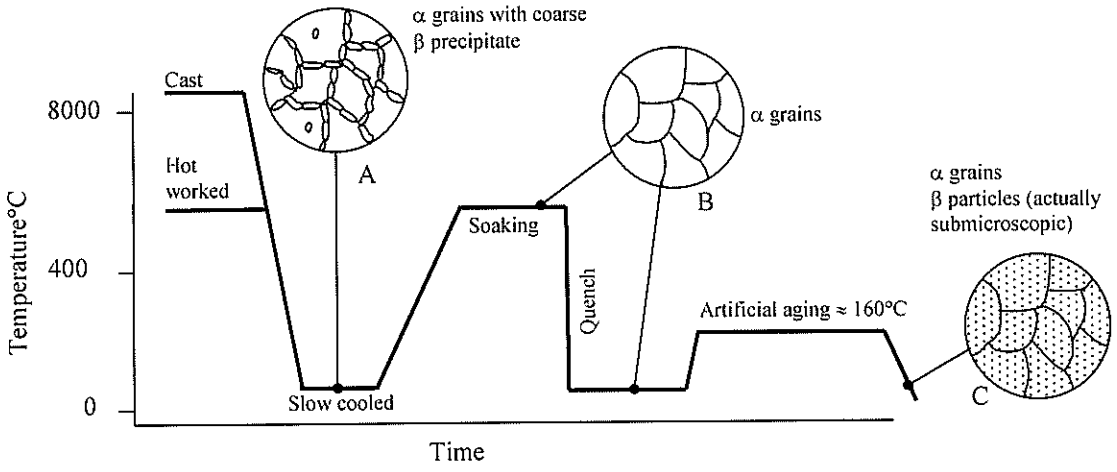


Fig. 2.33: *The precipitation hardening (age hardening) process for duralumin. After slow cooling the alloy is reheated, soaked and quenched. Then in this case it is artificially aged.*

Aluminium bronze (Cu/11%Al) may also be heat-treated. When quenched, it forms a hard structure that is similar to martensite in steels; hence it is named *β Martensite*. It can be tempered in a similar manner to steels forming a tough product.

Ceramics and Glasses

Ceramics

Ceramics offer a range of properties that make them desirable for transport applications.. Research has been conducted into manufacturing petrol and diesel motors from ceramic materials to improve thermal efficiency. Since ceramics can withstand higher temperatures than metal alloys used in engines they could, conceivably, run at higher operating temperatures, without cooling systems. As a cooling system is responsible for the loss of around 20% of the heat energy the motor uses, improved thermal efficiency and better fuel efficiency would be the results of a ceramic motor. Unfortunately these motors have never progressed beyond the experimental phase. Modern ceramics, such as partially stabilized Alumina and Zirconia, do not possess the brittleness normally associated with ceramics such as porcelain and china. They are also strong enough to withstand the forces and shock waves developed in an internal combustion engine. Ceramics are now used for high performance disc brakes because of their improved performance at the elevated temperatures which high performance car brakes experience.

While ceramic materials may not initially be thought to be important to transport systems but they have their place. For example, the humble spark plug relies on an alumina insulator for operation. The alumina offers insulation, thermal stability and resistance to vibration, properties hard to find away from ceramics.

Glass

Apart from cycling, all forms of transport rely on glass in one of its many forms. Glass may be defined as an inorganic fusion product that has failed to crystallise upon cooling. Glasses are generally not crystalline but amorphous (except for glass-ceramic, Volume 1 p28). The structure of glass does not allow for deformation. So, when glass is deformed, it is unable to dissipate the applied forces through a slip/dislocation mechanism. Once the bond resistance is exceeded, the structure fractures. The speed of fracture is very rapid. There are four general categories of glass used. These are listed below.

High Silica Glass is refined from borosilicate glass and is nearly entirely silica (SiO_2). These glasses are almost perfectly clear, and are used in situations where they experience elevated temperatures, such as in missile nose cones and space vehicle windows.

Soda Lime Glass is the most common glass. It contains large amounts of soda (Na_2O) and lime (CaO). The presence of soda will prevent devitrification (crystallisation), however, it also makes the glass water-soluble. The addition of lime alleviates this, hence the name Soda Lime Glass. Soda Lime Glasses soften at approximately 850°C, are easily formed to shape when hot, will not recrystallise, are water resistant, and cost effective. It is used for window and plate glass, bottles, tableware, electric light bulbs and windscreens.

Borosilicate Glasses are glasses with up to 20% boron and silica. This imparts good levels of chemical resistance and low thermal expansion, so these glasses have high resistance to fracture at elevated temperatures. Borosilicate glasses, one of which is known by the trade name "Pyrex", are extensively used in electrical insulation, gauge glasses for laboratory ware, and domestic cooking and ovenware.

Lead Glasses contain up to 40% lead. This lowers the softening temperature to well below the 850°C of soda lime glass. They have a high refractive index, which makes them optically clearer. They are used extensively for optical glass. They are also used for thermometer tubes and the tableware known as "crystal" which is a misleading name as they are not crystalline.

Ceramics as an Insulating Material

In terms of transport one of the best examples of a ceramics as an insulating material is in spark plugs. Spark plugs carry large voltages, but also are subjected to high temperatures caused by combustion; this makes polymers unsuitable as the insulator. So alumina is used as an insulator between the threaded metal body and the inner copper electrode. This stops the spark being earthed by the contact between the spark plug's metal thread and the engine block.

Ceramic insulators are also used in supporting power lines, such as the overhead wires for electric trains. Here glazed porcelain is used, although glasses have been used as well at times. A newer development in porcelain insulators has been mildly conductive glazes which use a small milliamp flow of electricity on the surface of the insulator to prevent contamination by dirt and dust, which minimises the chances of arcing.

Laminating and Heat Treatment of Glass

Glass is rarely used in its normal condition in transport devices. Normal glass is far too brittle to resist projectiles. As a consequence two processes, that strengthen glass, are used. Laminated and toughened glasses are outlined in Chapter 1 pages 58 and 54 respectively.

Polymers

Structure/Property Relationships and Applications

Thermoplastics (or thermosoftening polymers)

This type of polymer softens on the application of heat. It can also be re-melted and reformed. Thermoplastics have long linear chain structures, with the chains formed by covalent bonds. Weak van der Waal's forces hold the separate chains together. As a consequence, they are usually flexible and often transparent. When put under a tensile load, they stretch readily, as there is little resistance to the chains straightening or sliding over one another (refer to Volume 1, p 26). Examples are polyethylene, polystyrene, polytetrafluroethylene (PTFE), polymethylmethacrylate (acrylic), polypropylene, polyvinyl chloride (PVC) and acrylonitrile butadiene styrene (ABS). They find use in cable coating on bicycles and for car parts such as grilles, badges, door handles and window winders.

Thermosets (or thermosetting polymers)

This type of polymer undergoes a chemical change when heat is applied. The change is not reversible so these polymers do not soften when they are reheated. Thermoplastics have network structures, with covalent bonds, along the chains and across the chains. They tend to be more rigid than thermosoftening polymers. When they are put under tension, the cross-linking resists deformation. This makes them less flexible than a linear polymer. Epoxy resins, silicone, polyurethane and polyester resins are examples of thermosetting polymers. Epoxy resins find use in the aircraft industry for joining panels, and the silicones are often used in gasket manufacture for cars. Epoxy resisn also find extensive use in bicycles and racing cars for use as the matrix in carbon fibre reinforced polymers.

Rubber

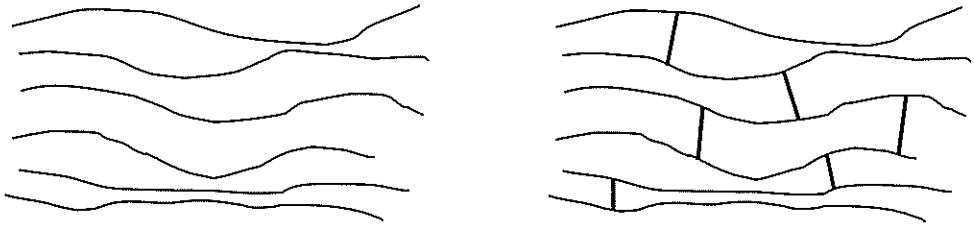


Fig 2.34: *Raw rubber, on the left, has polymer chains that have no linking. The vulcanised rubber structure, on the right, has cross-linking to improve rigidity.*

Rubber is a natural polymer. In synthetic form it has great use in transport. The tyres for cycles and cars use a modified rubber, which is called *vulcanised rubber*. Rubber in its natural form is a linear polymer that is too flexible for use in a tyre. By adding around 5% sulphur to the rubber mix and heating it to about 150°C, a moderate level of cross-linking is achieved. The vulcanised rubber is more rigid, but still flexible. To increase rigidity, tyres are usually constructed from a composite of rubber with cotton strands, while radial tyres for cars use steel wire and polyester cord.

Engineering Textiles

Engineering textiles are polymer resins that are drawn into threads and then woven into 'cloth' like sheets. They are an important field in the world of engineering. The engineering textiles are synthetic polymers and offer vast improvements over natural fibres. Various engineering textiles that are used today are discussed below.

Polyester is a synthetic fibre that is strong and resilient. It is also hydrophobic, i.e. resistant to water absorption. It is used in helium airships, and in the manufacture of some tyres. It is also used in the manufacture of various car parts, like fan belts and radiator hoses.

Nylon finds use in the engineering world in dry lubrication. It is now being replaced by PTFE (polytetrafluroethylene). It is resistant to acids, bases and oil.

Aramid fibres are extensively used in engineering. Nomex™ and Kevlar™ are the best known examples. These aromatic polyamid polymers are strengthened by a backbone

of benzene rings. They have excellent strength qualities but are limited to low temperature uses. They are used in aircraft manufacture and in bullet-proof vests.

Olefins are polyethylene or polypropylene fibres shaped into sheets. They are waterproof and find use in the manufacture of collapsible shelters and buildings.

PTFE (Teflon) fibres are fire resistant and will also stop water vapour, but not water. They are used for filters in engines.

Manufacturing Process For Polymer Components

Blow moulding

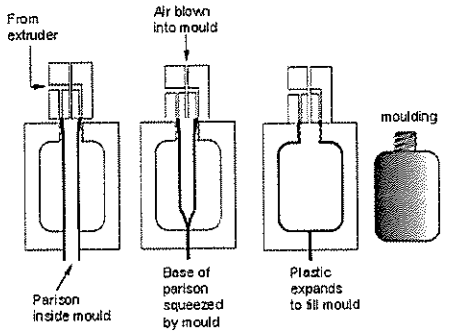


Fig 2.35: *Blow moulding (from http://tinyurl.com/blowmoulding)*

This is used to shape thermoplastics. A polymer tube is lowered into a mould, and air forces the tube to the shape of the mould. It is used to make plastic bottles and containers.

Extrusion

As for metals, polymers can be extruded, with the polymer taking the shape of the die through which it is extruded. The polymer granules are melted and the molten material is forced through the die. This process is only suitable for thermosoftening polymers. Polymer tubing is manufactured using this method. The outer covering of bicycle cables is coated with plastic in this way.

Thermoforming

This process is used in the manufacture of various thermoplastic containers. Heated thermoplastic sheets are placed over dies to produce the required shape. The forming can be done using matching dies, a vacuum or air pressure. Often, special thermoplastic sheets are used for this process.

Calendaring

A thermoplastic is poured into a cavity between two rollers, and the plastic is squeezed through the rollers. The rollers may be embossed with patterns or they may be smooth. Tiles, films and curtains can be made this way.

Rotational moulding

The molten polymer is poured into a mould and the centrifugal force throws the polymer to the walls of the mould, forming a hollow article.

Injection Moulding

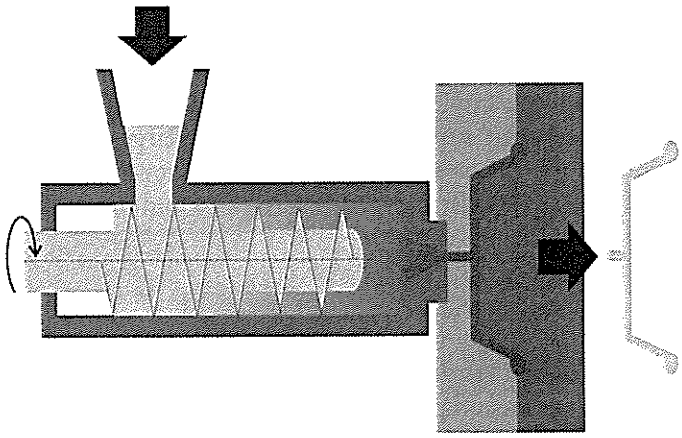


Fig. 2.36: Diagram representing the injection moulding process.

This is one of the most commonly used polymer forming procedures. Molten polymer is injected into a cavity in the shape of the finished article. When the polymer solidifies, it is ejected and the procedure starts again. Injection moulding lends itself readily to mass production. Many items are injection moulded and are often identified by a small nipple on the item and a split line where the polymer was injected into the mould. They are used in the manufacture of small thermoplastic mouldings for cars and bicycles.

It is strongly suggested that students refer to *Additional Resources* section to find information on plastics forming. The books by D. Askeland and W. Smith have excellent coverage and diagrams of plastics forming.

Engineering Electricity/Electronics

Power Generation/Distribution

Generation

In Australia electricity is produced in a number of ways. Some are listed below.

Coal burning power stations are by far the most common method of electricity generation. Here coal is used to produce steam that drives a steam turbine. The turbine connects to a generator that is spun, producing electricity. Because coal is a readily available resource this method is very popular in Australia. Unfortunately, this produces huge volumes of carbon dioxide, which contributes to the greenhouse effect.

Hydroelectric systems offer electricity without atmospheric pollution. They utilise water held in dams above the power station. The waters potential energy is converted to kinetic energy as it travels through pipes to the power station. At the power station,

the water drives a turbine connected to a generator. This produces electricity, and the water is fed back into water systems for irrigation. While this system sounds totally "clean", it has large impacts on the surrounding environment because of the damming and diverting of rivers. It is also only possible in mountainous regions. One unforeseen side-effect of the Snowy Mountains hydroelectric scheme has been rising salinity levels because of irrigation and farming in formerly dry regions.

Wind power is often touted as a solution to power demands. The wind drives a large turbine with blades far more efficient than the ancient windmills. The turbine drives a generator and produces electricity. While it is truly clean, to power larger towns and cities, a very large number of turbines are needed, so large tracts of land must be devoted to it. Wind, in many places, is intermittent and unreliable, which also limits this method. Wind "farms" exist in Perth (WA), in Goulburn (NSW) and in New Zealand.

Nuclear Power – a loaded gun or not?

One very contentious issue in Australia is nuclear power. In the recent Kyoto summit on global warning, nuclear fission power emerged as a method that does not contribute to global warming. The benefits of nuclear power are great. They operate like coal power stations, but instead of the water being turned to steam by coal powered boilers, the heat from the nuclear reaction is used. Only a small amount of nuclear fuel is required to produce large amounts of heat. It is non-polluting to the atmosphere but it presents other problems. The by-products of nuclear power generation are often contaminated for thousands of years, and there have been two accidents (Three Mile Island and Chernobyl), that occurred because of human error, proving that all the failsafe systems are still vulnerable. The recent Fukushima accident was caused by a large natural event overwhelming failsafe systems.

Until it is proven that nuclear fission is completely safe, it will remain contentious in Australia.

Distribution

In recent years many smaller power stations have been closed and replaced by larger ones in more remote locations. For example, in the Sydney and Wollongong areas, there are now no power stations. These areas are fed by the larger power stations of Eraring and Vales Point near Lake Macquarie, Mt Piper near Newcastle, Wallerawang in the Lithgow area, Bayswater and Liddell in the Hunter Valley and the Snowy Mountains Hydroelectric system. This means that the distribution and carriage of electrical energy becomes very important.

The power lines used to carry electricity are steel cored aluminium. The steel core provides the strength to let the wire support itself, and the aluminium provides the electrical conductivity. There are now cables being made that replace the steel core with a carbon and glass fibre reinforced polymer composite. Although aluminium has only 60% of the conductivity of copper, its vastly lower density means that per kilogram, aluminium is a better conductor. When copper conductors are used, as in the railway system, more poles are required because of its extra weight.

To reduce resistive loss in aluminium cable, power is transmitted along transmission lines at very high voltages, i.e. 333,000 volts and even 500,000 volts. Power loss is directly proportional to the voltage, and the current squared. Raising the voltage reduces the current for the same amount of power, thus reducing the resistive loss in the cable.

In addition to the power lines, various substations are used to reduce the high voltage to the 240 volts that is then fed to urban areas.

AC/DC Circuits

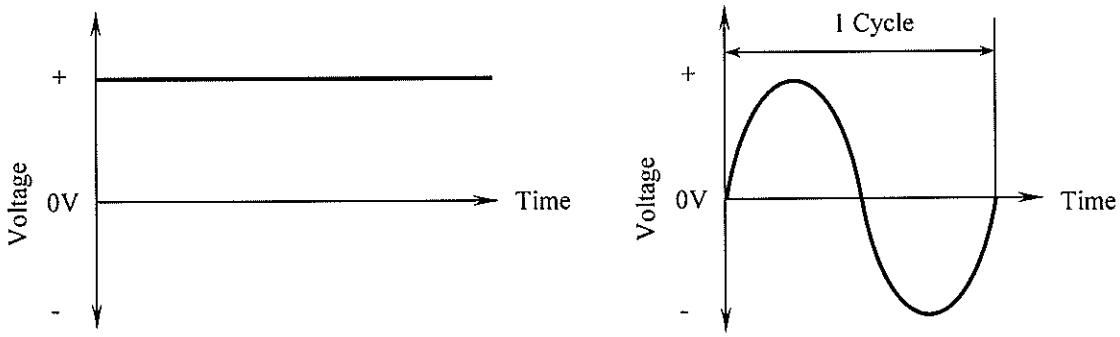


Fig. 2.37: *In (A) DC current maintains a constant potential. In (B) AC voltage flows in the shape of a sine wave.*

Alternating Current (AC) and Direct Current (DC) differ considerably. DC has a constant potential. AC has a constantly varying voltage. In Australia, the AC mains power varies from 0 to +240V, to 0 and –240V, every $1/50^{th}$ of a second.

It is possible to run a DC appliance on AC but the AC must be "converted" into DC. This is never fully effective but it is good enough to be used practically.

Rectification

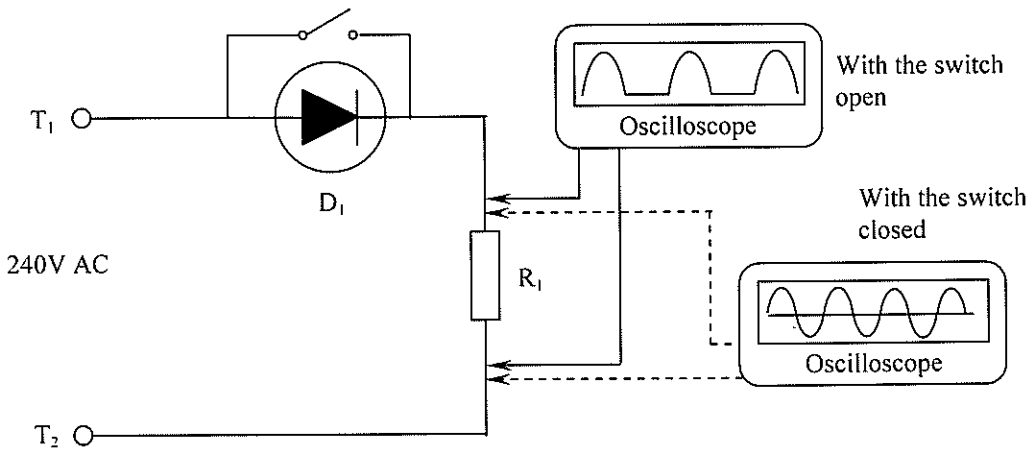


Fig 2.38: *The half wave rectification is gained by a single diode in the AC circuit. The diode will block current in one direction, so only the negative direction is blocked. With the switch closed, AC can flow and the oscilloscope shows an AC waveform.*

The conversion of AC to DC is called rectification. Half wave rectification occurs when one diode is used. This eliminates the current flowing the opposite way, so blocks the negative part of the waveform.

Full wave rectification can be achieved by using four diodes. This will allow all of the sine wave to pass but it will have all of the waves on the positive side, so they will travel in the same direction. The final waveform is not true DC, but a varying DC. The four diodes are connected as shown below.

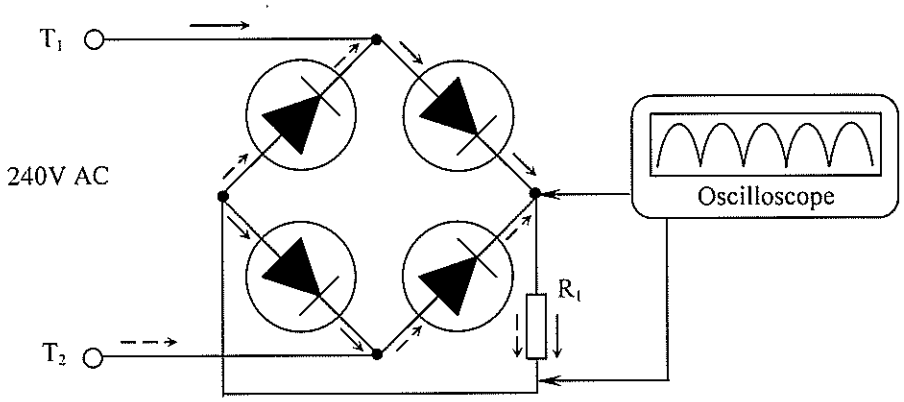


Fig. 2.39: *Full wave rectification is achieved by the four diodes connected as shown.*

This varying DC, produced by both the half and full wave rectifier circuits, is not ideal for most DC equipment. To achieve a better waveform, a capacitor is added to either rectifier circuit. The capacitor stores energy that can be used when the waveform reduces in voltage. The result is a nearly flat waveform, with the capacitor smoothing out the troughs.

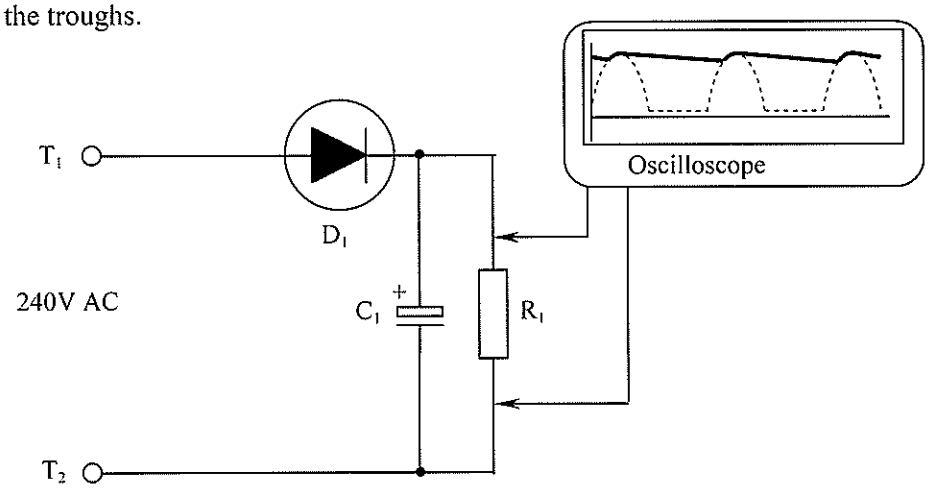


Fig 2.40: *The use of a smoothing capacitor can be used to flatten a half-rectified waveform.*

When T_1 is positive the diode (D_1) passes current and the capacitor (C_1) charges to a potential equal to the AC supply. Current will also be flowing through R_1. When the potential at T_1 falls then the capacitor releases its charge and the circuit uses power stored in the capacitor. There is a slight voltage rise when T_1 becomes positive again, which is minimised by ensuring the capacitor value times the resistor value is larger than the cycle of the AC which is $1/50^{th}$ (0.02) of a second. Stated mathematically,

$$C_1 \times R_1 > 0.02$$

For efficient smoothing of the DC output it is best if the product of the capacitance and the resistance is 5 times the AC cycle time.

Example 2.13

What will be the required capacitor value for an AC (cycle = 0.02 s) rectifier if the resistor used has a value of 390 Ω?

$$C_1 R_1 = 5(0.02)$$
$$C_1 = \frac{5(0.02)}{R_1}$$
$$C_1 = \frac{0.1}{390}$$
$$C_1 = 2.56 \times 10^{-4}$$
$$C_1 = 256\ \mu\text{F}$$

Therefore the capacitor required for this rectifier would be 256 μF.

In rectifier circuits the smoothing capacitor is usually an electrolytic capacitor and the value lies in the range from 470 μF to 2200 μF.

Electric Motors used in Transport Systems

Many years ago only DC motors were used in trains, partially because it was difficult to control AC motors when the generator driven by the diesel engine produced DC. Nowadays, however, there are AC induction motors used in diesel electric trains. In NSW older electric trains still run using DC power, while newer electric trains now use AC motors. See Volume 1, pp 77-79 to review how electric motors work.

DC Motors

Shunt wound motors are rarely used in locomotives. They have constant speed but low starting torque and therefore not suitable if there is a lot of stopping and starting.
Series wound motors offer excellent torque at slow speeds and will operate at high speed under light load. This makes them excellent for use in trains, and is a reason why they are widely used.
Compound motors combine the best features of the other two. Good starting torque and will not run away under no-load conditions.

AC Motors

The AC motors used in trains are generally induction motors. Their great advantage is a lack of commutator and brushes, which wear out over time. They rely on the frequency of the electricity and magnetic induction for their power and. Their function is helped by generators being replaced by alternators on diesel electric trains. Alternators produce AC so one would assume the use of an AC motor would be easier.

On the English Channel Tunnel, locomotives are powered by single phase AC voltage, which is rectified to DC voltage, passed through a control circuit. The voltage is the converted into AC for the motors.

The Millennium trains used on the Sydney rail network use 3 phase induction motors, but because the supply is 1500 volts DC the train uses an inverter to create AC from the supply DC voltage. Pulse Width Modulation (PWM) is used to control the motors.

This large field can be further researched through the Additional References section at the end of the chapter. The author also recommends this website for electric motors:
http://www.phys.unsw.edu.au/hsc/hsc/electric_motors.html

Pulse Width Modulation

Pulse Width Modulation (PWM) is a method of controlling electric motors in an electronic manner. Instead of varying the voltage or current levels in an analogue way, the supply is effectively switched on and off rapidly to replicate an average on time equal to a set level, e.g. 50% power created by being off and on 50% each. To understand this process imagine a standard light, and that it can be switched on and off at a rate faster than the human eye can detect. Now imagine that the light is off 50% of the time and on 50% of the time, the resultant effect will be the light seems to be dimmed by 50%. Or furthermore if we adjust the off to on time to be 75%-25% then the light will appear to be 25% of full light.

Control Technology

Control technology is the use of some type of mechanism or circuit to control the operation of an item. The simple float arm in a toilet controls the water delivery; this is an example of control technology.

In transport, one of the earliest control devices was the governor on steam engines. Centrifugal force caused arms to rise. This controlled a throttle system, which more-or-less maintained the engine at a constant speed under varying loads.

Cars often have electronic cut-outs that sense the speed of the crankshaft and cut the ignition to the car if the engine revs too fast. Control technology has been trialled for years on car suspension, trying to reduce the compromise by engineers between ride and handling, which exists with conventional suspension systems.

Control technology can be a simple mechanical linkage, a simple digital yes/no circuit, or a more complex circuit that reads various inputs to produce a variety of outputs.

In 1989 Citroën released the XM with *Hydractive* suspension. It was not a true active system, but a reactive system. It used inputs from the motor, the gearbox, suspension arms and the driver (throttle and brake) to decide how to control the six hydro-pneumatic springs and the four struts in the suspension system. The result was a car that could ride like a plush "boulevard cruiser" or change to a sportier handler with minimal body roll. This complex system has been further refined over the years.

Control technology is used to trigger safety devices like ABS brakes and airbags. As a wheel begins to lock, the computer reduces the brake pressure, preventing the wheel from locking and the car from skidding out of control. ABS now forms part of Dynamic Stability Control (DSC) on cars, where the DSC can operate individual brakes to improve the cars response to the driver's inputs.

Communication

Sectional Views

Summary of Sectioning Rules

- Half section means half the sectioned view is sectioned, while a full section means the section covers the whole of that view.
- Features likes ribs/webs are not sectioned, this avoids confusion about the final shape.
- Bolts, nuts, studs and shafts are not sectioned.
- Hatching lines are thin dark lines.

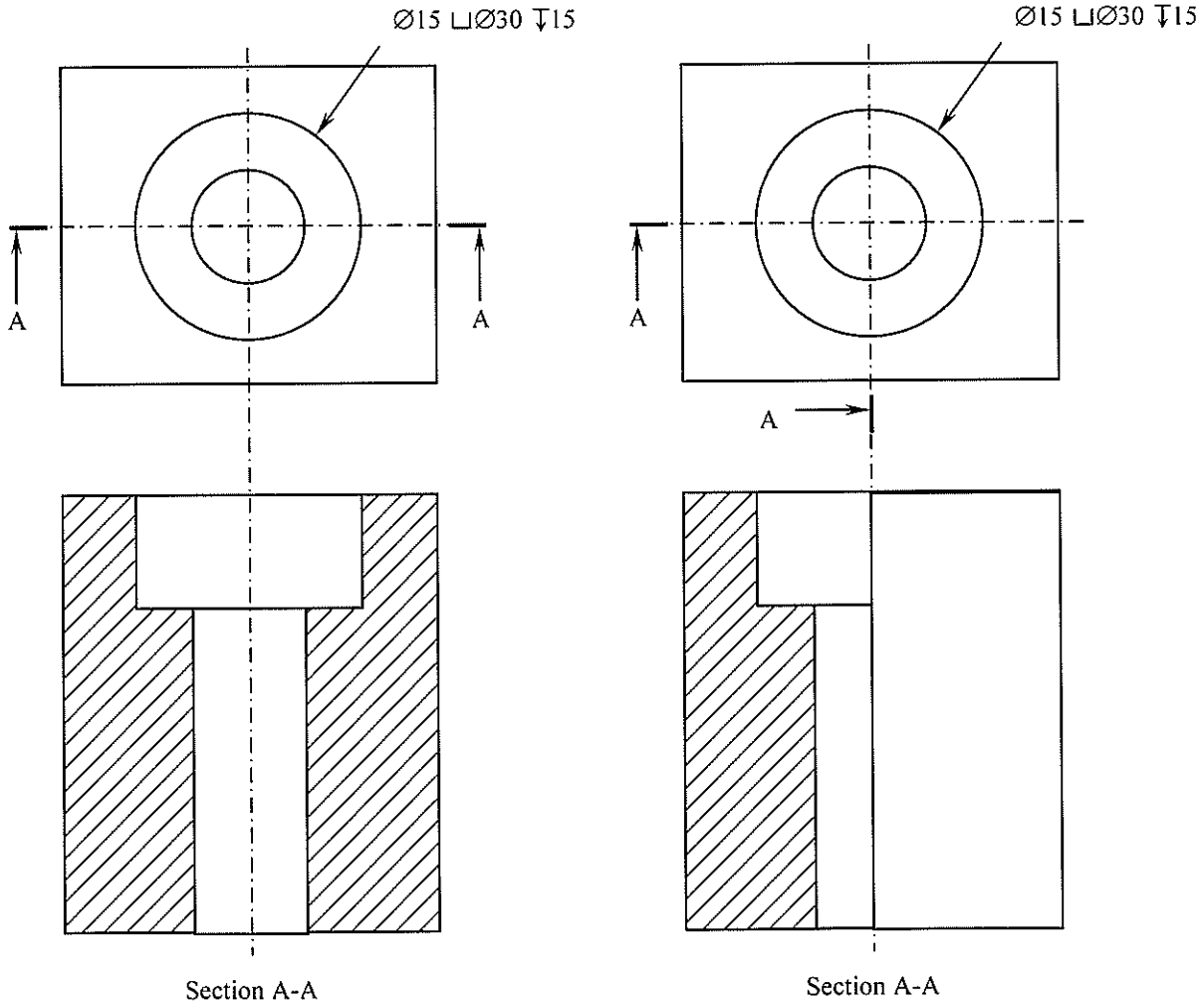


Fig. 2.41: *A through hole with counter bore, shown on the left as a full section, while the right shows a half section.*

- Hatching lines are usually at 45° unless one side of the shape is at 45°.
- If different parts are sectioned but are adjacent first use the opposite 45° orientation, then select other common angles such as 30° or 60°.
- Hidden detail should not be shown on the sectioned view

Sectioning threads

When threaded holes are sectioned specific rules come into play. Fig 2.42 displays three examples. First a through threaded hole is shown, then a blind threaded hole is shown. Note the 120° triangular bottom to show where the drill stopped. Finally a through threaded hole is shown but now a bolt with a full length thread is inserted with a washer. Note how the hatching lines don't pass through the thread where the bolt has engaged it, because we don't section bolts.

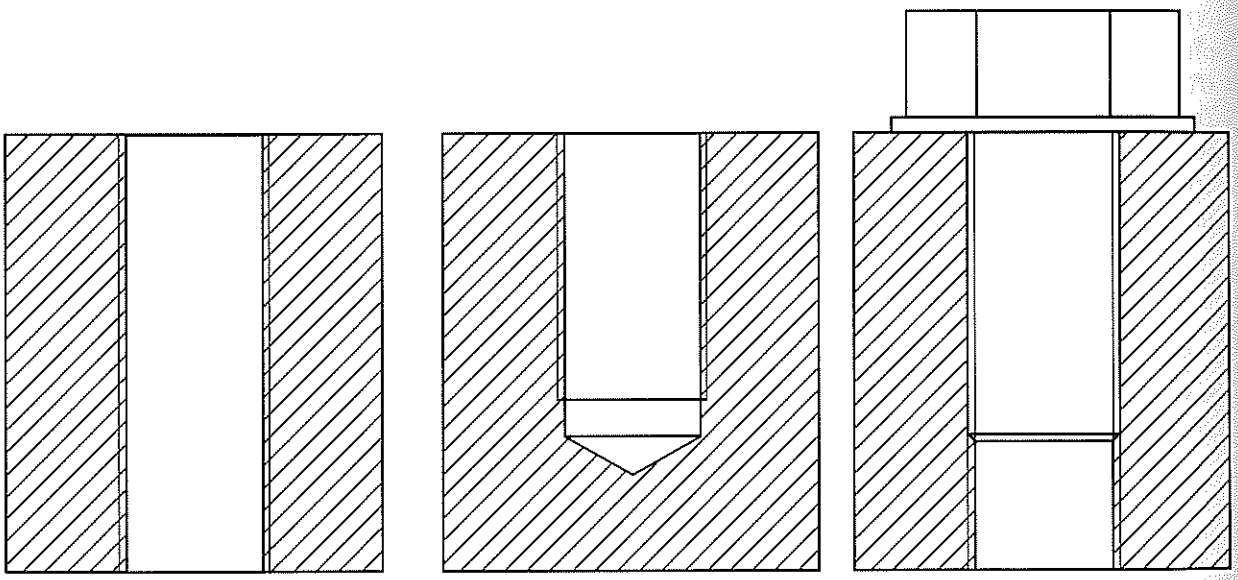


Fig. 2.42: *Three examples of sectioned threaded holes: left, a through hole; centre, a blind hole; and right, a threaded through hole with a bolt in place.*

Engineering Report

Using the outline from Chapter 1, develop an engineering report on the following topic.

Currently the bicycle industry has many choices of materials from which to manufacture their frames. Make a judgement on the best material for a small hand built bicycle company, currently making 500 bicycles a year. The materials to choose from are Reynolds 853, Titanium alloy, 7005 Aluminium alloy, Carbon fibre and 4130 chromium molybdenum steel.

Your report should consider the strengths of the materials, the fatigue resistance, its resistance to bending and the ease with which it can be joined.

Remember this is a small company, and cost is important, but so is weight, so the company wants to keep weight and cost down.

Additional References

This list of useful references allows students further investigation. Any highly recommended book or website has two stars next to it.

Historical and Societal Influence

Ballantine, R., (2000), *Richard's 21st Century Bicycle Book*, Pan Books, Bath.
Eardley, G. H. & Stephens, E. M., (2000), *The Shale Railways of New South Wales*, Australian Railway Historical Society, Redfern.
Hadland, T., (1994), *The Spaceframe Moultons*, Tony Hadland, Coventry.
Hicks, G. & O'Brien, D., (1999), *Shays in the Valley*, New South Wales Rail Transport Museum, Burwood.

Engineering Mechanics and Hydraulics

Ivanoff, V., (1996), *Engineering Mechanics*, McGraw Hill, Sydney.
**Schlenker, B. and McKern, D., (1976), *Introduction to Engineering Mechanics*, Jacaranda Wiley, Milton.

Engineering Materials

**Askeland, D., (1989), *The Science and Engineering of Materials*, Chapman and Hall, London.
**Avner, S., (1983), *Introduction to Physical Metallurgy 2nd Edition*, McGraw Hill, Tokyo.
**Higgins, R., (1997), *Materials for the Engineering Technician 3rd Edition*, Arnold, London.
Schlenker, B., (1974), *Introduction to Materials Science*, Jacaranda Wiley, Milton.
**Smith, W., (1990), *Principles of Materials Science and Engineering*, McGraw Hill, New York.

Engineering Electricity/Electronics

Cooper, M. and Vella, G., (1988), *Physics – The Core*, Brooks Waterloo, Melbourne.
**Plant, M., (1992), *Electronics*, Hodder and Stoughton, London.
Pollock, G., (1996), *Active Physics*. Science Press, Sydney.
Sams, H. (Ed.), (1973), *Basic Electricity and an Introduction to Electronics 3rd Ed.*, Howard Sams and Company (A division of Macmillan Inc.), New York.

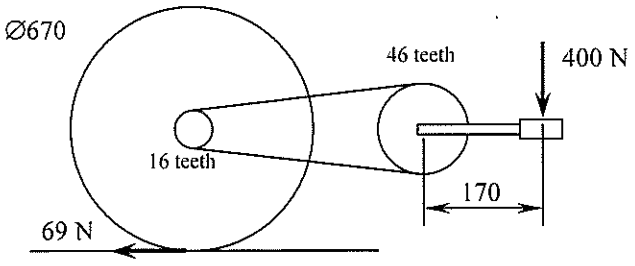
Communication

**Boundy, A., (1991), Engineering *Drawing*, McGraw Hill, Sydney.
Boundy, A. & Hass, I. L., (1996), *Graphics 2000 Book 2*, McGraw Hill, Sydney.
Mullins, R. K. & Cooper, D. A., (1990), *Programmed Technical Drawing Book 3*, Longman Cheshire, Melbourne.

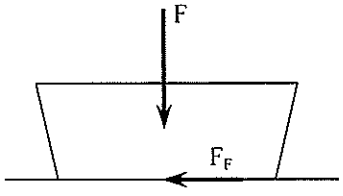
Park, A, Dodds, K and Bland, S., (1989), *Technology Drawing*, Longman Cheshire, Melbourne.

Review Questions

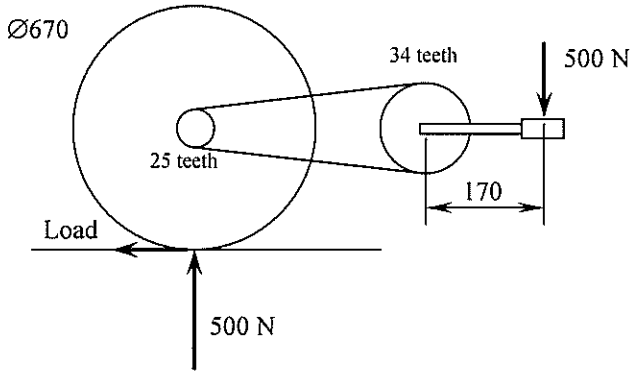
1. What was the first real bicycle?
2. Why did early bicycles use front wheel drive?
3. Why did the Old Ordinary get developed? What were its advantages and disadvantages?
4. Why was the Safety Bicycle such an important development?
5. What important development for transport devices occurred in 1888?
6. What is a mountain bike?
7. The Moulton bicycle and the Brompton folding bicycle use smaller wheels than standard bikes, what are the advantages and disadvantages of this feature?
8. What are the advantages and disadvantages of recumbent bicycles?
9. The Ford Model T is generally considered the most important car ever. Why is this so?
10. Why was the Mini such a revolutionary car design?
11. Why was Stephenson's *Rocket* important?
12. What were the advantages of the Garret locomotive?
13. What were Shay locomotives used for?
14. Why do diesel/electric trains use electricity to provide traction instead of a gearbox, like cars and trucks?
15. Why are newer alloy steels still desirable in bicycles?
16. Describe why are aluminium alloys used so extensively in bicycles?
17. Explain what makes carbon fibre composite frames desirable for racing bikes?
18. List two environmental impacts of the car?
19. Imagine that the car is banned tomorrow in Sydney. What would the effects be? What would the transport solution be?
20. A bicycle drive system is shown below. Determine the efficiency of the drive system for the load and effort shown.



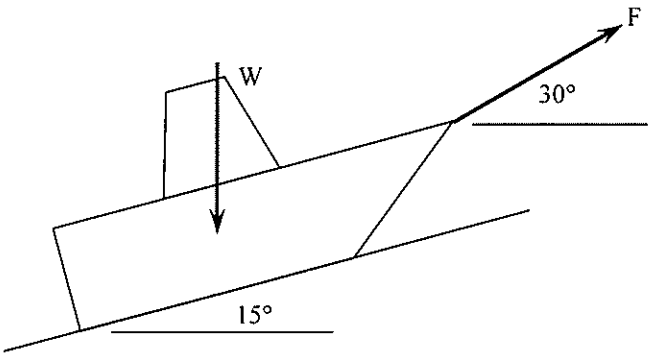
21. Define the normal reaction.
22. A bicycle brake pad is forced against the rim to stop the bicycle. What would the applied force be to produce a 100N braking force if μ =0.6?



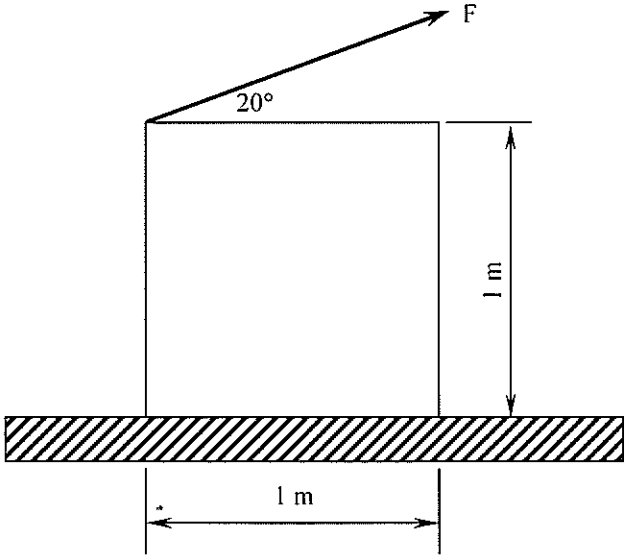
23. A bicycle drive system has an efficiency of 95%, determine the force at the back wheel if an effort of 500 N is applied to the pedal. If the coefficient of friction between the tyre and the road is 0.6, and the reaction between the road and the rear tyre is 500 N, will the wheel slip?



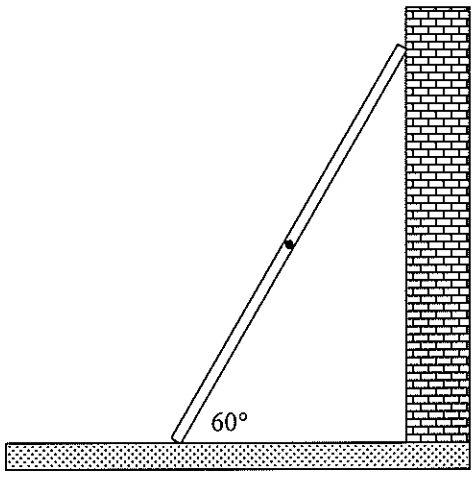
24. What force would be needed on the rope to (a) drag it up the slipway, or (b) stop the 1 tonne boat sliding down the slipway? (μ =0.2)



25. A 200 kg crate of vehicle parts is shown below. It is meant to be slid along the floor by the force, *F*, as shown. If the coefficient of friction between the crate and the floor is 0.3, calculate the minimum force that will make the box tip, and determine whether the box will tip or slide first.

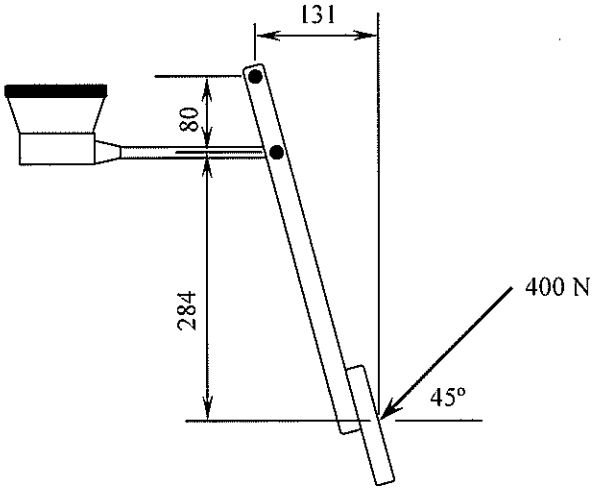


26. The 5 m ladder of mass 20 kg is shown and rests against a rough wall and a rough floor. If the coefficient of friction between the wall and the ladder is 0.2, determine the coefficient of friction between the ground and the ladder to keep the ladder in place.



27. A 1.5 tonne car rests on a hill 50 metres from the bottom, and the hill is at an angle of 5°. The handbrake of the car lets go and it starts to roll down the hill, determine the speed the car will be travelling at when it reaches at the bottom, if the average resistance to rolling is 200 N.
28. A 10 kg bicycle plus the 70 kg rider are travelling at 25 km/h at the top of a 100 m hill, the bicycle coasts down to the bottom, of the hill where the speed is now 45 km/h. Determine the total resistance to motion if the rider descended a vertical height of 9 m while travelling down the 100 m hill.

29. A brake pedal for a car is shown below; determine the reaction at the pivot pin and the load in the pushrod.



30. Define potential energy.
31. What is work in engineering terms?
32. Explain what is meant by the terms power and torque?

33. A motorcycle and rider of total mass 250 kg is ascending a hill of gradient 1 in 15 at 80 km/h. Determine the power needed by the motor if the motorcycle is to maintain this velocity against a total resistance of 200 N.
34. A 1 tonne car is travelling at 54 km/h on a flat road and is using 30 kW to maintain this velocity. If the engine stalls just as the car hits a hill of gradient 1 in 10, how far will the car travel up the hill until it stops.
35. A diesel-electric locomotive can provide a maximum tractive force of 4 MN, the loaded wagons weigh 60 tonnes. Rolling resistances are 50 N/tonne. Determine the maximum amount of wagons the train can pull on flat ground at 100 km/h, and how many it can pull up a 1 in 50 incline at the same speed.
36. How do process annealing and full annealing differ?
37. What is normalising?
38. Explain why is it possible to harden steels?
39. How is it that cooling in air can harden some steels?
40. What is Bainite?
41. What effect does forging have on the macrostructure of an item?
42. What are the advantages of cold rolling over hot rolling?
43. Describe the sand casting process.
44. How does direct and indirect extrusion differ?
45. Explain why powder forming is used.
46. What are the advantages of aluminium lithium alloys?
47. Explain what brass is? Where is it used?
48. Explain why, in terms of structure, cartridge brass is more ductile than Muntz metal.
49. Explain what is meant by the term aluminium bronze.
50. Describe the process of precipitation hardening?
51. List four types of glasses. What are their properties?
52. Explain the difference between thermoplastics and thermosets.
53. What are engineering textiles?
54. Why is Aramid important? Where is it used?
55. Explain the process of injection moulding.
56. How do coal and nuclear power generation differ? How are they similar?
57. What does the term rectification mean?
58. Explain how pulse width modulation is used to control motor speed.
59. Describe how control technology is used to make cars safer.

CHAPTER 3

Engineering Focus Module 1 – Aeronautical Engineering

Scope of the Profession

Nature and Scope of the Aeronautical Engineering Profession

Aeronautical engineers are responsible for the design and development of new aircraft. They are also employed in modifying and improving existing designs. The aeronautical engineer is required to take a set of requirements and develop an aircraft that can perform according to the design specifications. The requirements for a jet fighter are vastly different from those of a large commercial airliner so the engineer must be able to utilise the latest technologies to produce the best aircraft for the job.

Aeronautical engineers are close comrades to aerospace engineers. Aerospace engineers are more concerned, however, with space flight and its related requirements.

Current Projects and Innovations

Aeronautical engineers are constantly engaged in the design of new aircraft to improve air travel and air defence. Always, the aeronautical engineer will work with the materials engineer in an attempt to reduce the weight of the aircraft. For many years fighter jets have used composites, such as carbon fibre reinforced polymer, to reduce the weight of aircraft. With reduced empty weight, a fighter can either have a longer range, or carry a greater weapon load.

In civil aviation, one of the biggest developments in recent years has been the Airbus A380 series of aircraft. This large double deck aircraft can seat up to 525 people (3 class seating) and was the first commercial aircraft to rival the Boeing 747 for long haul air supremacy. Its development depended to a large degree on increased power, increased lift and reduced weight. Another development has been the Boeing 787 Dreamliner, which is the first commercial passenger aircraft with a carbon fibre composite fuselage. Boeing has faced significant challenges when developing this plane.

Health and Safety Matters

Aeronautical engineers are exposed to the risks of any other engineer. It depends, however, on where they are employed. Both civilian and military aeronautical engineers can be exposed to dangers.

Aeronautical engineers may be exposed to machinery when overseeing manufacture, or they may be exposed to risks if doing fieldwork. If the engineer only works in the design office the risk to safety and health is minimised greatly. It then becomes more a question of correct ergonomics for the engineer while he/she works.

Training for the Profession

To embark on a tertiary course in Aeronautical Engineering an HSC student should be studying subjects such as Mathematics, Physics, Chemistry and Engineering Studies. There are Aeronautical courses at some universities e.g. The University of Sydney and The University of New South Wales, and interstate at RMIT and Queensland University. The courses are Bachelor Degrees. Aeronautical engineers learn about physics, aerodynamics and the materials required by modern aircraft. The Australian Defence Force Academy (ADFA) also offers courses in Aeronautical Engineering.

Career Prospects

Although a vast country such as Australia has a great need for aircraft, our requirements have never been able to sustain an indigenous aircraft industry. During and after WWII, Australia had quite a strong aircraft industry with the Commonwealth Aircraft Corporation (CAC) producing the Wirraway and Boomerang fighters and the Woomera bomber. Nowadays there are a number of parts suppliers currently in operation. The greatest employer of aeronautical engineers are the Defence Forces. They have great needs for such engineers to deal with modifications and upgrades to the various aircraft used. Large companies, such as Hawker de Havilland (now owned by Boeing), have operations in Australia that successfully bid for overseas contracts, so they also need aeronautical engineers. Currently Hawker de Havilland is manufacturing parts for Airbus and Boeing aircraft. Organisations, such as the Royal Flying Doctor Service, also use aeronautical engineers, but usually on a sub-contract basis.

Unique Technologies in the Profession

The successful development of the turbine has been a technology that has stayed unique to aircraft, and the aeronautical engineer. Although they have used turbines for ships and power stations, the turbine has remained the of significant importance aircraft. This is fascinating considering it was developed in the late 1930s and was perfected by the British and Germans before the end of WWII.

Many material technologies remain unique to the profession, but through a trickle down effect some find use in other fields. Some aircraft alloys are used in bicycles, as cycle designers search for high strength to weight materials.

Generally there are four key material properties looked for in aeronautical engineering: strength to weight ratio, formability, durability and corrosion resistance. The property, however, that the aeronautical engineer requires most for engine design is stability at high temperature. Here titanium alloys and the nickel base super alloys such the Nimonic® alloys find great use. Nickel based super alloys, usually have additions of chromium and/or cobalt among other materials. They maintain their strength and are corrosion resistant at high temperatures; they also have great resistance to creep.

Extensive use of composite materials and adhesive technologies have been developed in this field. Composites offer good specific strengths and can be readily joined by adhesives. The Eurofighter Typhoon and Saab Gripen are examples of planes with carbon fibre parts joined with adhesives.

Legal and Ethical Implications

"Ethics" means the morality or the treatment of moral questions, or honourable actions, in any situation. Professional ethics should always govern the actions of the engineer. The field of engineering is one where new developments are often occurring. This brings us to the question of intellectual property. Who owns the idea or development?

To protect these new ideas the engineer or his/her company can patent them. Patenting involves registering a design so that it is legally recognised as being owned by somebody. If a design is patented, it is not ethical for another company or individual to copy that design. Designs may be used, however, by others under licence (that is by agreement), or a company may allow other companies to freely use the design in their own products without a licence. Volvo, for example, developed the three point car seatbelt that we all use today. After developing it they patented the design and then allowed all companies to use it. They considered that people's safety was more important than commercial advantage. Had they not allowed others to use it, other companies would have needed to design another type of restraint that did not infringe the Volvo patent.

Patents need not be granted only on a finished product. They may be granted on an idea or a series of annotated sketches and descriptions. Nonetheless, it protects the engineer's ideas from unscrupulous people who may use their ideas for their own gain.

While patents are there to protect a new development, some people have claimed that patents hamper development. I. K. Brunel claimed that patents had the potential to slow 19th Century engineering, as new developments could not be utilised by others. The early Watt steam engines, for example, could not use the crank mechanism to convert linear motion into rotary motion because someone else held a patent over this mechanism. Patents can be renewed but they do eventually expire. Once this occurs, anyone is free to use that design, without compensation to the original patentee.

While most people comply with registered patents, some use unethical methods to obtain and copy registered designs. This is often called industrial espionage. Companies may use industrial espionage to gain ideas or knowledge of developments, which make it easy for them to design something of their own that is similar.

Engineers as Managers

Engineers may be managers in two senses of the word. First, they may be the managers of the design process, overseeing the development of a project. An engineer in such a position will have other engineers working on the project, while he/she co-ordinates the teams and oversees all development.

In a large project, such as building an aircraft, a hierarchical structure is developed with various aeronautical engineers having different responsibilities. Some would be responsible for materials, some for airframe, some for avionics and some for power plants. Of course there would be other areas, and other types of engineers used, such as those expert in computing. Obviously, someone must oversee the whole project so engineers will take on management roles.

Secondly engineers also make good managers of companies. They have, through their training at university and on the job, good problem solving skills and a common sense, which often transfers well to the boardroom. It is also important that aircraft companies have a balance of engineers on the management team to ensure that the company does not lose its way in terms of aircraft requirements.

Relations with the Community

Aeronautical engineers have a good relationship with the community. Through their designs they provide people with safe, fast transport. Prior to the development of air transport, a trip from Sydney to London took six weeks by boat. Even when flying was developed it was, for a long time, extremely expensive. It was not until the last three decades that the average Australian could afford to fly overseas. With a country such as Australia, isolated from the rest of the world, the aeronautical engineer has played a vital role in bridging the distance.

In recent years, aeronautical engineers have produced commercial aircraft that are quieter than were previously available through the development of turbofan engines, which produce around 10% of the noise of a turbojet. Older aircraft are "hush kitted" so that they can reduce noise levels to match newer aircraft.

Even disarmament groups rarely vent their anger at aeronautical engineers for their development of fighter planes. This is because they correctly see that it is governments that decide how to use these devices; the aeronautical engineer is merely fulfilling a request from government authorities.

Historical and Societal Influences

Explanatory Note: It is impossible to cover every aircraft built, so the author has chosen to briefly cover the early years and then turn to the jet age. It is suggested that the student uses some references at the end of the chapter to gain any further historical information.

Historical Developments in Aeronautical Engineering

Flight has revolutionised transport in a way that we have not seen before. While the car and the boat made far-flung places accessible, the aircraft gets its passengers there far more quickly.

The Early Years of Heavier than Air Flight

Flight had been postulated for many years, with hot air balloons and people flapping wings. Aerodynamic testing had also been done with unmanned gliders but it was not until 1903 that humans successfully flew. Wilbur and Orville Wright had built a powered biplane (two wings) *Flyer I*, which Orville flew successfully on December 17 at Kitty Hawk, North Carolina.

On the other side of the Atlantic, the French were working hard to produce a viable flying machine. In fact, it was in France, that ailerons were first used. Initially banking, which is where the aircraft rolls into a side to turn, was achieved by warping the wing. Harry Farman, an Englishman living in France, successfully used ailerons.

Ailerons are a type of flap mounted on the outboard aft surface of each wing, which move up, varying the lift of each wing allowing the plane to bank. This was a significant development.

A plethora of aircraft designs appeared. Biplanes, triplanes (three wings) and the now common monoplane were developed. By WWI, the plane no longer looked like an unwieldy glider. The planes developed rapidly as German, British, French and American air forces continually sparred in the air. One air force would develop a new plane only to see it rapidly eclipsed by one from the other side. The Fokker Dr.I (The Red Baron's triplane), the Fokker D.VII (the only aircraft specifically listed in the armistice agreement) and the Sopwith Camel were three wartime planes that proved the potential of aircraft.

The Jet Age

Military Jets

By 1938, Englishman Frank Whittle had developed a working turbojet engine, which promised great improvements in speed and rate of climb. Once WWII began, it became a race between the British and the Germans to produce the first jet fighter. The first flight of a turbojet aircraft was by a German Heinkel He178 on the 27th August, 1939. Britain, however, was the first to have an operational turbojet fighter with the Gloster Meteor having its first flight on the 5th March, 1943 and entering operational duty in July 1944. The Meteor was a conventional looking aircraft with one jet in each wing. It was fast enough to destroy Hitler's flying V1 bombs, and this was first demonstrated in August, 1944. The Meteor, however, was not made in large numbers during WWII.

In Germany, the first operational jet fighter was the Messerschmitt Me262 Schwalbe (Swallow), which first flew in July, 1942, but did not enter service until late in 1944. A truly impressive aircraft, it had a jet under each wing, and swept wings which hinted at the future shape of jet aircraft wings. It ripped through Allied bomber formations, because of its impressive speed and four 30 mm cannons mounted in the nose. In fact, the airframe of this aircraft had been ready far in advance of the engines, which were not ready until 1944. Had the engines been ready earlier, the Me262 may well have altered the air war.

As the Allies thrust into Germany in the last days of WWII, and during the occupation, they acquired all the German jet technology that they could. The Me262 would become the basis for many jets, and the small Messerschmitt Me163 Komet rocket plane and the Arado bomber each showed how advanced German jet technology had become.

The British had developed a centrifugal flow turbojet, while the Germans used an axial flow turbojet. The problem with the centrifugal type was that it was more bulky, and eventually all jets were designed to use the axial flow system.

In Britain, de Havilland led the way with the Vampire twin boom jet, but it was the United States that had enough money to pour into fighter development. With a range of impressive fighters, they were at the forefront of jet development. The North American F86 Sabre was an impressive aircraft with swept wings and good

manoeuvrability. The USSR was not to be left behind though; it released its Mikoyan-Gurevich MiG 15, which was superior to the Sabre. The Sabre outmanoeuvred the MiG 15 in the Korean War, partly because of better pilot training and experience.

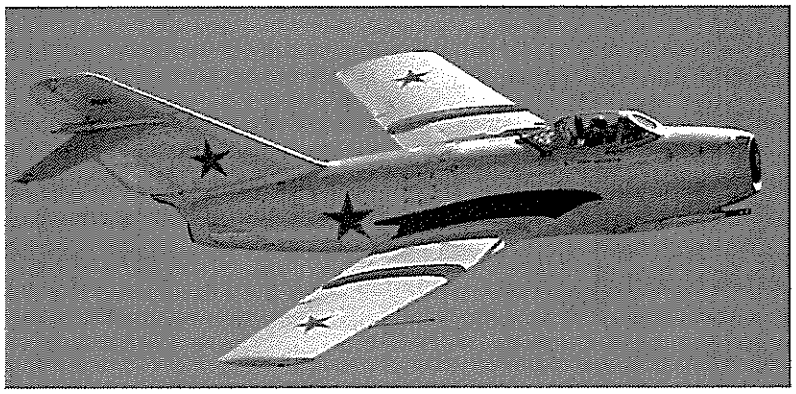


Fig 3.1: *Mikoyan-Gurevich MiG 15 fighter jet, a notable early Soviet jet fighter from the Cold war. Note the swept wings and tail plane.*

After the Korean War, jets continued to increase in performance, and by the Vietnam Conflict a variety of fighter jets existed that could travel at twice the speed of sound. An important jet of the 1960s was the General Dynamics F-111. This was the first aircraft that utilised variable geometry wing (swing-wing) technology. A big problem with jets is that they need swept wings to deal with the shock wave effect of high-speed travel. Swept wings have different aerodynamic characteristics which can reduce lateral control and performance at low speeds and increase stall speeds.

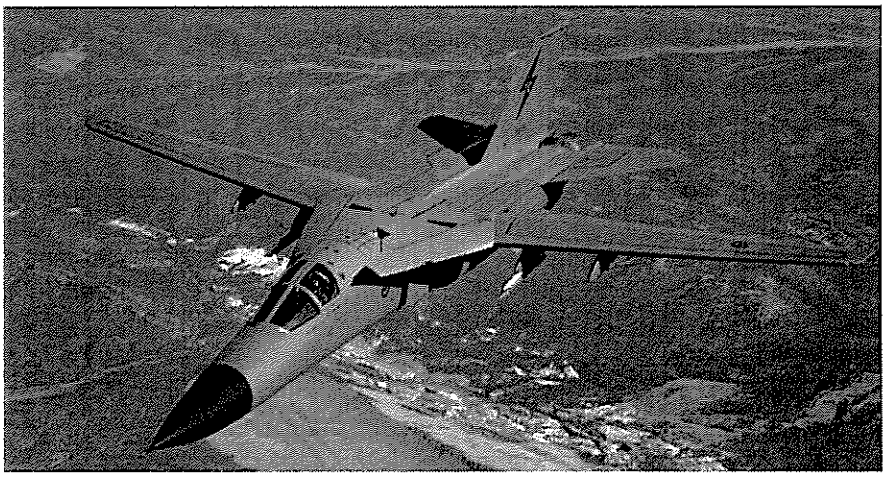


Fig 3.2: *An RAAF General Dynamics F-111 fighter-bomber.*

The F-111 countered this by having variable-geometry, or swing-wings. These are controlled by the pilot. For take-off, landing and low level flight the wings are swept forward to increase lift and to maximise low speed stability and control. At transonic speeds they are partially swept back, while at full Mach 2 speeds the wings are swept further back to produce a delta wing arrangement that is superb for high speed. This feature made the F-111 one of the most potent aircraft in the world. It could act as a bomber, a fighter and also a ground attack aircraft. Terrain following radar (TFR) enabled high speed operations at very low altitudes.

While the technology of variable geometry wings is impressive, it makes the wings heavy, costly, and maintenance intensive with shoulder joints and pivoting pylons for missiles, so they don't shoot off at odd angles. The Swedish Air Force had a problem that swing-wings could have solved. They needed a STOL (Short Take Off and Landing) jet aircraft that could fly fast and intercept Soviet aircraft straying into their airspace. Because of limited funds, they needed a cost effective aircraft, which ruled out variable geometry wings. The Saab Aircraft Company evolved a truly impressive aircraft that combined STOL performance with a rapid climb rate and Mach 2 performance while still maintaining fighter manoeuvrability. The aircraft was the Saab AJ37 Viggen (Thunderbolt).

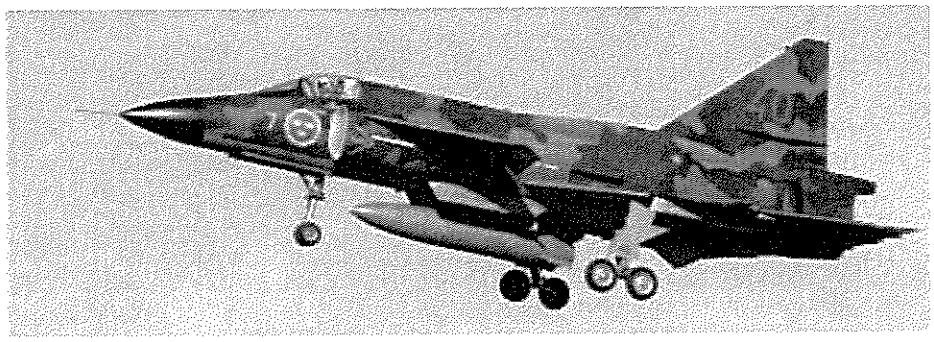


Fig 3.3: *Saab Viggen on approach. Note the canards behind the air intakes.*

The Viggen solved the problem of combining STOL and high performance, using a delta wing, good for high speed, in combination with forward canards (small wings). These improved lift and enabled excellent take-off performance. This was important in Sweden where the aircraft are required to operate from roads instead of established air bases. This system limits the chance of an attacker taking out a whole air force while on the ground, as Israel did to Egypt in the Six Day War (1967). The soundness of the Viggen's design was evident in the many variants that developed and how important it became to the Swedish Air Force. It was also very carefully designed in terms of maintenance, allowing easy access to most areas, which is necessary as many Swedish service people are low-skilled conscripts.

Another option for small airfield capability and operational versatility is VTOL (Vertical Take Off and Landing). Most VTOL aircraft development was done in the USA but the British are the only country outside Russia to design and manufacture a true VTOL fighter to production (although the design has been improved collaboratively in the USA). The British Aerospace Harrier was (and still is) truly revolutionary. It gave the fixed wing aircraft great versatility, by allowing it to take off and land vertically, hence removing the need for long airstrips. The Harrier uses a single turbofan with four outlet nozzles, two on each side, that direct the thrust through a 90 degree arc. When at maximum weight, the Harriers take off using a "ski jump", so they are a very effective STOL jet, while still having little need for long runways.

Originally the Harrier was designed for the strike or ground attack role, but the Royal Navy purchased about twenty Sea Harriers prior to the Falklands War against Argentina. Here, the Mach 1 Sea Harrier came up against the Mach 2 Dassault Mirages of Argentina. The Harrier, however, proved that speed is not essential in dog-fighting. With its adjustable nozzles it could easily out manoeuvre the faster but less agile Mirage.

Fig 3.4: *A Harrier performing a vertical take-off, note the forward nozzle. (http://en.wikipedia.org/wiki/File:Bae_harrier_gr9_zg502_takeoff_arp.jpg)*

Changes in technology have now seen a move towards highly manoeuvrable, multi-role fighters, which can create savings because they mean one aircraft can do a variety of jobs. They still offer good payload and excellent manoeuvrability, but may also be used for reconnaissance duties. A fighter, such as the Eurofighter Typhoon is an example of this. The Eurofighter Typhoon and the new US fighters like the Lockheed/Boeing F-22 Raptor, make extensive use of composites, are highly manoeuvrable, and use STOL features. These newer fighter aircraft are, however, quite expensive, which makes them difficult for a country like Australia to buy in large numbers.

The US has moved towards designs incorporating stealth technology. Through the use of radar absorbing materials and the design of surfaces to minimise radar reflection, stealth fighters and bombers offer a great deal in the modern war and strategic defence theatres. The B-2 "flying wing" stealth bomber cost a billion dollars US per plane and as a consequence it has sold in very few numbers.

Commercial Jets

In 1949 the turbojet was designed to fit into commercial aircraft. The US was leading the world in aircraft manufacture and Britain appeared to be languishing. Then in 1949, the British rolled out the de Havilland Comet. This 36 seat turbojet aircraft was revolutionary. Its features included swept wings, a pressurised cabin, four jets in the wing roots, an aluminium stressed skin airframe and a top speed of around 432 kt (800 km/h). Immediately American companies tried to tell airlines that turboprop airliners could match this and US jets would be there soon, but airlines wanted the new fast jet airliner. The Comet represented the image that post-war Britain needed, a prosperous and innovative nation.

The Comet entered service in May, 1952 and was performing well. Then on 26th October, 1952 a Comet ran off the end of the runway in Rome when attempting take-off. The reason was that the pilot had over rotated the aircraft (raised the nose too high), which meant the wings did not produce enough lift. Luckily there were no fatalities. Then in March 1953, over rotation during a take-off run at Karachi airport, caused the death of all eleven people on board. This was a Canadian Pacific Airways aircraft en route to Canada via Australia. To de Havilland it was to be a promotional

tour for the Comet in Australia. The pilot had raised the nose too high on the take-off run, but there seemed to be no attempt to abandon the take-off. The aircraft used the whole runway, then the overrun. At the end of this, the undercarriage struck a culvert, which swung the aircraft into a gully where it erupted in flames.

Two Comets had now suffered take off problems and the answer was to not rotate the aircraft until take off speed was reached. Then in March 1953 a Comet was lost at an altitude of 9,756 m (32,000 ft) when severe turbulence caused the tailplane to fail. The thirty-seven passengers and six crew all perished. The Comets was starting to look rather "patchy" in terms of safety, when the next disasters occurred.

Fig 3.5: *A Comet Mk1A (courtesy Mark Schaeffer; http://surf.to/comet).*

On January 10th 1954 a BOAC (British Overseas Airline Corporation) Comet left Rome en route to London. A minute after the aircraft passed 7,926 m (26,000 ft), the pilot started a radio transmission that was cut short. The aircraft was never heard from again, as it had exploded in mid-air. This mid-air explosion was followed by another in August of the same year when a BOAC Comet, chartered to South African Airways, took off from Rome en route to Cairo. During its climb to an altitude of 10,671 m (35,000 ft) it exploded in mid-air over the Mediterranean Sea. Both aircraft, somehow, had exploded as they climbed to their operating altitude. One of the Comets was brought up from the sea bottom and painstakingly examined for flaws. It proved that the Comet's body had exploded, but why? To determine the cause of the problem another Comet was tested inside a specially constructed tank. The wings and fuselage were subjected to pressurisation, flight and undercarriage loads on continuous cycles for twenty-four hours a day.

The fuselage of the Comet ruptured in the test and the fault was traced to two small fibreglass antenna windows at the top of the fuselage. It appeared that the rivet pattern and design of the windows promoted the propagation of fatigue cracks during pressurisation cycles. The result was catastrophic fuselage cracking and explosive decompression. The fault finding had cost 2 million pounds and the Comet had been grounded through this time. It was now up to de Havilland to fix the fault and to their credit they succeeded. They released the larger Comet 4 which carried up to eighty-one passengers and became the first passenger jet to make a regular transatlantic crossings. Out of the jaws of defeat, de Havilland had snatched a small victory. The aircraft that looked ready to secure the British aviation industry had nearly destroyed it.

While the Comet had its many problem solved, Boeing was putting the finishing touches to a passenger jet that would elevate them to the top of the aviation world. Although beaten by the Comet for a transatlantic service, the Boeing 707 was to become the most utilised aircraft of its time. Like the Comet it used four turbojet engines, but these were slung beneath the wings. The 707 could also carry more passengers with a typical load being around 180. The 707 was followed by the similar looking McDonnell Douglas DC-8. Between the two of them, they carried the vast majority of the world's commercial jet passengers.

Fig. 3.6: *A former Qantas Boeing 707 with its undercarriage down on approach for landing. (http://en.wikipedia.org/wiki/File:Boeing_707-138B_Qantas_Jett_Clipper_Ella_N707JT.jpg)*

Fig. 3.7: *The Boeing 747, the Jumbo jet, is one of the most successful commercial passenger planes ever designed. (http://commons.wikimedia.org/wiki/File:Qantas_Boeing_747-400_SYD_Gilbert-1.jpg)*

Passenger jets kept increasing in size. A huge step was taken with the development of the Boeing 747. Nicknamed the Jumbo jet, this was originally designed as a military transport for the US Air force. When Boeing lost the military contract they decided to develop their aircraft as a passenger jet. The 747 was revolutionary in design, with its four high bypass turbofan engines, advanced wing design and complex high lift devices. The original 747-100 would typically carry 397 passengers, more than double the capacity of the initial 707 version. In fact, the 747 has been so dominant that it was not until 2001 that a manufacturer released an aircraft (Airbus A380) larger than it in passenger carrying capacity. The 747 design still has much life left with the

latest model the 747-8 released in 2008. The 747-8 is currently the longest passenger aircraft in the world.

While military jets were pushing past Mach 1 and then Mach 2, all passenger jets were designed, because of size, cost and technology, to operate at subsonic speeds. The first supersonic transport to enter service was the Soviet Tupovlev Tu144. It bore a remarkable similarity to the Concorde but after a crash at an air show it was withdrawn. The British Aerospace/Aerospatiale Concorde, which first flew in 1969, remains the only supersonic transport to remain in regular service. The aircraft was revolutionary in utilising substantial military technology to produce fast commercial transport. Although the cabin is a little cramped most people flew on its regular service because of time savings. A person could have breakfast in London, board the Concorde and have breakfast again in New York because of the time difference and rapid speed of travel.

Fig. 3.8: *The unmistakeable shape of Concorde, its double delta wing created for both acceptable take off and landing performance, and for suitable performance at Mach 2. (http://commons.wikimedia.org/wiki/File:Concorde.planview.arp.jpg)*

It was not until 2000 that there was a devastating crash with the Concorde, which marred an otherwise good safety record (it has had many incidents but only one accident). Debris cut the tyre causing it to blow, resulting in tyre pieces hitting the underside of the wing fuel tank. This caused a significant fuel leak which ignited, engulfing the rear fuselage. Number 2 engine was shut down, number 1 lost power, the aircraft lost control and plunged into a small hotel. As a result of this and airframe issues the Concorde is no longer in service. The Concorde proved that while supersonic passenger transport was possible it was very costly and not as popular as larger planes like the 747 which travelled more slowly but provided much more cost effective travel for people.

One plane also worth a mention here is the McDonnell Douglas DC-10. The DC-10 was released not long after the Boeing 747. It is a smaller plane with a three-engine arrangement; one engine under each wing and one in the tail. The third engine was added to meet requirements to operate from existing runways. This arrangement gives fewer problems with a single engine out, as all thrust is not concentrated away from

the centreline. The DC-10 suffered from some early teething problems and then some crashes caused through inadequate maintenance. The DC-10 is still operated, and is also used by the US military as an in-flight tanker, which has proved the fundamental soundness of the design.

Fig. 3.9: *The Antonov An124, a large cargo aircraft, note the extensive undercarriage and the high mounted anhedral wing. (http://commons.wikimedia.org/wiki/File:Antonov_An-124_EFHK.jpg)*

In the late 1970s the Soviet Union aircraft company Antonov developed what for many years would be the largest series production cargo aircraft. The An124 is a four engine body with a heavily constructed fuselage complete with internal cranes to cope with loads. It also has the ability to lower it's opening nose, to reduce the ramp over angle for loads. The long wingspan demanded Soviet engineers to come up with extra long aluminium alloy wings spars which required special computer controlled cooling to ensure their strength was even along the entire spar.

Fig. 3.10: *The double deck Airbus A380 is the largest passenger plane to enter service. (http://commons.wikimedia.org/wiki/File:SIA_Airbus_A380,_9V-SKA,_SIN_3_resized.jpg)*

The 1980's and 1990's saw Boeing continue its domination of the commercial air industry. McDonnell Douglas is now owned by Boeing with the main competition for Boeing coming from a European consortium, Airbus, eighty percent owned by the European consortium, EADS and twenty percent by BAE Systems. The biggest development in the commercial jet world in recent years was the development of the Airbus A380 family of aircraft. This aircraft first flew in 2005 and entered service in 2007, and has eclipsed the Boeing 747 for passenger capacity on long haul routes.

The A380 offers 525 seats in a standard three-class layout and a capacity of 853 passengers is possible in an all economy layout. A freighter version was put on hold to concentrate on the passenger version.

Boeing based much of its corporate plan on the 777 family of planes. These have a new wing design, make extensive use of composites and have huge turbofan engines in the 444 kN (100,000 lb) class compared to the 747's 267 kN (60,000 lb) class. The Boeing 777-300 can seat 368 passengers and has a range of 5,720 nm (10,595 km). The huge reliability and thrust of modern turbofan engines used on planes such as the 777, made tri-jets like the DC-10 obsolete. Now only the very largest of commercial aircraft use four engines, with most designs using two turbofans. The latest Boeing design, the 787 Dreamliner has focussed not on size but on efficiency, trying to make it a highly fuel efficient and light design, with extensive use of composites in-lieu of aluminium alloys for the airframe.

Fig. 3.11: *The Boeing 787-8 Dreamliner, designed with efficiency in mind, its fuselage is made predominately of carbon fibre reinforced composites.*
(http://commons.wikimedia.org/wiki/File:All_Nippon_Airways_Boeing_787-8_Dreamliner_JA801A_OKJ.jpg)

The Effects of Aeronautical Engineering on People's Lives and Living Standards

The effect of the aeronautical engineers' work has been astounding. Prior to the advent of commercial passenger aircraft service, the trip from Sydney to London would take about six weeks; now it can be done in as little as twenty-four hours. In regional Australia, the aircraft has had a great influence. The Royal Flying Doctor Service allows access to medical facilities that could take days with road transport. Also many farmers on large outback properties utilise helicopters and light planes for checking their land, herding animals and crop spraying. Aircraft are also used for geological surveying and cartographic services.

Below is a list of some impacts that aeronautical engineering has had.
Aircraft:

- have made travel more accessible to people who have time constraints on holidays.
- particularly jets, have made overseas commerce much quicker.
- allow families who live overseas to be brought closer together.
- have reduced the time of overseas postal and freight services.

- for military purposes, have developed to a point where they may be used instead of ground troops and with less casualties, (to the attacking side at least).
- acting as ambulances save lives, by getting patients to hospitals more quickly.
- may be used to fight bushfires, hence protecting property and people.
- are used to gather images for news services quickly and efficiently.
- (both conventional and helicopter) are used to transport people into and out of isolated areas that otherwise may be difficult to reach.
- have boosted tourism, particularly to isolated countries like Australia.

Environmental Implications of Flight

The ability of humans to fly has had an impact on the environment in a variety of ways. Some of these are listed below:

Pollution is created by both piston and turbine engines that are now used. Aircraft pollute the environment with exhaust gases. There have, however, been advances in aircraft fuel technology and engine flight management aimed at dramatically reducing air pollution from engines. There have also been cases of fuel dumping , which is a serious form of pollution. Another type of pollution is noise pollution; this is a very contentious issue particularly with airports in urban areas. Much work has been done on turbine design to get noise levels down.

Insect spraying has become much easier and expanded to a large scale through the use of both conventional aircraft and helicopters. The ability of crop dusters to cover a wide swathe of land has meant far greater use of insecticides. These insecticides do protect crops, but they also have a major impact on the food chain. Concentrations of these poisons increase as we go up the food chain leading to a particular concern for the food we eat.

Destruction through the use of aerial bombing has been on the rise since WWII. The damage caused to British cities by Germany and to Germany by the Allies showed the true potential of the aircraft in warfare. Allied bombers destroyed German industry, leading to its eventual defeat. In The Vietnam Conflict, bombing was taken to a new level. Saturation bombing with chemicals saw huge areas of forest destroyed in Vietnam in an attempt to flush out the Viet Cong troops.

Prior to the aircraft, this level of destruction required artillery close to the target. The aircraft can now deliver destruction far away from its base with devastating results.

Engineering Mechanics and Hydraulics

Aircraft Geometry and Terms

Fuselage: houses flight crew, passengers and freight. It is pressurised (not in small aircraft) to allow people to survive at high altitudes where turbine aircraft are most efficient.

Wing: provides most (not all) of the lift to support an aircraft's weight. Different shapes are used depending on the role of the aircraft.

Ailerons: outboard trailing edge sections of the wing used to control roll.

Flaps: inboard trailing edge sections, used to generate additional lift during low speed manoeuvring and landing.

Horizontal stabiliser: (also called the tailplane) in level flight it provides (typically) downward force to keep aircraft level and stable.

Elevators: trailing edge sections of the horizontal stabiliser, used for pitch (fore and aft rotation) control.

Elevons: Some high performance delta wing aircraft have the wing and horizontal stabiliser merge at the back of the wing. Positioned on the outside of the delta wing's trailing edge, the elevons perform the role of both the elevator and the ailerons.

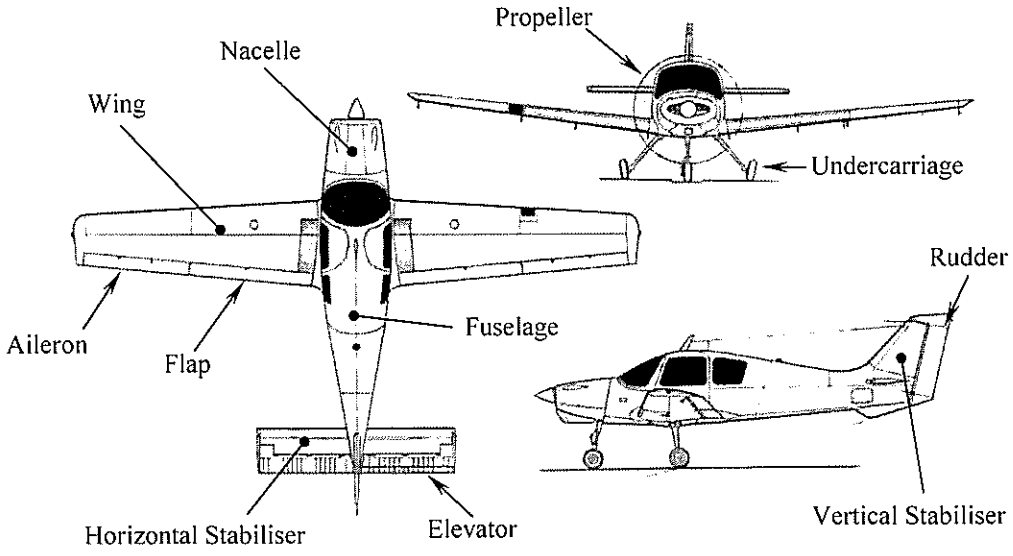


Fig 3.12: *A small four seat propeller plane with some key features labelled.*

Fin: (also called vertical stabiliser) used to provide left/right stability. Together with the rudder it provides yaw (left-right rotation) control.

Rudder: trailing edge section on the fin.

Undercarriage: provides a means of manoeuvring on the ground.

Flight Mechanics

1. Level Flight

In its simplest form an aircraft flying at constant velocity in level flight can be considered to have a force equilibrium of four concurrent forces:

Weight – gravitational force on the aircraft mass.
Lift – a force generated by airflow over the wing and tailplane.
Thrust – a forward force caused by the power plant.
Drag – a force resisting movement of the aircraft through the air.

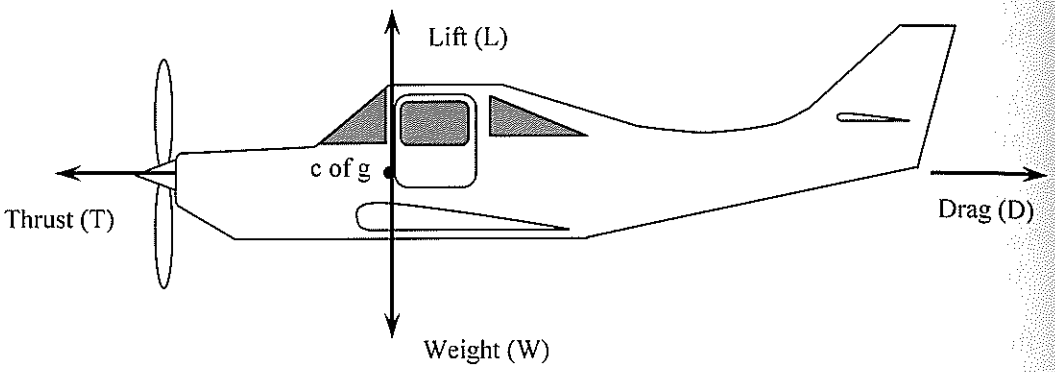


Fig. 3.13: *The four forces acting on an aircraft in level flight.*

Fig. 3.13 shows the simplest relationship of four forces, each acting through the centre of gravity of the aircraft.

$$+\uparrow \Sigma F_V = 0$$
$$0 = L - W$$
$$\therefore L = W \quad \text{——————} \quad (1)$$

$$\rightarrow + \quad \Sigma F_H = 0$$
$$0 = D - T$$
$$\therefore T = D \quad \text{——————} \quad (2)$$

Drag is made up of two components:

Induced drag – the drag developed as a result of lift production and,
Parasitic drag – the drag resulting from moving the aircraft shape through the air.

2. Level Flight (a more complex view)

When we consider the aircraft and its force system more closely we need to recognise a number of important facts:

(i) The weight force acts through the centre of gravity.
(ii) The line of the thrust force is inclined to the direction of flight – the angle of attack (α).
(iii) The aerodynamic forces of lift are generated at the aerodynamic centre of the wing and tailplane.
(iv) The centre of gravity of an aircraft moves during flight because of changes in passenger loading, cargo, and fuel usage, which makes analysis more complex.

Neglecting (iv) above, we can represent the aircraft force system as follows:

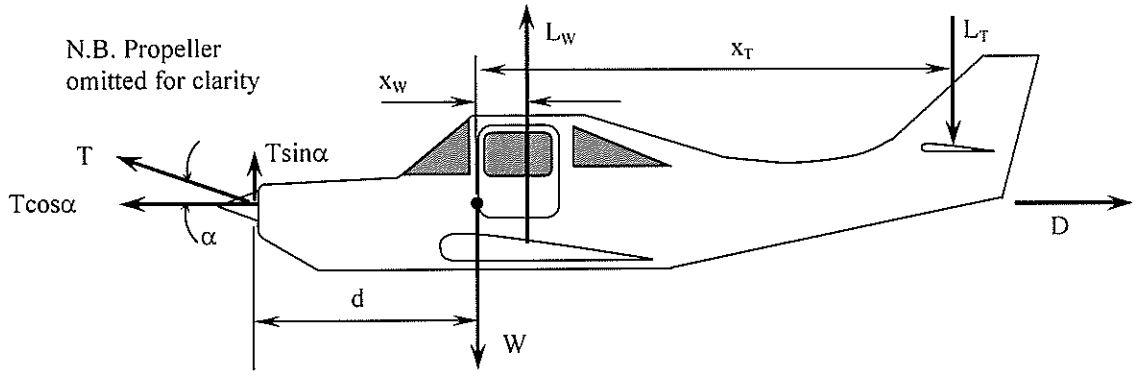


Fig. 3.14: *A more detailed analysis of the forces of level flight.*

This is still a simplified view, which neglects several more complex concepts such as:

(i) aerodynamic moments set up on the wing and tailplane
(ii) thrust moments
(iii) drag moments.

Nonetheless this force system (a non-concurrent system) allows us to get a good idea of the force relationship in flight.

$$+\uparrow \Sigma F_V = 0$$
$$0 = L_W - L_T + T\sin\alpha - W \qquad (3)$$

$$\xrightarrow{+} \Sigma F_H = 0$$
$$0 = D - T\cos\alpha \qquad (4)$$

$$\circlearrowleft^{+} \Sigma M_{cg} = 0$$
$$0 = L_W x_W - L_T x_T - T\sin\alpha d \qquad (5)$$

Most medium sized aircraft have the centre of gravity in front of the lift vector and therefore the lift on the tailplane must act downward (as shown in Fig 4.8) to maintain trim.

Equation 3 shows that the lift of the wing has to be greater than the aircraft weight to maintain level flight. Equation 5 shows that the moment generated by the wing is balanced by the moment generated by the tailplane. As α and d are small, the thrust moment will also be small compared to the tailplane moment.

3. Steady Climb

We can now consider the forces acting on an aircraft in steady climb, at constant velocity, with a climb angle θ, and assuming the direction of flight as equal to the thrust direction (a point that is not quite true).

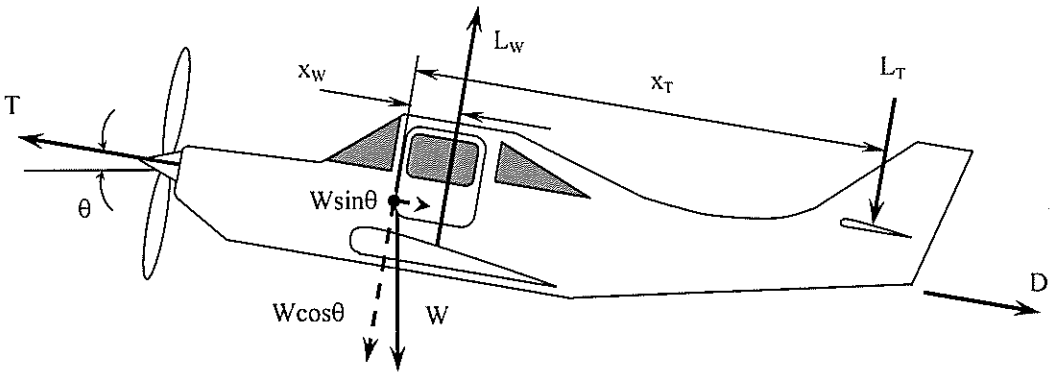


Fig. 3.15: *Forces on an aircraft in a steady climb at constant velocity*

We also assume that thrust (T) and Drag (D) pass through the centre of gravity.

By adding the force system parallel and perpendicular to the direction of flight we get:

$$+\uparrow \Sigma F_{perp} = 0$$

$$0 = L_W - L_T - W\cos\theta$$

$$L_W - L_T = W\cos\theta \qquad (6)$$

$$\xrightarrow{+} \Sigma F_{par} = 0$$

$$0 = D - T + W\sin\theta$$

$$W\sin\theta = T - D \qquad (7)$$

Taking moments about the centre of gravity we get:

$$\circlearrowleft(+)\Sigma M_{cg} = 0$$

$$0 = L_T x_T - L_W x_W$$

$$L_T x_T = L_W x_W \qquad (8)$$

From equation 8 we can see that a loss of wing lift (L_W) must be compensated with an increase in tailplane lift (L_T) to maintain trim. The sudden loss of lift, however, which occurs when the wing stalls, will cause the aircraft to drop suddenly (from equation 6, $W\sin\theta$ will exceed L_W - L_T) with the potential for a loss of control.
Solving equations 6 and 7 for θ (angle of climb):

$$\frac{W\sin\theta}{W\cos\theta} = \frac{T - D}{L_W - L_T} \qquad (9)$$

Given that L_W - L_T is the total lift for the system:

$$\tan\theta = \frac{T - D}{L} \qquad (10)$$

For very small angles $\tan\theta$ in *radians* is equal to θ, and L approximates W; therefore:

This formula is really only applicable to commercial planes with low angles of climb. Not suitable for use with fighter jets.

$$\theta = \frac{T - D}{W} \qquad (11)$$

Example 3.1

Calculate the angle of climb for an aircraft with a lift to drag ratio of 15 and a thrust to weight ratio of 0.25.

Using Equation 11

$$\theta = \frac{T - D}{W}$$

Can be rewritten as follows:

$$\theta = \frac{T}{W} - \frac{D}{L} \quad (\because W \approx L \text{ when } \theta \text{ is small})$$

$$\theta = 0.25 - \frac{1}{15}$$

$$\theta = 0.183 \text{ rad (Note 1 rad} = 57.29°)$$

$$\therefore \theta = 10.3°$$

Note: "rad" is the symbol for radian, the SI unit for angular measurement.

The angle of climb for the aircraft is 10.3°.

Example 3.2

An aircraft has a lift to drag ratio of 15. At an altitude of 1500 m the engine fails. Will the pilot be able to glide to an airport 16 km ahead?

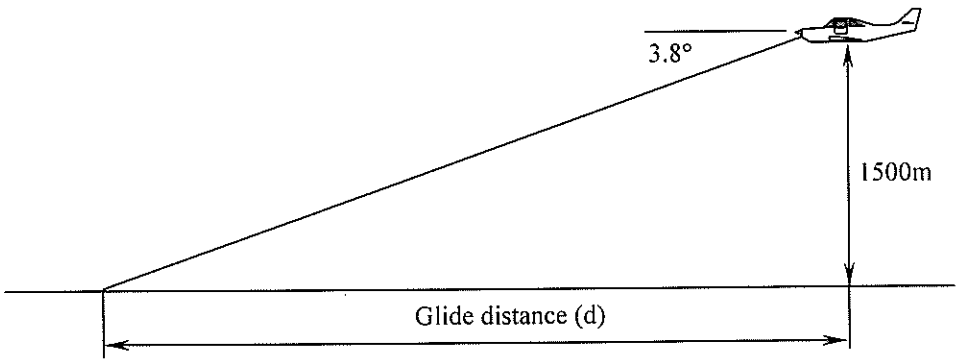
$$\theta = \frac{T}{W} - \frac{D}{L}$$

$$\theta = 0 - \frac{1}{15} \quad (\because T = 0)$$

$$\theta = -0.066 \text{ rad}$$

$$\therefore \theta = -0.066 \times 57.29 = 3.8°$$

The angle of the glide will be 3.8°.



$$\tan 3.8° = \frac{1500}{d}$$

$$d = \frac{1500}{\tan 3.8°}$$

$$\therefore d = 22.5 \text{ km}$$

Since the plane can glide 22.5 km it will reach the airport.

Basic Aerodynamics

The basic principle of flight is all centred around the aerofoil and the way it behaves as air passes over it. The aerofoil is the cross-sectional shape that a wing or a rotor blade has; an aerofoil is shown in Fig 3.16. The full explanation of lift generated by an aerofoil is complex and difficult to simplify, so often, simplified theories are presented to explain how lift is generated. Two such theories are presented, the classic theory and the Flow Turning theory

Fig 3.16: *A bare aerofoil is shown (left) along with another showing airflow (right).*

Flight Theory 1: Classic Theory

The classic theory has been accepted orthodoxy among many texts and educators for years. Airflow over the top surface must accelerate to follow the curved surface. This produces a region of very low pressure above the aerofoil (Bernoulli's principle). The difference in the pressure between the upper and lower surfaces acting on the aerofoil produces lift. Airflow along the lower surface of the wing decelerates, creating a region of slightly higher pressure, thus acting to lift the wing.

The speed at which the air passes over the aerofoil is crucial as is the angle the aerofoil makes with the oncoming air (angle of attack), which determines the amount of lift. The aircraft must be moving to get relative movement between the air and the wing, so a plane must travel along the runway until the required take off speed is reached.

Flight Theory 2: Lift from Flow Turning

The problem with the Classic Theory is it doesn't really fit with lift generation and relies on some questionable assumptions. An alternative view of flight theory is that the lift force is created by the action of turning the flow of air. Since F=ma and acceleration is the rate of change of velocity over time, by turning the flow of air the velocity changes *direction*, This changing velocity gives us an acceleration, which hence creates the lift force. A related but incorrect theory is that lift is created by the wing deflecting air. While this theory holds for hypersonic speeds, it is not quite correct for most aircraft because it ignores the role of the upper surface of the aerofoil.

For a good analysis of differing flight theories students should check out the NASA education site here: http://www.grc.nasa.gov/WWW/k-12/airplane/lift1.html

It should be noted that while the Flow Turning theory is more correct, in an HSC exam either of the two theories listed above would be accepted as an answer to lift creation by an aerofoil.

Stalling

Stalling can be a dangerous aspect of flight that is often confused with the engine stopping. When a plane stalls, it means that the wing no longer produces lift. This can be particularly dangerous at low altitudes, when the pilot has no time to recover the aircraft.

In level flight, lift is produced as above, however, the wing can stall if the airspeed is reduced below a certain point, or if the angle of attack is increased beyond a certain angle.

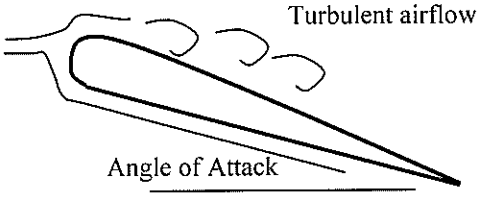


Fig 3.17: *An aerofoil undergoing a stall*

When the wing stalls, the air flowing over it cannot follow the surface, hence it breaks up and causes turbulence. This means a reduction in lift. Stalls may occur during landing as the aircraft has its nose up and is slowing down, but aircraft have stall-warning devices, to warn the pilot of an impending stall. As a plane lands, the pilot will reduce power, and raise the nose of the plane, to further reduce the airspeed (by increasing drag) and to ensure the main landing gear contacts the ground before the nose. This is desirable except, as airspeed falls and the angle of attack increases, a stall is more likely. To avoid this, flaps are placed on plane wings; their primary function is to increase the lift generated at a given angle of attack and airspeed. This allows the aircraft to produce the required amount of lift at a lower speed, which is obviously desirable. Flaps also produce additional drag which assists in slowing the aircraft down.

Slats are sometimes fitted to the leading edge of the wings, and these channel the air over the front of the wing and force the airflow over the wing. Slats are retractable and are fitted to large commercial aircraft to reduce the chance of stalling when flying at high angles of attack.

Stalling also may occur in manoeuvres when tight turns or steep climbs may cause the airflow over the wing to separate. Students should be aware that stalling has little to do with the engine. A plane can stall with the engine at full power. This is a danger with taking off in confined areas where the plane must climb and turn sharply immediately after take off. No matter how powerful the engine is, if the wings cannot provide lift, the plane cannot fly.

Lift to Drag Ratio

The lift to drag ratio is a measure of the amount of lift compared to the drag the plane has, engineers will want a high lift to drag ratio. The lift to drag ratio has a big effect on the range of a commercial aircraft because a low lift to drag ratio will mean the plane uses more fuel. In unpowered aircraft such as gliders, a high lift to drag ratio will allow a glider to travel further as it descends. The lift to drag ratio is related to the glide angle (θ) as follows.

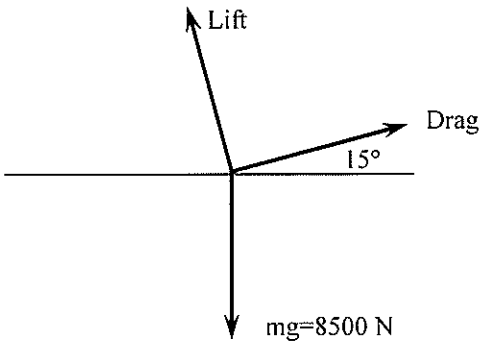
$$\tan\theta = \frac{D}{L}$$

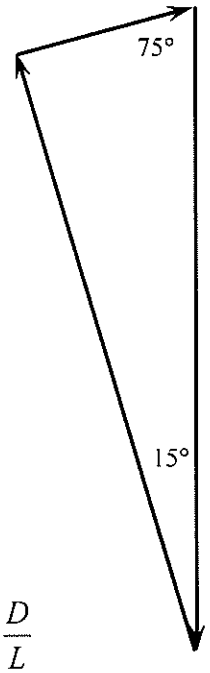
$$\frac{L}{D} = \frac{1}{\tan\theta}$$

Example 3.3

A glider and pilot have a combined mass of 850 kg is the glider is descending at an angle of 15° at constant velocity. Determine the lift of the glider and then calculate the lift to drag ratio of the glider.

Analysis: First we shall draw the FBD, note there is not thrust with the glider. The lift vector is perpendicular to the glider's flight path, while the drag force is in line with the flight direction. Then to the right we have added the three vectors, since we are descending at constant velocity ΣF=0.





$$\cos 15° = \frac{L}{W}$$

$$L = W\cos 15°$$

$$L = 8500 \times \cos 15°$$

$$L = 8210.4 \text{ N}$$

$$L \approx 8.2 \text{ kN}$$

$$\tan 15° = \frac{D}{L}$$

$$\frac{L}{D} = \frac{1}{\tan 15°}$$

$$\frac{L}{D} = 3.73$$

The glider's lift force is approximately 8.2 kN, and its lift to drag ratio is 3.73.

Effect of Angle of Attack

The angle of attack (aoa or α) as mentioned earlier refers to the angle the aerofoil makes with the oncoming air. Higher angle of attacks will crate greater lift up to a point. If the angle of attack is to great then the stalling may occur and then lift falls away sharply.

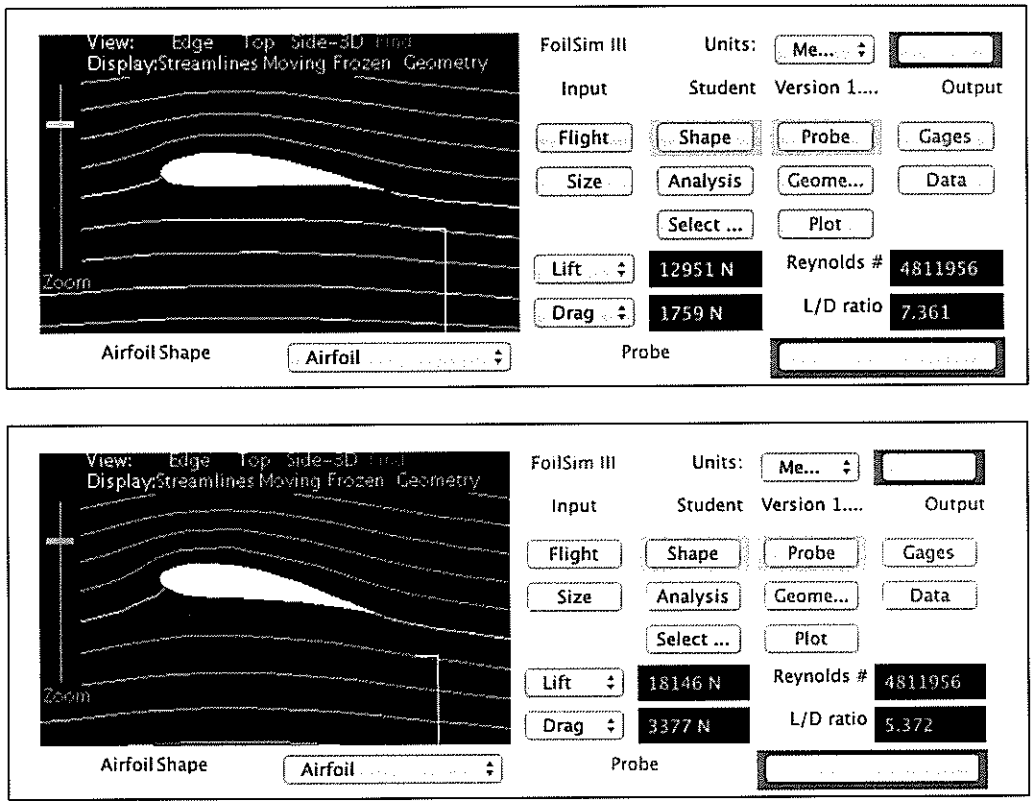


Fig. 3.18: *Two screenshots of the from the FoilSIM aerofoil simulator. The upper shot has a 5.2° aoa while the lower shot has an aoa of 10°.*

In Fig. 3.18, NASA's FoilSIM III aerofoil simulator is used to compare the lift and drag forces that occur at two different angles of attack, first 5.2° then 10°. You can see both lift and drag increase. For the aerofoil shown maximum lift occurred at 14.4° after that the lift force fell away. As mentioned earlier in the chapter both flaps and slats are used to allow greater lift at high angles of attack.

A good exercise is to access FoilSIM and compare a range of aerofoil cross sections with different angles of attack. It is also possible to alter the camber of the aerofoil and its thickness. Foil SIM can be accessed here:

http://www.grc.nasa.gov/WWW/k-12/airplane/foil3.html

Bernoulli's Principle

Consider an amount of fluid being pushed through a tube by a piston (Fig. 3.19). When we first see the piston it is moving at a speed v_1, at a distance y_1 from the floor. A short time later it is moving at speed v_2, and is now a distance y_2 from the floor.

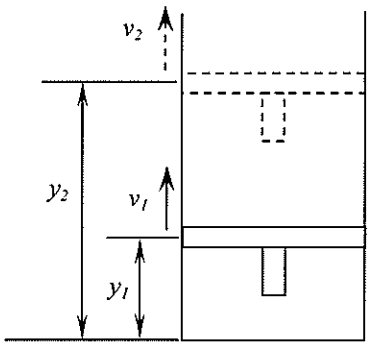


Fig. 3.19: *The vertical motion of a piston through a fluid.*

Applying the conservation of energy theory, the work done on the fluid equals the change in potential energy plus the change in kinetic energy. That is:

$$\text{Net work} = \text{change in PE} + \text{change in KE} \quad \text{——— (1)}$$

Now the work done on the fluid is related to the change in pressure at the two piston locations above, multiplied by the volume (V) of the fluid:

$$\text{Net work} = (P_1 - P_2)V \quad \left(\because\ W = \frac{F}{A} \times Ay = Fy\right) \quad \text{——— (2)}$$

Considering the potential energy change, we know that when we lift an object, its increase in PE is given by:

$$PE = mgh$$
$$PE = mg(y_2 - y_1) \quad \text{——— (3)}$$

Since mass is equal to the product of density and volume, we can rewrite equation (3) to become:

$$\Delta PE = \rho Vg(y_2 - y_1) \quad \text{——— (4)}$$

The kinetic energy of the system will be given by:

$$KE = \tfrac{1}{2}mv^2 \quad \text{——— (5)}$$

The change in KE of the fluid will be given by this equation. Considering the change in velocities, this becomes:

$$\Delta KE = \tfrac{1}{2}m(v_2^2 - v_1^2) \quad \text{——— (6)}$$

Substituting $m = \rho V$ into the KE equation as we did with the PE equation gives:

$$\Delta KE = \tfrac{1}{2}\rho V(v_2^2 - v_1^2) \quad \text{——— (7)}$$

Putting all three energy equations (②, ④ and ⑦) into equation ⑧ we get:

$$(p_1 - p_2)V = \rho V g(y_2 - y_1) + \tfrac{1}{2}\rho V(v_2^2 - v_1^2) \quad \text{——} \; ⑧$$

We can divide by V to give the equation in a "per unit volume" form:

$$(P_1 - P_2) = \rho g(y_2 - y_1) + \tfrac{1}{2}\rho(v_2^2 - v_1^2) \quad \text{——} \; ⑨$$

Expanding this equation and gathering all the terms relating to point 1 and point 2 on opposite sides gives:

$$P_1 + \rho g y_1 + \tfrac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \tfrac{1}{2}\rho v_2^2 \quad \text{——} \; ⑩$$

Because the system has its energy conserved, the right hand side is a constant. So referring to the fluid at any point (drop the subscripts) we get Bernoulli's equation.

$$P + \rho g y + \tfrac{1}{2}\rho v^2 = \text{constant} \quad \text{——} \; ⑪$$

Venturi Effect

If we consider fluid flowing through a pipe with a constricted section where there is no difference between heights, then Bernoulli's equation will become:

$$P + \tfrac{1}{2}\rho v^2 = \text{constant}$$

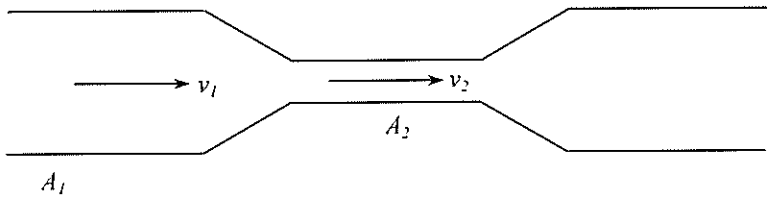


Fig. 3.20: *The Venturi effect illustrated by a constricted pipe. When the area reduces the velocity increases. This means the pressure falls to keep the equation constant.*

With a Venturi, mass must be conserved, so when a fluid of certain density is flowing at a velocity through a pipe, when the pipe's area reduces, only a smaller volume of fluid can flow through, but to keep mass conserved the velocity must increase. So whenever a flowing fluid reaches a constricted section the rate of flow, i.e. the velocity will increase.

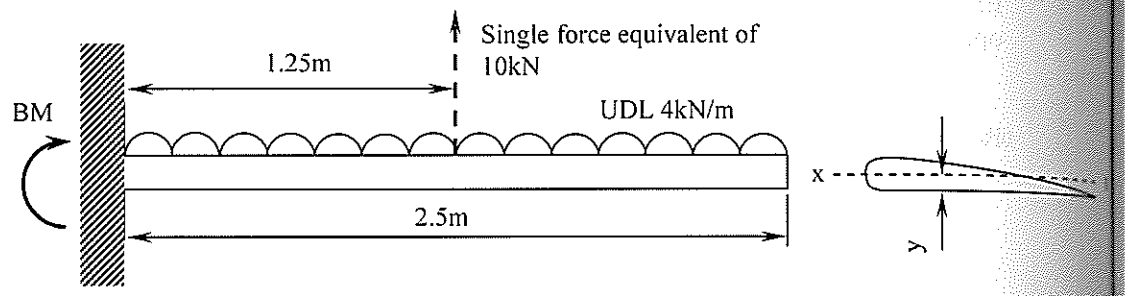
This is used in carburettors on some aircraft with piston engines, as air flows through the restricted section in the carburettor the airflow increases and then draws in fuel. The Venturi effect may also be used to increase the velocity of air as it passes through slats on the front of the wing.

Bending Stress (Airframes)

Bending stress as a concept was first visited in Chapter 1 in relation to beams. This type of bending also occurs when an aircraft is in flight. To see this, consider the wing in Example 3.4.

Example 3.4

A simple wing is loaded as shown. Find the bending moment at the wing root and the bending stress in the lower surface if $I_{xx} = 2\times10^{-6}$ m^4 and y = 20 mm.



First we must calculate the bending moment that occurs at the wing root.

$$\circlearrowleft \Sigma M = 0$$
$$0 = 4\times2.5\times1.25 + BM$$
$$BM = 12.5 \text{ kNm (positive bending moment)}$$

Because the bending moment tends to act in a clockwise direction, we call it a positive bending moment. In effect, the wing is just like a cantilevered beam.

The bending moment at the wing root will be 12.5 kNm.

To find the bending stress we must now consider the values for I_{xx} and y.

$$\sigma_b = \frac{My}{I}$$
$$\sigma_b = \frac{12500\times0.02}{2\times10^{-6}}$$
$$\sigma_b = 125000000 \text{ Pa}$$
$$\sigma_b = 125 \text{ MPa}$$

The bending stress in the wing will be 125 MPa.

Propulsion Systems

Propulsion systems are required on aircraft to provide thrust. Aircraft either rely on a propeller or a turbine to provide thrust.

Piston Engines

Flat (horizontally opposed) piston engines are usually used in small commercial aircraft. These are similar to some car engines (Subaru and Porsche use flat engines), however, some plane engines still use a carburettor whereas virtually all new cars are fuel injected. The piston engine does not offer the same performance as a turbine, but they are powerful enough for small commercial applications and are also cost effective for small planes. Some manufacturers, to extract impressive performance, will place a large piston engine in a small aircraft.

To improve the performance of piston engines they may be turbocharged or supercharged. Many high performance piston engine planes use turbochargers. Turbochargers and superchargers both force air into the engine under greater pressure, boosting power. The turbo is driven off the exhaust gases of the engine, while a supercharger is belt or gear driven off the crankshaft.

Piston engine planes use a dual independent ignition system running in parallel, which provides more efficient fuel combustion and greater safety. Carburettor planes also have a special heater in the carburettor to stop fuel icing up.

Jet Engines

After WWII, the jet engine became increasingly popular, originally in its turbojet form. This was because there was a demand for larger and faster airliners, and in military circles the new technology allowed for much faster fighters and bombers. Essentially the jet engine comes in four types.

Turbojet

The turbojet is the original form of the jet engine. A description of the turbojet, from the inlet to the exhaust is as follows,

1. *Inlet* – air enters and is compressed slightly.
2. *Compressor* – sets of turning blades which compress the air, hence heating it.
3. *Combustor/Burner* – fuel is injected into the combustion chambers where it is mixed with the compressed air and burned.
4. *Turbine* – burnt gases expand through the turbine stages, which are rotated by the gas flow. The turbine is connected on the same shaft as the compressor and provides the energy to drive the compressor. The turbine only extracts enough energy to keep the compressor going.
5. *Nozzle* – the gas is expanded through the nozzle and exits the engine as very hot, very fast gas, providing the thrust.

The operation of the turbojet may be summarised as Suck-Squeeze-Bang-Blow. That is: *suck* in air, compress (*squeeze*) the air, introduce fuel and ignite it (hence *bang*), and finally *blow* the exhaust out the back.

Turboprop

The turboprop has the same operation as the turbojet, but with the important distinction that the turbine is used to power the compressor *and* drive the propeller.

The majority of the energy of the gas is used to drive the turbine (and hence the propeller) leaving only a very small amount of the exhaust energy to provide thrust.

Turbofan

The turbofan has the same operation as the turbojet, but with the additional fan stage mounted inside the nacelle, forward of the compressor (i.e. aft of the inlet). An additional turbine stage drives the fan. Although turbofans were used prior to the Boeing 747, the turbofans fitted to the 747 were the first *high bypass* turbofans used. They are known as a bypass engine, because the majority of the air entering the engine nacelle passes *around*, not through the engine (approximately 5:1 by weight). The fan acts as a ducted propeller producing thrust from the bypass air, while the engine still produces thrust as per the turbojet. A majority of the net thrust is produced by the fan. The advantages of turbofans is that the bypass air shields the hot engine core gases, reducing the noise significantly, moreover most of the air is lower velocity bypass air which also contributes to noise reduction. They are more efficient than turbojets, particularly at the transonic speeds of commercial jet aircraft.

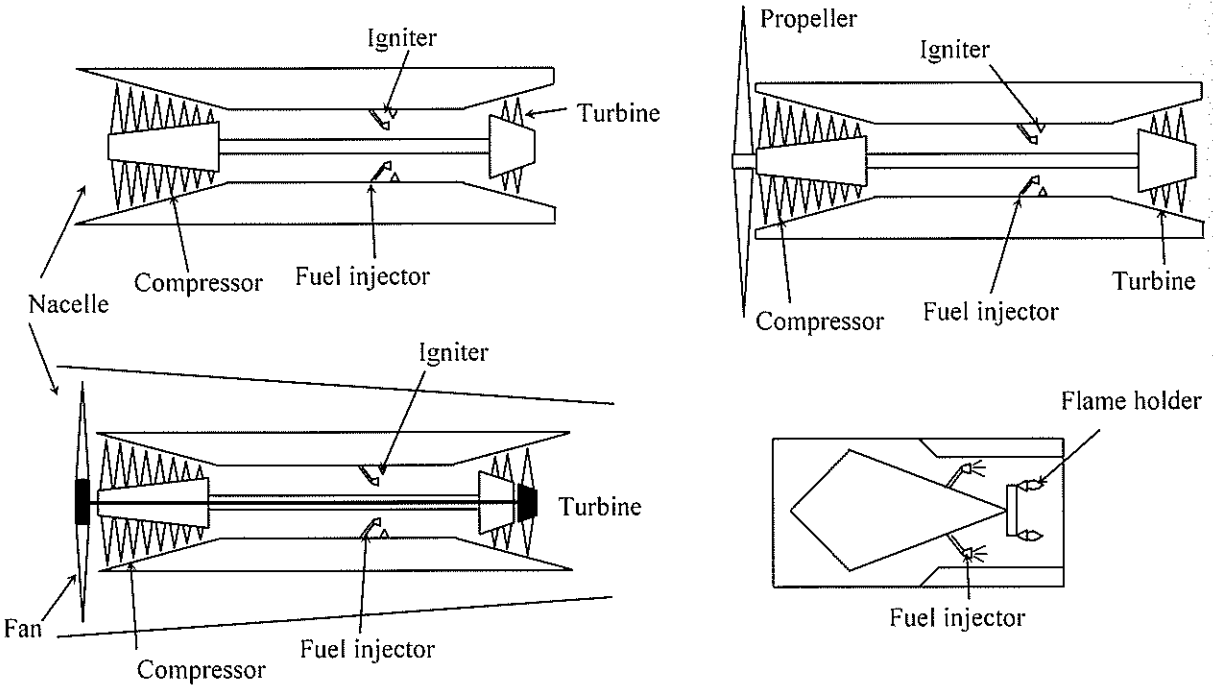


Fig. 3.21: *Four types of gas turbine engine, the turbojet, upper left, turboprop, upper right, the turbofan, lower left, and the ramjet lower right.*

Ramjet

The ramjet is simplicity itself, as it has no moving parts. The shape of the engine compresses the air entering. This compressed, and hence heated, air has fuel injected and then a flame holder ignites the fuel air mix. The expanding gases are forced out the nozzle to produce thrust. The ramjet, however, needs to reach high speeds to work that makes it useless for take-off, landing or low speed work.

Scramjet

A scramjet is a supersonic combustion ramjet. All of the preceding jet engines require the air entering the motor to be slowed to subsonic speeds. A scramjet operates similarly to the ramjet but the airflow through the scramjet will be supersonic. This allows scramjet to operate effectively above speeds of Mach 3.

Fluid Mechanics

Fluid mechanics is the study of fluids in motion and at rest. Fluids are both liquids and gases.

Pascal's Principle

This concept was first covered in Chapter 3 of Volume 1. Pascal's principle is as follows:

> *Pressure applied to an enclosed liquid is transmitted undiminished to every point in the fluid and to the walls of the container.*

A common use for Pascal's principle is in sealed hydraulic systems, such as simple hydraulic systems. In a sealed hydraulic system pressures must be constant. We can express this as forces and areas below:

$$\frac{F_2}{A_2} = \frac{F_1}{A_1}$$

$$F_2 = \frac{F_1 A_2}{A_1}$$

Hydrostatic and Dynamic Pressure

Hydrostatic pressure is pressure that results from a static fluid. Air pressure is an example of static pressure. In aerodynamics, hydrostatic pressure is called *static pressure*. A still gas or fluid will produce a static pressure, either due to the mass of the fluid (liquid) acting on the container or through the random movement of molecules, as in a gas.

If an object is placed in a fluid, the fluid pressure will act upon the entire surface of the object at right angles to the surface (Fig. 3.22). The pressure may be defined as the force per unit area, thus:

$$P = \frac{F}{A}$$

P = pressure (Pa)
F = force (N)
A = area (m^2)

The cross sectional size or shape of a container does not affect hydrostatic pressure, but the depth does. The deeper something is under the surface of a fluid the greater the pressure exerted on the item. For example, the deeper a diver is, the greater the water pressure on him/her.

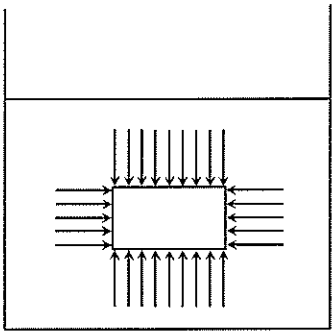


Fig. 3.22: *Hydrostatic pressure acts at right angles to the surface of the immersed object*

We can determine the hydrostatic pressure at a specified depth of a container if we know the pressure at the top (or the surface). The formula is:

$$P = P_0 + \rho g h$$

P = pressure at depth h (Pa)
P_0 = pressure at the top (Pa)
ρ = density of the fluid [constant] (kgm^{-3})
g = acceleration due to gravity (ms^{-2})
h = depth below top (m)

Example 3.5

The gauge float in a fuel tank is jammed at the bottom of the tank. The float is 300 mm below the surface of the diesel, and the density of diesel is 800 kgm^{-3}. What will be the pressure on the float if the pressure at the surface is 101 kPa?

$$P = P_0 + \rho g h$$
$$P = (101 \times 10^3) + (800 \times 10 \times 0.3)$$
$$P = (101 \times 10^3) + 2400$$
$$P = 103400 \text{ Pa}$$
$$P = 103.4 \text{ kPa}$$

The pressure on the jammed float is 103.4 kPa.

Dynamic pressure is pressure that occurs from a moving fluid. This moving fluid tends to create pressure because of ordered velocity. With dynamic pressure the movement is ordered, and we consider the velocity of the fluid and its density to determine the dynamic pressure.

$$P_D = \tfrac{1}{2} \rho v^2$$

P_D = dynamic pressure
ρ = density
v = velocity

Applications to Aircraft Components

Hydrostatic and dynamic pressures have a large effect on aircraft structures. Many aircraft are pressurised to allow them to fly at elevated heights. When this occurs the cabin is pressurised. Therefore there is a static pressure difference between inside and out, which the structure must withstand. When an aircraft climbs to height and descends again, pressure in the cabin is increased and decreased. This static pressure differential stresses the cabin each time, meaning there is a potential for metal fatigue.

All aircraft are subjected to dynamic pressure, as the aircraft encounters the air in flight. If we change our frame of reference we can consider the aircraft to be still and the air moving past it at the same speed as the aircraft (analogous to standing on the ground watching an aircraft move across the sky, or sitting on an aircraft's wing in flight). The airframes of supersonic aircraft (aircraft travelling above Mach 1), in particular are subjected to very large amounts of dynamic pressure because of their high speed.

Jet engines are subjected to large amounts of dynamic pressure through the intake of air and the thrust produced. Many jet aircraft use reverse thruster vanes, which guide the jet thrust forward to aid in braking following landing. Such a device undergoes immense amounts of dynamic pressure, as it is actually changing the momentum and direction of the fluids, which puts large dynamic pressures onto them.

Applications to Aircraft Instruments

Aircraft instruments rely on the principles of aerodynamics to convey flight information to the pilot. One of the simplest and best examples is the pitot (pronounced pee-toe) tube. This small tube is placed under the wing or on the nose to measure the speed of the aircraft. Its operation is based on difference between static and dynamic pressure.

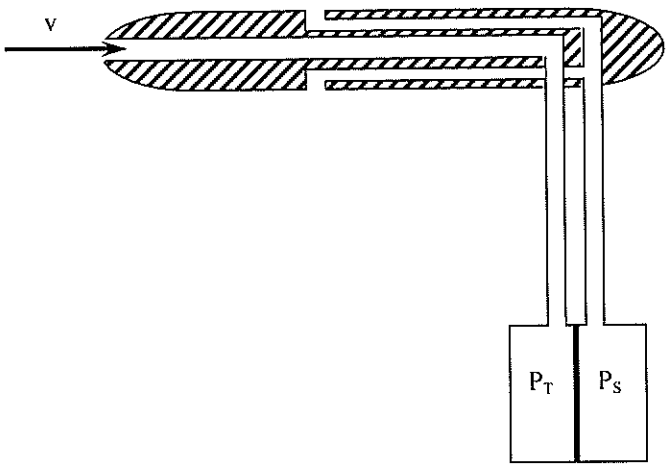


Fig. 3.23: *Schematic diagram of a pitot static tube.*

When the aircraft is flying the air is forced into the pitot tube, relative to the aircraft this air has velocity and hence dynamic pressure. The total pressure is registered by the pitot tube. Around the circumference of the pitot tube are a number of orifices that measure the static pressure. The respective pressures act on two areas that are

separated by a pressure transducer. A strain gauge is used at the transducer to detect movement in the transducer to determine the pressure differential. Velocity is determined using the formula below.

$$P_S + \tfrac{1}{2}\rho v^2 = P_T$$
$$v^2 = \frac{2(P_T - P_S)}{\rho}$$
$$v = \sqrt{\frac{2(P_T - P_S)}{\rho}}$$

By measuring the static and total pressure the pitot tube can read the aircraft's forward airspeed. Pitot tubes have some limitations, however most aircraft have stall speeds between 50-100 kt (93-185 km/h), and at these crucial speeds the pitot tube works just fine. At supersonic speeds this form of Bernoulli's equation does not apply, so pitot tubes for supersonic aircraft are differently designed.

The Airspeed Indicator

Many airspeed indicators are actually mechanical devices. In such a device the total pressure entering the pitot tube acts on the inside of a diaphragm. The outside of the diaphragm is surrounded by static pressure. The diaphragm is connected to a linkage that controls the airspeed indicator. The diaphragm positions itself according to the difference between total pressure (P_T) and static pressure (P_S), i.e. the dynamic pressure ($\frac{1}{2}\rho v^2$). This is shown in Fig. 3.24.

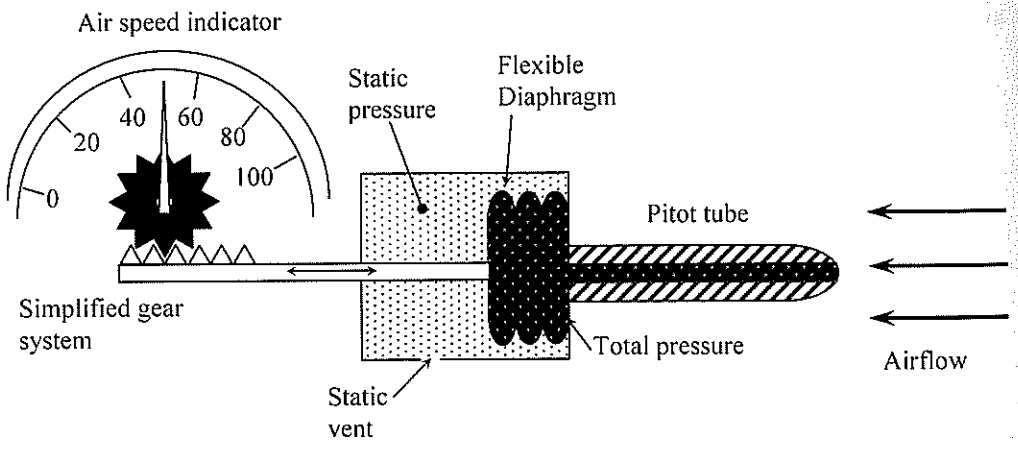


Fig. 3.24: *A mechanical airspeed indicator*

Altimeter

The altimeter is used to measure the altitude of the aircraft. It works on the principle of having a small expandable vessel of air, called an aneroid, surrounded by static air pressure. As the aircraft ascends, the static air pressure falls which allows the aneroid to expand. This expansion acts on a linkage system, which controls the needles on the altimeter.

Engineering Materials

Specialised Testing of Aircraft Materials

The aircraft requires extensive materials testing to ensure that the airframe is in good order. As a consequence a variety of non-destructive tests are used to ensure an airframe is sound. Three of these processes, X-ray, ultrasonic and dye penetrant testing were covered in Chapter 1. Magnetic particle testing is another that is used.

Magnetic Particle Method

Also known as the magnetic dust method, this process is used to find cracks in a part. The piece of metal is placed across two magnetic poles, or a magnetic field is induced in it, and it is then sprinkled with a magnetic powder. The excess powder is removed and cracks are revealed by magnetic powder sticking to the area each side of the crack. This occurs because each side of the crack becomes a magnetic pole.

The major limitation of his method is that it will only work with magnetic materials, which makes it useless with aluminium and titanium alloys, which find extensive use in aircraft.

Gamma Ray

This is similar in principle to X-ray testing but uses gamma rays, which have far greater penetration power. Great care must be taken due to the harm they can have on humans.

Aluminium and its Alloys Used in Aircraft

Aluminium and its alloys are used in aircraft because of their desirable low density. The aluminium family of alloys was covered in Chapter 2, along with precipitation hardening of duralumin, an important alloy for aeronautical engineering. One important point as an addendum to the information in Chapter 2. It is possible to *refrigerate* duralumin after Stage 1 of the precipitation hardening process (p 130), so it will not age. This is done so it can be naturally aged in service, hence receiving max strength whilst in service.

Aluminium Alloys for aircraft

Aluminium Copper (2xxx Series)

This range of aluminium copper alloys are the most common family of alloys used in aircraft today. Formerly duralumin (2017) was widely used but the 2024 alloy, which offers higher tensile strengths, has largely replaced this. The 2000 series of alloys have better resistance to crack formation and thus find use in the lower surface of wings and the fuselage where they are most likely to suffer from tensile loads and fatigue cracking. They are usually used in clad form due to them having poorer corrosion resistance than pure aluminium.

Aluminium Silicon

This type of aluminium alloy is the basis for most cast aluminium alloys, the silicon improves the castability of the alloy and they exhibit excellent corrosion resistance.

Aluminium Silicon Magnesium (6000 series)

Used in both the wrought form and as a casting alloy, these alloys are readily heat-treated by precipitation hardening. Although not generally as strong as the 2000 series alloys, they are more easily worked and offer better corrosion resistance.

Aluminium Zinc Magnesium alloys (7000 series)

These alloys find extensive use in aeronautical applications at various parts of the airframe. These aluminium zinc magnesium alloys at times have additions of copper. They are more readily heat treatable than the 2000 series but stress corrosion cracking is a challenge, in part addressed by the addition of copper.

Aluminium Lithium Alloys

These high strength alloys offer lower densities than standard aluminium alloys, but also present some challenging properties. They have found extensive use in military aircraft and space vehicles. The final version of the external fuel tank of the Space Shuttle was made using aluminium lithium alloys, which yielded a weight saving of around 3.4 tonnes.

Alclad

Worthy of mention here is Alclad. This is duralumin (or other 2000 series alloy) that is clad with pure aluminium. This is done because duralumin is not as corrosion resistant as pure aluminium. The duralumin substrate and pure aluminium sheet are rolled together; pressure welding joins the layers together. The aluminium sheet generally accounts for 1% of the total thickness of the two.

Thermosetting Polymers

Polymer technology is increasingly important in aircraft as a solution to keeping weights down. Thermosetting polymers have covalent bonding across the polymer chains as well as along them. They cannot be re-softened by the application of heat. Thermosetting polymers such as epoxy resins find extensive use as the matrix material for fibre reinforced composite construction used in aircraft.

Compression moulding

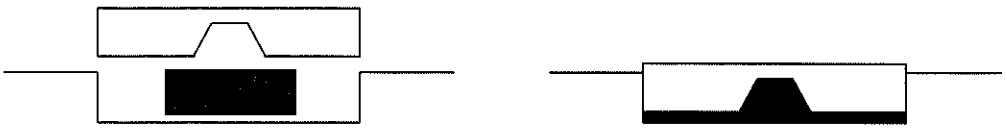


Fig 3.25: *The compression moulding process. Melting polymerising and moulding happens in the same place.*

This process takes an unpolymerised preform and compresses it in a mould with heat. The heat and the pressure form the shape and polymerise the polymer. The finished moulding is then ejected. It is used for making plugs, switches and casters.

Transfer moulding

This is similar to compression moulding but instead of the polymerising happening in the mould, it happens in the adjacent cavity. The molten polymer is transferred via a sprue to the actual mould. It is also used for moulding thermosets.

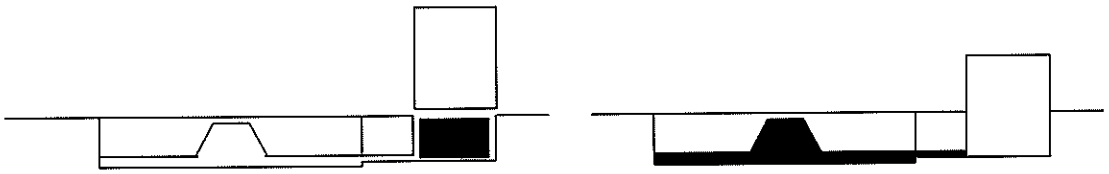


Fig 3.26: *The transfer moulding process. Melting and polymerising is remote from where moulding occurs.*

Hand Lay-up

This is where fibres are placed into a mould and the resin is manually applied onto the fibres. First a gel coat is applied, and then the fibre reinforcement is added in the form of mat or cloth. The resin is poured, brushed or sprayed on, and then rollers are used to force the resin to impregnate the fibre mat. In areas where extra thickness is needed more fibre mat is placed down and the resin rolled into that area again. It is essential that the resin bonds properly to the fibres.

An alternative to hand lay-up is the spray-up process where a special gun sprays the resin and continuous strand roving (glass fibre) which is chopped and sprayed at the same time. This chopped roving and resin mix in the mould. It is then rolled to remove air and improve impregnation of the resin into the fibres.

Vacuum Lay-up

The vacuum bag autoclave process is usually used with epoxy resin based composites. The material used is called *prepreg* and is uni directional carbon fibres within a partially cured epoxy resin. The material is arranged in layers in the mould and then placed in a bag. A vacuum is applied to the bag and this removes any air within the laminates. The bag is then placed in an autoclave at an appropriate temperature and elevated pressure where curing is then completed. For epoxy resins the autoclave will be set to a temperature of around 190°C and at a pressure of around 690 kPa (6.9 bar).

This vacuum bag process finds use in forming composite aircraft panels such as wings and other large surfaces.

Composites

Composites are one of the newest and most important areas for aircraft manufacture. Composite materials offer good specific strength, high resistance to cyclic stresses and weather resistance. Composites may be degraded by UV light, but making them

heavily pigmented reduces this. Composites may also be manufactured to be stronger in one direction than another. A composite may be manufactured to be *isotropic* (equal strength in all directions), or *anisotropic* (strength in one direction).

Types and Applications in Aircraft

Glass Fibre Reinforced Polymers

Glass Fibre Reinforced Polymers (GFRP), also known by the trade name Fibreglass, are fine glass fibres embedded in a resin matrix. Polyester, vinyl ester and epoxy resins are all possible resin materials. The glass fibre provides tensile strength while the resin provides toughness. Although not quite as strong as other fibres and also heavier, they are a cost effective option that has also found aeronautical use in Fibre Metal Laminates (FML).

Carbon Fibre Reinforced Polymer (CFRP)

This is one of the most important composites in the aircraft industry. Carbon fibre reinforced composites are a composite of carbon fibres embedded in a resin matrix, which in aeronautical applications is normally epoxy resin. The combination of the two components produces the desirable strength and toughness. Typically, carbon fibre reinforced polymers consist of 62% by volume carbon fibres. The overall properties are lightweight, very high specific strength and high modulus of elasticity. Particularly important is the resistance to cyclic stress that carbon fibre exhibits. It is these properties that make it so desirable for aircraft. Not only are fighters such as the Saab Gripen and Eurofighter Typhoon using CFRP, but CFRP is also used in the Boeing 737 and Airbus control surfaces and wingtips. The Boeing 787 Dreamliner has its fuselage predominately made from CFRP.

One disadvantage of CFRP is that fails like a brittle material with no plastic failure first.This contrasts with a metal like steel, which progressively fails and usually shows warnings such as cracks or bending. Damage detection and repair are also much more complex.

Aramid Fibre

Aramid fibre reinforced composites are very similar to carbon fibre composites. The difference is that Aramid fibres are used in lieu of the carbon fibres in an epoxy resin matrix. Aramid, better known under the trade name Kevlar, offers similar properties to carbon fibre, but is more resistant to impacts and abrasion. This is important in some battlefield conditions where shrapnel or debris could shorten the life of a carbon fibre component by producing surface defects.

Fibre Metal Laminates (FML)

These are laminated structures of metal and fibre reinforced composites, the most common one being GLARE (Glass Laminate Aluminium Reinforced Epoxy) which is used in the Airbus A380. Because they are laminated with metals they perform more like metal sheeting than fibre reinforced composite sheets, but they offer enhanced properties to a non-laminated metal sheet. Advantages include, improved impact resistance, better fatigue and corrosion resistance, and weight advantages. They can

also have their strength properties directional for specific locations in the fuselage unlike a standard metal sheet.

Metal Matrix Composites (MMC)

A lot of research has been done into metal matrix composites, with an intention to use them in advanced high-speed jets. MMCs are similar to the two above except that the matrix is not a polymer but a metal. An example would be boron fibre aluminium, a composite using an aluminium matrix with boron fibres to reinforce it. This improves the tensile strength of the aluminium alloy by a huge factor. For example aluminium alloy 6061 has a tensile strength of 310 MPa. With a boron/6061 composite with 51% boron, the tensile strength is increased to 1417 MPa.

MMC's exist in three forms, continuous fibre reinforced MMCs, discontinuous fibre reinforced MMCs and particulate reinforced MMCs. The continuous fibre type gives far greater tensile strength but is also more difficult to make. The discontinuous fibre type is manufactured using powder metallurgy methods (see Chapter 2).

MMCs are capable of withstanding higher temperatures, as the metal matrix can withstand elevated temperatures better than the epoxy matrix. They are, however, heavier than polymer matrix composites.

Adhesives

Epoxy adhesives are used to join composite aircraft surfaces to the base structure. Aircraft such as the Eurofighter Typhoon utilise epoxy adhesives in this way.

Structure/Property Relationships

The structure of fibre reinforced composites has a big impact on their behaviour. This is because the way that the fibres are arranged will affect how the loads are carried. Unidirectional fibres are fibres that all travel parallel to one another. These include materials like continuous MMCs. These materials are very effective at carrying a load parallel to the fibres. They are, however, more likely to suffer separation of the fibres if bending occurs. Such materials are termed anisotropic, i.e. their properties differ in different directions.

It is possible to arrange shorter fibres in random direction; this will lower the maximum tensile strength but increases the resistance to separation if bending occurs.

Another option is to arrange the fibres at angles of 0°, 45° and 90° to one another in alternate layers. This is similar to plywood in principle and will produce a material that maximises strength in desired directions. This does not, however, make it isotropic, as isotropic materials have the same strength in *all* directions. It is also possible to produce a three-dimensional structure of fibres that will give the required strength along three axes.

Corrosion

Corrosion is a great problem in aircraft design. The frames and skin of aircraft are already highly stressed, so weakening via corrosion is a major concern. Although

composites offer resistance to electrochemical corrosion, UV light and the weather may degrade them.

A large majority of aircraft still use metal airframes and skins, so corrosion is still a problem. Corrosion mechanisms are outlined in Chapter 1 and the student should refer to this section.

Pit and Crevice Corrosion

This is a concentration cell that occurs because of different oxygen levels at the top and bottom of a crevice. The example in Chapter 1 of a car door is similar to crevice corrosion, but crevice corrosion usually involves a very fine gap that traps a fluid at the base. This type of corrosion is a concern, as metal aircraft are often riveted with fine gaps where the skins join. Moisture forms in these gaps each time the aircraft crosses the "dew point". The dew point is a particular combination of pressure, temperature and moisture content in the air, where water precipitates onto a surface (e.g. misting of a car windscreen).

Aircraft always have moisture on them because of:

- condensation on the airframe from temperature changes in the atmosphere [20°C at sea level, -53°C at 10671 m (35000 ft)]
- airborne moisture accumulating
- rain

Stress Corrosion Cracking (SCC)

This is a particularly dangerous form of corrosion, where the action of tensile stress and corrosive environment conspire to bring about crack formation. The stresses that cause SCC can be residual stresses or applied. On their own neither would cause failure but with the two-combined failure occurs. Different alloys are affected by different environments; stainless steels are susceptible to SCC in chloride environments but safe in ammonia-based environments, while brasses are the reverse. Some aluminium alloys may suffer SCC in seawater or in the presence of air and water vapour.

In SCC the combined action involves the tensile stress causing pits or cracks to form and then the corrosion occurring with the tip of the crack becoming anodic. It has been shown that applying cathodic protection will stop SCC occurring. But once the protection is removed the SCC continues.

SCC can be reduced or prevented though lowering stress levels, removing the corrosive environment if possible, changing alloys to one not affected by the environment present or applying cathodic protection.

Communication

Developments

In the engineering world many items are made out of sheet metal fabricated into a variety of shapes. Often things such as exhaust hoods, air-conditioning ducts and dust

extractors require sheet metal to be planned out and shaped according to a specific design before construction.

Simple Developments

Simple developments are something you have probably seen in mathematics. A development is when a shape is laid out flat with any fold lines shown. It is then possible to cut out the shape from sheet metal and fabricate the required shape.

Developments can be simple, such as the development of the cube below (Fig. 3.27). In this case, the cube shape is rolled out so that a flat two-dimensional view of the cube material is shown.

Development by Triangulation

This method of producing a development is used in the development of transition pieces. These are shapes that involve a change from one shape to another, such as square to circle or square to rectangle. In Engineering Studies it is not necessary to do circular transition pieces.

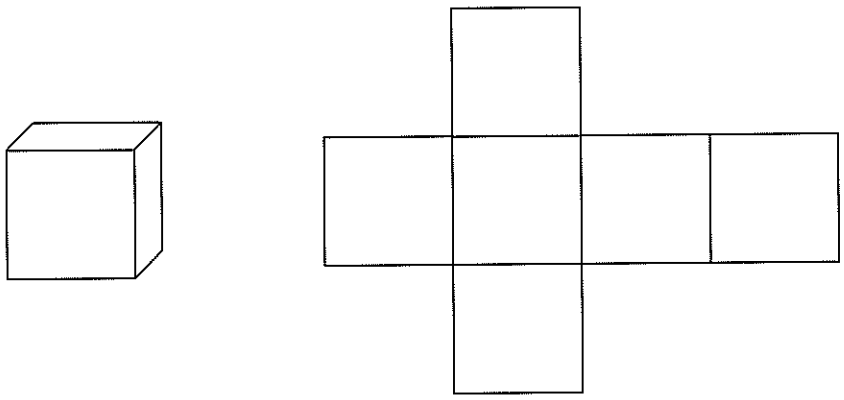


Fig. 3.27: *The cube on the left can be folded flat to produce a development.*

When doing developments by triangulation, the shapes or surfaces must be broken up into triangles, hence the name of the procedure. The following two examples show how this has been done.

One challenge in producing transition pieces is that they often have lines that are not true lengths when drawn in orthogonal projection. In a front and top view of a doubly inclined line the true length of the line cannot be seen, because in neither view is it perpendicular to the viewer. To overcome this we must find the true lengths of the lines by constructing a true length diagram as is shown in Examples 3.6-3.8. (See below.)

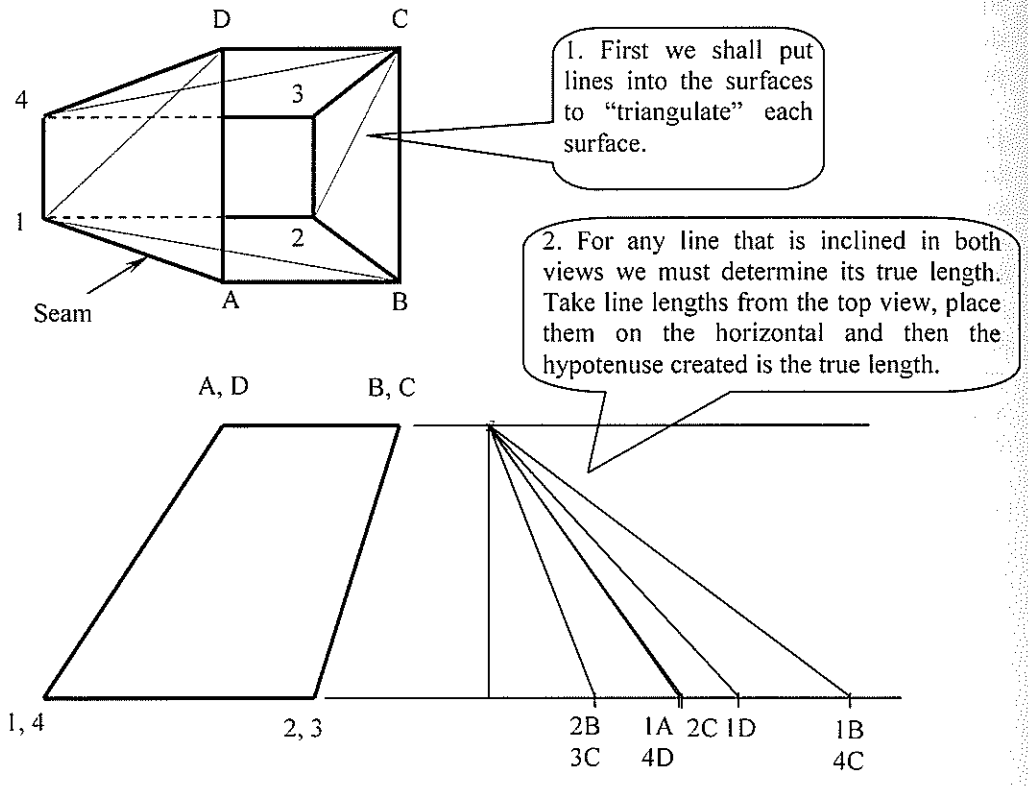
Two rules that may help you complete a transition piece drawing where true lengths are not obvious are:

1. If, in one view, the line is seen as a point it is a true length in the other view.
2. If, in one view, the line is perpendicular to the viewer, in the other view it is a true length.

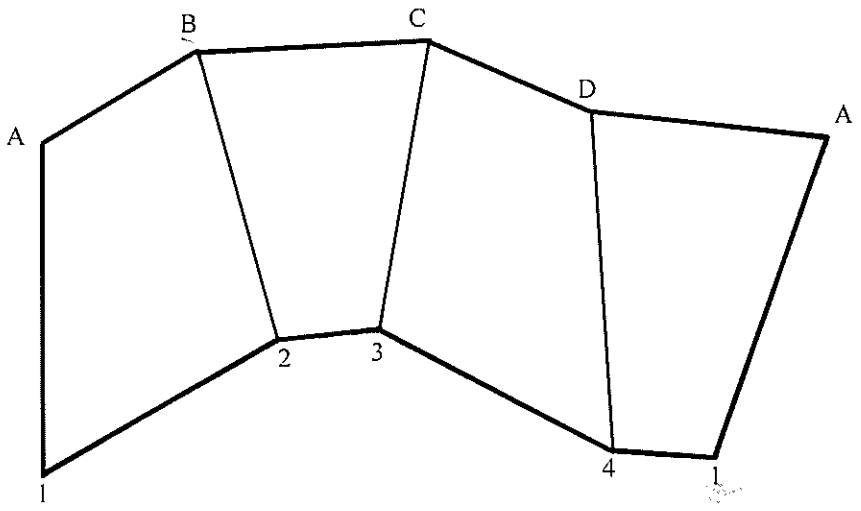
Examples of transition piece development are on pages 188-190.

Example 3.6

Construct the full development for the transition piece shown.

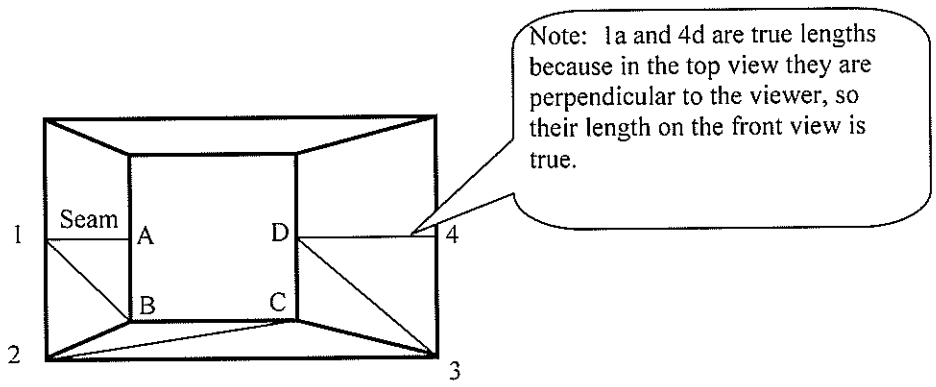


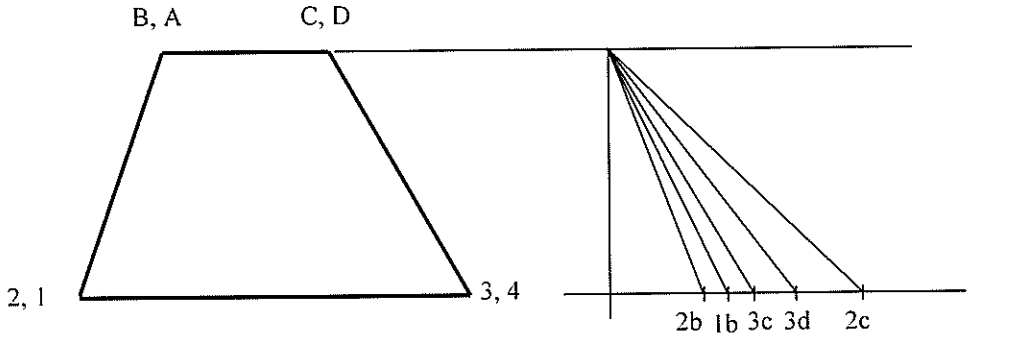
Now we can use the true lengths to help us create the development by triangulating ou way from point to point. Start at the seam A 1.

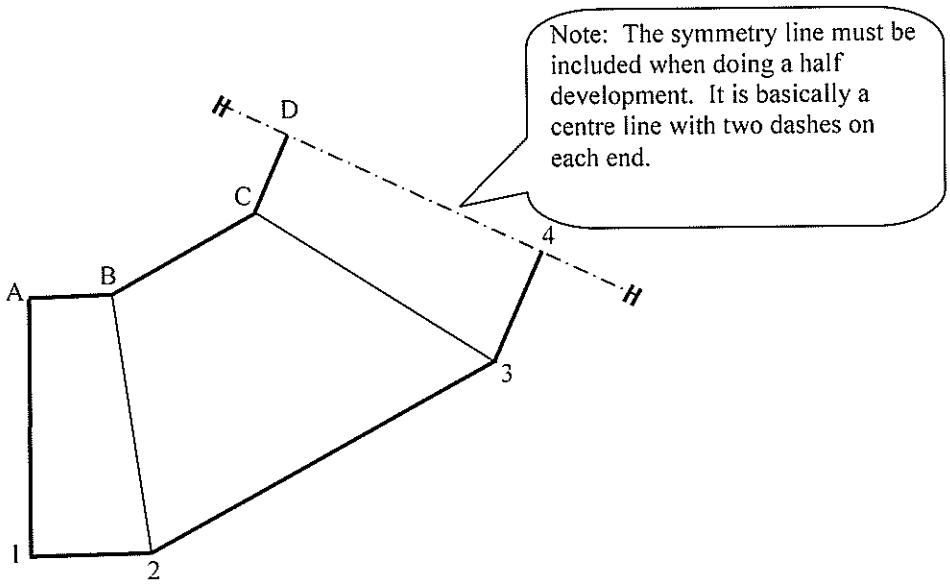


Example 3.7

Construct the half development for the transition piece shown.



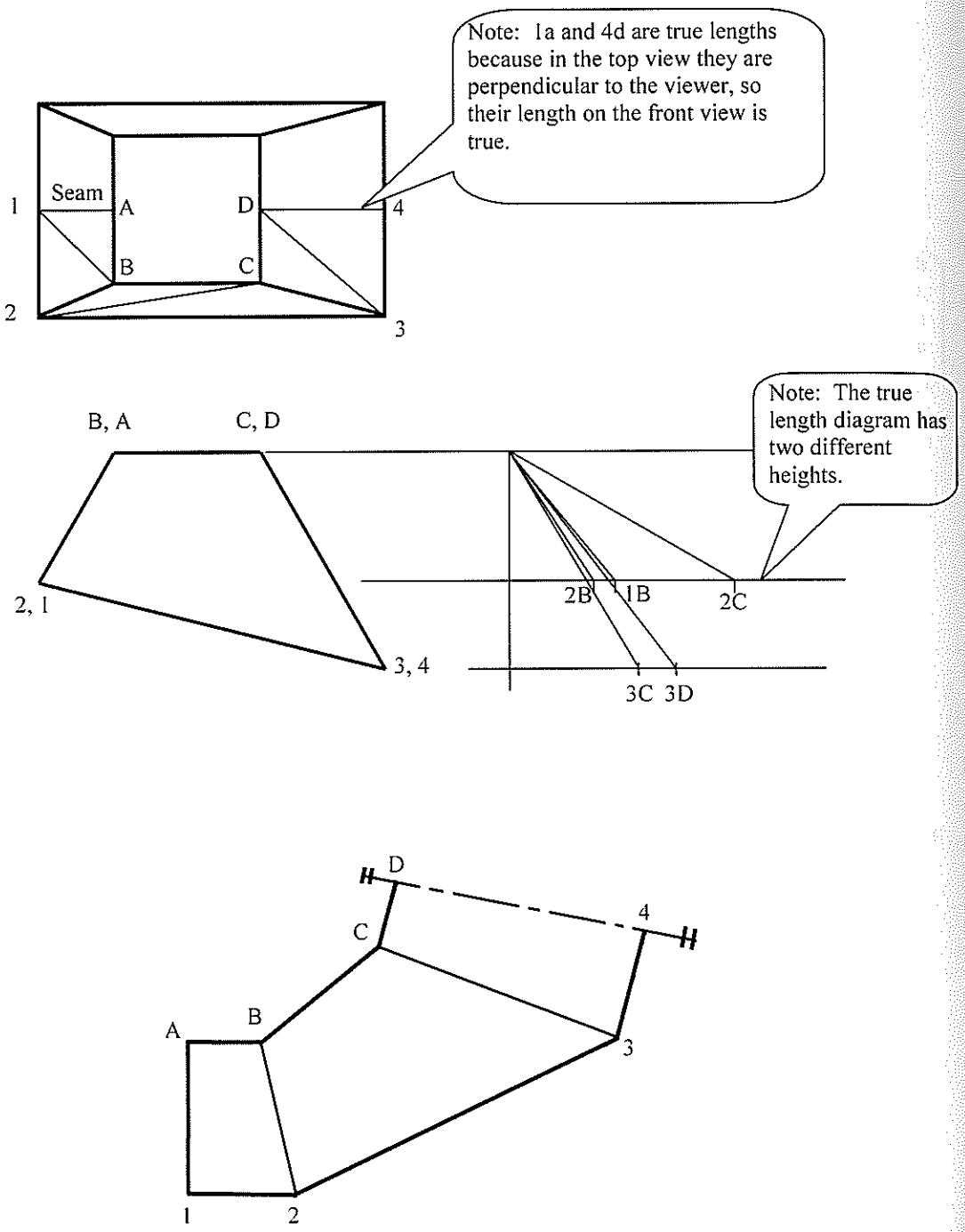




It should be noted that if the front view has different heights from one side to the other then the true length diagram will have different heights depending on which points the true lengths are between.

Example 3.8

Construct the half development for the transition piece shown.



Engineering Report

PLC Aeronautics has just employed you as an aeronautical engineer. They are planning a new single engine propeller plane called the Vortex. This is to be a small four seat high power, high performance plane, and to fit with this image, composites

are being considered. The company is undecided whether to use a carbon fibre reinforced composite or a 2024 aluminium alloy. Determine the properties of each and their suitability. Also compare their costs as a secondary issue.

If each wing is to be 4.8m long and the lift force acts at 45% of the distance from the wing root, what will be the bending moment if the lift force is 9 kN? What will be the bending stress if $I_{xx} = 1.8 \times 10^{-6}$ m^4 and y = 20 mm?

Additional References

This list of useful references allows students further investigation. Any highly recommended book or website has two stars next to it.

Historical and Societal Influences

Frawley, G., (2001), *The International Directory of Civil Aircraft 2001/2002*, Aerospace Publications Pty Ltd, Canberra.
Job, M., (1996), *Air Disaster Vol 1*, Aerospace Publications Pty Ltd, Canberra.
Wilson, S., (1998), *Boeing 707, Douglas DC-8 & Vickers VC10*, Aerospace Publications Pty Ltd, Canberra.

Engineering Mechanics and Hydraulics

**Gordon, J. E., (1978), *Structures – or Why things don't fall down*, Da Capo Press, New York.
Ivanoff, V., (1996), *Engineering Mechanics*, McGraw Hill, Sydney.
McCormick, B. W., (1995), *Aerodynamics, Aeronautics and Flight Mechanics*, John Wiley and Sons Inc, New York.
**Schlenker, B. and McKern, D., (1976), *Introduction to Engineering Mechanics*, Jacaranda Wiley, Milton.
Thorn, T., (1990), *Basic Aeronautical Knowledge*, Aviation Theory Centre, Williamstown.

Engineering Materials

**Askeland, D., (1989), *The Science and Engineering of Materials*, Chapman and Hall, London.
**Avner, S., (1983), *Introduction to Physical Metallurgy 2nd Edition*, McGraw Hill, Tokyo.
DeGarmo, E. P., Black, J. T. and Kohser, R. A., (1988), *Materials and Processes in Manufacturing 7th Edition*, Macmillan Publishing Company, New York.
**Gordon, J. E., (1974), *The New Science of Strong Materials – or Why you don't fall through the floor*, Penguin, Middlesex.
Higgins, R., (1997), *Materials for the Engineering Technician 3rd Edition*, Arnold, London.
Schlenker, B., (1974), *Introduction to Materials Science*, Jacaranda Wiley, Milton.
**Smith, W., (1990), *Principles of Materials Science and Engineering*, McGraw Hill, New York.

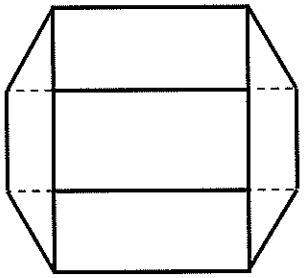
Communication

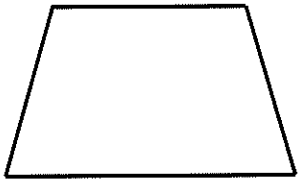
**Boundy, A., (1991), Engineering *Drawing*, McGraw Hill, Sydney.
Boundy, A. & Hass, I. L., (1996), Graphics *2000 Book 2*, McGraw Hill, Sydney.
Park, A, Dodds, K and Bland, S., (1989), *Technology Drawing*, Longman Cheshire, Melbourne.

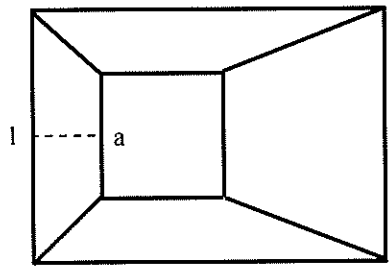
Review Questions

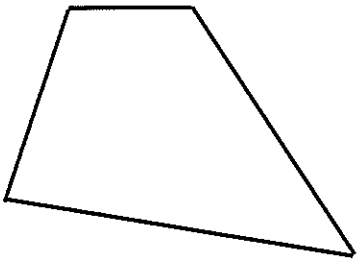
1. What is a patent?
2. What was the name of the first manned aircraft to fly?
3. What are ailerons? Who developed them? How were planes turned before ailerons were developed?
4. What was the first turbojet aircraft to fly?
5. Name the British fighter jet to enter service during WWII.
6. The F-111 was the first production jet to use what technology? Why was this technology used?
7. What do STOL and VTOL stand for?
8. What aerodynamic feature did the Saab Viggen use to gain STOL capabilities?
9. What makes the Harrier such an important jet?
10. What material family is made extensive use of in the Eurofighter Typhoon.
11. Why was the de Havilland Comet such an important passenger aircraft?
12. What aircraft, made by Boeing, set them up for a prosperous future with passenger jets?
13. The de Havilland Comet, Tupovlev Tu104 and Boeing 707 are all early commercial jets. Compare their design similarities and differences.?
14. What makes the Boeing 747 such an important plane in the development of passenger jets?
15. The Antonov An124 is a large cargo plane, how does its wing placement differ from the large Airbus A380?
16. The Concorde was first and only commercial supersonic airliner to enter regular service. Outline some of the key design differences between the Concorde and a regular commercial jet airliner.
17. Explain the term, fuselage.
18. What is the vertical stabiliser?
19. Explain in terms of moments, why the force from the tail plane must usually have a downward direction.
20. Describe what the movements of pitch, roll and yaw mean. Outline which control surfaces are used to control these movements.
21. Explain with reference to the flow turning theory, how an aerofoil creates lift.
22. A 650 kg glider is descending at an angle of 15° at constant velocity. Determine the mass of the pilot if the glider has a lift to drag ratio of 5, and the drag force is 1900 N.
23. What will be the lift to drag ratio of a 3 tonne aircraft that has a climb angle of 9° and experiences a drag force of 5 kN?
24. What is meant by the term, stalling?
25. Explain what is the Venturi effect is?
26. Find the bending moment at the wing root and the bending stress in the lower surface if the wing is 5 m long, has lift UDL of 2 kN/m and $I_{xx} = 2.2\times10^{-6}$ m^4 and y = 20 mm.
27. How does a turbofan differ from a turbojet?

28. Explain the operation of an airspeed indicator and an altimeter.
29. What properties of carbon fibre make it desirable in aircraft construction?
30. Explain how composite materials such as CFRP can be made to have preferred directional properties.
31. What is a metal matrix composite?
32. Why do aircraft always have moisture on them?
33. Explain pit or crevice corrosion.
34. Explain what stress corrosion cracking is.
35. Complete half development of the transitions pieces below.









CHAPTER 4

Engineering Focus Module 2 – Telecommunications

Scope of the Profession

Nature and Scope of Telecommunications Engineering

Telecommunication engineers are responsible for the design and development of communication equipment and infrastructure. They are also employed in modifying and improving existing designs. The telecommunication engineer is responsible for developing equipment that makes use of the various communication technologies to improve communication for society.

Telecommunications include telephones (fixed or mobile), satellite systems, television and radio networks.

Health and Safety Matters

Telecommunication engineers work on the management and development of telecommunications equipment. They oversee the manufacture and installation of broadcast, telephony, satellite and cable systems. Much of their work is carried out in office locations where occupation health and safety (OH&S) concerns relating to the ergonomics, lighting and good housekeeping are a priority. At other times their work takes them into manufacturing industry where the hazards associated with machinery and equipment may be relevant. In this environment they have to be aware of numerous industrial hazards such as electrocution.

Overseeing field installation of equipment like optical cable networks or satellite dishes may expose telecommunications engineers to danger from falling equipment, lightning strikes, electrical hazards or poor safety standards on the job site.

Training for the Profession

To embark on a tertiary course in telecommunications engineering, an HSC student should be looking at subjects such as 2 Unit Mathematics, Physics, Chemistry and Engineering Studies. There are telecommunication engineering courses at some universities e.g. The University of Sydney, The University of Technology, Sydney and The University of Newcastle. The courses are Bachelor degrees. Telecommunication engineers learn about electronics, computers and also must be familiar with networks and associated equipment. In fact, in some courses the first two years of a telecommunications engineering degree will be common with electrical and computer engineering.

Career Prospects

Australia often looks to newer industries such as telecommunications to become a core industry. With large telecommunications companies such as Telstra, Optus and others, actively looking to expand, the future prospects for telecommunication engineers should be good. It must be remembered that with the technology and communication revolution, Australia will need trained engineers in this field to ensure it does not fall behind the rest of the developed world.

The expansion of digital technology into all facets of telecommunication requires the design and redesign of equipment, hence providing new opportunities for qualified telecommunication engineers.

Relations with the Community

The infrastructure developed by telecommunication engineers is designed to provide services to the consumer that probably far exceed the needs of most people. That this technology is accepted by the community in Australia is quite clear. For example, Australia has one of the World's highest levels of mobile telephone ownership. The work of the telecommunication engineer, and the technology they develop, is not without serious community based controversy. Radiation from satellite transmitters, and mobile phones has led to community fears of tumour and cancer, particularly in children exposed to the radiation.

Technologies Unique to the Profession

Many of the technologies required for successful telecommunications are not unique to the profession. For example, the industry is heavily dependant on semiconductor technology, but the use of these semiconductors allows telecommunication engineers to create unique technologies, like digital television, FM radio and mobile telephones. All of these are pivotal to modern communication and are really the province of the telecommunication engineer.

Satellite technology is almost exclusively in the realm of telecommunications. There are a huge number of communication satellites utilised by modern telecommunication systems. The use of satellites has allowed the telecommunication revolution to occur. Satellites are essential in the broadcast of world news and communication from one country to another. They are also being used with special mobile phones to ensure worldwide mobile coverage.

Whilst fibre optic cable is used in applications to transmit light, such as for medical examination and to illuminate pipes for TV cameras, the great bulk of optical fibre is used to carry communications signals. It is therefore, almost unique to the profession.

Legal and Ethical Implications

For ethics and patent law, refer to Chapter 3.

Engineers as Managers

Telecommunications engineers may be managers in two senses of the word. First they may be the managers of the design process, overseeing the design and development of a project. An engineer in such a position will have other engineers working on the project, while he/she co-ordinates the teams and oversees all development.

Secondly, on large projects such the establishment of a telecommunications network, or the development of a satellite system, a team structure would develop with various telecommunications engineers having various management responsibilities for completion of their own segment.

They should have a logical method of dealing with problems, good problem-solving skills and good communication skills. These attributes often equip the engineer well for a move into the management team of a company.

Current Projects and Innovations

The telecommunications field is so diverse that many projects are continually occurring. Perhaps one of the most important projects currently underway is the mobile telephone network. It is continually expanding to cover more and more of Australia, with the aim that all areas will eventually have access to a mobile telephone service.

Digital television has now replaced analogue TV in Australia, and in capital cities digital radio is offered in addition to analogue AM and FM signals.

Mobile telephony has grown to allow large amounts of data transfer where once phone and SMS were the only means of mobile communication. Now modern smartphones utilise 3G or LTE networks to browse the Internet, access social media and they also use GPS for personal guidance and fitness applications.

Historical and Societal Influences

Explanatory note: There are a number of forms of telecommunication that could be discussed in this course. Among them are the telegraph, television, the video, satellite communication and the facsimile machine. In this volume we have chosen to look at two of the earliest forms of modern communication, the *telephone* and the *radio*, both of which have maintained their relevance to society for over a century.

There are similarities between these two forms of communication technology. Today, in fact, the two technologies have merged in the creation of the mobile telephone.

Historical Developments within the Telecommunications Industry

The Telephone and its History

The need for people to communicate with one another quickly and over ever increasing distances was an outcome of the industrial revolution. As the industrial

world opened up, people in remote areas demanded better travel and better communications. The spread of the railways brought with it a need to convey timetable information between centres. The electric telegraph was a solution for the time. The telegraph worked by sending electrical impulses of different lengths from the sender, along overhead cables to the receiver. The "dots" and "dashes" were coded to represent letters of the alphabet; the code (Morse code), was developed by Samuel Morse in 1838. Hence messages could be sent between people.

By 1851, a telegraph cable was run under the English Channel, marking the beginning of the international telegraph system. From the Crimean War, which took place in Eastern Europe, in the 1850's, messages could reach Britain the same day. Previously they would have taken weeks.

We should be aware that the electronic telegraph had been in use for nearly forty years before Alexander Graham Bell's landmark experiment. The telephone was invented almost by accident. The Scottish born, Alexander Graham Bell, was in America experimenting on the invention of a machine that would enable one to converse with the deaf. The principle on which Bell worked was that air from the lungs makes the vocal chords vibrate, causing speech. These waves in the air, 'sound waves', when striking a diaphragm of thin metal, will in turn, set it vibrating.

Bell expanded on the telegraph by considering that he could send a message down an electrified line. In his case, however, the vibrations he sent were caused by the voice. By speaking, a disc was made to vibrate. This vibrating disc moved a coil of wire in a magnetic field, thus inducing current in the coil. These currents were sent to the receiver where they were to be received by the deaf mute through a diaphragm held against the ear. He/she would then be taught to understand speech by feeling the vibrations rather than hearing them.

The original experiment was set up, with a mouth piece in one room, joined by a wire to a receiver in the next. Bell's young assistant, Watson, was startled, when he picked up the receiver, to clearly hear Bell's voice. The telephone had been invented. The year was 1876.

Early Improvements

The telephone we use today has not changed much in principle from the one invented by A.G. Bell. It was simply the first invention to turn sounds into electrical signals and back into sounds again. Bell's first telephone, however, produced such weak currents that it was unsuitable for use over long distances because the electrical resistance of the wires caused such severe reductions in the level of the original speech. Important improvements had to be made to make it commercially viable.

It was Thomas Edison, in 1877, who first suggested the addition of the granules behind the vibrating disc in the telephone. Edison used graphite. In 1878, Henry Hunnings, an English clergyman, found that granular carbon was superior. The microphone had now been invented. Thomas B. Doolittle, in 1877, found that copper wire was superior to iron wire for telecommunication because it offered less resistance to the transmitted speech. The transmitter and receiver functions were then separated, so phones now had a separate transmitter (microphone) and receiver (ear piece). This reduced noise and interference and further reduced resistance in the system. With

these improvements by 1885, a successful line was opened between New York and Boston.

The first telephones were connected to an exchange, where the call was connected by an operator to the correct receiver. This was satisfactory while there were not too many telephones. As the telephone became more popular, the 'person operated' exchange became inadequate. (It should be noted, that until recently, long-distance calls still went through an operator.)

In 1889, a man called Almon Stowger invented a switch that could transfer one's call automatically. The "Stowger Switch" gave rise to the automatic telephone exchange, which has become the norm in telephone communication. Stowger switches, however, were quite slow in making connections and they were replaced by transistor technology, which was quicker and more efficient. Computer based automatic exchanges have now largely replaced the earlier electronic switched exchanges. These enable calls to be switched through a number of different exchanges if one exchange happens to be engaged. Modern technology is continuing to make faster telephone connections to all parts of the world.

Later Improvements

The spread of the telephone produced huge numbers of overhead wires until they could no longer be accommodated. The practice began of binding groups of wires into single cables which were carried on poles or in underground tunnels. Originally, about 50 pairs of wires could be bundled together. In modern times, this has increased to 2,000 pairs. Even so, in the United States today, there are 56 million kilometres (35 million miles) of cables covering 416,000 km (260,000 miles) of telephone route.

The Public Telephone

The Post Master General's Department in London made possible, in the 1880's, the first public pay phones in the world. By 1884, in Britain there were only about 13,000 private phones in operation. These were confined to larger businesses and wealthy households. The payphone made the technology available to the general public. Early phones were in shops, post offices and railway stations.

There were various designs for call boxes. Some didn't have a coin box; the user paid to enter the box, while others had an attendant to collect the money. Some were decorative, with thatched roofs or rustic woodwork.

It was not until 1921, in Britain, that a standard, easily recognisable design was used: the concrete kiosk with the red door. It was rather ugly and various later 'telephone boxes' were architect designed. The bright red, easily recognisable box, with glass panels became a feature of British streets. It is appropriate to mention the British call box because it became the pattern for public telephones across the world. Indeed, they were very much a part of the Australian city-scape for decades. They have now generally been replaced by the more open style call box.

The public telephone still plays a minor role in modern communications. This is due to the facts:

- that most homes, in developed countries, now contain a private telephone,

- that there has been a proliferation in the ownership of mobile phones.

The Changing Shape of the Telephone

The telephone has undergone some wonderful changes in style and shape since its inception. The original telephone needed only a handle to turn or some other signal to connect the user to the exchange, where the caller could request his/her number.

Later telephones had dials with finger holes. The telephone 'number' was a series of letters, A, B, C, etc, followed by numerals. The user rotated the dial, using the finger holes corresponding to the letters/numerals of the required number. This slow method of phoning was replaced, in the 1980's, by the touch telephone, with buttons rather than dials.

Fig. 4.1: *A modern cordless telephone*

We now also have redial functions and such advances as being able to program our often used numbers into the telephone. There is also the option of having an automatic answering machine attached. The 'cordless' telephone (Fig. 4.1) allows the user to carry the handset around the house, instead of rushing from another room, or in from the garden to answer a call. In spite of all these changes, the telephone still employs the same basic system developed by Alexander Graham Bell.

The History of Telephones in Australia

In a country as vast as Australia, a communication system linking the colonies in the nineteenth century, would be essential before there could be any thought of unification. Melbourne and, to a lesser extent Sydney, were the leading cities in the middle of the century. Any communication with Western Australia and the Northern Territory, in particular, took weeks by ship or overland transport before the introduction of the telegraph lines.

Flushed with wealth from the Victorian gold fields in the 1850's, Melbourne was the first city to take the initiative. In 1854, a Mr S.W. McGowan completed the first telegraph line in Australia between Melbourne and Williamstown. McGowan, who had studied under Samuel Morse of Morse Code fame, became the General

Superintendent of Telegraph for Victoria and by 1855 had connected Geelong, Queenscliff and Sandridge to the original line.

In the next five years the telegraph links were made between the colonies of Victoria, NSW, South Australia and Queensland. An attempt to run a cable across Bass Strait in 1858 was unsuccessful. Tasmania was finally linked to the mainland in 1869. In the same year, the colony of Western Australia, which had developed much more slowly than the eastern states, built its first telegraph line from Perth to Freemantle.

A major boost to Australia's international communications came in 1871. In that year, the Java to Port Darwin undersea telegraph cable was completed. This not only linked Australia with Indonesia, but also with Europe, because continuous lines stretched all the way to Britain. In 1872, the Overland Telegraph Line was completed from Darwin to Adelaide, which effectively connected most of Australia to London. The system was completed in 1877, when Perth finally hooked up with Adelaide.

The Coming of the Telephone

The first telephone line in Australia was installed by a private enterprise. Melbourne hardware importers, McLean Bros and Rigg, put in a line between their Elizabeth and Spencer Street stores in 1878, just two years after Bell's invention. In the same year long distance telephone calls were demonstrated in NSW (225 km from Sydney to Maitland), South Australia (386km from Semaphore to Port Augusta) and Victoria (113 km from Melbourne to Ballarat).

Telephone exchanges were first started, by private enterprise, in Melbourne and Sydney, in the early 1880's. There were in fact two in Sydney by 1881, one private and one government. Both the NSW and Victorian governments soon decided to acquire and control their respective telephone systems.

In the same decade, exchanges were opened with a handful of subscribers (10 in Hobart), in all the Australian capital cities. In 1893 the first public telephone was installed at Sydney's GPO. Others soon followed throughout Australia.

For some time, communication between the Australian colonies was carried on by telegraph, not the telephone. In 1897 the Morse Code system for all the colonies was unified. Slowly, with the laying of suitable cables, the telephone replaced the telegraph, until the last Morse-code telegram was sent from Bombala to Sydney in 1962. Indeed, Australians were the world's greatest users of the telegram. In 1950, for example, 35 million telegrams were sent in a year. This declined to less than 20% of that number by the 1980's with the improvement of telephone services.

Trunk line telephone services between the states started in 1902. From this we realise just how poor interstate communications were at the time of Federation in 1901. Automatic exchanges did not even begin to appear until 1912 and these were only able to be used for local calls.

Throughout the early 20th century, Australia was linked to most other countries by telegraph and then telephone lines. From 1905, wireless or radio telegraphy began to be employed and by 1911 we were contacting ships at sea from the roof of Sydney's Hotel Australia.

From the 1960's, with the advent of space travel, satellite technology began to take over. Now, most international telecommunication is carried via orbiting satellites.

The Radio and its History

The big advantage of radio is that it transmits sound without wires. Indeed, the early radio set/ receiver, taking up a prominent place in suburban lounge-rooms, was called a wireless.

The telephone depended on expensive cabling that was slow to install, while radio equipment was simple. Telephone cables could break down, and in wartime, they could be cut. The radio could also be used where cabling was impossible, to contact ships at sea and later, aeroplanes.

The Italian, Guglielmo Marconi (1874 – 1937) has been credited with the invention of radio in 1894/95. His invention showed that it was not enough for the scientist or engineer to produce an idea. It needed to be proven to be useful and commercially viable.

Marconi drew on the previous work of the British scientist, James Clerk Maxwell and the German, Heinrich Hertz. In a lecture in 1864, Maxwell suggested that when an electric current is passed through a wire, the current produces electrical disturbances, which are thrown off in the form of invisible electromagnetic waves. In 1887, Hertz showed, by experiment, that radio waves existed. He proved that a spark of electricity also sent out a radio wave. It was like the ripple on a pond when one throws in a stone. He also discovered that waves could be bent and reflected like light and that they travelled at the speed of light, 300,000 km per second.

Alas, neither of these men could see the practical use of their knowledge. Marconi could, but he was not able to convince the Italian authorities to back him. He brought his work to England. The British government too, was slow in making up its mind but Marconi managed to interest commercial customers.

The first of these was the insurance giant, Lloyds of London. Marconi's new Wireless Telegraph and Signal Company, installed radios for them in lighthouses off the Irish coast in 1898, to report on the progress of transatlantic shipping. Radio was also installed on the East Goodwin Lightship, in the English Channel. In the same year, Marconi transmitted the results of a regatta to the Dublin Daily Express from a steam tug. These early transmissions were not by voice but by Morse Code.

One of the most important innovations to help in the advancement of early radio was the invention of wavelength tuning in 1900. Before that, transmitters produced short bursts of uncoordinated radio waves. Signals were being picked up by any receiver in the neighbourhood. Marconi worked out a way to store energy and release it in a sustained train of waves. The waves were all of the same length. In this way different wave-lengths could be allotted to each transmitter. Today, when we turn the dial on our radio, we are selecting the wave length of the station we want.

One of the most important radio services in the early years was the international carriage of telegrams. In 1907, Marconi set up a service across the Atlantic, using stations in Ireland and Canada. The messages were called Marconigrams.

The first human voice, broadcast by radio, was believed to be that of Professor Reginald Fessenden in America, in 1906. This demonstrated the possibilities for radio telephony but broadcasts were only over short distances at first. Morse transmission remained the norm and it was not until 1928 that a transatlantic telephone service was established.

It was the ship owners and navies that became Marconi's biggest customers. By 1903, dozens of merchant ships were kept in touch with shore stations using wireless telegraphy. In 1905, a Royal Navy fleet in Gibraltar, was kept in constant touch with the admiralty in London. In 1916, during World War 1, at the Battle of Jutland, for the first time, the fleets of both navies were under single command because of radio communications.

Lives were saved, because of radio transmission, after the sinking of the *Titanic*, in 1912. 803 passengers were rescued out of 2,206. There would have been far more survivors if the radio operator of the nearest ship had been at his post. From that time on, large vessels were compelled to keep a 24 hour radio watch.

Citizen Band Radio

The early mobile telephone, or two-way radio was fitted to vehicles like police cars, taxis, fire engines and delivery vans. Messages were transmitted to a base station, which then routed the call on the telephone's system to the person receiving the call. Also, in some cities, private citizens could use mobile phones on which they could call local, long distance or overseas numbers. This system depended on a subscriber being issued with a particular wavelength that did not interfere with other services.

The modern mobile phones that have proliferated in the last ten years operate on a similar system. How, you may ask, with there now being so many mobile phones, can every person be allotted a wavelength that does not interfere with others? This question will be answered later in the chapter.

Entertainment Radio

Radio, in the early years, was almost totally devoted to serious communication. The entertainment radio industry started slowly. The first full-scale operation was started by the Westinghouse Company. They started station KDKA in 1920 in Pittsburgh, Pennsylvania and it is still operating today. This company was the manufacturer of, among other things, wireless receivers. The station was funded by the sale of wireless sets.

A consortium, which included Marconi himself, set up the BBC (British Broadcasting Company) in London. This was, of course, to become the publicly funded, voice of Britain. It was, in the early years, also funded by licences, paid by subscribers. It was not so many years ago that Australian radio and TV owners had to pay a licence fee to help fund the service.

It was in America that the idea was born to fund radio programs using advertising. This, of course, has become the norm. The first station, thus funded, was WEAF in New York, in 1922.

Up to the 1960's, the large, often ornate, wireless cabinet was the centre piece of many lounge areas across the world. Families would sit together, listening to the evening radio programs. Then came television and there was a great decline in the popularity of the radio. This decline was halted by innovative programming, including "talk back", and more so by the invention of the transistor. This tiny invention replaced the much larger, less efficient thermionic valve.

The transistor made possible the portable, battery operated radio, nick-named the "Tranny" in Australia. The transistor was also the basis of the car radio from the 1960's. The radio stations now had a new audience: people on the move.

Important Events in Australian Radio

- 1905 – The Wireless Telegraphy Act was passed to regulate experiments in wireless communication.
- 1906 – The first wireless station was opened in Sydney at Mosman.
- 1910 – The Wireless Institute of Australia made radio contact with two trains.
- 1913 – AWA (Amalgamated Wireless Australasia Ltd) was formed by Ernest Fisk and established the Marconi School of Wireless.
- 1915 – The Navy took control of broadcasting during WWI.
- 1918 – The first direct radio message between England and Australia was received in Sydney.
- 1920 – The Postmaster General's Department regained control of radio.
- 1921 – AWA ran wireless concerts in Melbourne then in Sydney.
- 1923 – Sealed station radios were developed. People subscribed to a single station on an un-tuneable radio. The first station was Sydney's 2SB, (now 702 ABC)
- 1924 – Commercial Station 2FC opened in Sydney, while in Melbourne 3AR began broadcasting. Sealed set radios were not selling so new sets with an A and B station were developed. The A station was subscribed to while the B station gained its fees through advertising.
- 1925 – Australia's oldest B class station 2UE began broadcasting, under the name of 2EU (Electrical Utilities). Later that year, the Labor Party owned 2KY began broadcasting as another B station.
- 1930 – AWA began operating various stations for licensees.
- 1932 – The ABC was formed and took over the operation of A class stations across the country.
- 1934 – Cricket games were broadcast over the radio from cables, with the announcer reading out the state of play and simulating the bat hitting the ball.
- 1938 – The ABC network included two radio stations in every capital city in Australia.
- 1939 – Emergency radio regulations came into force because of WWII.
- 1941 – The Postmaster General revoked the licence of four radio stations, because they could obstruct the war effort.
- 1947 – The ABC's independent news service became operational.
- 1951 – The School of the Air broadcasts, using the Royal Flying Doctor Service network, commenced.

- 1952 – Some experimental FM stations were established.
- 1956 – ABC radio covered the Melbourne Olympics.
- 1961 – Experimental FM stations were ordered to close.
- 1962 – To counter the popular new TV stations, radio stations increased the number of news and community broadcasts.
- 1973 – Commercial radio stations were required to have 10% Australian music as content. The ABC also adopted this figure.
- 1974 – FM licences were granted to 2MBS-FM in Sydney and 3MBS-FM in Melbourne.
- 1975 – 2JJ Sydney became the first station to broadcast 24 hours when it opened.
- 1976 – The ABC began FM broadcasts in Adelaide, Sydney, Melbourne and Canberra.
- 1978 – 2WS began broadcasting; it was the first new AM station in Sydney since the 1930s.
- 1979 – 2JJ converted to FM and became 2JJJ. The station promoted (and still does) youth issues.
- 1980 – Commercial FM stations began to open in the major capitals.

The Effect of Telecommunication Engineering Innovation on People's Lives

Telephone

- Telegraph lines in Australia pre-date the telephone and were responsible for reducing the "tyranny of distance".
- Australia was linked to Europe and Asia by telegraph in 1871. This allowed more rapid communication with Britain.
- The introduction of the telephone allowed a small group of people to communicate more directly.
- The introduction, to Sydney, of public telephones in 1893, brought the telephone to the NSW masses.
- The public telephone became a regular feature in post offices.
- The telephone expanded in use in the early half of the 20th Century, with more private subscribers getting them. Greater communication was now available for these people.
- By 1930 all of Australia was linked by telephone, reducing the time for cross continent communication.
- The introduction of operator dialling in 1945, over trunk lines, allowed further access to people in other capital cities.
- By 1953 there were 1 million subscribers to the telephone. This drastically improved the time for communication.
- By 1962 Canberra, Sydney and Melbourne were linked by coaxial cable, which ensured more effective telephone communication.
- By 1976 Australian telephones were linked by submarine cables to the USA, Japan, and Singapore. This improved communication for Australian companies and people with interests abroad.
- By the 1980s virtually every home had a telephone. This made communication rapid and instant compared with slower postal services.

- The arrival of the Internet was helped by the humble telephone, as modems make use of existing telephone lines.
- The mobile telephone revolutionised personal communication making a person contactable in most places in the developed world. Some people view this as a bad thing as they see the telephone intruding on their privacy.
- The mobile telephone initially created a sub-culture with new terms and symbols for communicating using text messages. This form of communication, is now embraced by the majority of people.
- The Internet used the telephone system to allow the dissemination of all types of information across the globe.
- The smartphone revolution takes off first with the Apple iPhone, which took PDA functionality, mixed it with mobile telephony and used a large full size touchscreen with software based keyboard.
- The smartphone market has grown to two dominant platforms vying for market share and driving along developments which continually push the abilities of smartphones further and further.
- The smartphone now allows communication using social media via data networks in addition to SMS and voice calls.

Radio

- Marconi developed radio, but few realised the potential it had for the many facets of late 19th Century life.
- Lloyds of London backed Marconi and his wireless as they saw it as a means to protect ships they had insured. This decision undoubtedly improved the safety of travel off the coast of Ireland where the first shipping radios were used.
- The ability to tune the radio was a boon for shipping. Merchant vessels were able to transport goods more safely with the radio.
- The Royal Navy also embraced radio as a means for co-ordinating ships and movements.
- The Titanic tragedy forced all ships to monitor radios 24 hours; this made life on the oceans safer for crews and passengers.
- The establishment of radio stations for entertainment provided families with a new form of pleasure.
- Prior to the TV, families sat around radios to catch up on news and for pleasure as they listened to the serialised programs. These were called "soapies" as soap companies were the original sponsors.
- The radio became a vital link for all people in the outback for communication and news.
- Radio was vital to the safety of soldiers during WWII as it was used to communicate between units.
- After WWII the radio was still the primary form of family entertainment and disseminator of news.
- The radio provided lessons to isolated children, ensuring their educational needs were fulfilled.
- People looked to television for their pleasure, hence severely affecting station viability.
- Radio remodelled itself into a provider of news. Talk back radio allowed the "average person" to put forward his/her views on various, sometimes contentious, topics. It gave people a feeling of empowerment.

- The radio became important for drivers as a form of information and entertainment.
- Radio stations tailored their music and news to suit a particular audience. This resulted in very parochial followings for radio stations far more than was the case with television.
- The radio was an essential component in all aircraft, thus improving the safety of air travel.

Engineering Materials

Specialised Testing

Voltage, Current and Insulation

To test whether a material is suitable for use in telecommunications its resistivity (Volume 1 Ch 2) must be assessed and checked. A material with a high resistivity will be suitable as an insulator, while a material with low resistivity will make an excellent conductor. To assess a material's resistivity, current at a certain potential will be passed through the material. Using Ohm's Law, resistance may be found and with information, such as the length and cross-sectional area of a conductor, it will be possible to find the resistivity. Conversely Ohm's Law may be used to determine how much current will flow in a conductor when resistance and potential difference are known.

Various testing devices such as multi-meters and Cathode Ray Oscilloscopes (CRO) are used to test electrical quantities. Multi-meters are useful test devices that allow multiple test function to be woven into a single device. Voltage, current, resistance, capacitance and transistor gain are all possible test functions on these devices.

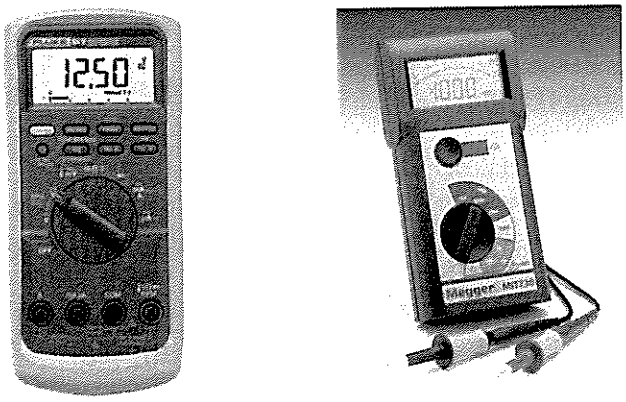


Fig 4.2: *Left, a large digital multi-meter; right, an insulation tester.*

In telecommunication applications it may be necessary at times to ensure that insulation has not broken down which would allow leakage of current. Insulation testers such as "Megger" testers allow insulation to be tested by passing through voltages and checking there is no leakage. Any low resistance reading will indicate problems with the insulation surrounding the conductor.

Conductivity

Conductivity is an important property in electrical and telecommunication applications. Conductivity is the reciprocal of the resistivity ($\sigma=1/\rho$) which is a constant value for a material at a given temperature. Hence a poor resistor makes a good conductor. The SI unit for conductivity is Siemens per metre (Sm^{-1}). Industrially the conductivity is often expressed as a percentage of the International Annealed Copper Standard (IACS). This means some near pure coppers will have a conductivity of 101% IACS while silver is 106% IACS.

Copper and its Alloys, Used in Telecommunications

Copper, in terms of electrical conductivity, is second only to silver. It is, however, far more cost effective than silver. Copper for telecommunications is used in two forms, unalloyed and alloyed. Copper and its alloys are also used because they are ductile, hence allowing them to be drawn into wires; they are easily annealed and good for soldering.

Pure Copper

Industrially *electrolytic tough pitch copper* (ETP) is the most widely used copper alloy. It has low oxygen levels (0.02-0.04%) and being produced through electrolytic processes it also has exceedingly low levels of impurities such as sulphur. This ensures it has excellent electrical conductivity. A purer form of copper used is *oxygen free high conductivity copper* (OFHC). This copper is 99.99% pure, and has the same excellent electrical conductivity as ETP. But OFHC is used in some environments where hydrogen embrittlement affects ETP.

Small amounts of some impurities can lower the conductivity by a huge amount, so keeping the impurity level low is essential. For example, by adding only 0.2% phosphorus, the conductivity will be 25% that of pure copper.

Copper Beryllium (Cu/Be) Alloy

Beryllium is a light metal that offers excellent strength to weight ratio and stiffness. But its rarity and its poisonous oxide make its industrial uses small. Beryllium is used as an alloying element with copper with amounts of between 0.5-4% Be. The addition of beryllium results in significant strength over pure copper but still the alloy has good electrical and thermal conductivities. In fact copper beryllium alloys offer tensile strength and hardness around alloy steels yet they have better electrical conductivity, are non-magnetic and non-sparking. The highest strength alloys have lower conductivity, so if maximum conductivity is desired then the trade off is slightly lower maximum tensile strength.

Copper beryllium alloys are able to be hardened through precipitation hardening (age hardening). They are solution annealed and quenched to trap the beryllium rich precipitates in solution. They are then worked or formed and after those forming processes are completed they are precipitation hardened artificially. The beryllium rich precipitates lead to their high strength over most other copper alloys.

While these alloys have been used for non-sparking tools and electrical applications in the past, in terms of telecommunications these alloys now find use because of their combination of high strength, good electrical conductivity and excellent fatigue resistance which makes them useful as electrical contacts and spring contacts.

Copper Cadmium (Cu/Cd) Alloy

Pure annealed copper is a dense material with relatively low strength. These qualities limit the span over which cables can be strung between support towers. For fine cables, the problem is over come to some degree by using cold drawn cable. But where heavy cables have to be used, as in overhead train/tram lines, solid solution hardening alloys replace pure copper. Cu/1%Cd has good wear resistance (train pantographs), and good strength, thus reducing the number of support towers per kilometre of overhead cable. High transmission voltages (1500 V on Sydney trains) are used to reduce the resistive power loss in the cable.

The microstructure of annealed Cu/1%Cd alloy is an equiax single-phase structure where all the cadmium is dissolved in the solid copper to produce an α solid solution.

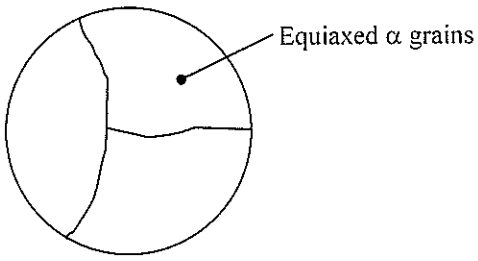


Fig. 4.3: *The microstructure of annealed Cu/1%Cd alloy*

Table 4.1: *Tensile Strengths and Conductivities of selected Copper Alloys*

Copper Alloy	**Composition**	**UTS (MPa)**	**Conductivity (% IACS)**[1]
Electrolytic Tough Pitch Copper (C11000)	Cu 99.97%	234 (Soft) 455 (Hard)	101
Oxygen Free Copper (C10200)	Cu 99.99%	215 (Soft) 455 (Hard)	101
Brass (C22000)	Cu 90% Zn 10%	280 (Soft) 510 (Hard)	44
Copper Cadmium (C16200)	Cu 98.8-99.2% Cd 0.8-1.2%	241 (Annealed) 689 (Hard)	90
Copper Beryllium (C17500)	Cu 97% Be 0.4-0.7% Co 2.4-2.7%	240 (Annealed) 965 (Aged)	50-60

[1] IACS stands for International Annealed Copper Standard and measures the conductivity of a material as a percentage against annealed copper which is classed as 100%.

Copper Zinc (Cu/Zn) Alloy (Brass)

Copper zinc alloy or brass is often used as a connector for telecommunications. They are harder and stronger than pure copper and while being a relatively good conductor their conductivity is well below that of OFHC copper. While having good corrosion resistance they will tarnish and can be affected by ammonia environments, so the connectors may be plated to resist conductivity issues caused by oxidation.

Semiconductors

Semiconductors are essentially poor conductors. They are usually made up of an insulating material that is doped with another atom to give a surplus of electrons, or a deficiency in electrons. If the semiconductor has more electrons than are needed it is termed an n-type, as it is slightly negative. A semiconductor with a deficiency in electrons is called p-type, as it is slightly positive.

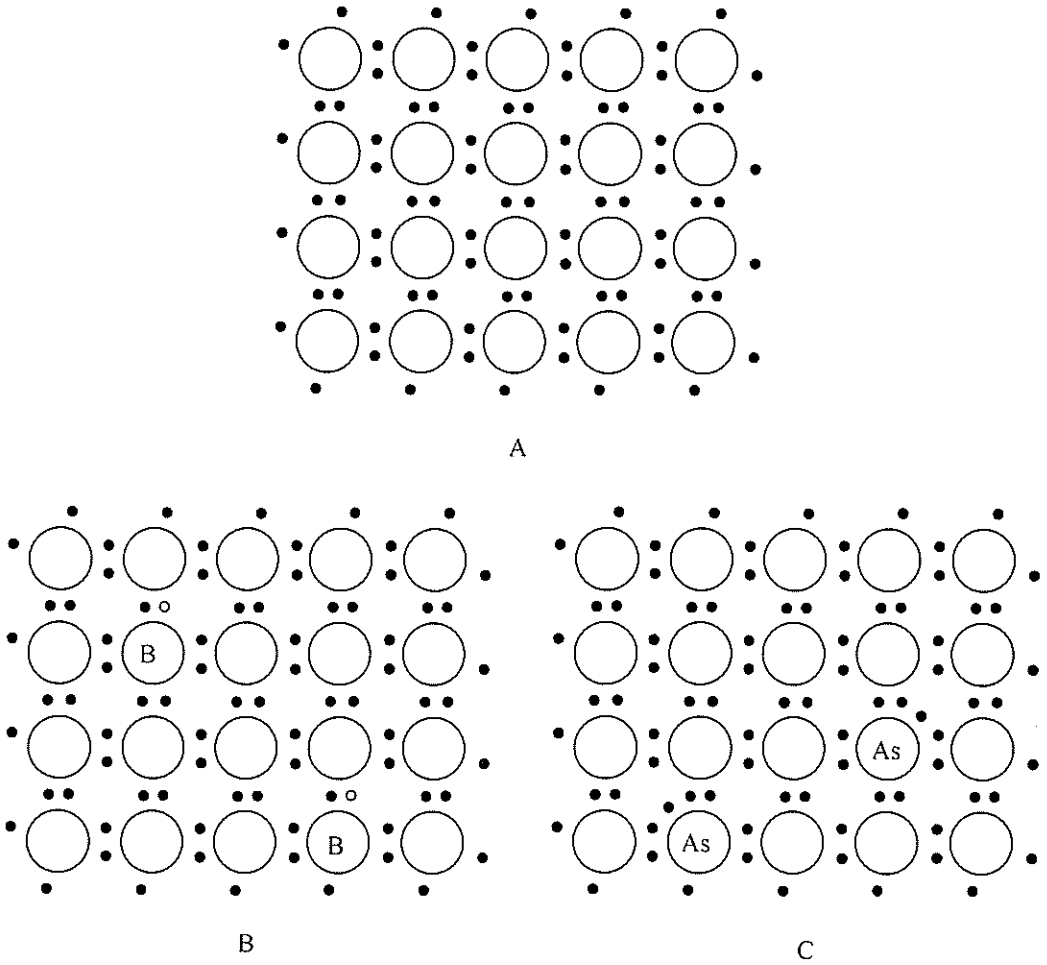


Fig. 4.4: *The structure of covalently bonded silicon or germanium is shown in A. B illustrates a p-type semiconductor doped with boron, while C displays an n-type semiconductor doped with arsenic.*

Silicon and germanium compounds are most commonly used for semiconductors as they both have four electrons in their outer shell. To create n-type material, they are doped with arsenic, as it has 5 electrons in its outer shell. To create p-type it is doped with boron as it only has 3 electrons in its outer shell. With extra electrons, the material can now let current flow, because the free electrons move through the structure. With less electrons, the material has holes which let electrons fill them. As one electron moves to fill a hole it creates a new hole, letting the positive hole move through the material.

One of the most important things semiconductors are used for is a p-n junction. This is where a layer of p-type material and n-type material are butted against one another. When this occurs, a one-way component is created, which will allow electricity to flow one way and will block it in the other direction. This occurs because of the *depletion* zone. The depletion zone is the gap between the bulk of the positive and negative charges of the semiconductor materials. If current flows in one direction, the unlike particles attract, which reduces the depletion zone and allows current to flow. This is called forward bias. If the current flows the opposite way, the charges repel, the depletion layer grows and current will not flow. This is called reverse bias. The principle is illustrated in Fig. 4.5.

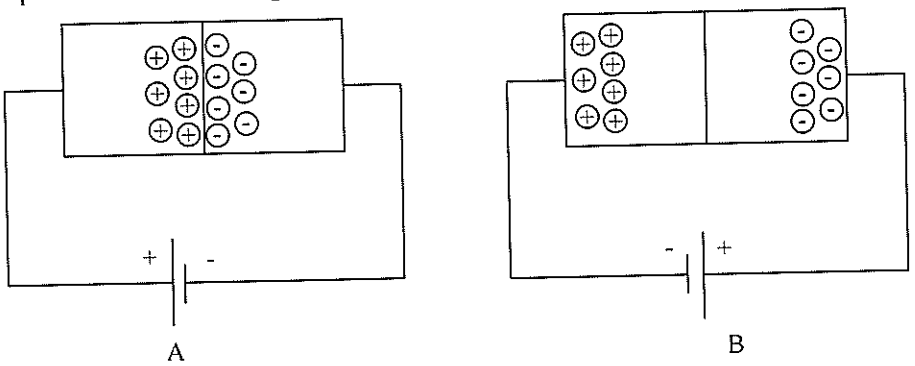


Fig 4.5: *When the p-n junction is connected with forward bias (A), the current will flow as the depletion zone disappears. If it is connected with reverse bias (B), the depletion zone grows, inhibiting current flow.*

It is also possible to have a double p-n junction, creating a transistor. In this case, applying a current to the sandwiched material allows current to flow between the outer materials. The sandwiched material acts like a gate. The true operation of a bipolar transistor is difficult to explain. Simply put, by supplying a small current to the base of the transistor it reduces the depletion zone of the collector/base junction and allows current to flow. If the small base junction is removed, current will stop flowing, because the collector/base depletion zone will increase. Bipolar transistors are either NPN or PNP. The letters signify whether it is n-type, sandwiched between two p-type layers or vice versa.

Another type of transistor, often used in computers, is the field-effect transistor (FET). FETs are smaller, cheaper and draw less power than bipolar transistors. They work by applying a voltage to a gate to attract electrons to an area between a source and a "drain". The electron bridge allows current to flow. The most common FET in use is the MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor). For further reading about these concepts refer to the Additional References at the end of the chapter.

Semiconductors are essential in our modern lives. They are at the heart of most modern electronic devices, including computers and smartphones. Semiconductors form the basis for almost all modern telecommunication systems. The world is truly dependent on them. It is strongly suggested that you consult references at the end of the chapter to expand your knowledge of these important materials.

Types and Uses in Telecommunications

Semiconductors are used in nearly all forms of telecommunications. Below is a list of some such components.

Diodes are used in devices as one-way components. The basic diode is the simplest semiconductor, but more complex ones exist.

Zener diodes are diodes that act as a one-way component up to a certain voltage. Beyond this voltage, current begins to flow in both directions through the diode.

The *Light Emitting Diode (LED)* is a special diode with which most people will be familiar. These diodes give off light, and are used extensively as indicators for telecommunication equipment. The little red indicator on the television, to tell us that it is on standby, is an LED. Like standard diodes, current can only flow in one direction, but as it does, visible radiation is given off, as white, red, green, orange or blue light. LED's are used, as they offer greater reliability and less power consumption than conventional lamps.

The *transistor*, developed in 1947 by John Bardeen, Walter H. Brattain, and William B. Shockley, working at the Bell Telephone Laboratory in the USA, is the father of modern communications and the computer age. The transistor has been pivotal in the miniaturisation of modern electrical equipment. The basic transistor can act as either a switch or an amplifier. It is the basis for all modern digital systems.

Integrated Circuits are a complex miniaturised circuit containing various semiconductors and other electronic components, in a tiny external package. Various types are available, with each one performing a differing function. For example the very common 555 IC is a timer.

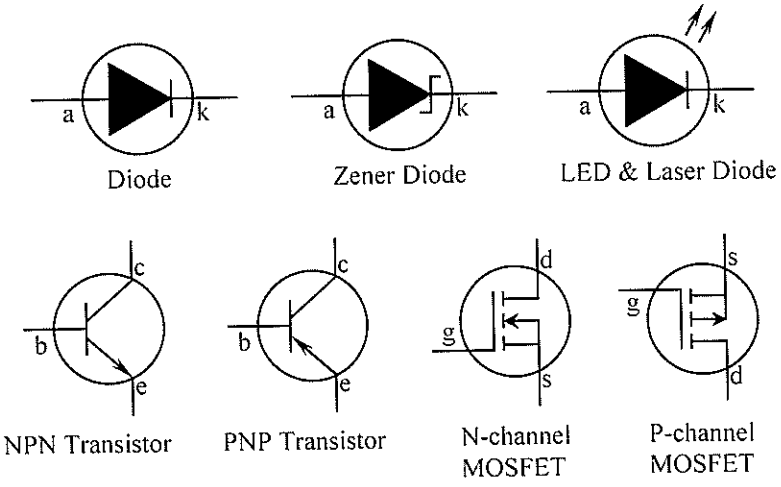


Fig 4.6: *Circuit symbols for some key semiconductors.*

Lasers are another type of semiconductor. Originally lasers were made using materials like ruby and composites of ruby, yttrium, aluminium and garnet. Semiconductor, or *diode lasers* are the smallest lasers in the world. The function of these is quite complex and beyond the scope of this book, however, we shall briefly describe their operation. By creating two PN junctions, and passing a large forward bias current through them, the electron holes recombine and give off photons. These lasers are used in DVD players and as signal transmitters for fibre optic systems.

Polymers

Polymers find extensive use in the world of telecommunications engineering. They are used, not only for insulation but also as casings for various devices. By utilising polymers for radios and televisions it is possible to double insulate them, which removes the need for an earth on the casing.

Manufacturers make extensive use of polymer casings for mobile and fixed telephones. Polymers offer insulation, shock resistance and also lend themselves to mechanised production by injection moulding. Telephone casings tend to be made from acrylonitrile-butadiene-styrene (ABS) or polycarbonates or a mixture of the two.

Many smartphones tend to use a polycarbonate housing in conjunction with the glass face for the touchscreen. A notable exception is the Apple iPhone 5 which uses an anodised 6000 series aluminium alloy uni-body case instead of a polymer.

Fibre Optics

Fibre optics is an important field in the world of telecommunications. They function around the concept of transmitting electronic information, encoded onto a light beam. A fibre optic system essentially consists of three parts: a device to convert electric current into light, the cable to carry the light and a receiver to convert the light back into electric current. A simplified fibre optic system is shown in Fig 4.7.

Fibre optic cabling is lighter and less expensive than copper cables and, as the signal is transmitted as light, it is less affected by interference from external sources. There is also no risk of induction in the cable. But the primary reason for using fibre optic cabling is that it can carry hundreds of times more information than can be carried on a copper wire.

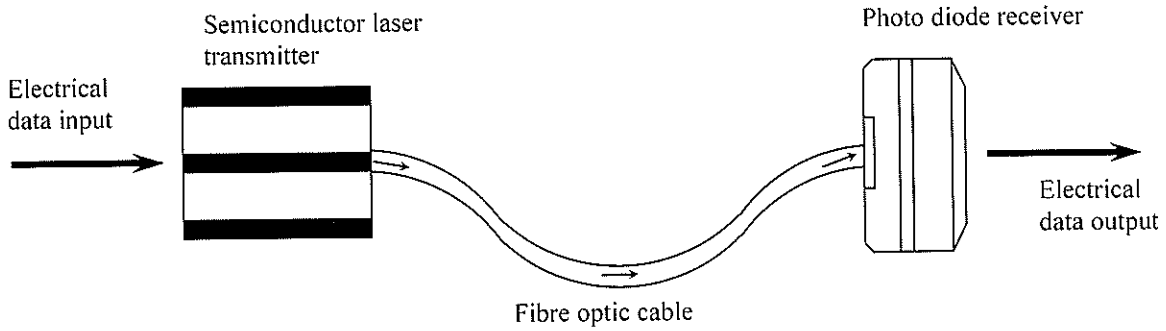


Fig. 4.7: *The basic components in a fibre optic system.*

Types and Applications

There are two basic families of fibre optic cabling, single mode and multimode. Single mode fibres have a central core diameter of 8 μm, with the outer cladding up to 125 μm in diameter. The narrow core means that light passes along a single light ray path (Fig 4.8 a). By having a sharp difference in refractive index between the core and the cladding, light waves cannot escape from the cable sides.

Multimode cables have a larger inner core diameter, the light reflects off the cladding and is reflected internally through the cabling. They come in two types, graded-index and stepped-index. Stepped-index cabling utilises a sharp difference in the refractive index of the core and cladding to reflect light waves. With the larger diameter core, light waves can take differing paths, which in turn means they take differing times to reach their destination. The graded-index core cable uses a refractive index that progressively decreases from the centre of the core to the cladding. This allows the light waves to "flow", through the cable instead of reflecting along it.

Single mode fibre is favoured for long distance telecommunications as it has the highest bandwidth and suffers the least from losses of light. It is, however, more costly to manufacture. When less bandwidth is required, and information carrying is needed, multimode cables are used. Stepped-index cabling is the least used.

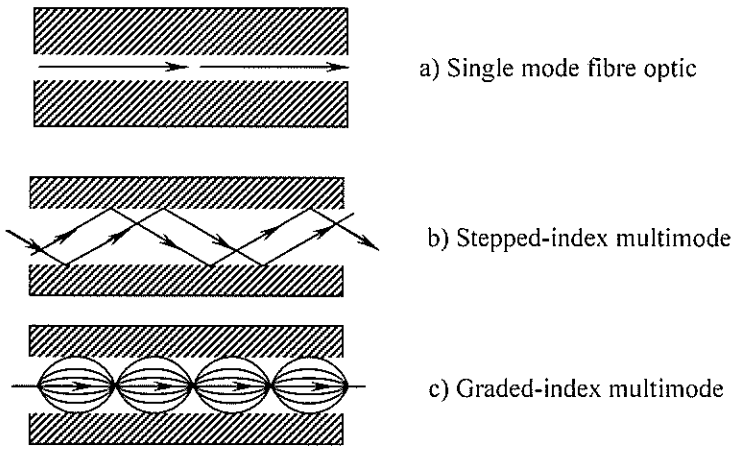


Fig 4.8: *The three different fibre optic cables.*

Engineering Electricity/Electronics

Telecommunications

Analogue and Digital Systems

From its inception most telecommunications equipment utilised analogue technology. Analogue electrical systems utilise a continually varying range of values. When one speaks the voice changes in amplitude (loudness) and frequency (tone). The amplitude varies continuously, thus producing a signal that we call an analogue signal (Fig. 4.9). The telephone system transmits this varying amplitude as a continually varying voltage.

A digital signal is one that is either on or off. The on/off signal can be interpreted as a binary coded number, that is, a number made up only of '0' or '1'. Digital data can be converted into binary numbers, and these can be transmitted as digital electrical signals. These signals can be processed by digital circuits and decoded to meaningful quantities. Digital systems are extremely effective in data transmission and for computerised systems; unfortunately many of the inputs or outputs required must be analogue. As a consequence we need converters to turn one signal into another.

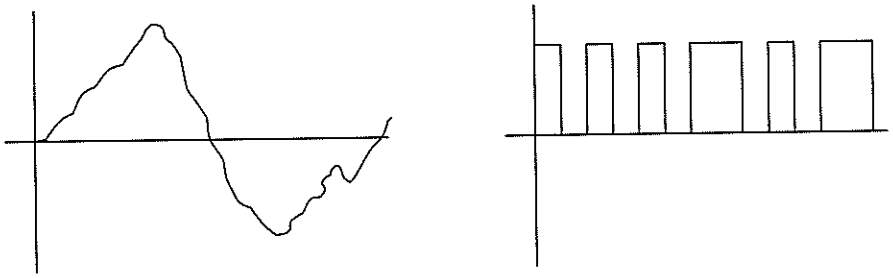


Fig 4.9: *The continually varying values of an analogue signal (left), while the digital signal is merely on or off (right).*

A true advantage of digital signals, used in mobile phones and digital television, is that they can be more easily compressed than analogue signals. This means an increase in *bandwidth*, which allows more information to be carried. Bandwidth is a measure of the amount of space available for carrying information; the more bandwidth, the more information that can be transmitted.

Table 4.2: *Bit Depth in Decimal and Binary*

Bit Depth	Number of Digits	Decimal Range	Binary Range
2^1	2	0-1	0-1
2^2	4	0-3	00-11
2^3	8	0-7	000-111
2^4	16	0-15	0000-1111
2^5	32	0-31	00000-11111
2^6	64	0-63	000000-111111
2^7	128	0-127	0000000-1111111
2^8	256	0-255	00000000-11111111

Digital signals are more than simply on or off, because we usually need to be able to send more information than a simple on (1) or off (0) can convey. But if we put together many binary digits then more information can be conveyed by representing the data as strings of large binary digits. Binary systems are described as the amount of bits available and they are based on powers of two. So 1-bit binary has 2^1 digits, i.e. 2, (0 and 1). Where as 3-bit binary has 2^3 digits available, that is 8.

Analogue to Digital Conversion

An analogue signal is continuously variable and conveys a lot of information, some of which may be superfluous, and some may be static with noise that is unnecessary. To take the complex wave and convert it into a digital signal then we need to *sample* the

signal wave at regular time intervals. The rate of sampling is called the *sampling rate*, and the rate must be sufficient to accurately represent the waveform we are trying to convert.

Now in conjunction with the rate at which we sample is the accuracy with which we record the amplitude of the signal wave. This is the sampling precision of quantisation level. The process of *quantisation* is where we approximate the amplitude of the analogue wave to the nearest whole number that our chosen quantisation accuracy allows. If we chose 1 bit binary we could only assign the values of 0 or 1, if we chose 8 bit binary we could assign any value between 0 and 255. As you can see the greater the binary depth chosen the more accurate our quantisation will be.

The sampling rate and quantisation levels are important because we will eventually need to convert this digital signal back into an analogue signal so it can be consumed, so if our sampling rate was too low or our quantisation level lacking in depth, then we will lose important information, that we cannot recover.

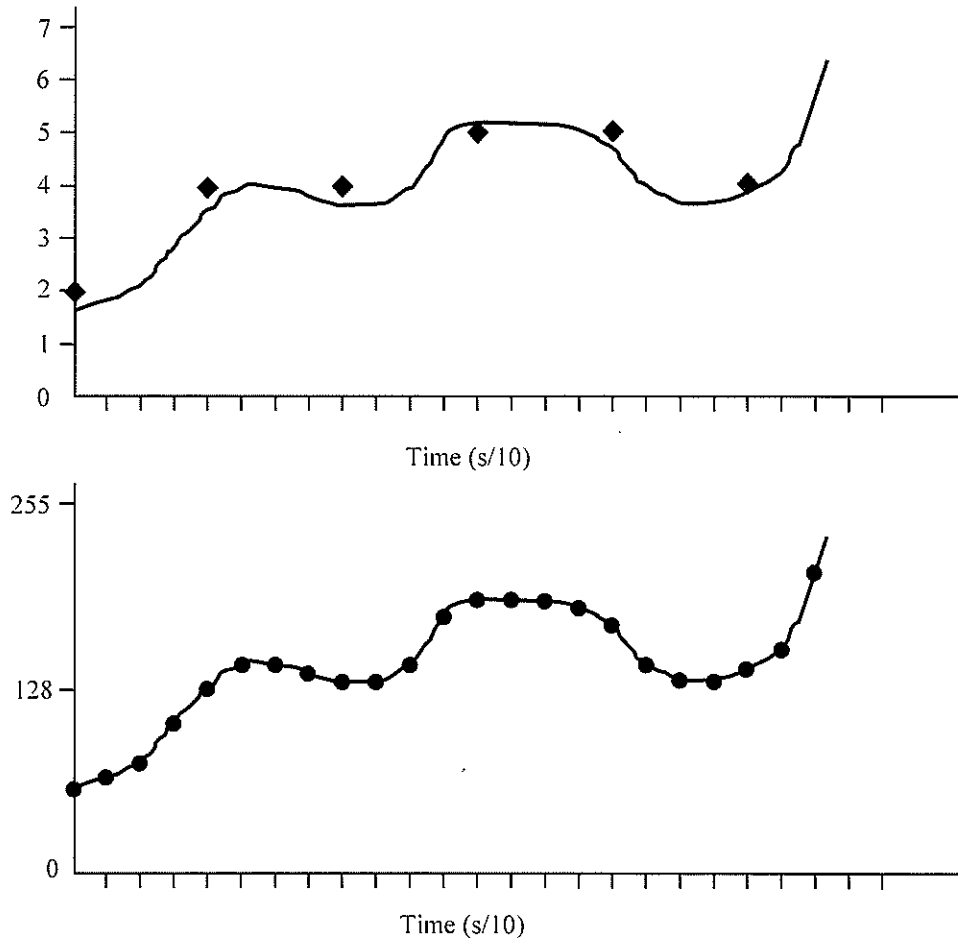


Fig 4.10: *Two sampling rates and quantisation levels are tested on the same analogue wave. The top takes a sample every 2/5ths of a second, while the bottom one takes a sample every 1/10th of a second. The quantisation level of the top is 3 bit, while the bottom is 8 bit. The bottom sampling rate and quantisation levels will yield a far more accurate digital representation of the original wave.*

Let us consider the following example to illustrate digital to analogue conversion. Imagine you are walking along a winding path for 10 metres, and you have a friend photograph it with a still image camera. The rate at which the photos are taken is analogous to the sampling rate. If you take too few photos it will not accurately represent your movement along the path. If we take photos at a high rate the path will be presented more accurately but the trade off is we will have more images, or snapshots of data, to store.

Now coupled with this is the quantisation level. Imagine you are wearing clothes that are red and black. If we chose only low quality black and white photos all of this colour information will not be collected and this is similar to low quantisation levels. If we photograph using high quality colour then all the details of your outfit and the surrounding area will be recorded. Like the photography rate the more detail in the photo comes at a cost in the form of more data being stored at each snapshot so all the extra details can be described.

A good example of digital and analogue conversion is a modern stereo system recording a voice on tape. If one takes a microphone and sings into it, the signal from the voice is analogue. It is then passed through a recording amplifier. The system converts this into a digital signal to be placed onto the tape. When the tape is played, the digital information is converted back to analogue and then passed through an amplifier and sent to a loudspeaker.

The compact disc (CD) is another device where a digital to analogue converter is used to listen to music. It has the information stored on the disc in the form of "bumps", which a laser collects as digital information. This is converted to analogue, amplified and sent to a loudspeaker. MP3 players work in a similar manner, but the information is stored as bytes in a solid state memory. It is collected by the processor from the memory byte by byte, is decompressed, then is converted to analogue and sent to the loudspeaker. The operation of analogue to digital conversion for CDs is explained at (http://electronics.howstuffworks.com/analog-digital.htm). Pay close attention to concepts such as sampling rate and sampling precision.

Modulation and Demodulation

During early experiments on the transmission of electromagnetic waves it was found that it was easier at high frequencies which require less power for greater distance. To achieve this, it was found that it was possible to transfer the low frequency signals representing speech/music onto a high frequency carrier wave. The most suitable waveform to achieve this was the analogue sine wave. Modulation is the process of using a carrier wave to transmit information. Once the signal is sent, it is necessary to demodulate the signal back to the original form. Here a demodulator reverts the modulated sine wave back to its original form and the signal is separated from it, before converting it back to sound waves.

There are three common forms of analogue modulation: Amplitude Modulation (AM), used for some radio stations and for the picture for television transmission and Frequency Modulation (FM), used by radio and for the television sound wave, and Phase Modulation (PM). AM and FM are outlined next in radio transmission.

Multiplexing is another term that deserves mention. This is the process of placing multiple signals along a single communication channel. Using this, it is possible to increase the amount of information carried. A multiplexer achieves this. Once the signal is received, a demultiplexer separates the multiple signals from one another.

Radio Transmission

Electromagnetic wave transmission or radio transmission is one of the most important parts of telecommunications. The transmission and reception of radio waves is essential to the existence of radio stations, television, wireless communication, mobile phones and space communication.

How Radio Waves Travel

When a radio signal is transmitted it may travel through space in a few ways. Waves passing through the ground serve those close to the transmitter. They are, however, quickly absorbed by poor conductors like sand and hence do not travel very far. Airwaves travel up to a part of the atmosphere called the ionosphere. This area of the atmosphere is charged with positive ions and negative electrons. If the radio waves are below a frequency of 30MHz they will be reflected from the ionosphere back to earth. The power of the transmitter determines how far the radio wave will travel. For example, mobile telephone systems use a weak low power transmitter, while shipping requires far more powerful transmitters.

Fig 4.11: *Radio waves travelling around the globe. Note the reflected wave from the ionosphere.*

For outer space communication the frequencies must exceed 30 MHz for it to travel through the ionosphere to orbiting craft. In practice the frequencies used exceed 1000 MHz (1 GHz).

Amplitude modulation

To transmit the signal wave it is necessary to modulate it onto a carrier wave that allows it to be sent far greater distances. One way of doing this is through amplitude modulation. Here the amplitude of the wave is varied to transmit the information. The wave produced by a voice, is modulated onto the carrier wave and as the signal wave's amplitude increases the amplitude of the modulated wave increases, while the frequency of the carrier wave is not altered. (Fig. 4.6)

Amplitude modulation is used for some radio stations and for transmitting the visual signal for television. The disadvantage of AM is that induced "noise" reduces the signal quality because the noise affects the amplitude of the modulated wave. Noise is an unwanted signal, such as that induced by electrical storms.

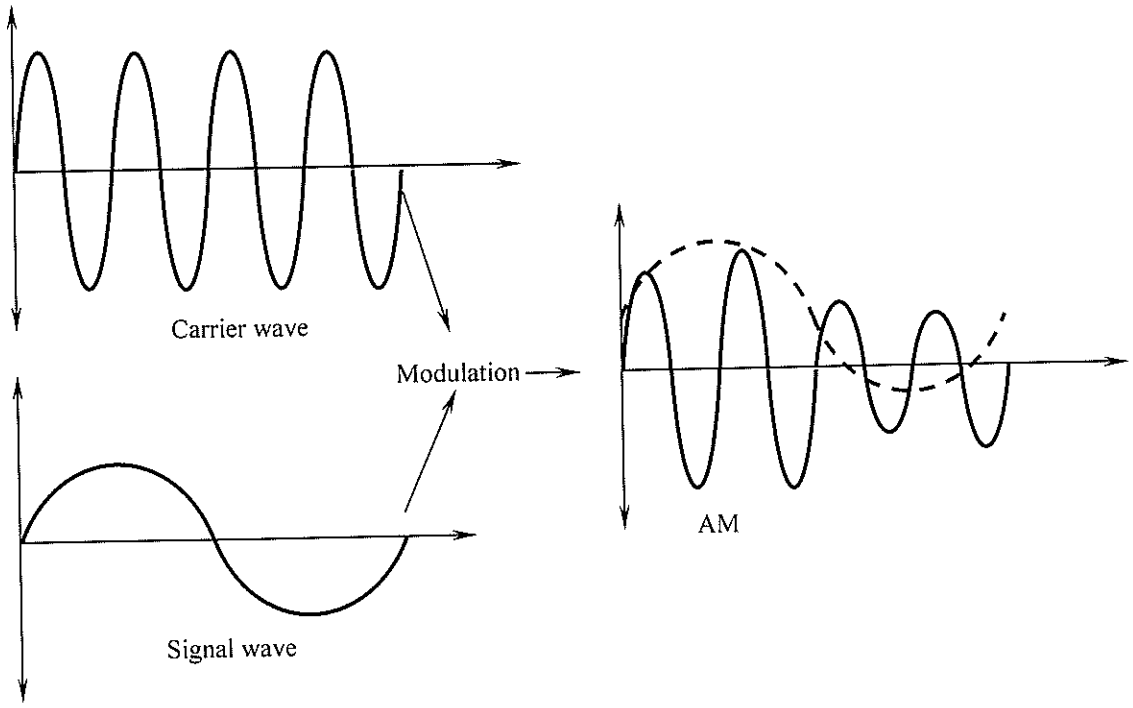


Fig. 4.12: *Amplitude modulated transmission*

When the AM waves are sent out by the transmitter they are eventually received by a radio receiver. To receive the required signal, the radio must be tuned to the required carrier frequency. The radio wave is then detected and passed through a demodulator, which removes the carrier frequency waves while the signal remains. The signal is then amplified and fed into a loud speaker. The variations in amplitude are converted into sound by the loudspeaker.

Frequency Modulation (FM)

This method of modulation is more common now than AM for radio transmission. Unlike AM, the FM system modulates the frequency of the carrier wave and not the amplitude (Fig. 4.13). As the amplitude of the signal wave increases the frequency of the FM wave increases, while a reduction in amplitude decreases the frequency.

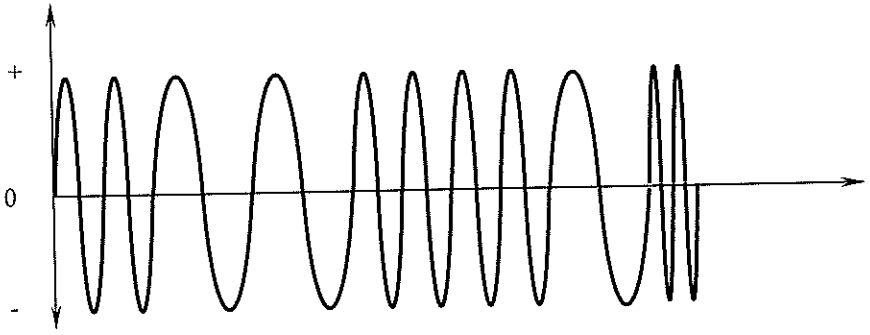


Fig 4.13: *Frequency modulated transmission*

The advantage of frequency modulation over amplitude modulation is that the quality of signal is not affected as much by induced "noise", because noise normally affects

the amplitude of the wave and the variations in amplitude are not carrying the information of the signal wave.

FM transmissions go through the same basic process as AM waves when received. They are received by the aerial, amplified, detected, demodulated to remove the carrier frequency, then amplified and sent to the loudspeaker. The main difference is that it is the variations in frequency that are converted into sound.

Demodulation at its simplest – Crystal Radio Receiver

A simple radio receiver can be made using the simple crystal radio. Here we need an antenna, a coil, a variable capacitor and a diode. If the coil were of a fixed number of turns this receiver would be fixed to a set frequency that it would resonate at. But if the coil is made variable we can tune the circuit to receive our desired frequency. The variable capacitor is used to make the fixed coil variable, so we can tune to a certain frequency. The antenna picks up the modulated radio wave, the coil resonates at our chosen frequency so picks up the modulated wave at the frequency we want. The modulated signal is a sine wave but the diode will now cut off the negative half, which leaves behind a fluctuating half wave rectified signal. The "fluctuation envelope" is the signal wave that was modulated onto the carrier wave, and the ear piece can now move according to this variation

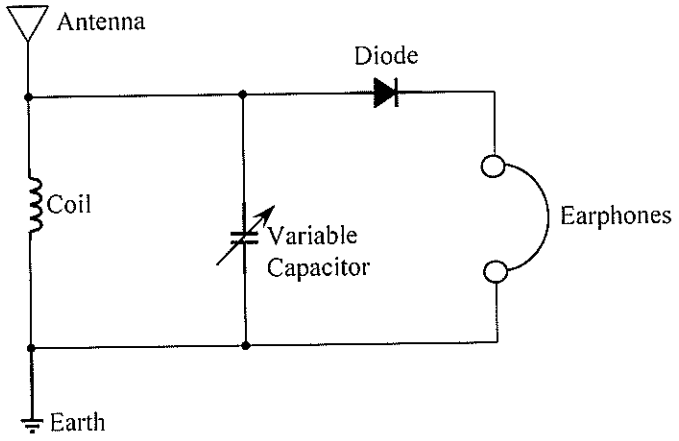


Fig. 4.14: *A classic crystal radio circuit, a tuned circuit that forms a very simple demodulator.*

Television Transmission

Television transmission relies on radio waves to transmit information, as was outlined in the preceding section. These radio waves are either very high frequency (VHF) or ultra high frequency (UHF) waves. But how is the television signal created and then received?

Black and White Television

It is best for us to start our discussion with black and white television. A camera is used to create a video signal of the situation. A black and white camera uses optics and electronics to display variations in brightness. The vidicom tube fires electrons at a screen, which varies in resistance, depending on the amount of light present. The beam scans across the image in a zig-zag fashion that produces a *raster*. Once the

image is scanned, it is sent to the transmitter after being modulated when received. At the television, it is demodulated and then the process of producing an image begins.

The heart of the television is the cathode ray tube (CRT). It works by firing electrons from a cathode at an anode placed behind the glass screen. The anode is coated with phosphor, a luminescent material. When the electrons strike the anode, the phosphor glows, creating a bright part. The kinetic energy of the electron decides how bright the phosphor glows. The electron beam scans the screen and produces a raster image. To fool the human eye there must be at least 16 images per second.

At the same time an audio signal is transmitted with the video signal. The recording sound system produces an audio signal that is modulated, sent to the transmitter, received, demodulated and reproduced by the television's sound system.

Colour Television

The principle behind colour television is similar to the black and white television except both the television camera and the television set have added features to deal with the colour. Whereas black and white TV uses a single electron beam, the colour system has three beams that react on red, blue and green colour phosphors on the front surface of the CRT. Other colours are produced by varying combinations of these three. To produce white, the three beams illuminate adjacent phosphors which appear as white to the human eye, while for black none are fired at all.

In addition to this there is a mask, which has millions of holes in it between the screen and the electron gun. The mask determines the position of the dot pattern for each phosphor. The Sony Trinitron television uses a single gun firing three beams, which pass through a special grill to produce vertical phosphor lines.

Digital Television

By the end of 2013 all free to air TV will be digital TV. Digital TV involves using digital modulation techniques and multiplexing to prepare a signal for transmission as opposed to analogue methods.

The advantage is improved resolution and the ability to carry greater amounts of channels within a given bandwidth. Television stations can run various programs off one frequency at reduced resolution (i.e. equal to current analogue resolution) then drop back to one station for movies and large sporting events that require high resolution. Another clear advantage is an improvement in sound quality. Currently all free to air television stations have multiple channels allowing them to offer stations tailored to particular markets or to have continuous news services.

Digital TV also allows the easy combination of video and data systems, thus giving access to interactive systems on one machine. Many televisions are now termed Smart TVs where they have some level of Internet connectivity allowing them to gain some content from the Internet as well as digital free to air services.

Products such as Apple TV is an example of a product that delivers media content via the Internet to a television. Movies and television shows can be rented or subscribed to instead of watching them via free to air TV.

Television Display Technology

For many years the Cathode Ray Tube (CRT) was the dominant television display technology, but since the turn of the century thinner and lighter displays have become popular using either LCD (liquid crystal display) or Plasma displays.

LCD TVs use a colour liquid crystal display like that used for computer monitors; they rely on filtering a white back light. This backlight shines through millions of LCD shutters that are arranged in a grid. On earlier models the back light used a cold cathode fluorescent tube but more modern LCD TVs now use LEDs for the backlight, with the trend now being to edge mounted LEDs. There is some work in the development of organic LED (OLED) displays but as of 2013 these were restricted to only small displays, such as media players and some smartphones.

Plasma TVs use a plasma display panel for visual display. They use electricity to excite a mixture of noble gases and a small amount of mercury in cells which then strikes a phosphor producing the appropriate colour and light. They arguably produce better contrast than LCD TVs but produce more heat and they are only suitable for large displays.

3D TVs are television displays that use some mechanism to create depth perception in the viewing experience. Currently most displays require the use of glasses to achieve the 3D display but work is in progress to create displays that obviate the need for glasses.

Telephony – Fixed and Mobile

How the Fixed Telephone Works

The mouthpiece of the telephone is a simple microphone. The ear-piece is a simple loudspeaker. The mouth piece uses a small electric condenser microphone.

The current is fed down the telephone line to the exchange, then on to the receiver's telephone. In the earpiece of the telephone, the current travels up a coil around a magnet. The coil is connected to a diaphragm of thin metal or foil. The alterations in the current, made by the voice at the other end, make the diaphragm move, recreating the speech patterns of the caller.

When one lifts the hand piece off the cradle, a dial tone is heard that tells you the telephone is connected to the network. When a number is dialled, an automatic computerised switchboard connects the caller to the "dialled" number.

Often a voice on the telephone sounds different from how the person sounds in real life. This is because telephones only allow frequencies from around 400Hz up to 3400Hz to pass, to increase the number of calls that can be multiplexed on a single line. Any voice pattern above or below this range is hard to hear or lost entirely.

How Mobile Phones Work

The mobile telephone is essentially a computerised low power radio system. It has little in common with the telephone except the principle of a person-to-person

conversation. Originally radio telephones, were limited to a small number per city. They required a sensitive receiver and powerful transmitter and as such were usually fitted to cars. Radio telephones were limited by the reduced number of allocated frequencies in the VHF and UHF bands. It was therefore impossible to have a lot of radio telephones, as there were not enough frequencies available.

The secret behind the mobile telephone is the cellular system. For mobile phones a city or area is broken up into small cells. Each cell has a certain amount of frequencies (more than for radio phones), but because there are more cells, there can be more phones. Why is it that radio frequencies from one cell do not interfere with the same frequencies from other cells? This is so because the signal from a mobile telephone is weak, as it only has to transmit a short distance.

When a mobile telephone is turned on it sends a signal to the service provider to inform it that the telephone is on. The network computer registers where it is and periodically sends out tracing signals to keep track of the phone's position as the user might be moving about. When someone rings the telephone, the call goes to the network's central system, which knows in which cell the required telephone is to be found. The network rings the telephone on a frequency it has allocated, and when the person answers the phone it automatically selects a frequency on which to talk. When talking on a mobile phone, two frequencies are used, one to talk on and another to listen on. This is called duplex signalling.

Fig 4.14: *Three mobile phones with the oldest at left and the newest at right. The oldest phone is a feature phone while the other two are smartphones.*

If one is travelling in a car and talking on a telephone, he/she might move out of the original cell. As the user gets to the boundary of the current cell, the network computer registers that the signal is getting weaker, while the computer in the next cell will detect a growing signal. The network finds an available frequency in the next cell and allocates the call to it. Sometimes this does not work properly and it is one case when the mobile "drops out".

A cell is a relatively small area that is served by a low-powered transmitting tower. A large city may be made up of hundreds of individual cells. By breaking up a city into cells it is possible to duplicate radio frequencies in each cell. Because the tower and the mobile phones are only low-powered transmitters and receivers, they will not interfere with adjacent cells. When a very large number of people in a crowd are

using their phones, it may be difficult for some people to use their phones because all of the available frequencies within the local cell could be in use.

The mobile telephone is a low powered transmitter and receiver. This is why such a small telephone can be used on battery power for so long. If it required a stronger transmitter then the battery power would be used more quickly.

Of course most mobile phones now do much more than voice calls. Obviously test messages are still important. But now a key feature is mobile data connections using technologies such as 3G and now LTE to deliver Internet hosted services to handsets.

Satellite Phones – Calling from Antarctica

Like landline phones, cellular phones are limited to areas where the network is in place. When entering an area without a network there will be "no service". Satellite phones are true mobile phones in that they can be used anywhere. They make use of orbiting satellites to provide a network that allows a user to use a telephone in truly remote locations.

The user dials a number and the telephone makes a connection to a network of low earth orbiting satellites. The satellites then relay the signal back to an a ground based receiving station, which then links the signal to either a mobile or fixed telephone network.

Transmission Media

Cables

Oxygen free, high conductivity copper and electrolytic tough pitch (or copper alloy) cabling is used in many of the telecommunication systems. For example, telephone lines make use of copper cabling to transmit information. In some applications they are being replaced by fibre optics which have greater bandwidth and are unaffected by magnetic induction.

Twisted pair cables, such as Category 5 or 6 data cables, involves taking pairs of wires and twisting them together. This greatly reduces the chance on induced noise (or crosstalk) in the wires. Category 5 and 6 cables are still used extensively in local area network (LAN) connections.

Microwave

Microwave transmission makes use of microwaves for transcontinental communication. Microwaves are line-of-sight waves, so at about every 40 km there must be repeater towers. Originally used with analogue transmissions, it is now also used with digital signals. Microwaves are extensively used in communication satellites.

Fibre Optics

The principle of fibre optics was outlined earlier in the chapter. The use of fibre optics has revolutionised communications, as it allows greater information carrying

capacity because of greater use of the bandwidth. Fibre optics are used in computer and telephone communication networks. They have also improved the operation of cable television systems.

Coaxial Cables

These use a copper conductor, surrounded by a polymer dielectric (often polyethylene). Around the dielectric is a braided copper cable and the whole unit is coated with a vinyl insulator. UHF carrier waves, with encoded sound and television signals, are used to route cable television from transmission centres to people's homes.

Satellite Communication Systems

Satellites are one of the most important new technologies for worldwide communication. They provide us with communication (voice, data and television), weather information, scientific research and are also used in more covert roles e.g. spying and military use.

The artificial satellite age (natural satellites are celestial objects like the Moon, orbiting a planet) began in 1957 when the USSR launched Sputnik 1. From this time on the superpowers underwent a space race, which saw the earth surrounded by various satellites, primarily for military purposes. Australia did not enter the satellite age until the early 1980's when the first of three AUSSAT satellites were launched (although they were not launched from Australia).

Satellites are used with a variety of communication systems, particularly linking radio, data and television broadcasts from all over the world. Satellites are also used by some cable television companies for transmitting their programs. Telephone companies also utilise satellites for the relaying of data and conversations.

There are three types of orbits that satellites use:

Geostationary, also termed geosynchonous or synchronous, orbits occur when the satellite orbits in such a way that it is always over the same point of the earth. Most communication satellites fall into this category. The orbit altitude is 35,785 km, and usually somewhere over the equator. This area is very congested with satellites so the satellite must be correctly placed to ensure it functions properly without interference.

Medium Earth Orbit occurs at an altitude between 2000 km and 35,000 km. The most common satellites here are for navigation and communication. The most common altitude is 20,200 km and it here that most Global Positioning System (GPS) satellites orbit.

Low Earth Orbit occurs between altitudes of 160 km and 2000 km. At this altitude manned craft such as the International Space Station orbit, so to does the Hubble Space Telescope.

Asynchronous orbits occur when the satellite passes over parts of the globe at varying times. The altitude is much lower, at only 644km. When the Space Shuttle orbits the earth it uses an asynchronous orbit.

Polar orbits are low earth orbits in which satellites pass over the polar caps. Because the earth rotates under them, they gain excellent coverage. They are used by photography and mapping satellites. The Iridium satellite telephone system uses 66 polar satellites in orbits at an altitude of 780 km. This ensures excellent coverage of the globe for their users.

How Satellites Work

Originally communication satellites (sometimes called comsats) were passive devices. The signal was beamed to them and they reflected it back to earth. Nowadays communication satellites are active. They capture the signal, amplify it and relay it back to earth. They may also relay it to another satellite, which then relays it back to earth.

A *ground station* sends a signal in the form of low frequency microwaves, as these are less affected by the ionosphere and weather than high frequency microwaves. The satellite receives these microwaves, and then uses a transponder to boost the signal strength, before relaying it to either another satellite or another ground station.

Satellites are extensively used in broadcasting television signals all over the world. Whenever we see a reporter "live" in another part of the world, satellite technology is in use. The signal from the reporter's camera is sent to the nearest television broadcaster or to a transportable satellite system (e.g. truck), they channel the signal to a satellite, which transfers it to the destination television station, for broadcast. Big events, like the Olympic Games, can be shown live anywhere in the world. The Gulf War, in 1991 was the first war to be shown live around the world.

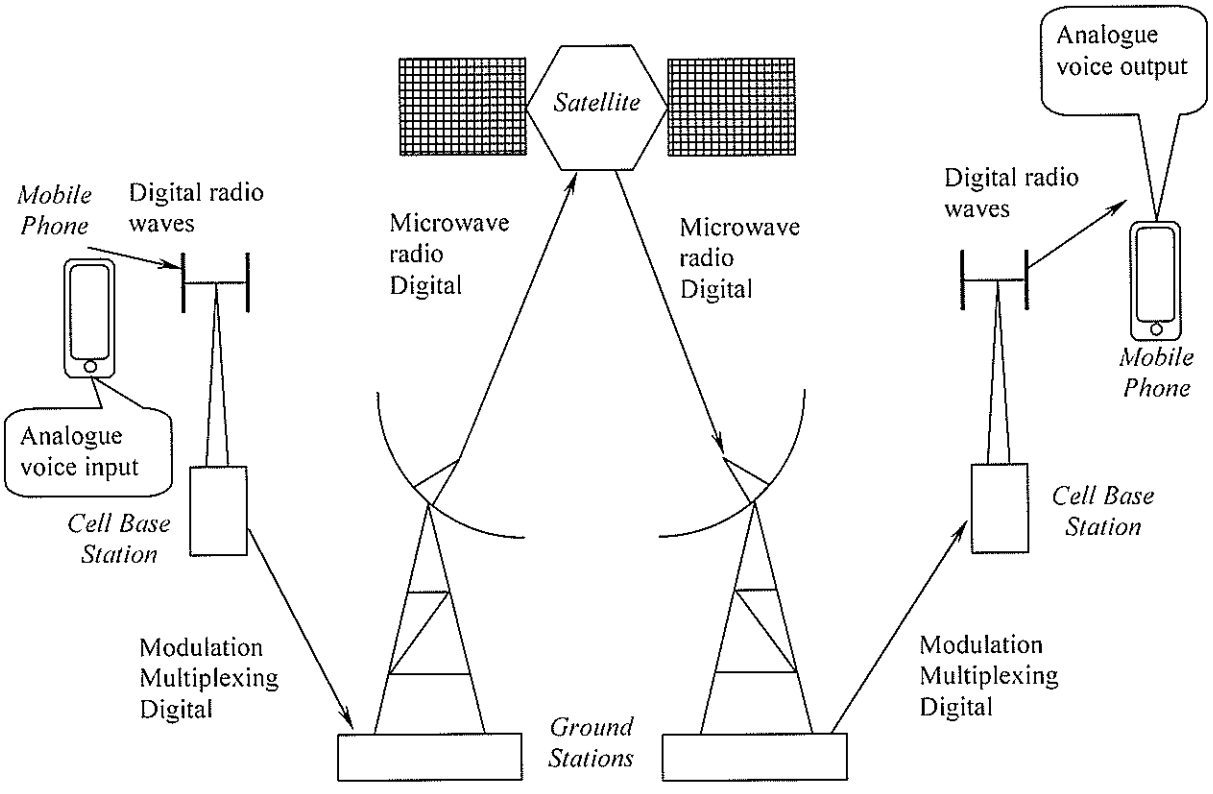


Fig. 4.15: *The process of a mobile-to-mobile telephone call. Labels in italics represent the hardware; labels in normal text are describing the transmissions.*

For international telephone calls, whether mobile or landline, satellite communication is used. A landline call in analogue is modulated, multiplexed and converted to digital. It is then sent to the ground station and to the satellite, which relays it to the destination ground station. The signal travels to the local network where it is demultiplexed and demodulated then converted back to analogue. The network then transfers it to the person receiving the call. There is usually a short delay (around a second or two) in the conversation because of the time taken for this to happen.

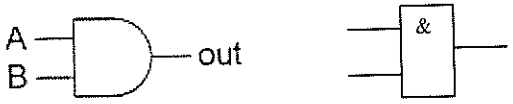
For mobile international calls, the same process is used, but the signal from the two mobiles is carried by the cellular network to which they are connected. The satellite used is the same for both types of call. One advantage of digital phones is that the signal conversion from analogue to digital occurs in the telephone, not at the network base station. Fig. 5.9 shows the flow chart for a mobile-to-mobile telephone call overseas.

Logic Gates

In the Preliminary course we first looked at Logic gates. Here we recap the main gates that are used.

AND gate

Symbol:

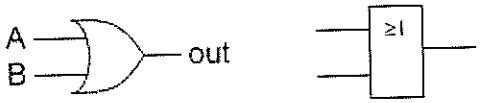


The output of an AND gate is high (1) when both the inputs are high (1).

Input	Input	Output
A	B	out
1	1	1
1	0	0
0	1	0
0	0	0

OR gate

Symbol:

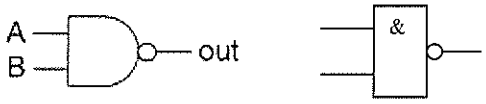


The output of an OR gate is high (1) when any input is high (1).

Input	Input	Output
A	B	out
1	1	1
1	0	1
0	1	1
0	0	0

NAND gate

Symbol:

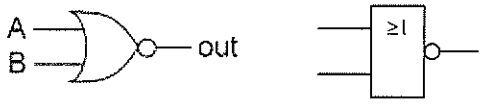


A NAND gate is the inverse of an AND gate: the output of a NAND gate is low (0) when both the inputs are high (1). NAND gates are one of the most useful logic gates. Using combinations of NAND gates it is possible to simulate all other logic gates.

Input	Input	Output
A	B	out
1	1	0
1	0	1
0	1	1
0	0	1

NOR gate

Symbol:



A NOR gate is the inverse of an OR gate: the output is low (0) when either of the inputs is high (1).

Input	Input	Output
A	B	out
1	1	0
1	0	0
0	1	0
0	0	1

NOT gate or inverter

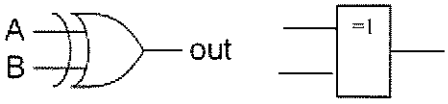
Symbol:

The output is always in the opposite state to the input.

Input	Output
A	out
1	0
0	1

XOR or exclusive OR gate

Symbol:



In an XOR gate the output is high when the inputs are opposite to each other. This makes it a true OR gate, unlike the standard OR gate which operates as an AND/OR gate.

Input	Input	Output
A	B	out
1	1	0
1	0	1
0	1	1
0	0	0

Communication

You are a telecommunications engineer for CallSat. They are a new mobile telephone company. They require a report that is to be presented to the board of directors that answers the following questions.

1. Why is a polar orbit desirable as opposed to a geosynchronous orbit?
2. What type of polymer casing should be used for the handsets?
3. To connect the satellite telephone system to terrestrial telephone systems it is necessary to connect the satellite receivers with existing telephone networks. What type of fibre optic cabling will be used?
4. Draw a flow chart of the way a satellite telephone will connect to an existing landline telephone. (Hint check www.iridium.com)
5. Explain the difference between analogue and digital signals. Explain the advantage to transmission, of digital signals.

Additional References

This list of useful references allows students further investigation. Any highly recommended book or website has two stars next to it.

Engineering Materials

**Askeland, D., (1989), *The Science and Engineering of Materials*, Chapman and Hall, London.
DeGarmo, E. P., Black, J. T. and Kohser, R. A., (1988), *Materials and Processes in Manufacturing 7th Edition*, Macmillan Publishing Company, New York.
Gordon, J. E. (1974), *The New Science of Strong Materials – or Why you don't fall through the floor*, Penguin, Middlesex.
Higgins, R., (1997), *Materials for the Engineering Technician 3rd Edition*, Arnold, London.
Schlenker, B., (1974), *Introduction to Materials Science*, Jacaranda Wiley, Milton.
**Smith, W., (1990), *Principles of Materials Science and Engineering*, McGraw Hill, New York.

Engineering Electricity/Electronics

Cooper, M. and Vella, G., (1988), *Physics – The Core*, Brooks Waterloo, Melbourne.
Fletcher, R. & Warner, N., (1997), Introducing Design in Electronics.
**Plant, M., (1992), *Electronics*, Hodder and Stoughton, London.

Pollock, G., (1996), *Active Physics*, Science Press, Sydney.
Sams, H. (Ed.)., (1973). *Basic Electricity and an Introduction to Electronics 3rd Ed.* Howard Sams and Company (A division of Macmillan Inc.), New York.

Review Questions

1. What historical revolution prompted people to demand better communications?
2. What device for sending electrical impulses pre-dated the telephone?
3. Who invented the telephone?
4. How did Thomas Edison modify the telephone?
5. Why was the public telephone important?
6. List three features on a modern telephone that makes it easier to use than older telephones.
7. When was Australia linked to the world by telegraph cable?
8. When and where was the first public telephone installed in Australia?
9. Who is credited with the invention of the radio?
10. Why is the transistor so important to the development of the radio?
11. What is a multi-meter?
12. What is resistivity?
13. Compare a copper beryllium allow with pure copper. What are the advantages and disadvantages in properties?
14. Explain why a copper cadmium alloy may be used in lieu of pure copper for overhead train wiring.
15. How does a zener diode differ from a standard diode?
16. Who developed the transistor? Explain why is has revolutionised the electronic world.
17. What type of laser is the smallest?
18. Outline which polymers are typically used in phone casings?
19. What are the advantages of fibre optic cabling?
20. What are the two families of fibre optic cabling?
21. How does an analogue signal differ from a digital signal?
22. After reading http://electronics.howstuffworks.com/analog-digital.htm explain what the terms sampling rate and sampling precision mean.
23. What does modulation mean?
24. Explain multiplexing.
25. What section of the atmosphere is used to reflect radio waves back to the earth's surface?
26. Explain the differences between AM and FM?
27. Explain how a black and white television produces an image. How is this different to colour television?
28. Compare the operation of LCD and Plasma televisions.
29. How does a cellular mobile phone work?
30. Compare smartphones to feature phones. How have they changed the way we use mobile telephony?
31. How does a satellite phone differ from a cellular mobile phone?
32. Explain how a coaxial cable is constructed.
33. Outline how three different satellite orbits are used?
34. What is a ground station?
35. Why is there a slight delay with international telephone calls?

APPENDIX 1
Answers to Mathematical Questions

Chapter 1

9. a) $R_A = 521.09$ kN 65.98°
   $R_B = 131.91$ kN ↑

   b) $R_A = 814.78$ N 27.3°
   $R_B = 1039.25$ N ↑

10. T = 26 kN
    $R_A = 17$ kN ↑

11. $R_A = 131.05$ kN 51.12°
    $R_B = 95.26$ kN →
    $F_{AB} = 35$ kN (tension)

12. $R_A = 47.6$ kN 50.9°

    $R_C = 160.92$ kN ↑
    $F_{BC} = 104$ kN (comp)
    $F_{BD} = 30$ kN (tension)

13. $R_A = 159.8$ kN ↑
    $R_B = 176.1$ kN 50°

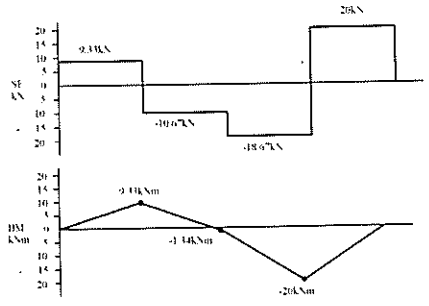
    $F_{DC} = 50$ kN (tension)

14. $R_A = 13.8$ kN 18.43°

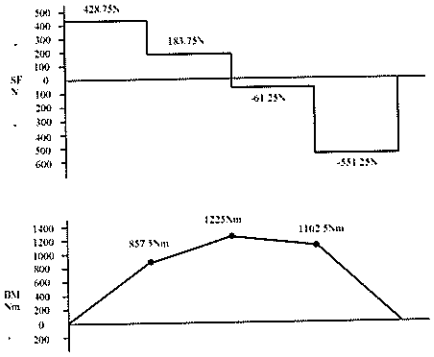
    $R_B = 4.64$ kN 51.4°

15. $R_A = 5.8$ kN 52.6°
    $R_B = 3.9$ kN ↑
    $F_{DE} = 6.5$ kN (tension)
    $F_{CE} = 9.2$ kN (tension)

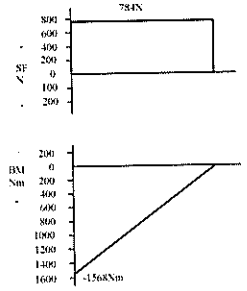
16.



17.



18.



19. Rect 1 = 2123.8 N
    Rect 2 = 606.67 N
    Triangle = 1496.87 N
    Circle = 962.09 N

20. Bar = 1764.71 MPa
    Tube = 588.7 MPa
    Tube preferred, less material and lower stress levels.

21. $\sigma_t = 636.6$ MPa
22. 11.5 mm
23. $\sigma_s = 157.2$ MPa
24. $\sigma_s = 8.7$ MPa
28. P = 9.424 kN
29. d=5.046 mm

Chapter 2

20. $\eta = 97.7\%$
22. F = 166.67 N
23. Load = 177.23 N
    Wheel will not slip
24. (a) F = 4.44 kN
    (b) F = 718 N
25. Force to tip ≈ 760 N
    Box will slide
26. $\mu_G \approx 0.27$
27. v = 31 km/h
28. F = 28.76 N
29. $L_P = 1750$ N
    $R_{PP} = 1494$ N 11°

33. P = 8.14 kW
34. s = 37.56 m
35. On flat = 48 wagons
    On incline = 9 wagons

Chapter 3

22. Mass = 84.1 kg
23. L:D = 5.92
25. BM = 25 kNm
    σ_b = 227.3 MPa